

**NOTES ON
M.SC. STATISTICS (PART I)
DESIGN AND ANALYSIS OF
EXPERIMENTS
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Notes Are Available in Courier Services.



Experimental Design:

"Experimental design is a set of rules or plans to collect the data relevant to the problem under investigation in such a way as to provide a basis for valid and objective inferences about the stated problem."

This plan generally consists of the selection of treatment, The specification of experimental layout, allocation of treatment and collection of observations for analysis

"Hence experimental design is simply a complete sequence of steps taken ahead of time to ensure that appropriate data will be obtained in a way that permits an objective analysis leading to valid inferences with respect to the problem."

Purpose:

The following purposes of experimental design are:

- To provide maximum amount of information relevant to the problem under investigation for a minimum cost in the minimum possible time i.e. to

conduct investigation as efficient as possible,

- To help reduce the experimental error in the data produced.
- To ignore spurious effects, if any.
- To investigate and examine the outcomes critically and logically.

Basic

Three design

- F
- F
- L

detailed

Steps Of Experimental Design:

The steps of experimental design are given below:

- Statement to the problem.
- Formulation of the hypothesis.
- Choice of response or dependent variable.
- Selection of factors to be vary.
- Choice of the level of these factors (qualitative, quantitative, fixed, random).
- How factors are to be combined.
- No. of observations to be taken.
- Method of randomization to be used.
- Mathematical Model to describe the experiment.
- Data Collection and processing.
- Application of statistical techniques.
- Drawing conclusions with measure of reliability.
- Evaluation of whole experiment/ investigation with other similar investigations.

Design nature and by allocation units.

Comple

CR design are as completely is

This design Number not

Applic

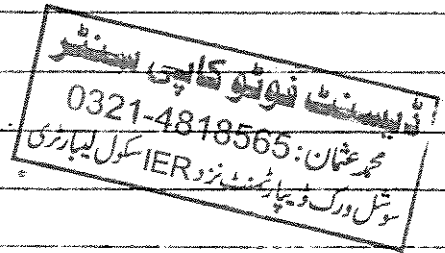
to be where;

Basic Principles Of Experimental Design:

Three basic principles of experimental design are:

- Randomization
- Replication
- Local Control

detailed = Assignment.



Designs are usually characterized by the nature of grouping of experimental units and by the procedure of random allocation of treatments to the experimental units.

Completely Randomized Design:

CR design may be defined as a design in which the treatments are assigned to the experimental units completely at random i.e. Randomization is done without any restriction.

This design is completely flexible i.e. Number of units per treatment need not to be equal.

Application:

A CR design is considered to be more useful in situations where;

- The experimental units are homogenous.
- The experiments are small such as laboratory experiments.
- Some experimental units are likely to be destroyed or failed to response.

An ex-
four
repeated

1
1
1

Experimental Layout:

Suppose we have "k" treatments and experimental material is divided into "n" experimental units. We shall then assign "k" treatments to the "n" experimental units in such a way that the treatments

α_j ($j = 1, 2, 3, \dots, k$) is applied

" r_j " times with $\sum_{j=1}^k r_j = n$. When each

treatment is applied same number of times then $r_1 = r_2 = \dots = r_k = r$ and $\sum_{j=1}^k r_j = n$

Advantage

- The are given
- The easily
- It analysis
- It freedom
- The de any n No. of

Observations	(K x j) Treatments					
	1	2	...	j	...	k
1	Y_{11}	Y_{12}	...	Y_{1j}	...	Y_{1k}
2	Y_{21}	Y_{22}	...	Y_{2j}	...	Y_{2k}
...	...	...	...	...	...	...
i	Y_{i1}	Y_{i2}	...	Y_{ij}	...	Y_{ik}
...	...	...	...	...	...	...
r_j	Y_{r_1}	Y_{r_2}	...	Y_{r_j}	...	Y_{r_k}
Total	$T_{.1}$	$T_{.2}$	...	$T_{.j}$	...	$T_{.k}$
Mean	$\bar{Y}_{.1}$	$\bar{Y}_{.2}$	...	$\bar{Y}_{.j}$	...	$\bar{Y}_{.k}$

Disadvantage

The de number These whole

An example of CR design layout using four treatment A, B, C, D. Each repeated 3 times is given below.

likely to
response

C	A	B	D
C	B	A	D
A	D	B	C

tments and
l into
shall then
the "n"

Advantages:

The core advantages of CR design are given below.

- The design is very simple and easily laid out.
- It has the simplest statistical analysis.
- It provides the maximum degree of freedom for the residual sum of square.
- The design is flexible i.e. any number of treatment and any No. of replication can be used.

Disadvantages:

The design is applicable to a small number of treatment.

There is a possibility of entering the whole variation among the experimental

y_{ik}

y_{1k}

y_{2k}

y_{ik}

y_{rk}

$\bar{y}_{.k}$

$\bar{y}_{..}$

$T_{..}$

$\bar{y}_{..}$

units into experimental error as the randomization is not restricted in any direction

Fixed

- i) "k" choice
- ii) α_j
- iii) α_j
- iv) $H_0: \mu_1 = \mu_2 = \dots = \mu_k$
- v) Conclusion i.e. (Reject H_0)

Statistical Model:

In CR design, the statistical model is:

$$Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

$$i = 1, 2, \dots, r$$

$$j = 1, 2, \dots, k$$

where;

μ = overall Mean (fixed, does not depend on any factor)

α_j = Effect of j th treatment

ϵ_{ij} = Residual term $\sim NII(0, \sigma^2)$
i.e.

$$E(\epsilon_{ij}) = 0$$

$$E(\epsilon_{ij}^2) = \sigma^2$$

$$E(\epsilon_{ij} \epsilon_{jh}) = 0 \quad \text{if } i \neq h$$

$$E(\epsilon_{ij} \epsilon_{gh}) = 0 \quad \text{if } j \neq h$$

Random

- i) "k" process
- ii) α_j is variability
- iii) α_j is variability
- iv) $H_0: \mu_1 = \mu_2 = \dots = \mu_k$
- v) Conclusion the mean (i.e. μ)

te: ANOVA Model

$\therefore NII$ (Normal distribution identically and independently).

$\therefore$ When less same).

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Fixed Effect Model:

i) "k" treatments are selected by the choice of experimental.

ii) α_j is fixed.

iii) α_j is estimated.

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iv) $H_0: \mu_1 = \mu_2 = \dots = \mu_k$

v) Conclusions are only for selected treatment i.e. (Results can not be generalized).

Random Effect Model:

i) "k" treatments are selected by random process from the population of treatments.

ii) α_j is a random variable

iii) Variability of treatment is estimated say σ^2

iv) $H_0: \sigma^2 = 0$

v) Conclusions are applicable for all the treatment of population (i.e. Results can be generalized)

Note: ANOVA is same for both of the Models.

∴ When F is not large, then there is less variation (average values are almost same).

Prove That:

$$E(MSE) = \sigma^2$$

In CR Design

Balanced Problem
(equal size)

Unbalanced Problem
(unequal size)

Fixed

Random

Fixed

Random

Fixed

In
fixed

Correctio

$$G =$$

$$G = \sum$$

$$G =$$

$$G = 1$$

Squaring

$$G^2 =$$

$$G^2 = r^2 k$$

°

$$G^2 = r^2 k$$

Dividin

$$\frac{G^2}{rk} = r^2 k$$

Proof # 1:

Balanced and fixed effect Model

Prove That $E(MSE) = \sigma^2$

In CR Design the observations Y_{ij} is expressed as:

$$Y_{ij} = \mu + \alpha_j + E_{ij} \quad \begin{matrix} i = 1, 2, \dots, r. \\ j = 1, 2, \dots, k. \end{matrix}$$

with assumptions

$$E_{ij}'s \sim NII(0, \sigma^2)$$

i.e

$$E(E_{ij}) = 0$$

$$E(E_{ij}^2) = \sigma^2$$

$$E(E_{ij} E_{gh}) = 0 \quad \text{for } \begin{matrix} i \neq g \\ j \neq h \end{matrix}$$

Fixed Effect Model:

In this case the effect of α_j is fixed such that

$$\sum_{j=1}^k \alpha_j = 0$$

balanced Problem
(unequal size)

Correction Factor = C.F. = $\frac{G^2}{rk}$ where $G = \sum_j \sum_i Y_{ij}$

Random

$$G = \sum_j \sum_i (Y_{ij})$$

$$G = \sum_j \sum_i (\mu + \alpha_j + \epsilon_{ij})$$

sel Model

$$G = r k \mu + r \sum_j \alpha_j + \sum_j \sum_i \epsilon_{ij} \quad \because \sum_j \alpha_j = 0$$

$$G = r k \mu + \sum_j \sum_i \epsilon_{ij}$$

tions Y_{ij}

Squaring on both sides.

$$G^2 = (r k \mu + \sum_j \sum_i \epsilon_{ij})^2$$

= 1, 2, ..., kr.

= 1, 2, ..., k.

$$G^2 = r^2 k^2 \mu^2 + \left(\sum_j \sum_i \epsilon_{ij} \right)^2 + 2 r k \mu \sum_j \sum_i \epsilon_{ij}$$

$$\because \left(\sum_j \sum_i \epsilon_{ij} \right)^2 = \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh}$$

$$G^2 = r^2 k^2 \mu^2 + \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh} + 2 r k \mu \sum_j \sum_i \epsilon_{ij}$$

Dividing by "rk" on both sides.

$$\frac{G^2}{rk} = r k \mu^2 + \frac{1}{rk} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{rk} \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh} + 2 \mu \sum_j \sum_i \epsilon_{ij}$$

Applying expectation on both sides

$$\sum_j \sum_i Y_{ij}^2 = r$$

$$E(C.F) = E\left(\frac{G^2}{rk}\right) = rk\mu^2 + \frac{1}{rk} \sum_j \sum_i E(\varepsilon_{ij}^2) + \frac{1}{rk} \sum_{j \neq g} \sum_{i \neq h} E(\varepsilon_{ij} \varepsilon_{gh}) + 2\mu \sum_j \sum_i E(\varepsilon_{ij})$$

$$\left(\sum_j \sum_i Y_{ij}^2\right) =$$

by

$$\because E(\varepsilon_{ij}) = 0, \quad E(\varepsilon_{ij} \varepsilon_{gh}) = 0$$

$$E(T.S.S) =$$

$$E(\varepsilon_{ij}^2) = \sigma^2$$

$$E(T.S.S) =$$

$$E(T.S.S) =$$

$$E(C.F) = rk\mu^2 + \frac{rk}{rk} \sigma^2 + 0 + 0$$

$$E(C.F) = rk\mu^2 + \sigma^2 \quad (1)$$

Sum of

Total Sum of Squares:

S.S.B =

$$T.S.S = \sum_j \sum_i Y_{ij}^2 - C.F \quad \text{where}$$

$$Y_{ij} =$$

$$Y_{ij} = \mu + \alpha_j + \varepsilon_{ij}$$

$$Y_{ij} = r$$

$$Y_{ij}^2 = (\mu + \alpha_j + \varepsilon_{ij})^2$$

Squaring

$$Y_{ij}^2 = \mu^2 + \alpha_j^2 + \varepsilon_{ij}^2 + 2\mu\alpha_j + 2\mu\varepsilon_{ij} + 2\alpha_j\varepsilon_{ij}$$

$$Y_{ij}^2 = (r\mu$$

$$\sum_j \sum_i Y_{ij}^2 = rk\mu^2 + r \sum_j \alpha_j^2 + \sum_j \sum_i \varepsilon_{ij}^2 + 2\mu r \sum_j \alpha_j + 2\mu \sum_j \sum_i \varepsilon_{ij} + 2 \sum_j \sum_i \alpha_j \varepsilon_{ij}$$

$$Y_{ij}^2 = r^2 \mu^2$$

$$2r\mu$$

$$\because \sum_j \alpha_j = 0$$

Taking Expectation on both sides

Now Di
both

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$$R) \quad \left(\sum_j \sum_i Y_{ij}^2 \right) = rku^2 + r \sum_j \alpha_j^2 + \sum_j \sum_i E(E_{ij}^2) + 2u \sum_j \sum_i E(E_{ij})$$

$$E \left(\sum_j \sum_i Y_{ij}^2 \right) = rku^2 + r \sum_j \alpha_j^2 + rk\sigma^2 + 0 \quad \text{--- (II)}$$

$$\sum_{j \neq h} \sum_{i \neq h} (E_{ij} E_{jh}) + \text{by subtracting eq (II) and (I)}$$

$$E(T.S.S) = E \left(\sum_j \sum_i Y_{ij}^2 \right) - E(C.F)$$

$$E(T.S.S) = \cancel{rku^2} + r \sum_j \alpha_j^2 + rk\sigma^2 - \cancel{rku^2} - \sigma^2$$

$$E(T.S.S) = r \sum_j \alpha_j^2 + (rk-1)\sigma^2 \quad \text{--- (III)}$$

Sum of Square b/w Treatments:

$$S.S.B = \sum_j \frac{Y_{.j}^2}{r} - C.F \quad \text{where}$$

$$Y_{.j} = \sum_i Y_{ij} = \sum_i (u + \alpha_j + E_{ij})$$

$$Y_{.j} = ru + r\alpha_j + \sum_i E_{ij}$$

Squaring On both sides.

$$Y_{.j}^2 = \left(ru + r\alpha_j + \sum_i E_{ij} \right)^2$$

$$Y_{.j}^2 = r^2 u^2 + r^2 \alpha_j^2 + \sum_i E_{ij}^2 + \sum_{j \neq h} \sum_{i \neq h} E_{ij} E_{jh} + 2r^2 u \alpha_j + 2ru \sum_i E_{ij} + 2r\alpha_j \sum_i E_{ij}$$

$$\because \sum_j \alpha_j = 0$$

Now Dividing by "r" and Apply $\sum_j$ on both sides.

$$\sum_j \frac{Y_{ij}^2}{r} = rku^2 + r \sum_j \alpha_j^2 + \frac{1}{r} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{r} \sum_j \sum_i \epsilon_{ij} \epsilon_{ij} + 2ru \sum_j \alpha_j + 2u \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \epsilon_{ij}$$

By Taking Expectation on both sides

$$E\left[\sum_j \frac{Y_{ij}^2}{r}\right] = rku^2 + r \sum_j \alpha_j^2 + \frac{rk}{r} \sigma^2 + 0 + 0 + 0 + 0$$

$$= rku^2 + r \sum_j \alpha_j^2 + k\sigma^2 \quad \text{--- (iv)}$$

Subtracting eq (iv) by eq (i)

$$E(SSB) = rku^2 + r \sum_j \alpha_j^2 + k\sigma^2 - rku^2 - \sigma^2$$

$$E(SSB) = r \sum_j \alpha_j^2 + (k-1)\sigma^2 \quad \text{--- (v)}$$

Error Sum of Square:

$$E(\text{Error SS}) = E(\text{Total SS}) - E(\text{B/w SS})$$

$$\text{eq (iii)} - \text{eq (v)}$$

$$E(\text{Error SS}) = r \sum_j \alpha_j^2 + (rk-1)\sigma^2 - r \sum_j \alpha_j^2 - (k-1)\sigma^2$$

$$= (rk-1-k+1)\sigma^2$$

$$E(\text{Error SS}) = (rk-k)\sigma^2$$

Dividing both sides By $(rk-k)$: we get:

$$E(MSE) =$$

$$E(MSE) =$$

Proof:

gn C
written

with as

ϵ_{ij}^2
i.e. E

because
Therefore,
with as

i.e.

and α_j

$$E(MSE) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sigma^2$$

$$E(\text{MSE}) = \sigma^2 \quad \text{Hence Proved.}$$

Balanced and Random Effect Model.

In CR design, the observations y_{ij} is written as:

$$i = 1, 2, \dots, r$$

$$j = 1, 2, \dots, k$$

with assumptions are:

$$\begin{aligned} \epsilon_{ij}'s &\sim NII(0, \sigma^2) \\ \text{i.e. } E(\epsilon_{ij}) &= 0, \quad E(\epsilon_{ij}^2) = \sigma^2 \end{aligned}$$

$$E(\epsilon_j \epsilon_{gh}) = 0 \quad \text{if } i \neq g \text{ or } j \neq h$$

because it is in Random Effect Model
Therefore, α_j is a random variable
with assumptions:

$$(k-1)\sigma^2 \quad \alpha_j's \sim NII(0, \sigma_\alpha^2)$$

i-e $E(\alpha_j) = 0, \quad E(\alpha_j^2) = \sigma_\alpha^2$

$$E(\alpha_j \alpha_m) = 0 \quad j \neq m$$

$\therefore$ we get: and α_j and ε_{ij} are independent
i.e. $E(\alpha_j \varepsilon_{ij}) = 0$.

Correction Factor = C.F. = $\frac{G^2}{rk}$ where $G = \sum_j \sum_i y_{ij}$ Total

$$G = \sum_j \sum_i (\mu + \alpha_j + \epsilon_{ij})$$

T.S.S. =

$$G = rku + r \sum_j \alpha_j + \sum_j \sum_i \epsilon_{ij}$$

$$y_{ij}^2 = (\mu + \alpha_j + \epsilon_{ij})^2$$

Squaring on both sides:

$$y_{ij}^2 = \mu^2 + \alpha_j^2 + \epsilon_{ij}^2 + 2\mu\alpha_j + 2\mu\epsilon_{ij} + 2\alpha_j\epsilon_{ij}$$

$$G^2 = \left(rku + r \sum_j \alpha_j + \sum_j \sum_i \epsilon_{ij} \right)^2$$

$$\sum_j \sum_i y_{ij}^2 = \sum_j \sum_i \mu^2 + \sum_j \sum_i \alpha_j^2 + \sum_j \sum_i \epsilon_{ij}^2 + 2\mu \sum_j \sum_i \alpha_j + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \sum_i \alpha_j \epsilon_{ij}$$

$$G^2 = r^2 k^2 \mu^2 + \left(r \sum_j \alpha_j \right)^2 + \left(\sum_j \sum_i \epsilon_{ij} \right)^2 + 2rku \sum_j \alpha_j + 2rku \sum_j \sum_i \epsilon_{ij} + 2r \sum_j \alpha_j \sum_i \epsilon_{ij}$$

$$\therefore \left(r \sum_j \alpha_j \right)^2 = r^2 \sum_j \alpha_j^2 + r^2 \sum_{j \neq m} \alpha_j \alpha_m$$

By Taking

$$\therefore \left(\sum_j \sum_i \epsilon_{ij} \right)^2 = \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{jh}$$

$$E \left(\sum_j \sum_i y_{ij}^2 \right) = rku^2 + r \sum_j E(\alpha_j^2) + r \sum_{j \neq m} E(\alpha_j \alpha_m) + \frac{1}{rk} \sum_j \sum_i E(\epsilon_{ij}^2) + \frac{1}{rk} \sum_{j \neq h} \sum_{i \neq g} E(\epsilon_{ij} \epsilon_{jh}) + 2\mu \sum_j E(\alpha_j) + 2\mu \sum_j \sum_i E(\epsilon_{ij}) + \frac{2}{rk} \sum_j \sum_i E(\alpha_j \epsilon_{ij})$$

Now Dividing by rk and taking expectation on Both sides.

$$+ 2\mu$$

$$E \left(\frac{G^2}{rk} \right) - E(C.F.) = rku^2 + r \sum_j E(\alpha_j^2) + r \sum_{j \neq m} E(\alpha_j \alpha_m) + \frac{1}{rk} \sum_j \sum_i E(\epsilon_{ij}^2) + \frac{1}{rk} \sum_{j \neq h} \sum_{i \neq g} E(\epsilon_{ij} \epsilon_{jh}) + 2\mu \sum_j E(\alpha_j) + 2\mu \sum_j \sum_i E(\epsilon_{ij}) + \frac{2}{rk} \sum_j \sum_i E(\alpha_j \epsilon_{ij})$$

$$= rku$$

$$= rk$$

$$+ 2\mu \sum_j E(\alpha_j) + 2\mu \sum_j \sum_i E(\epsilon_{ij}) + \frac{2}{rk} \sum_j \sum_i E(\alpha_j \epsilon_{ij}) \quad \text{eq (ii)}$$

$$= rku^2 + r \sum_j \sigma_{\alpha}^2 + 0 + rk \sigma^2 + 0 + 0 + 0 + 0$$

$$E(T.S.S.) =$$

$$= rku^2 + r \sum_j \sigma_{\alpha}^2 + 0 + rk \sigma^2 + 0 + 0 + 0 + 0$$

$$= rku^2 +$$

$$E(C.F.) = rku^2 + r \sigma_{\alpha}^2 + \sigma^2 \quad (1)$$

$$E(S.S.) = (rk - r)$$

le $G = \sum_j \sum_i y_{ij}$

Total Sum of Squares:

T.S.S = $\sum_j \sum_i y_{ij}^2 - C.F.$ where
 $y_{ij} = \mu + \alpha_j + \epsilon_{ij}$

$$y_{ij}^2 = (\mu + \alpha_j + \epsilon_{ij})^2$$

$$y_{ij}^2 = \mu^2 + \alpha_j^2 + \epsilon_{ij}^2 + 2\mu\alpha_j + 2\mu\epsilon_{ij} + 2\alpha_j\epsilon_{ij}$$

$$\sum_j \sum_i y_{ij}^2 = r k \mu^2 + r \sum_j \alpha_j^2 + \sum_j \sum_i \epsilon_{ij}^2 + 2\mu r \sum_j \alpha_j + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \epsilon_{ij}$$

By Taking Expectation on both sides.

$$\begin{aligned} E\left(\sum_j \sum_i y_{ij}^2\right) &= r k \mu^2 + r \sum_j E(\alpha_j^2) + \sum_j \sum_i E(\epsilon_{ij}^2) + 2\mu r \sum_j E(\alpha_j) \\ &\quad + 2\mu \sum_j \sum_i E(\epsilon_{ij}) + 2 \sum_j \sum_i E(\alpha_j \epsilon_{ij}) \\ &= r k \mu^2 + r k \sigma_\alpha^2 + r k \sigma^2 + 0 + 0 + 0. \\ &= r k \mu^2 + r k \sigma_\alpha^2 + r k \sigma^2 \quad \text{--- (II)} \end{aligned}$$

eq (II) - eq (I)

$$E(T.S.S) = E\left(\sum_j \sum_i y_{ij}^2\right) - E(C.F)$$

$$= r k \mu^2 + r k \sigma_\alpha^2 + r k \sigma^2 - r k \mu^2 - r \sigma_\alpha^2 - \sigma^2$$

$$T.S.S = (r k - r) \sigma_\alpha^2 + (r k - 1) \sigma^2 \quad \text{--- (III)}$$

Sum of Square between Treatments:

$$SSB = \sum_j \frac{Y_{.j}^2}{r} - C.F. \quad \text{where,}$$

$$Y_{.j} = \sum_i Y_{ij} = \sum_i (\mu + \alpha_j + \varepsilon_{ij})$$

$$Y_{.j} = r\mu + r\alpha_j + \sum_i \varepsilon_{ij}$$

$$Y_{.j}^2 = \left(r\mu + r\alpha_j + \sum_i \varepsilon_{ij} \right)^2$$

$$Y_{.j}^2 = r^2\mu^2 + r^2\alpha_j^2 + \sum_i \varepsilon_{ij}^2 + \sum_j \sum_{i \neq g} \varepsilon_{ij} \varepsilon_{ig} + 2r\mu\alpha_j +$$

$$2r\mu \sum_i \varepsilon_{ij} + 2r\alpha_j \sum_i \varepsilon_{ij}$$

Now Dividing by r and apply $(\sum_j)$ on both sides

$$\sum_j \frac{Y_{.j}^2}{r} = rk\mu^2 + r\sum_j \alpha_j^2 + \frac{1}{r} \sum_j \sum_i \varepsilon_{ij}^2 + \frac{1}{r} \sum_j \sum_i \varepsilon_{ij} \varepsilon_{ig} + 2k\mu\sum_j \alpha_j + 2\mu\sum_j \sum_i \varepsilon_{ij} + 2\alpha_j \sum_j \sum_i \varepsilon_{ij}$$

By Applying expectation both both sides.

$$E\left[\sum_j \frac{Y_{.j}^2}{r}\right] = rk\mu^2 + rk\sigma_\alpha^2 + k\sigma^2 + 0 + 0 + 0 + 0.$$

$$= rk\mu^2 + rk\sigma_\alpha^2 + k\sigma^2 \quad (IV)$$

$$\text{eq (IV)} - \text{eq (I)}$$

$$E(SSB) =$$

$$E(SSB) =$$

$$\text{Error}$$

$$E(\text{Error})$$

$$E(\text{Error SS})$$

$$= (r$$

$$E(ESS) =$$

Dividing both side

$$E(MSE) =$$

$$E(M$$

nts: $E(SSB) = \cancel{r.k.\mu^2} + r.k.\sigma_\alpha^2 + k.\sigma^2 - \cancel{r.k.\mu^2} - r.k.\sigma_\alpha^2 - \sigma^2$

$$E(SSB) = (r.k - r)\sigma_\alpha^2 + (k-1)\sigma^2 \quad \text{--- (v)}$$

Error Sum of Square:

$$E(\text{Error SS}) = E(TSS) - E(BSS).$$

$$\text{eq (ii)} - \text{eq (v)}$$

$$\begin{aligned} E(\text{Error SS}) &= \cancel{(r.k-r)}\sigma_\alpha^2 + (r.k-1)\sigma^2 - \cancel{(r.k-r)}\sigma_\alpha^2 - (k-1)\sigma^2 \\ &= (r.k-1-k+1)\sigma^2 \end{aligned}$$

$$E(ESS) = (r.k - k)\sigma^2$$

Dividing by its degree of freedom $(r.k - k)$ on both sides.

$$E(MSE) = \left(\frac{r.k - k}{r.k - k} \right) \sigma^2$$

$$E(MSE) = \sigma^2 \quad \text{Hence Proved.}$$

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Proof # 3:

Unbalanced and Fixed Effect Model

Prove That $E(MSE) = \sigma^2$.In CR design, the observations Y_{ij} is expressed as:

$$Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

with assumptions:

$$\begin{matrix} i=1, 2, \dots, r_j \\ j=1, 2, \dots, R \end{matrix}$$

$$\epsilon_{ij}'s \sim NIT(0, \sigma^2)$$

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2$$

$$E(\epsilon_{ij} \epsilon_{gh}) = 0 \quad \text{for } \begin{matrix} i \neq g \\ j \neq h \end{matrix}$$

Fixed Effect Model:

The effect of treatment α_j is fixed such that

$$\sum_j r_j \alpha_j = 0$$

Correction Factor:

$$C.F. = \frac{G^2}{\sum_j r_j} = \frac{G^2}{R} \quad \text{where } R = \sum_j r_j$$

$$G = \sum_i$$

$$G = \sum_j \sum_i$$

$$G = \sum_j r_j$$

Squaring

$$G^2 = (R \mu)^2$$

$$G^2 = R^2 \mu^2$$

Now $\sum$
on μ

$$E\left(\frac{G^2}{R}\right) = E$$

$$= R \mu$$

$$= F$$

$$E(C.F.) = F$$

Total

$$T.S.S. =$$

$$Y_{ij} =$$

$$Y_{ij}^2 =$$

Effect Model

$$G = \sum_j \sum_i y_{ij} = \sum_j \sum_i (\mu + \alpha_j + \epsilon_{ij})$$

$$G = \sum_j \sum_i (\mu + \alpha_j + \epsilon_{ij})$$

$$\because \sum_j r_j \alpha_j = 0$$

$$G = \sum_j r_j \mu + \sum_j r_j \alpha_j + \sum_j \sum_i \epsilon_{ij} = \sum_j r_j \mu + \sum_j \sum_i \epsilon_{ij} =$$

is

$$\text{Squaring on both sides.} \quad = R\mu + \sum_j \sum_i \epsilon_{ij}$$

$$G^2 = (R\mu + \sum_j \sum_i \epsilon_{ij})^2$$

$$\begin{matrix} 2, & \dots & r_j \\ 2, & \dots & k \end{matrix}$$

$$G^2 = R^2 \mu^2 + \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh} + 2R\mu \sum_j \sum_i \epsilon_{ij}$$

Now Dividing By R and applying Expectation on both sides.

$$E\left(\frac{G^2}{R}\right) = E(C.F) = R\mu^2 + \frac{1}{R} \sum_j \sum_i E(\epsilon_{ij}^2) + \frac{1}{R} \sum_{j \neq h} \sum_{i \neq g} E(\epsilon_{ij} \epsilon_{gh}) + 2\mu \sum_j \sum_i E(\epsilon_{ij})$$

$$= R\mu^2 + \frac{1}{R} \sum_j r_j \sigma^2 + 0 + 0$$

$$\because \sum_j r_j = R$$

$$\alpha_j \text{ is}$$

$$= R\mu^2 + \frac{R}{R} \sigma^2$$

$$E(C.F) = R\mu^2 + \sigma^2 \quad \text{--- (1)}$$

Total Sum of Squares:

$$T.S.S = \sum_j \sum_i y_{ij}^2 - C.F \quad \text{where,}$$

$$R = \sum_j r_j$$

$$y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

$$y_{ij}^2 = (\mu + \alpha_j + \epsilon_{ij})^2$$

$$Y_{ij}^2 = \mu^2 + \alpha_j^2 + E_{ij}^2 + 2\mu\alpha_j + 2\mu E_{ij} + 2\alpha_j E_{ij}$$

$$\sum_j \sum_i Y_{ij}^2 = \sum_j r_j \mu^2 + \sum_j r_j \alpha_j^2 + \sum_j \sum_i E_{ij}^2 + 2\mu \sum_j r_j \alpha_j + 2\mu \sum_j \sum_i E_{ij} + 2 \sum_j \sum_i \alpha_j E_{ij}$$

By Taking Expectation on both sides

$$E\left[\sum_j \sum_i Y_{ij}^2\right] = \sum_j r_j \mu^2 + \sum_j r_j \alpha_j^2 + \sum_j r_j (\sigma^2) + 0 + 0 + 0$$

$$\because \sum_j r_j = R$$

$$E\left(\sum_j \sum_i Y_{ij}^2\right) = R\mu^2 + R\alpha_j^2 + R\sigma^2 \quad \text{--- (II)}$$

$$\text{eq (II)} - \text{eq (I)}$$

$$E(TSS) = E\left(\sum_j \sum_i Y_{ij}^2\right) - E(C.F)$$

$$= R\mu^2 + R\alpha_j^2 + R\sigma^2 - R\mu^2 - \sigma^2$$

$$E(TSS) = R\alpha_j^2 + (R-1)\sigma^2 \quad \text{--- (III)}$$

Between Sum of Square:

$$BSS = \sum_j \frac{Y_{.j}^2}{r_j} - C.F \quad \text{where}$$

$$Y_{.j} = \sum_i (Y_{ij}) = \sum_i (\mu + \alpha_j + E_{ij})$$

$$Y_{.j}^2 = r_j \mu^2 + r_j \alpha_j^2 + \sum_i E_{ij}^2$$

$$Y_{.j}^2 = (r_j \mu + \dots)$$

$$Y_{.j}^2 = r_j^2 \mu^2 + \dots$$

$$2r_j \mu \sum_i \dots$$

$$\sum_j \frac{Y_{.j}^2}{r_j} = \sum_j \dots$$

$$2 \sum_j r_j \dots$$

By Taking

$$\left[\sum_j \frac{Y_{.j}^2}{r_j} \right] = \dots$$

$$= R$$

$$\text{eq (I)}$$

$$E(SSB) = \dots$$

$$E(SSB) = \dots$$

Error S

$$E(\text{Error S})$$

$$y_j = r_j \mu + r_j \alpha_j + \sum_i \epsilon_{ij}$$

$$y_j^2 = r_j^2 \mu^2 + r_j^2 \alpha_j^2 + \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_i \epsilon_{ij} \epsilon_{jh} + 2r_j^2 \mu \alpha_j + 2r_j \mu \sum_i \epsilon_{ij} + 2r_j \alpha_j \sum_i \epsilon_{ij}$$

$$\sum_j \frac{y_j^2}{r_j} = \sum_j r_j \mu^2 + \sum_j r_j \alpha_j^2 + \frac{1}{r_j} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{r_j} \sum_j \sum_{j \neq h} \epsilon_{ij} \epsilon_{jh} + 0 + 0 + 0 + 0$$

$$\because \sum_j r_j = R \quad 2 \sum_j r_j \mu \alpha_j + 2 \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \sum_i \alpha_j \epsilon_{ij}$$

By Taking Expectation on both sides

$$E\left[\sum_j \frac{y_j^2}{r_j}\right] = \sum_j r_j \mu^2 + \sum_j r_j \alpha_j^2 + \frac{1}{r_j} \sum_j r_j E(\epsilon_{ij}^2) + 0 + 0 + 0 + 0$$

$$= R\mu^2 + R\alpha_j^2 + k\sigma^2 \quad \text{--- (iv)} \quad \because \sum_j r_j = R$$

$$\text{eq (iv)} - \text{eq (i)}$$

$$E(SSB) = R\alpha_j^2 + R\alpha_j^2 + k\sigma^2 - R\mu^2 - \sigma^2$$

$$E(SSB) = R\alpha_j^2 + (k-1)\sigma^2 \quad \text{--- (v)}$$

Error Sum of Squares:

$$E(\text{Error S.S}) = E(TSS) - E(BSS)$$

$$= R\alpha_j^2 + (R-1)\sigma^2 - R\alpha_j^2 - (k-1)\sigma^2$$

$$E(\text{Error SS}) = (R - 1 - k + 1) \sigma^2$$

$$E(\text{Error SS}) = (R - k) \sigma^2$$

Dividing by its degree of freedom.

$$E(\text{MSE}) = \sigma^2 \text{ Hence Proved}$$

Proof # 4:

Unbalanced and Random Effect Model.

Prove That $E(\text{MSE}) = \sigma^2$.

In CR Design, the observations Y_{ij} is written as:

$$Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

with assumptions:

$$\epsilon_{ij}'s \sim NII(0, \sigma^2)$$

i.e.

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2$$

$$E(\epsilon_{ij} \epsilon_{gh}) = 0 \quad \begin{matrix} i \neq g \\ j \neq h \end{matrix}$$

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in effect assumption

i.e.

$$E(Y_{ij} \alpha_j)$$

and

Correction

C.F

$$G = \sum_j$$

$$G = \sum_j$$

$$G^2 = \left(\sum_j \right)^2$$

$$G^2 = \sum_j r_j^2 \mu^2$$

$$2 \sum_j r_j^2 \mu \alpha$$

Dividing

$$\left(\frac{G^2}{R} \right) = E(C.F)$$

$$E(C.F) = R$$

In random effect Model, The treatment effect α_j is a random variable with assumptions:

of freedom. α_j 's $\sim NII(0, \sigma_\alpha^2)$

i.e

$$E(\epsilon_{ij} \alpha_j) = 0, \quad E(\alpha_j^2) = \sigma_\alpha^2$$

$$E(\alpha_j \alpha_m) = 0 \quad \text{for } j \neq m$$

and α_j and ϵ_{ij} are independent Therefore

$$E(\alpha_j \epsilon_{ij}) = 0$$

Correction Factor:

$$C.F = \frac{G^2}{R} \quad \text{where} \quad R = \sum_j r_j$$

as Y_{ij} is

$$G = \sum_j \sum_i Y_{ij} = \sum_j \sum_i (\mu + \alpha_j + \epsilon_{ij})$$

$i=1, 2, \dots, r_j$
 $j=1, 2, \dots, K$

$$G = \sum_j r_j \mu + \sum_j r_j \alpha_j + \sum_j \sum_i \epsilon_{ij}$$

$$G^2 = \left(\sum_j r_j \mu + \sum_j r_j \alpha_j + \sum_j \sum_i \epsilon_{ij} \right)^2$$

$$G^2 = \sum_j r_j^2 \mu^2 + \sum_j r_j^2 \alpha_j^2 + \sum_{j \neq m} r_j \alpha_j \alpha_m + \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq n} \sum_{i \neq q} \epsilon_{ij} \epsilon_{qn} + 2 \sum_j r_j^2 \mu \alpha_j + 2 \sum_j \sum_i r_j \mu \epsilon_{ij} + 2 \sum_j \sum_i r_j \alpha_j \epsilon_{ij}$$

Dividing by R and Taking expectation

$$\left(\frac{G^2}{R} \right) = E(C.F) = \sum_j r_j \mu^2 + \sum_j r_j E(\alpha_j^2) + 0 + \frac{1}{R} \sum_j r_j E(\epsilon_{ij}^2) + 0 + 0 + 0 + 0$$

$$E(C.F) = R\mu^2 + R\sigma_\alpha^2 + \frac{R\sigma_\epsilon^2}{R} \quad \text{--- (1)}$$

Total Sum of Square:

$$TSS = \sum_j \sum_i Y_{ij}^2 - C.F. \quad \text{where}$$

$$Y_{ij} = \mu + \alpha_j + E_{ij}$$

$$Y_{ij}^2 = (\mu + \alpha_j + E_{ij})^2$$

$$Y_{ij}^2 = \mu^2 + \alpha_j^2 + E_{ij}^2 + 2\mu\alpha_j + 2\mu E_{ij} + 2\alpha_j E_{ij}$$

$$\sum_j \sum_i Y_{ij}^2 = \sum_j r_j \mu^2 + \sum_j r_j \alpha_j^2 + \sum_j \sum_i E_{ij}^2 + 2\mu \sum_j r_j \alpha_j + 2\mu \sum_j \sum_i E_{ij} + 2 \sum_j \sum_i \alpha_j E_{ij}$$

$$2 \sum_j \sum_i \alpha_j E_{ij}$$

Taking Expectation on both sides

$$E\left[\sum_j \sum_i Y_{ij}^2\right] = \sum_j r_j \mu^2 + \sum_j r_j E(\alpha_j^2) + \sum_j r_j E(E_{ij}^2) + 0 + 0 + 0$$

$$\sum_j r_j = R$$

$$E\left(\sum_j \sum_i Y_{ij}^2\right) = R\mu^2 + R\sigma_\alpha^2 + R\sigma^2 \quad \text{--- (II)}$$

$$\text{eq (II)} - \text{eq (I)}$$

$$E(TSS) = R\mu^2 + R\sigma_\alpha^2 + R\sigma^2 - R\mu^2 - R\sigma_\alpha^2 - \sigma^2$$

$$E(TSS) = (R-1)\sigma^2 \quad \text{--- (III)}$$

Sum of Square B/w Treatment:

$$SSB = \sum_j \frac{Y_{.j}^2}{r_j} - C.F.$$

where, $Y_{.j}$

$$Y_{.j} = r_j \mu$$

Squaring

$$Y_{.j}^2 = r_j^2 \mu^2$$

$2r_j \mu$

Dividing

sides.

$$\sum_j \frac{Y_{.j}^2}{r_j} = \sum_j \mu^2$$

Taking

$$E\left[\sum_j \frac{Y_{.j}^2}{r_j}\right] =$$

$$E(SSB) =$$

$$E(SSB) =$$

where, $y_j = \sum_i (y_{ij})$

$$y_j = \sum_i (\mu + \alpha_j + \varepsilon_{ij})$$

$$y_j = r_j \mu + r_j \alpha_j + \sum_i \varepsilon_{ij}$$

Squaring On both sides.

$$y_j^2 = r_j^2 \mu^2 + r_j^2 \alpha_j^2 + \sum_i \varepsilon_{ij}^2 + \sum_{j \neq h} \sum_i \varepsilon_{ij} \varepsilon_{jh} + 2 r_j^2 \mu \alpha_j +$$

$$2 \alpha_j \varepsilon_{ij}$$

$$2 r_j \mu \sum_i \varepsilon_{ij} + 2 r_j \alpha_j \sum_i \varepsilon_{ij}$$

$$\alpha_j + 2 \mu \sum_i \varepsilon_{ij}$$

Dividing By r_j and apply $\sum_j$ on both sides.

$$\sum_j \frac{y_j^2}{r_j} = \sum_j r_j \mu^2 + \sum_j r_j \alpha_j^2 + \frac{1}{r_j} \sum_i \sum_j \varepsilon_{ij}^2 + \frac{1}{r_j} \sum_{j \neq h} \sum_i \varepsilon_{ij} \varepsilon_{jh} +$$

$$2 r_j \mu \sum_j \alpha_j + 2 \mu \sum_j \sum_i \varepsilon_{ij} + 2 \sum_j \sum_i \alpha_j \varepsilon_{ij}$$

$$+ 0 + 0 + 0$$

ii)

Taking Expectation On both sides

$$E \left[\sum_j \frac{y_j^2}{r_j} \right] = R \mu^2 + R \sigma_\alpha^2 + k \sigma^2 + 0 + 0 + 0 + 0$$

$$\frac{1}{\sigma_\alpha^2} - \sigma^2$$

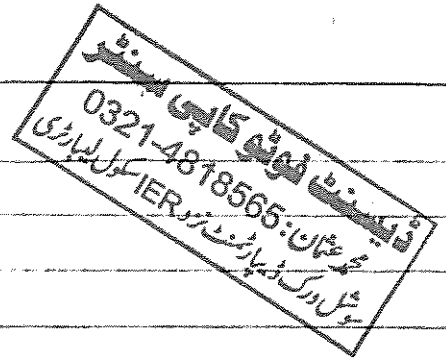
$$= R \mu^2 + R \sigma_\alpha^2 + k \sigma^2 \quad \text{--- (iv)}$$

$$\text{eq (iv)} - \text{eq (i)}$$

$$E(SSB) = R \mu^2 + R \sigma_\alpha^2 + k \sigma^2 - R \mu^2 - R \sigma_\alpha^2 - \sigma^2$$

$$E(SSB) = (k-1) \sigma^2 \quad \text{--- (5)}$$

nt:



Error Sum of Square:

$$E(\text{Error SS}) = E(\text{T.S.S}) - E(\text{S.S.B})$$

$$E(\text{Error SS}) = (R-1)\sigma^2 - (k-1)\sigma^2$$

$$= (R-1-k+1)\sigma^2$$

$$E(\text{Error SS}) = (R-k)\sigma^2$$

Dividing by its degree of freedom:

$$E(\text{MSE}) = \sigma^2 \text{ Hence Proved}$$

$$rk = k$$

$$\sum_j r_j = k r_j$$

$$\sum_j r_j = \sum_j r_j$$

$$R = R$$

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Multiple Comparison Test:

3).

If F-test after the analysis of variance rejects our null hypothesis, all we can conclude is that the k population means are not equal. This conclusion might not be sufficient to satisfy the experimenter, rather we would like to know which means might differ significantly from each other.

For this purpose, several tests based on different view points, have been developed to make comparisons b/w pairs of means.

Such tests, used as a follow-up to F-test are called the multiple comparison tests.

$$n_k = k$$

$$\sum_{j=1}^k r_j = k r_j$$

$$\sum_{j=1}^k r_j = \sum_{j=1}^k r_j$$

$$R = R$$

- The commonly used tests includes
- Least Significant Difference Test (LSD Test)
 - The Student's Newman-Kuel's M. R. T
 - Duncan's Multiple Range Test.
 - The Scheffe's Test
 - Tukey's T-Method.

Least Significant Difference Test:

When the null hypothesis of equal means is rejected by F-Test after the ANOVA, we can test the significance of differences b/w means of k samples by computing the smallest difference that would be judged significant and by comparing the

absolute values of all differences of means with it. This smallest difference is called the least significant difference or LSD and is given by

$$LSD = t_{(\alpha/2, v)} \sqrt{\frac{2MSE}{r}} \quad (\text{For equal case})$$

$$LSD = t_{(\alpha/2, v)} \sqrt{MSE \left[\frac{1}{r_i} + \frac{1}{r_j} \right]} \quad (\text{For unequal case})$$

where;

MSE = Mean Square Error

r = Sample (Treatment) size.

$t_{\alpha/2}$ = is the value of t at $\alpha/2$ level.

v = degree of freedom of error.

The test criterion that uses the least significant difference is called LSD Test.

$$\text{If } |\bar{X}_i - \bar{X}_j| < LSD \quad (\text{means are equal})$$

$$\text{If } |\bar{X}_i - \bar{X}_j| > LSD \quad (\text{means are unequal})$$

Note:

The LSD test is only applied when the null hypothesis (H_0) is rejected in the ANOVA.

Duncan

In 19 comparison range is number compared. appropriate obtained significant

for the significant and

R_i

To perform means complete ranges R_i

the larger with difference second is significant all the difference

Duncan's Multiple Comparison Test:

In 1955, Duncan introduced a multiple comparison test based on the studentized range statistic $q(p, v)$ where p is the number of steps b/w two means being compared. But the values of $q(p, v)$ for appropriate values of p and v are obtained from Duncan's Table for significant ranges.

The $\alpha\%$ critical ranges for the p means called a least significant Range and denoted by R_p and is given by:

$$R_p = q_{\alpha}(p, v) \sqrt{\frac{MSE}{r}} ; p = 2, 3, 4, 5, \dots, k.$$

To perform the test, arrange the sample means in increasing order of magnitude and compute a set of $k-1$ least significant ranges R_p .

Compare the observed difference b/w the largest mean and the smallest mean with the least significant Range R_k ; the difference b/w the largest mean and the second smallest mean with least significant Range R_{k-1} , and so on until all have been compared with the largest mean. Next, compare the difference of the second largest mean

and the smallest mean with LSR R_{k-1} , the difference of the 2nd largest mean with the 2nd smallest mean with LSR R_{k-2} . The process continues until the difference of all possible $k(k-1)/2$ pair of means are considered. A pair of means is declared to be significantly different when the observed difference exceeds the corresponding least significant Range. This test is quite powerful and is very popular.

Note:

In case of unequal sample size (n_i) the Harmonic Mean r_h of all r_i

$$r_h = \frac{k}{\sum_{i=1}^k \left(\frac{1}{r_i} \right)}$$

is used in place of " r ".

Range

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Randomized Complete Block Design:

A randomized complete block design may be defined as one in which:

i) The experimental material is divided into groups or blocks in such a manner that the experimental units within a particular block are relatively homogeneous.

ii) Each block contains a complete set of treatments.

iii) The treatments are assigned at random to the experimental units within each block which means that the randomization is restricted within blocks.

i.e. A new randomization is made for every block.

The purpose of this type of arrangement is to control one source of variation in the experimental material. RCB design is perhaps the most widely used design.

Experimental Layout:

Suppose that there are " k " treatments and " r " blocks in a RCB design then each block contains " k " homogeneous plots, one for each treatment, an experimental layout for such a design using 6 treatments A, B, C, D, E, F in

Three blocks might be as follows:

Block I

D	B	A	C	F	E	/
---	---	---	---	---	---	---

Block II

C	B	E	F	D	A	/
---	---	---	---	---	---	---

Block III

C	F	B	D	A	E	/
---	---	---	---	---	---	---

or treatment may be randomized in other way.

Statistical Model & Analysis:

The statistical Model is:

$$Y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij}$$

where;
 $i = 1, 2, \dots, r$
 $j = 1, 2, \dots, k$

μ = overall Mean

α_j = effect of j^{th} treatment

β_i = effect of i^{th} block.

ϵ_{ij} = Random Error $\sim N(0, \sigma^2)$

Y_{ij} = The i^{th} observation in the j^{th} treatment

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The analysis of RCB design contains of a two way analysis of variance where the test of hypothesis is may be:

E ☒

A ☒

E ☒

$H_0: \beta_i = 0$ No Block Effect.

$H_0: \alpha_j = 0$ NO Treatment Effect.

ized

The computations are performed as usually and F statistic is used to test the hypothesis related to block effect and treatment effect.

Analysis:

Advantages:

The advantages are as follow:

- The source of extraneous variation is controlled by grouping the experimental material and Hence the estimate of experimental error is reduced.
- The design is flexible i.e. any no. but not less than 2 of replication may be run and any no. of treatment may be tested.

5²)

- The experiment can be set up easily.
- The statistical analysis is simple and straight forward.

the

→ It is easy to adjust for missing observations.

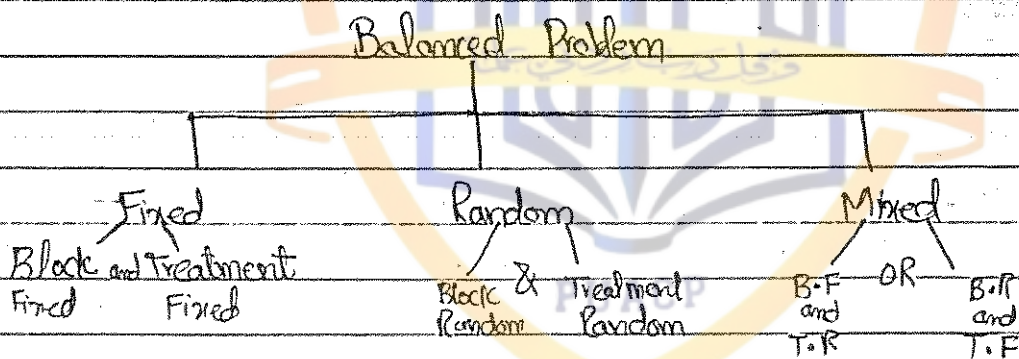
Disadvantages

→ It controls variability only in one direction.

→ It is not a suitable design when the number of treatments are very large or the blocks are not homogeneous.

Proves:

In RCB Design.



Proof # 1:

Fixed Effect Model:

Prove That

$$E(MSE) = \sigma^2$$

Proof:

In RCB design, the observations y_{ij} is expressed as:

Core

C.F

$G =$

$G =$

$G^2 =$

$G^2 =$

$\frac{G^2}{rk} =$

$$Y_{ij} = \mu + \beta_i + \alpha_j + \epsilon_{ij}$$

with assumptions

$$i = 1, 2, \dots, r$$

$$j = 1, 2, \dots, k$$

$$\epsilon_{ij} \sim \text{NII}(0, \sigma^2)$$

i.e.

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2, \quad E(\epsilon_{ij} \epsilon_{gh}) = 0$$

for $i \neq g$
 $j \neq h$

In case of fixed effect Model, the treatment effect and block effect both are fixed such that

$$\sum_j \alpha_j = 0, \quad \sum_i \beta_i = 0$$

Corection Factor:

$$C.F = \frac{G^2}{rk} \quad \text{where} \quad G = \sum_j \sum_i Y_{ij}$$

$$G = \sum_j \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$G = rk\mu + r \sum_j \alpha_j + k \sum_i \beta_i + \sum_j \sum_i \epsilon_{ij}$$

$$\because \sum_j \alpha_j = 0 \quad \text{and} \quad \sum_i \beta_i = 0$$

$$G^2 = \left(rk\mu + \sum_j \sum_i \epsilon_{ij} \right)^2$$

$$G^2 = r^2 k^2 \mu^2 + \sum_j \sum_i \epsilon_{ij}^2 + \sum_{i \neq g} \sum_{j \neq h} \epsilon_{ij} \epsilon_{gh} + 2rk\mu \sum_j \sum_i \epsilon_{ij}$$

$$\frac{G^2}{rk} = rk\mu^2 + \frac{1}{rk} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{rk} \sum_{i \neq g} \sum_{j \neq h} \epsilon_{ij} \epsilon_{gh} + 2\mu \sum_j \sum_i \epsilon_{ij}$$

Mixed
OR
B.F
and
T.F

variations

By Taking expectation on both sides

$$E(C.F) = E\left[\frac{Q^2}{rk}\right] = \frac{rku^2}{rk} + \frac{rk\sigma^2}{rk} + 0 + 0 + 0 + 0$$

$$E(C.F) = rku^2 + \sigma^2 \quad (1)$$

Total Sum of Square:

$$T.S.S = \sum_j \sum_i Y_{ij}^2 - C.F$$

where

$$Y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij}$$

$$Y_{ij}^2 = (\mu + \alpha_j + \beta_i + \epsilon_{ij})^2$$

$$Y_{ij}^2 = \mu^2 + \alpha_j^2 + \beta_i^2 + \epsilon_{ij}^2 + 2\mu\alpha_j + 2\mu\beta_i + 2\mu\epsilon_{ij} + 2\alpha_j\beta_i + 2\alpha_j\epsilon_{ij} + 2\beta_i\epsilon_{ij}$$

$$\sum_j \sum_i Y_{ij}^2 = rku^2 + r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + \sum_j \sum_i \epsilon_{ij}^2 + 2\mu r \sum_j \alpha_j + 2\mu k \sum_i \beta_i + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \beta_i + 2 \sum_j \alpha_j \sum_i \epsilon_{ij} + 2 \sum_i \beta_i \sum_j \epsilon_{ij}$$

$$2 \sum_j \alpha_j \sum_i \epsilon_{ij} + 2 \sum_i \beta_i \sum_j \epsilon_{ij}$$

$$2 \sum_j \alpha_j \sum_i \epsilon_{ij}$$

By Taking expectation on both sides

$$E\left[\sum_j \sum_i Y_{ij}^2\right] = rku^2 + r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + rk\sigma^2 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0$$

$$E\left[\sum_j \sum_i Y_{ij}^2\right] - E(C.F) = rku^2 + r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + rk\sigma^2 - rku^2 - \sigma^2$$

$$T.S.S = r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + (rk-1)\sigma^2 \quad (11)$$

Treatment

T.S.S =

$Y_{ij} =$

=

$Y_{ij}^2 =$

$Y_{ij}^2 = r^2$

2μ

$\sum_j \frac{Y_{ij}^2}{r} =$

2μ

By ta

$E\left[\sum_j \frac{Y_{ij}^2}{r}\right]$

$E\left[\sum_j \frac{Y_{ij}^2}{r}\right] -$

$E(T.S.S)$

Block

Block S

Treatment Sum of Square:

$$Tr. S.S = \sum_j \frac{Y_{.j}^2}{r} - C.F. \quad \text{where}$$

$$Y_{.j} = \sum_i (Y_{ij}) = \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$= r\mu + r\alpha_j + \sum_i \beta_i + \sum_i \epsilon_{ij}$$

$$\therefore \sum_i \beta_i = 0$$

$$Y_{.j}^2 = (r\mu + r\alpha_j + \sum_i \epsilon_{ij})^2$$

$$Y_{.j}^2 = r^2\mu^2 + r^2\alpha_j^2 + \sum_i \epsilon_{ij}^2 + \sum_{i \neq g} \epsilon_{ij}\epsilon_{ig} + 2r^2\mu\alpha_j +$$

$$2r\mu \sum_i \epsilon_{ij} + 2r\alpha_j \sum_i \epsilon_{ij}$$

$$\sum_j \frac{Y_{.j}^2}{r} = r k \mu^2 + r \sum_j \alpha_j^2 + \frac{1}{r} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{r} \sum_{h \neq j} \sum_{i \neq g} \epsilon_{ij}\epsilon_{ig} +$$

$$2r\mu \sum_j \alpha_j + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \epsilon_{ij}$$

By taking Expectation On Both sides

$$E\left[\sum_j \frac{Y_{.j}^2}{r}\right] = r k \mu^2 + r \sum_j \alpha_j^2 + k \sigma^2 + 0 + 0 + 0 + 0$$

$$E\left[\sum_j \frac{Y_{.j}^2}{r}\right] - E(C.F) = r k \mu^2 + r \sum_j \alpha_j^2 + k \sigma^2 - r k \mu^2 - \sigma^2$$

$$E(Tr. S.S) = r \sum_j \alpha_j^2 + (k-1) \sigma^2 \quad \text{--- (III)}$$

Block Sum of Square:

$$Block SS = \sum_i \frac{Y_{i.}^2}{k} - C.F.$$

where;

$$Y_i = \sum_j Y_{ij} = \sum_j (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$Y_i = k\mu + \sum_j \alpha_j + k\beta_i + \sum_j \epsilon_{ij}$$

$$\therefore \sum_j \alpha_j = 0$$

$$Y_i^2 = (k\mu + k\beta_i + \sum_j \epsilon_{ij})^2$$

$$= k^2\mu^2 + k^2\beta_i^2 + \sum_j \epsilon_{ij}^2 + \sum_{j \neq k} \epsilon_{ij} \epsilon_{jk} + 2k^2\mu\beta_i +$$

$$2k\mu \sum_j \epsilon_{ij} + 2k\beta_i \sum_j \epsilon_{ij}$$

$$\sum_i \frac{Y_i^2}{k} = r k \mu^2 + k \sum_i \beta_i^2 + \frac{1}{k} \sum_i \sum_j \epsilon_{ij}^2 + \frac{1}{k} \sum_i \sum_{j \neq k} \epsilon_{ij} \epsilon_{jk} +$$

$$+ 2k\mu \sum_i \beta_i + 2\mu \sum_i \sum_j \epsilon_{ij} + 2\beta_i \sum_i \sum_j \epsilon_{ij}$$

By taking Expectation on Both sides.

$$E\left[\sum_i \frac{Y_i^2}{k}\right] = r k \mu^2 + k \sum_i \beta_i^2 + r \sigma^2 + 0 + 0 + 0 + 0$$

$$E\left[\sum_i \frac{Y_i^2}{k}\right] - E(C.F) = r k \mu^2 + k \sum_i \beta_i^2 + r \sigma^2 - r k \mu^2 - \sigma^2$$

$$= k \sum_i \beta_i^2 + (r-1) \sigma^2 \quad \text{--- (iv)}$$

Error Sum of Square:

Error SS = Total SS - Treatment SS - Block SS.

$$E(\text{error SS}) = r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + (rk-1) \sigma^2 - r \sum_j \alpha_j^2 - (k-1) \sigma^2 - k \sum_i \beta_i^2 - (r-1) \sigma^2$$

E(Error

E(Error

Dividing

E[MSE

Proof #

Prove

Proof:

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ϵ_{ij}

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treatment

random.

α_j^2

i.e.

E(

$$\begin{aligned}
 E(\text{Error SS}) &= (rk - 1 - k + 1 - r + 1) \sigma^2 \\
 &= (rk - k - r + 1) \sigma^2 \\
 &= k(r-1) - 1(r-1) \sigma^2 \\
 E(\text{Error SS}) &= (r-1)(k-1) \sigma^2
 \end{aligned}$$

Dividing Both sides by its df $[(r-1)(k-1)]$

$$E[\text{MSE}] = \sigma^2 \quad \text{Hence Proved.}$$

Proof # 2:

Random Effect Model:

Prove That $E(\text{MSE}) = \sigma^2$

Proof:

In RCB Designs, the observations y_{ij} is expressed as:

$$y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij}$$

with assumptions

$$i = 1, 2, \dots, r.$$

$$j = 1, 2, \dots, k.$$

$$\epsilon_{ij}'s \sim \text{NII}(0, \sigma^2)$$

i.e.

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2, \quad E(\epsilon_{ij} \epsilon_{gh}) = 0$$

$$i \neq g \\ j \neq h$$

In random effect model, both treatment effect and block effect are random and independent such that

$$\alpha_j's \sim \text{NII}(0, \sigma_\alpha^2)$$

i.e.

$$E(\alpha_j) = 0, \quad E(\alpha_j^2) = \sigma_\alpha^2, \quad E(\alpha_j \alpha_h) = 0$$

$$j \neq h$$

SS

$$= (k-1) \sigma^2$$

$$\beta_i's \sim NII(0, \sigma_\beta^2).$$

i.e

$$E(\beta_i) = 0, \quad E(\beta_i^2) = \sigma_\beta^2, \quad E(\beta_i \beta_g) = 0 \quad i \neq g$$

and Also treatment effect and block effect are independent to the error term

i.e

$$E(\alpha_j \varepsilon_{ij}) = 0 \quad \text{and} \quad E(\beta_i \varepsilon_{ij}) = 0$$

Correction Factor:

$$C.F = \frac{G^2}{rk} \quad \text{where}$$

$$G = \sum_j \sum_i y_{ij} = \sum_j \sum_i (\mu + \alpha_j + \beta_i + \varepsilon_{ij})$$

$$= rk\mu + r \sum_j \alpha_j + k \sum_i \beta_i + \sum_j \sum_i \varepsilon_{ij}$$

$$G^2 = (rk\mu + r \sum_j \alpha_j + k \sum_i \beta_i + \sum_j \sum_i \varepsilon_{ij})^2$$

$$G^2 = r^2 k^2 \mu^2 + r^2 \sum_j \alpha_j^2 + r^2 \sum_{j \neq h} \alpha_j \alpha_h + k^2 \sum_i \beta_i^2 + k^2 \sum_{i \neq g} \beta_i \beta_g +$$

$$\sum_j \sum_i \varepsilon_{ij}^2 + \sum_{i \neq g} \sum_{j \neq h} \varepsilon_{ij} \varepsilon_{gh} + 2r^2 k \mu \sum_j \alpha_j + 2rk^2 \mu \sum_i \beta_i +$$

$$2r.k\mu \sum_j \sum_i \varepsilon_{ij} + 2rk \sum_j \alpha_j \sum_i \beta_i + 2r \sum_j \alpha_j \sum_i \varepsilon_{ij} + 2k \sum_i \beta_i \sum_j \varepsilon_{ij}$$

Dividing By rk .

$$\frac{G^2}{rk} = rk\mu^2 + \frac{r}{k} \sum_j \alpha_j^2 + \frac{r}{k} \sum_{j \neq h} \alpha_j \alpha_h + \frac{k}{r} \sum_i \beta_i^2 + \frac{k}{r} \sum_{i \neq g} \beta_i \beta_g +$$

$$\frac{1}{rk} \sum_j \sum_i \varepsilon_{ij}^2 + \frac{1}{rk} \sum_{i \neq g} \sum_{j \neq h} \varepsilon_{ij} \varepsilon_{gh} + 2r\mu \sum_j \alpha_j + 2k\mu \sum_i \beta_i +$$

$$2\mu \sum_j \sum_i \varepsilon_{ij} + 2 \sum_j \sum_i \alpha_j \beta_i + \frac{2}{k} \sum_j \sum_i \alpha_j \varepsilon_{ij} + \frac{2}{r} \sum_i \sum_j \beta_i \varepsilon_{ij}$$

By Taking

$$E\left(\frac{G^2}{rk}\right)$$

$$E(C.F)$$

Total

Total

$$y_{ij}^2 =$$

$$y_{ij}^2 = \mu^2$$

$$2\alpha_j$$

$$\sum_j \sum_i y_{ij}^2 =$$

$$2\mu.k$$

$$2$$

By Ta

$$E\left(\sum_j \sum_i y_{ij}^2\right)$$

$$E\left(\sum_j \sum_i y_{ij}^2\right)$$

$$E(\text{Total SS})$$

ڈیسٹ فوٹو کاپی سنٹر

محمد عثمان: 0321-4818565

سوشل ورک ڈیپارٹمنٹ نزد IER سکول لیبارری

or term

Total Sum of Square:

$$Y_{ij}^2 = (\mu + \alpha_j + \beta_i + \epsilon_{ij})^2$$

$$2\alpha_j \beta_i + 2\alpha_j \varepsilon_{ij} + 2\beta_i \varepsilon_{ij}$$

$$2\mu_k \sum_i \beta_i + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \beta_i + 2 \sum_j \alpha_j \sum_i \epsilon_{ij} +$$

$$2 \sum_i \sum_j \beta_i \epsilon_{ij}$$

By Taking Expectation On Both Sides.

$$E\left[\sum_j \sum_i y_{ij}^2\right] = r k \mu^2 + r k \sigma_\alpha^2 + r k \sigma_\beta^2 + r k \sigma^2 + 0 + 0 + 0 + 0 + 0 + 0.$$

$$E\left[\sum_i \sum_j y_{ij}^2\right] - E(C.F) = \cancel{rku^2} + r_k \sigma_\alpha^2 + r_k \sigma_\beta^2 + r_k \sigma^2 - \cancel{rku^2} - r_k \sigma_\alpha^2 - k \sigma_\beta^2 = \sigma^2$$

$$E(\text{Total SS}) = (rk - r) \sigma_\alpha^2 + (rk - k) \sigma_\beta^2 + (rk - 1) \sigma^2 \quad (11)$$

Treatment Sum of Square:

$$Tr. S.S = \sum_j \frac{Y_{.j}^2}{r} - C.F$$

$$Y_{.j} = \sum_i (Y_{ij}) = \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$Y_{.j}^2 = \left(r\mu + r\alpha_j + \sum_i \beta_i + \sum_i \epsilon_{ij} \right)^2$$

$$Y_{.j}^2 = r^2 \mu^2 + r^2 \alpha_j^2 + \sum_i \beta_i^2 + \sum_{i \neq g} \beta_i \beta_g + \sum_i \epsilon_{ij}^2 + \sum_{i \neq g} \epsilon_{ij} \epsilon_{ig} +$$

$$2r^2 \mu \alpha_j + 2r\mu \sum_i \beta_i + 2r\mu \sum_i \epsilon_{ij} + 2r\alpha_j \sum_i \beta_i +$$

$$2r\alpha_j \sum_i \epsilon_{ij} + 2 \sum_j \sum_i \beta_i \epsilon_{ij}$$

$$\sum_j \frac{Y_{.j}^2}{r} = r k \mu^2 + r \sum_j \alpha_j^2 + \frac{1}{r} k \sum_i \beta_i^2 + \frac{1}{r} \sum_j \sum_{i \neq g} \beta_i \beta_g +$$

$$\frac{1}{r} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{r} \sum_{h \neq j} \sum_{i \neq g} \epsilon_{ij} \epsilon_{ig} + 2r\mu \sum_j \alpha_j + 2\mu k \sum_i \beta_i +$$

$$2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \beta_i + 2 \sum_j \alpha_j \sum_i \epsilon_{ij} + \frac{2}{r} \sum_j \sum_i \beta_i \epsilon_{ij}$$

By Taking Expectation on Both sides

$$E \left[\sum_j \frac{Y_{.j}^2}{r} \right] = r k \mu^2 + r k \sigma_\alpha^2 + \frac{r k}{r} \sigma_\beta^2 + 0 + \frac{r k}{r} \sigma^2 + 0 \dots + 0$$

$$E \left[\sum_j \frac{Y_{.j}^2}{r} \right] - E(C.F) = E(Tr. S.S)$$

$$= r k \mu^2 + r k \sigma_\alpha^2 + k \sigma_\beta^2 + k \sigma^2 - r k \mu^2 - r \sigma_\alpha^2 - k \sigma_\beta^2 - \sigma^2$$

$$= (rk - r) \sigma_\alpha^2 + (k-1) \sigma^2 \quad \text{--- (III)}$$

Block S

Block SS

where,

$$Y_{i.}^2 = \left(\sum_j Y_{ij} \right)^2$$

$$Y_{i.}^2 = k^2$$

$$2rk\mu \sum_i \alpha_i$$

$$2rk\beta$$

$$\sum_i \frac{Y_{i.}^2}{rk} =$$

$$\frac{1}{rk}$$

$$2\mu$$

By Tak

$$E \left[\sum_i \frac{Y_{i.}^2}{rk} \right] =$$

$$E \left[\sum_i \frac{Y_{i.}^2}{rk} \right] -$$

$$=$$

Block Sum of Square:

$$\text{Block SS} = \sum_i \frac{Y_i^2}{k} - \text{C.F.}$$

$$\text{where } Y_i = \sum_j (Y_{ij}) = \sum_j (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$= k\mu + \sum_j \alpha_j + k\beta_i + \sum_j \epsilon_{ij}$$

$$Y_i^2 = \left(k\mu + \sum_j \alpha_j + k\beta_i + \sum_j \epsilon_{ij} \right)^2$$

$$Y_i^2 = k^2 \mu^2 + \sum_j \alpha_j^2 + \sum_{j \neq h} \alpha_j \alpha_h + k^2 \beta_i^2 + \sum_j \epsilon_{ij}^2 + \sum_{j \neq k} \epsilon_{ij} \epsilon_{ik} +$$

$$2k\mu \sum_j \alpha_j + 2k^2 \mu \beta_i + 2k\mu \sum_j \epsilon_{ij} + 2k\beta_i \sum_j \alpha_j + 2 \sum_j \alpha_j \sum_j \epsilon_{ij} +$$

$$2k\beta_i \sum_j \epsilon_{ij}$$

$$\sum_i \frac{Y_i^2}{k} = r k \mu^2 + \frac{1}{k} r \sum_j \alpha_j^2 + \frac{1}{k} \sum_i \sum_{j \neq h} \alpha_j \alpha_h + k \sum_i \beta_i^2 +$$

$$\frac{1}{k} \sum_i \sum_j \epsilon_{ij}^2 + \frac{1}{k} \sum_i \sum_{j \neq k} \epsilon_{ij} \epsilon_{ik} + 2\mu r \sum_j \alpha_j + 2\mu \sum_i \beta_i +$$

$$2\mu \sum_i \sum_j \epsilon_{ij} + 2 \sum_i \beta_i \sum_j \alpha_j + \frac{2}{k} \sum_i \sum_j \alpha_j \epsilon_{ij} + 2 \sum_i \sum_j \beta_i \epsilon_{ij}$$

By Taking Expectation On Both Sides

$$E \left[\sum_i \frac{Y_i^2}{k} \right] = r k \mu^2 + \frac{r k}{k} \sigma_\alpha^2 + 0 + r k \sigma_\beta^2 + \frac{r k}{k} \sigma^2 + 0 + \dots + 0$$

$$= r k \mu^2 + r \sigma_\alpha^2 + r k \sigma_\beta^2 + r \sigma^2$$

$$E \left[\sum_i \frac{Y_i^2}{k} \right] - E(\text{C.F.}) = r k \mu^2 + r \sigma_\alpha^2 + r k \sigma_\beta^2 + r \sigma^2 - r k \mu^2 -$$

$$r \sigma_\alpha^2 - k \sigma_\beta^2 - \sigma^2$$

$$= (r k - k) \sigma_\beta^2 + (r - 1) \sigma^2 \quad \text{--- (iv)}$$

Error Sum of Square:

$$\text{Error SS} = \text{Total SS} - \text{Treatment SS} - \text{Block SS.}$$

$$\text{eq (i)} - \text{eq (ii)} - \text{eq (iii)}$$

$$E(\text{Error SS}) = (rk-r) \sigma_a^2 + (rk-k) \sigma_b^2 + (rk-1) \sigma^2$$

$$(rk-r) \sigma_a^2 - (k-1) \sigma^2 - (rk-k) \sigma_b^2 - (r-1) \sigma^2$$

$$\begin{aligned} E(\text{Error SS}) &= (rk-1 - k + r - r + 1) \sigma^2 \\ &= (rk - k - r + 1) \sigma^2 \\ &= k(r-1) - 1(r-1) \sigma^2 \end{aligned}$$

$$E(\text{Error SS}) = (r-1)(k-1) \sigma^2$$

Dividing By its df:

$$E(\text{MSE}) = \sigma^2 \quad \text{Hence Proved.}$$

PROOF # 3:

Mixed Effect Model

Treatments effect →

Fixed

Blocks effect →

Random

Prove that

$$E(\text{MSE}) = \sigma^2$$

Proof:

In RCB Design, the observations y_{ij} is expressed as:

$$y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij}$$

with assumptions

$$\epsilon_{ij}'s \sim NII(0, \sigma^2)$$

$$i = 1, 2, \dots, r$$

$$j = 1, 2, \dots, k$$

$$2rk^2 \mu \sum \beta_i$$

i.e.

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2, \quad E(\epsilon_{ij} \epsilon_{gh}) = 0$$

The effect of treatment is fixed such that $i \neq g$
 $j \neq h$

$$\sum_j \alpha_j = 0$$

(-1) σ^2 and the effect of blocks is random such that

$$\beta_i \text{'s} \sim NII(0, \sigma_\beta^2)$$

i.e.

$$E(\beta_i) = 0, \quad E(\beta_i^2) = \sigma_\beta^2, \quad E(\beta_i \beta_g) = 0$$

and

β_i and ϵ_{ij} are independent such that $i \neq g$.

$$E(\beta_i \epsilon_{ij}) = 0.$$

Correction Factor:

$$C.F = \frac{G^2}{nr} \quad \text{where} \quad G = \sum_j \sum_i Y_{ij}$$

$$G = \sum_j \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$G = r k \mu + r \sum_j \alpha_j + k \sum_i \beta_i + \sum_j \sum_i \epsilon_{ij}$$

$$\therefore \sum_j \alpha_j = 0$$

$$G^2 = (r k \mu + k \sum_i \beta_i + \sum_j \sum_i \epsilon_{ij})^2$$

$$G^2 = r^2 k^2 \mu^2 + k^2 \sum_i \beta_i^2 + \sum_{i \neq g} \beta_i \beta_g + \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh} +$$

$$2 r k^2 \mu \sum_i \beta_i + 2 r k \mu \sum_j \sum_i \epsilon_{ij} + 2 k \sum_i \beta_i \sum_j \epsilon_{ij}$$

Dividing By rk on both sides.

$$\frac{G^2}{rk} = rku^2 + \frac{k}{r} \sum_i \beta_i^2 + \frac{1}{rk} \sum_{i \neq g} \beta_i \beta_g + \frac{1}{rk} \sum_{j,i} \epsilon_{ij}^2 + \frac{1}{rk} \sum_{i \neq j} \epsilon_{ij} \epsilon_{ji} + 2ku \sum_i \beta_i + 2u \sum_{j,i} \epsilon_{ij} + \frac{2}{r} \sum_{j,i} \beta_i \epsilon_{ij}$$

By Taking Expectation On Both Sides.

$$E(C.F) = E\left[\frac{G^2}{rk}\right] = rku^2 + \frac{k}{r} r \sigma_\beta^2 + 0 + \frac{rk}{rk} \sigma^2 + 0 \dots + 0$$

$$= rku^2 + k\sigma_\beta^2 + \sigma^2 \quad \text{--- (1)}$$

Total Sum of Squares:

Total SS = $\sum_j \sum_i Y_{ij}^2 - C.F$
where;

$$Y_{ij}^2 = (\mu + \alpha_j + \beta_i + \epsilon_{ij})^2$$

$$Y_{ij}^2 = \mu^2 + \alpha_j^2 + \beta_i^2 + \epsilon_{ij}^2 + 2\mu\alpha_j + 2\mu\beta_i + 2\mu\epsilon_{ij} +$$

$$2\alpha_j\beta_i + 2\alpha_j\epsilon_{ij} + 2\beta_i\epsilon_{ij}$$

$$\sum_j \sum_i Y_{ij}^2 = rku^2 + r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + \sum_{j,i} \epsilon_{ij}^2 + 2ur \sum_j \alpha_j +$$

$$2uk \sum_i \beta_i + 2u \sum_{j,i} \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \beta_i + 2 \sum_j \alpha_j \sum_i \epsilon_{ij} +$$

$$2 \sum_i \beta_i \sum_j \epsilon_{ij}$$

By Taking Expectation on both sides.

$$E\left[\sum_j \sum_i Y_{ij}\right]$$

$$E\left[\sum_j \dots\right]$$

$$= rku^2$$

$$= r \sum_j \dots$$

Trends

Tr.S.,
where;

$$Y_{ij}^2 =$$

$$Y_{ij}^2 = r^2 u$$

$$2r^2 u \alpha_j$$

$$2$$

$$\sum_j \frac{Y_{ij}^2}{r} =$$

$$\frac{1}{r} \sum_j \dots$$

$$2u \sum_j \dots$$

By Tak

$$E\left[\sum_j \frac{Y_{ij}^2}{r}\right]$$

$$E\left[\sum_j \sum_i Y_{ij}^2\right] = rku^2 + r\sum_j \alpha_j^2 + rk\sigma_p^2 + rk\sigma^2 + 0 + \dots + 0$$

$$E\left[\sum_j \sum_i Y_{ij}^2\right] - E(C.F.) = E(\text{Total SS})$$

$$= rku^2 + r\sum_j \alpha_j^2 + rk\sigma_p^2 + rk\sigma^2 - rku^2 - k\sigma_p^2 - \sigma^2$$

$$= r\sum_j \alpha_j^2 + (rk - k)\sigma_p^2 + (rk - 1)\sigma^2 \quad \text{--- (11)}$$

Treatment Sum of Square:

$$Tr. S.S = \sum_j \frac{Y_{.j}^2}{r} - C.F.$$

where;

$$Y_{.j} = \sum_i Y_{ij} = \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$Y_{.j}^2 = (r\mu + r\alpha_j + \sum_i \beta_i + \sum_i \epsilon_{ij})^2$$

$$Y_{.j}^2 = r^2\mu^2 + r^2\alpha_j^2 + \sum_i \beta_i^2 + \sum_{i \neq g} \beta_i \beta_g + \sum_i \epsilon_{ij}^2 + \sum_{i \neq g} \epsilon_{ij} \epsilon_{ig} +$$

$$2r^2\mu\alpha_j + 2r\mu\sum_i \beta_i + 2\mu r\sum_i \epsilon_{ij} + 2r\alpha_j\sum_i \beta_i + 2r\alpha_j\sum_i \epsilon_{ij} +$$

$$2\mu\sum_i \epsilon_{ij} +$$

$$2\sum_i \beta_i \epsilon_{ij}$$

$$\sum_j \frac{Y_{.j}^2}{r} = rku^2 + r\sum_j \alpha_j^2 + \frac{1}{r} k \sum_i \beta_i^2 + \frac{1}{r} \sum_j \sum_{i \neq g} \beta_i \beta_g +$$

$$\frac{1}{r} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{r} \sum_j \sum_{i \neq h} \epsilon_{ij} \epsilon_{jh} + 2r\mu\sum_j \alpha_j + 2\mu k \sum_i \beta_i +$$

$$2\mu\sum_j \sum_i \epsilon_{ij} + 2\sum_j \alpha_j \sum_i \beta_i + \frac{2}{r} \sum_j \sum_i \alpha_j \epsilon_{ij}$$

By Taking Expectations on Both Sides.

$$E\left[\sum_j \frac{Y_{.j}^2}{r}\right] = rku^2 + r\sum_j \alpha_j^2 + \frac{rk}{r} \sigma_p^2 + 0 + \frac{rk}{r} \sigma^2 + 0 + \dots + 0$$

$$E\left[\frac{\sum_{j=1}^r Y_{ij}^2}{r}\right] = rku^2 + r \sum_{j=1}^r \alpha_j^2 + k\sigma_p^2 + k\sigma^2$$

$$= E\left[\frac{\sum_{j=1}^r Y_{ij}^2}{r}\right] - E(C.F.)$$

$$= rku^2 + r \sum_{j=1}^r \alpha_j^2 + k\sigma_p^2 + k\sigma^2 - rku^2 - k\sigma_p^2 - \sigma^2$$

$$= r \sum_{j=1}^r \alpha_j^2 + (k-1)\sigma^2 \quad (iii)$$

Block Sums of Square:

$$B.S.S = \frac{\sum_i Y_i^2}{k} - C.F.$$

where;

$$Y_i = \sum_j (Y_{ij}) = \sum_j (u + \alpha_j + \beta_i + \epsilon_{ij})$$

$$Y_i = ku + \sum_j \alpha_j + k\beta_i + \sum_j \epsilon_{ij}$$

$$\therefore \sum_j \alpha_j = 0$$

$$Y_i^2 = \left(ku + k\beta_i + \sum_j \epsilon_{ij}\right)^2$$

$$Y_i^2 = k^2u^2 + k^2\beta_i^2 + \sum_j \epsilon_{ij}^2 + \sum_{j \neq k} \epsilon_{ij} \epsilon_{ik} + 2k^2u\beta_i +$$

$$2k u \sum_j \epsilon_{ij} + 2k\beta_i \sum_j \epsilon_{ij}$$

$$\frac{\sum_i Y_i^2}{k} = rku^2 + k \sum_i \beta_i^2 + \frac{1}{k} \sum_i \sum_j \epsilon_{ij}^2 + \frac{1}{k} \sum_i \sum_{j \neq k} \epsilon_{ij} \epsilon_{ik} +$$

$$2ku \sum_i \beta_i + 2u \sum_i \sum_j \epsilon_{ij} + 2 \sum_i \beta_i \sum_j \epsilon_{ij}$$

By Taking Expectation On Both Sides.

$$E\left[\frac{\sum_i Y_i^2}{k}\right]$$

$$= E\left[\frac{\sum_i Y_i^2}{k}\right]$$

$$= rku^2 +$$

=

Error

Error SS

E(Error SS)

(k-

E(1

Proof #

Treat
Block

Proof:

is on

$$\begin{aligned}
 E\left[\frac{\sum_i Y_{ij}^2}{k}\right] &= r k \mu^2 + r k \sigma_p^2 + \frac{r k}{k} \sigma^2 + 0 \dots + 0 \\
 &= E\left[\frac{\sum_i Y_{ij}^2}{k}\right] - E(C.F) \\
 &= \cancel{r k \mu^2} + r k \sigma_p^2 + r \sigma^2 - \cancel{r k \mu^2} - \cancel{k \sigma_p^2} - \sigma^2 \\
 &= (r k - k) \sigma_p^2 + (r - 1) \sigma^2 \quad \text{--- (iv)}
 \end{aligned}$$

Error Sums of Squares:

$$\text{Error SS} = \text{Total SS} - \text{Tr. SS} - \text{B. SS.}$$

$$\begin{aligned}
 E(\text{Error SS}) &= r \sum_j \alpha_j^2 + (r k - k) \sigma_p^2 + (r k - 1) \sigma^2 - r \sum_j \alpha_j^2 - \\
 &\quad (k - 1) \sigma^2 - (r k - k) \sigma_p^2 - (r - 1) \sigma^2 \\
 &= (r k - k - k + 1 - r + 1) \sigma^2 \\
 &= k(r - 1) - 1(r - 1) \sigma^2 \\
 &= (r - 1)(k - 1) \sigma^2 \\
 E(\text{MSE}) &= \sigma^2 \quad \text{Hence Proved.}
 \end{aligned}$$

Proof # 4:

Mixed Effect Model.
 Treatment Effect → Random
 Block Effect → Fixed
 Prove That $E(\text{MSE}) = \sigma^2$.

Proof:

In RCBD Design, the observations Y_{ij} is expressed as:

$$Y_{ij} = \mu + \alpha_j + \beta_i + E_{ij}.$$

$$i = 1, 2, \dots, r.$$

$$j = 1, 2, \dots, k.$$

with assumptions

$$\epsilon_{ij}'s \sim NII(0, \sigma^2)$$

i.e

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2, \quad E(\epsilon_{ij} \epsilon_{gh}) = 0$$

The effect of block is fixed such $i \neq g, j \neq k$

$$\sum_i \beta_i = 0$$

But the effect of treatment is random such that

$$\alpha_j \beta_i's \sim NII(0, \sigma_\alpha^2)$$

i.e

$$E(\alpha_j) = 0, \quad E(\alpha_j^2) = \sigma_\alpha^2, \quad E(\alpha_j \alpha_h) = 0$$

and α_j and ϵ_{ij} are independent such that

$$E(\alpha_j \epsilon_{ij}) = 0$$

Correction Factor:

$$C.F. = \frac{G^2}{rk}$$

where;

$$G = \sum_j \sum_i Y_{ij} = \sum_j \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$G = r k \mu + r \sum_j \alpha_j + k \sum_i \beta_i + \sum_j \sum_i \epsilon_{ij}$$

$$\therefore \sum_i \beta_i = 0$$

$$G^2 = (r k \mu + r \sum_j \alpha_j + \sum_j \sum_i \epsilon_{ij})^2$$

$$G^2 = r^2 k^2 \mu^2$$

$$2 r k \mu \sum_j \alpha_j$$

Dividing

$$\frac{G^2}{rk} = rk \mu^2$$

$$\frac{1}{rk} \sum_j \sum_i Y_{ij}^2$$

By

$$E\left[\frac{G^2}{rk}\right] =$$

$$E(C.F.) =$$

Total

Total

where;

$$Y_{ij}^2 = \mu^2$$

$$2 \alpha_j \beta_i$$

$$\sum_j \sum_i Y_{ij}^2 = r$$

$$2 \mu \sum_j \alpha_j$$

By

$$G^2 = r^2 k^2 \mu^2 + r^2 \sum_j \alpha_j^2 + r^2 \sum_{j \neq h} \alpha_j \alpha_h + \sum_j \sum_i \epsilon_{ij}^2 + \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh} +$$

$$2rk\mu \sum_j \alpha_j + 2rk\mu \sum_j \sum_i \epsilon_{ij} + 2r \sum_j \alpha_j \sum_i \epsilon_{ij}$$

$\epsilon_{jh} = 0$
 $i \neq g$
 $j \neq h$

Dividing By rk .

$$\frac{G^2}{rk} = r\mu^2 + \frac{r}{k} \sum_j \alpha_j^2 + \frac{r}{k} \sum_{j \neq h} (\alpha_j \alpha_h) + \frac{1}{rk} \sum_j \sum_i \epsilon_{ij}^2 +$$

ndom

$$\frac{1}{rk} \sum_{j \neq h} \sum_{i \neq g} \epsilon_{ij} \epsilon_{gh} + \frac{2r\mu}{k} \sum_j \alpha_j + \frac{2\mu}{k} \sum_j \sum_i \epsilon_{ij} + \frac{2}{k} \sum_j \sum_i \alpha_j \epsilon_{ij}$$

By Taking Expectation On both sides

$= 0$

$$E\left[\frac{G^2}{rk}\right] = E(C.F) = r\mu^2 + \frac{r}{k} \sigma_\alpha^2 + 0 + \frac{r}{rk} \sigma^2 + 0 \dots + 0$$

$j \neq h$

such that

$$E(C.F) = r\mu^2 + r\sigma_\alpha^2 + \sigma^2 \quad (1)$$

Total Sums Of Squares:

$$\text{Total SS} = \sum_j \sum_i Y_{ij}^2 - C.F.$$

where;

$$Y_{ij}^2 = (\mu + \alpha_j + \beta_i + \epsilon_{ij})^2$$

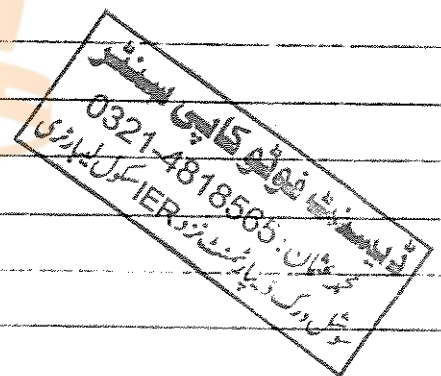
$$Y_{ij}^2 = \mu^2 + \alpha_j^2 + \beta_i^2 + \epsilon_{ij}^2 + 2\mu\alpha_j + 2\mu\beta_i + 2\mu\epsilon_{ij} +$$

$$2\alpha_j\beta_i + 2\alpha_j\epsilon_{ij} + 2\beta_i\epsilon_{ij}$$

$$\sum_j \sum_i Y_{ij}^2 = r\mu^2 + r \sum_j \alpha_j^2 + k \sum_i \beta_i^2 + \sum_j \sum_i \epsilon_{ij}^2 + 2\mu r \sum_j \alpha_j +$$

$$2\mu k \sum_i \beta_i + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \beta_i + 2 \sum_j \sum_i \alpha_j \epsilon_{ij} + 2 \sum_i \beta_i \sum_j \epsilon_{ij}$$

By Taking Expectation on both sides.



$$E\left[\sum_j \sum_i Y_{ij}^2\right] = rku^2 + rk\sigma_\alpha^2 + k \sum_i \beta_i^2 + rk\sigma^2 + 0 \dots + 0.$$

$$E\left[\sum_j \sum_i Y_{ij}^2\right] - E(C.F.) = rku^2 + rk\sigma_\alpha^2 + k \sum_i \beta_i^2 + rk\sigma^2 - rku^2 - r\sigma_\alpha^2 - \sigma^2.$$

$$E(\text{Total SS}) = (rk-r)\sigma_\alpha^2 + k \sum_i \beta_i^2 + (rk-1)\sigma^2 \quad \text{②}$$

Treatment Sums of Square:

$$Tr.S.S = \sum_j \frac{Y_{.j}^2}{r} - C.F.$$

$$Y_{.j} = \sum_i (Y_{ij}) = \sum_i (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$Y_{.j}^2 = (r\mu + r\alpha_j + \sum_i \beta_i + \sum_i \epsilon_{ij})^2$$

$$\because \sum_i \beta_i = 0$$

$$Y_{.j}^2 = (r\mu + r\alpha_j + \sum_i \epsilon_{ij})^2$$

$$Y_{.j}^2 = r^2\mu^2 + r^2\alpha_j^2 + \sum_i \epsilon_{ij}^2 + \sum_{i \neq g} (\epsilon_{ij} \epsilon_{ig}) + 2r^2\mu\alpha_j +$$

$$2r\mu \sum_i \epsilon_{ij} + 2r\alpha_j \sum_i \epsilon_{ij}$$

$$\sum_j \frac{Y_{.j}^2}{r} = rku^2 + r \sum_j \alpha_j^2 + \frac{1}{r} \sum_j \sum_i \epsilon_{ij}^2 + \frac{1}{r} \sum_j \sum_{i \neq g} \epsilon_{ij} \epsilon_{ig} +$$

$$2r\mu \sum_j \alpha_j + 2\mu \sum_j \sum_i \epsilon_{ij} + 2 \sum_j \alpha_j \sum_i \epsilon_{ij}.$$

By Taking Expectation on both sides.

$$E\left[\sum_j \frac{Y_{.j}^2}{r}\right] = rku^2 + rk\sigma_\alpha^2 + \frac{rk}{r} \sigma^2 + 0 + \dots + 0.$$

$$= E\left[\sum_j \frac{Y_{.j}^2}{r}\right]$$

$$= rku^2$$

$$E(Tr.SS) =$$

Block

$$B.S.S =$$

where;

$$Y_{i.}$$

$$Y_{i.} =$$

$$Y_{i.}^2 = (r\mu +$$

$$Y_{i.}^2 = r^2\mu^2$$

$$+ 2r\mu$$

$$2 \sum_j$$

$$\sum_j \frac{Y_{.j}^2}{r} = r$$

$$\frac{1}{r} \sum_j$$

$$2\mu \sum_j$$

By Taking

$$E\left[\sum_j \frac{Y_{.j}^2}{r}\right] =$$

+0...+0.

$$= E\left[\sum_j \frac{Y_{ij}^2}{k}\right] - E(C.F.)$$

+rk\sigma^2-

$$= rk\mu^2 + rk\sigma_\alpha^2 + k\sigma^2 - rk\mu^2 - rk\sigma_\alpha^2 - \sigma^2$$

)\sigma^2 (II)

$$E(Tr.SS) = (rk - r)\sigma_\alpha^2 + (k-1)\sigma^2 \quad (III)$$

Block Sums of Squares:

$$B.S.S = \frac{\sum_i Y_{ij}^2}{k} - C.F.$$

where;

$$Y_{ij} = \sum_j (\mu + \alpha_j + \beta_i + \epsilon_{ij})$$

$$Y_{ij} = k\mu + \sum_j \alpha_j + k\beta_i + \sum_j \epsilon_{ij}$$

$$Y_{ij}^2 = \left(k\mu + \sum_j \alpha_j + k\beta_i + \sum_j \epsilon_{ij}\right)^2$$

$$Y_{ij}^2 = k^2\mu^2 + \sum_j \alpha_j^2 + \sum_{j \neq h} \alpha_j \alpha_h + k^2\beta_i^2 + \sum_j \epsilon_{ij}^2 + \sum_{j \neq k} \epsilon_{ij} \epsilon_{ik}$$

$$+ 2k\mu \sum_j \alpha_j + 2k^2\mu\beta_i + 2k\mu \sum_j \epsilon_{ij} + 2\sum_j \alpha_j k\beta_i +$$

$$2\sum_j \alpha_j \epsilon_{ij} + 2k\beta_i \sum_j \epsilon_{ij}$$

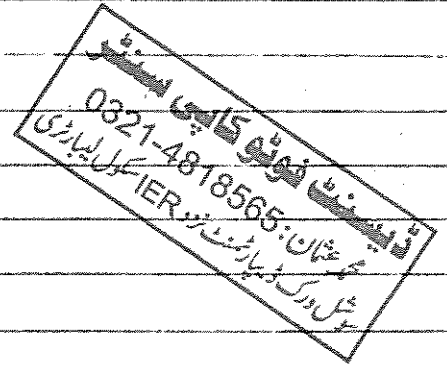
$$\frac{\sum_i Y_{ij}^2}{k} = rk\mu^2 + \frac{1}{k} \sum_i \sum_j \alpha_j^2 + \frac{1}{k} \sum_i \sum_{j \neq h} \alpha_j \alpha_h + k \sum_i \beta_i^2 +$$

$$\frac{1}{k} \sum_i \sum_j \epsilon_{ij}^2 + \frac{1}{k} \sum_i \sum_{j \neq k} \epsilon_{ij} \epsilon_{ik} + 2\mu r \sum_j \alpha_j + 2k\mu \sum_i \beta_i +$$

$$2\mu \sum_i \sum_j \epsilon_{ij} + \frac{2}{k} k \sum_i \beta_i \sum_j \alpha_j$$

By Taking Expectation On both sides.

$$E\left[\frac{\sum_i Y_{ij}^2}{k}\right] = rk\mu^2 + \frac{rk}{k} \sigma_\alpha^2 + 0 + k \sum_i \beta_i^2 + \frac{rk}{k} \sigma^2 + 0 \dots + 0$$



$$E\left(\sum_i \frac{Y_i^2}{k}\right) - E(C.F) = \cancel{rkd\mu^2} + r\sigma_\alpha^2 + k \sum_i \beta_i^2 + r\sigma^2 - \cancel{rkd\mu^2} - \cancel{r\sigma_\alpha^2} - \sigma^2$$

$$E(B.SS) = k \sum_i \beta_i^2 + (r-1)\sigma^2 \quad (10)$$

Error Sums of Square:

$$E(\text{Error SS}) = E(\text{Total SS}) - E(\text{Tr. SS}) - E(\text{Block SS})$$

$$E(\text{Error SS}) = \cancel{(rk-r)}\sigma_\alpha^2 + k \sum_i \beta_i^2 + \cancel{(rk-r)}\sigma^2$$

$$\cancel{(rk-r)}\sigma_\alpha^2 - \cancel{(k-1)}\sigma^2 - k \sum_i \beta_i^2 - \cancel{(r-1)}\sigma^2$$

$$E(\text{Error SS}) = (rk-r-k+r+1)\sigma^2$$

$$= k(r-1) - 1(r-1)\sigma^2$$

$$E(\text{Error SS}) = (r-1)(k-1)\sigma^2$$

Dividing by its df $(r-1)(k-1)$.

$$E(MSE) = \sigma^2 \quad \text{Hence Proved}$$

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Estimate

in R.C

Case -

In
expresses

let
in the
And

we know

Error SS =

Error SS = $\sum_i$

$\left\{ \frac{A}{r} \right\}$

ESS = $\sum \sum Y$

$\frac{1}{r} \sum_j$

Estimation of Missing Observations

in RCB Design:

Case I:

One Missing Observation:

In RCB design the observations y_{ij} is expressed as

$$y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij} \quad \begin{matrix} i=1,2,\dots,r \\ j=1,2,\dots,k \end{matrix}$$

$$\hat{y}_{ij} = \hat{\mu} + \hat{\alpha}_j + \hat{\beta}_i$$

Let the missing observation is $\hat{y}_{cd}$ (i.e. missing in the c th block and d th column).

And the effected totals are:

$$G = G' + \hat{y}_{cd}, \text{ where } G' = rk-1 \text{ observations}$$

$$B_c = B'_c + \hat{y}_{cd}, \text{ where } B'_c = (k-1) \text{ observations}$$

$$T_d = T'_d + \hat{y}_{cd}, \text{ where } T'_d = (r-1) \text{ observations}$$

we know that

$$\text{Error SS} = \text{Total SS} - \text{Block SS} - \text{Treat SS}.$$

$$\text{Error SS} = \sum_{i,j} \hat{y}_{ij}^2 + \hat{y}_{cd}^2 - C \cdot F - \left[\frac{1}{k} \left\{ \sum_i B_i^2 + (B'_c + \hat{y}_{cd})^2 \right\} - C \cdot F \right] - \left[\frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{y}_{cd})^2 \right\} - C \cdot F \right]$$

$$\text{ESS} = \sum \hat{y}_{ij}^2 + \hat{y}_{cd}^2 - C/F - \frac{1}{k} \left\{ \sum_i B_i^2 + (B'_c + \hat{y}_{cd})^2 \right\} - C/F - \frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{y}_{cd})^2 \right\} + C \cdot F$$

$$ESS = \sum_i \sum_j Y_{ij}^2 + \hat{Y}_{cd}^2 - \frac{1}{rk} \left\{ \sum_i B_i^2 + (B'_c + \hat{Y}_{cd})^2 \right\} - \frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{Y}_{cd})^2 \right\} + C.F$$

where;

$$C.F = \frac{(G' + \hat{Y}_{cd})^2}{rk}$$

To minimize the error sum of square, we use the method of least square and taking the partial derivative with respect to unknown value i.e. $\hat{Y}_{cd}$, we get

$$\frac{\partial \text{error SS}}{\partial \hat{Y}_{cd}} = 0 + 2\hat{Y}_{cd} - \frac{1}{rk} \left\{ 0 + 2(B'_c + \hat{Y}_{cd}) \right\} - \frac{1}{r} \left\{ 0 + 2(T'_d + \hat{Y}_{cd}) \right\} + \frac{2}{rk} (G' + \hat{Y}_{cd})$$

$$= 2\hat{Y}_{cd} - \frac{2}{rk} (B'_c + \hat{Y}_{cd}) - \frac{2}{r} (T'_d + \hat{Y}_{cd})$$

Again taking partial derivative w.r.t $\hat{Y}_{cd}$, we have:

$$\frac{\partial^2 \text{error SS}}{\partial \hat{Y}_{cd}^2} = 2 - \frac{2}{r} - \frac{2}{rk} + \frac{2}{rk}$$

$$= 2 + \frac{2 - 2r - 2k}{rk}$$

$$\text{As } \frac{2 - 2r - 2k}{rk} > 0 \quad \text{For } r, k > 1$$

is

$$\frac{\partial^2 \text{error SS}}{\partial \hat{Y}_{cd}^2} > 0$$

so

put

$\frac{\partial \text{error}}{\partial}$

$2\hat{Y}_{cd} -$

Multip

$rk\hat{Y}_{cd} -$

$(rk - r)$

$\hat{Y}_{cd}$

where;

2_{cd}

Case

There

- 1) In d
- 2) In son
- 3) In di

$\frac{\partial \text{errorss}}{\partial \hat{y}_{cd}} = 0$, we have.

$$2\hat{y}_{cd} - \frac{2}{rk} (B'_c + \hat{y}_{cd}) - \frac{2}{r} (T'_d + \hat{y}_{cd}) + \frac{2}{rk} (G' + \hat{y}_{cd}) = 0$$

Multiply both sides by $\frac{rk}{2}$, we get

$$rk\hat{y}_{cd} - r(B'_c + \hat{y}_{cd}) - k(T'_d + \hat{y}_{cd}) + (G' + \hat{y}_{cd}) = 0$$

$$(rk - r - k + 1)\hat{y}_{cd} = rB'_c + kT'_d - G'$$

$$2(T'_d + \hat{y}_{cd})^2$$

$$\hat{y}_{cd} = \frac{rB'_c + kT'_d - G'}{(r-1)(k-1)} = \boxed{\frac{Z_{cd}}{F}}$$

where;

$$F = (r-1)(k-1)$$

$$Z_{cd} = rB'_c + kT'_d - G'$$

Case II:

Two Missing Observations:

There can be three sub cases of this case.

- 1) In different blocks and for different treatment.
- 2) In same block but for different treatment.
- 3) In different blocks but for same treatment.

1) In Different blocks and for different

Treatments:

In RCB design, the observations Y_{ij} can be expressed as:

$$Y_{ij} = \mu + \alpha_j + \beta_i + E_{ij} \quad \begin{matrix} i=1,2,\dots,r \\ j=1,2,\dots,k \end{matrix}$$

Suppose that the two missing observations are in different blocks and for different treatments are Y_{cd} and Y_{ef} .

Now, the effected totals are:

$$B_c = B'_c + \hat{Y}_{cd} +$$

$$B_e = B'_e + \hat{Y}_{ef}$$

$$T_d = T'_d + \hat{Y}_{cd}$$

$$T_f = T'_f + \hat{Y}_{ef}$$

we know that

$$\text{Error SS} = \text{Total SS} - \text{Block SS} - \text{Treatment SS.}$$

$$\begin{aligned} &= \sum_i \sum_j Y_{ij}^2 + \hat{Y}_{cd}^2 + \hat{Y}_{ef}^2 - C \cdot F - \frac{1}{k} \left\{ \sum_i B_i^2 + (B'_c + \hat{Y}_{cd})^2 + \right. \\ &\quad \left. (B'_e + \hat{Y}_{ef})^2 \right\} + C \cdot F - \frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{Y}_{cd})^2 + (T'_f + \hat{Y}_{ef})^2 \right\} \\ &\quad + C \cdot F \end{aligned}$$

where

To find method of derivative $\frac{\partial}{\partial Y_{cd}}$, $\frac{\partial}{\partial Y_{ef}}$

$\frac{\partial}{\partial \text{Error SS}}$
 $\frac{\partial}{\partial Y_{cd}}$

As

So put

$2\hat{Y}_{cd} - \frac{2}{k} \left(\sum_i B_i + B'_c + \hat{Y}_{cd} \right)$
Multiplied

$r k \hat{Y}_{cd} - r \left(\sum_i B_i + B'_c + \hat{Y}_{cd} \right)$

$(rk - r - 1) \hat{Y}_{cd}$

where

E
 Z_{cc}

where;

$$C.F = \frac{(G' + \hat{Y}_{cd} + \hat{Y}_{ef})^2}{rk}$$

To minimize error sums of square, we use the method of least square and take the partial derivatives w.r.t unknown values $\hat{Y}_{cd}$ and $\hat{Y}_{ef}$, we get;

$$\frac{\partial \text{Error SS}}{\partial \hat{Y}_{cd}} = 0 + 2\hat{Y}_{cd} + 0 - \frac{2}{R}(\beta'_c + \hat{Y}_{cd}) - \frac{2}{r}(T'_d + \hat{Y}_{cd}) + \frac{2}{rk}(G' + \hat{Y}_{cd} + \hat{Y}_{ef})$$

$$\text{As } \frac{\partial^2 \text{error SS}}{\partial \hat{Y}_{cd}^2} > 0$$

So put

$$\frac{\partial \text{Error SS}}{\partial \hat{Y}_{cd}} = 0, \text{ we have}$$

$$2\hat{Y}_{cd} - \frac{2}{R}(\beta'_c + \hat{Y}_{cd}) - \frac{2}{r}(T'_d + \hat{Y}_{cd}) + \frac{2}{rk}(G' + \hat{Y}_{cd} + \hat{Y}_{ef}) = 0$$

Multiply both sides by $\frac{rk}{2}$, we get

$$rk\hat{Y}_{cd} - r(\beta'_c + \hat{Y}_{cd}) - k(T'_d + \hat{Y}_{cd}) + (G' + \hat{Y}_{cd} + \hat{Y}_{ef}) = 0$$

$$(rk - r - k + 1)\hat{Y}_{cd} + \hat{Y}_{ef} = r\beta'_c + kT'_d - G'$$

$$E\hat{Y}_{cd} + \hat{Y}_{ef} = Z_{cd} \quad \text{--- (1)}$$

where

$$E = (r-1)(k-1)$$

$$Z_{cd} = r\beta'_c + kT'_d - G'$$

Now, derivative w.r.t $\hat{Y}_{eq}$, we have:

$$\frac{\partial \text{error}_{ss}}{\partial \hat{Y}_{eq}} = 2\hat{Y}_{eq} - \frac{2}{k}(\beta'_e + \hat{Y}_{eq}) - \frac{2}{r}(T'_g + \hat{Y}_{eq}) + \frac{2}{rk}(G' + \hat{Y}_{cd} + \hat{Y}_{eq})$$

As $\frac{\partial^2 \text{error}_{ss}}{\partial \hat{Y}_{eq}^2} > 0$

so

put $\frac{\partial \text{error}_{ss}}{\partial \hat{Y}_{eq}} = 0$

$$2\hat{Y}_{eq} - \frac{2}{k}(\beta'_e + \hat{Y}_{eq}) - \frac{2}{r}(T'_g + \hat{Y}_{eq}) + \frac{2}{rk}(G' + \hat{Y}_{cd} + \hat{Y}_{eq}) = 0$$

Multiply both sides by $\frac{rk}{2}$, we get

$$rk\hat{Y}_{eq} - r(\beta'_e + \hat{Y}_{eq}) - k(T'_g + \hat{Y}_{eq}) + (G' + \hat{Y}_{cd} + \hat{Y}_{eq}) = 0$$

$$(rk - r - k + 1)\hat{Y}_{eq} + \hat{Y}_{cd} = r\beta'_e + kT'_g - G'$$

$$(r-1)(k-1)\hat{Y}_{eq} + \hat{Y}_{cd} = r\beta'_e + kT'_g - G'$$

$$E\hat{Y}_{eq} + \hat{Y}_{cd} = Z_{eq} \quad \text{--- (1)}$$

where

$$E = (r-1)(k-1)$$

$$Z_{eq} = r\beta'_e + kT'_g - G'$$

To find the values of $\hat{Y}_{cd}$ and $\hat{Y}_{eq}$, we solve eq (1) and eq (2) simultaneously.

For This
Eq (2)

(E

$\hat{Y}_{cd}$

by

$\frac{F^2 Z_{cd}}{E}$

(E

$\hat{Y}_{eq}$

$\hat{Y}_{eq}$

$\hat{Y}_{eq}$

2) Im

Tr

Y_{ij} is

For this multiply eq ① by E and subtract
Eq ② from this results, we get

$$\begin{aligned} E^2 \hat{Y}_{cd} + E \hat{Y}_{ef} &= E Z_{cd} \\ \pm \hat{Y}_{cd} \mp E \hat{Y}_{ef} &= \pm Z_{ef} \end{aligned}$$

$$(E^2 - 1) \hat{Y}_{cd} = E Z_{cd} - Z_{ef}$$

$$\hat{Y}_{cd} = \frac{E Z_{cd} - Z_{ef}}{(E^2 - 1)}$$

by putting this result in eq ①, we have.

$$\frac{E^2 Z_{cd} - E Z_{ef}}{(E^2 - 1)} + \hat{Y}_{ef} = Z_{cd}$$

$$\hat{Y}_{ef} = Z_{cd} - \frac{E^2 Z_{cd} - E Z_{ef}}{E^2 - 1}$$

$$\hat{Y}_{ef} = \frac{E^2 Z_{cd} - Z_{cd} - E^2 Z_{cd} + E Z_{ef}}{E^2 - 1}$$

$$\hat{Y}_{ef} = \frac{E Z_{ef} - Z_{cd}}{E^2 - 1}$$

2) In Same Blocks But For Different

Treatments:

In RCB design, the observation
 Y_{ij} is written as:

$$Y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij} \quad \begin{aligned} i &= 1, 2, \dots, r \\ j &= 1, 2, \dots, k \end{aligned}$$

Suppose that the two missing observations are in some blocks but for different treatments is Y_{cd} and Y_{cf} .

Now the effected Totals are:

$$G = G' + \hat{Y}_{cd} + \hat{Y}_{cf}$$

$$B_c = B'_c + \hat{Y}_{cd} + \hat{Y}_{cf}$$

$$T_d = T'_d + \hat{Y}_{cd}$$

$$T_f = T'_f + \hat{Y}_{cf}$$

we know that:

$$\text{Error SS} = \text{Total SS} - \text{Treat SS} - \text{Block SS}$$

$$= \sum_{i,j} Y_{ij}^2 + \hat{Y}_{cd}^2 + \hat{Y}_{cf}^2 - C.F - \frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{Y}_{cd})^2 + (T'_f + \hat{Y}_{cf})^2 \right\} + C.F - \frac{1}{k} \left\{ \sum_i B_i^2 + (B'_c + \hat{Y}_{cd} + \hat{Y}_{cf})^2 \right\} + C.F$$

where;

$$C.F = \frac{(G' + \hat{Y}_{cd} + \hat{Y}_{cf})^2}{rk}$$

To minimize error SS, we use method of least square, take partial derivative w.r.t $\hat{Y}_{cd}$ and $\hat{Y}_{cf}$.

$$\frac{\partial \text{Error SS}}{\partial \hat{Y}_{cd}} = 0 + 2\hat{Y}_{cd} + 0 - \frac{2}{r} (T'_d + \hat{Y}_{cd}) - \frac{2}{k} (B'_c + \hat{Y}_{cd} + \hat{Y}_{cf}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{cf})$$

tions are
treatments

$$\text{As } \frac{\partial^2 \text{errorss}}{\partial \hat{Y}_{cd}^2} > 0$$

so put

$$\frac{\partial \text{ESS}}{\partial \hat{Y}_{cd}} = 0$$

$$\frac{\partial \text{Errorss}}{\partial \hat{Y}_{cd}} = 2\hat{Y}_{cd} - \frac{2}{rk} (\beta'_c + \hat{Y}_{cd} + \hat{Y}_{cf}) - \frac{2}{r} (T'_d + \hat{Y}_{cd}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{cf}) = 0$$

Multiply by $\frac{rk}{2}$ on both sides.

$$= rk\hat{Y}_{cd} - r(\beta'_c + \hat{Y}_{cd} + \hat{Y}_{cf}) - k(T'_d + \hat{Y}_{cd}) + G' + \hat{Y}_{cd} + \hat{Y}_{cf} = 0$$

$$G' + \hat{Y}_{cd} + \hat{Y}_{cf} = 0$$

$$(rk - r - k + 1)\hat{Y}_{cd} + r\hat{Y}_{cf} + \hat{Y}_{cf} = r\beta'_c + kT'_d - G'$$

$$(r-1)(k-1)\hat{Y}_{cd} + (1-r)\hat{Y}_{cf} = r\beta'_c + kT'_d - G'$$

it can be written as

$$E\hat{Y}_{cd} + F\hat{Y}_{cf} = Z_{cd}$$

nd of
w.r.t

Also

$$\frac{\partial \text{ESS}}{\partial \hat{Y}_{cf}} = 0 + 0 + 2\hat{Y}_{cf} - 0 - \frac{2}{rk} (\beta'_c + \hat{Y}_{cd} + \hat{Y}_{cf}) - 0 - 0 - \frac{2}{r} (T'_d + \hat{Y}_{cd}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{cf})$$

(\beta'_c + \hat{Y}_{cd} + \hat{Y}_{cf})

$$\frac{2}{r} (T'_d + \hat{Y}_{cd}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{cf})$$

$$\text{As } \frac{\partial^2 \text{Errorss}}{\partial \hat{Y}_{cf}^2} > 0$$

so

put

$$= 2\hat{Y}_{cf} - \frac{2}{rk} (B'_i + \hat{Y}_{cd} + \hat{Y}_{cf}) - \frac{2}{r} (T'_f + \hat{Y}_{cf}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{cf}) = 0$$

Multiply by $rk/2$ on both sides.

$$= rk\hat{Y}_{cf} - r(B'_i + \hat{Y}_{cd} + \hat{Y}_{cf}) - rk(T'_f + \hat{Y}_{cf}) +$$

$$G' + \hat{Y}_{cd} + \hat{Y}_{cf} = 0$$

$$(rk - r - k + 1)\hat{Y}_{cf} - r\hat{Y}_{cd} + \hat{Y}_{cd} = rB'_i + kT'_f - G'$$

$$(r-1)(k-1)\hat{Y}_{cf} + (1-r)\hat{Y}_{cd} = rB'_i + kT'_f - G'$$

$$E\hat{Y}_{cf} + (k-1)F\hat{Y}_{cd} = Z_{ef}$$

To find the estimated value of $\hat{Y}_{cd}$, $\hat{Y}_{cf}$, we solve eq (I) and eq (II) simultaneously. For this multiply eq (I) by E and eq (II) by F and subtracting the results, we get

$$\begin{aligned} E^2\hat{Y}_{cd} + EF\hat{Y}_{cf} &= EZ_{cd} \\ \pm F^2\hat{Y}_{cd} \pm EF\hat{Y}_{cf} &= \pm FZ_{ef} \end{aligned}$$

$$(E^2 - F^2)\hat{Y}_{cd} = EZ_{cd} - FZ_{ef}$$

$$\hat{Y}_{cd} = \frac{EZ_{cd} - FZ_{ef}}{E^2 - F^2}$$

put this value in eq (II), we have:

$$\frac{E}{E^2 - F^2} (EZ_{cd} - FZ_{ef}) + F\hat{Y}_{cf} = Z_{cd}$$

$$\frac{E^2 Z_{cd}}{E^2}$$

$$F\hat{Y}_{cf}$$

$$F\hat{Y}_{cf} =$$

$$F\hat{Y}_{cf} =$$

$$\hat{Y}_{cf}$$

3) In
Treatm

expressed

Suppose
in dif
treatme
Now

$$\frac{1}{2} (G' + \hat{Y}_{cd} + \hat{Y}_{cf}) = 0$$

$$\frac{E^2 Z_{cd} - EF Z_{cf} + F \hat{Y}_{cf}}{E^2 - F^2} = Z_{cd}$$

$$F \hat{Y}_{cf} = Z_{cd} - \frac{E^2 Z_{cd} - EF Z_{cf}}{E^2 - F^2}$$

$$F \hat{Y}_{cf} = \frac{E^2 Z_{cd} - F^2 Z_{cd} - E^2 Z_{cd} + EF Z_{cf}}{E^2 - F^2}$$

$$F \hat{Y}_{cf} = \frac{-F Z_{cd} + E Z_{cf}}{E^2 - F^2}$$

$$\hat{Y}_{cf} = \frac{E Z_{cf} - F Z_{cd}}{E^2 - F^2}$$

we solve
multiply

3) In Different blocks but for same Treatment

In RCB Design, the Y_{ij} is expressed as:

$$Y_{ij} = \mu + \alpha_j + \beta_i + E_{ij} \quad \begin{matrix} i=1, 2, \dots, r \\ j=1, 2, \dots, k \end{matrix}$$

Suppose that two missing observations are in different blocks but for same treatment are $\hat{Y}_{cd}$ & $\hat{Y}_{ed}$.

Now, the effected totals are:

$$G = G' + \hat{Y}_{cd} + \hat{Y}_{ed}$$

$$B_c = B'_c + \hat{Y}_{cd}$$

$$B_e = B'_e + \hat{Y}_{ed}$$

$$T_d = T'_d + \hat{Y}_{cd} + \hat{Y}_{ed}$$

are:

We know that:

$$\text{Error SS} = \text{Total SS} - \text{Block SS} - \text{Treat SS.}$$

$$= \sum_i \sum_j y_{ij}^2 + \hat{Y}_{cd}^2 + \hat{Y}_{ed}^2 - C \cdot F - \frac{1}{rk} \left\{ \sum_i \beta_i^2 + (\beta'_c + \hat{Y}_{cd})^2 + (\beta'_e + \hat{Y}_{ed})^2 \right\} \\ + C \cdot F - \frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{Y}_{cd} + \hat{Y}_{ed})^2 \right\} + C \cdot F$$

where $C \cdot F = \frac{(G' + \hat{Y}_{cd} + \hat{Y}_{ed})^2}{rk}$

To minimize the error SS, we use method of least square and take partially derivative w.r.t $\hat{Y}_{cd}$ & $\hat{Y}_{ed}$. We have

$$\frac{\partial \text{error SS}}{\partial \hat{Y}_{cd}} = 0 + 2\hat{Y}_{cd} - \frac{1}{rk} \left\{ 2(\beta'_c + \hat{Y}_{cd}) \right\} - \frac{1}{r} \left\{ 2(T'_d + \hat{Y}_{cd} + \hat{Y}_{ed}) \right\} \\ + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{ed}) = 0$$

Multiply both sides by $\frac{rk}{2}$, we get

$$rk\hat{Y}_{cd} - r(\beta'_c + \hat{Y}_{cd}) - rk(T'_d + \hat{Y}_{cd} + \hat{Y}_{ed}) + (G' + \hat{Y}_{cd} + \hat{Y}_{ed}) = 0$$

$$(rk - r - k + 1)\hat{Y}_{cd} - k\hat{Y}_{ed} + \hat{Y}_{ed} = r\beta'_c + kT'_d - G'$$

$$E\hat{Y}_{cd} + F\hat{Y}_{ed} = Z_{cd} \quad \text{--- (1)}$$

where

$$E = (r-1)(k-1)$$

$$F = 1-k$$

$$Z_{cd} = r\beta'_c + kT'_d - G'$$

Also;

$$\frac{\partial \text{err}}{\partial \hat{Y}_{ed}}$$

put

$$2\hat{Y}_{ed} -$$

Multi

$$rk\hat{Y}_{ed} - r$$

$$(rk - r -$$

$$(r-1)(k$$

F

where

To find we solve

purpose by F

Also;

$$\frac{\partial \text{errorss}}{\partial \hat{Y}_{ed}} = \frac{\partial \hat{Y}_{ed}}{\partial \hat{Y}_{ed}} - \frac{2}{rk} (\beta'_e + \hat{Y}_{ed}) - \frac{2}{r} (T'_d + \hat{Y}_{cd} + \hat{Y}_{ed}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{ed})$$

put $\frac{\partial \text{ess}}{\partial \hat{Y}_{ed}} = 0$ as $\frac{\partial^2 \text{ESS}}{\partial \hat{Y}_{ed}^2} > 0$

$$\frac{\partial \hat{Y}_{ed}}{\partial \hat{Y}_{ed}} - \frac{2}{rk} (\beta'_e + \hat{Y}_{ed}) - \frac{2}{r} (T'_d + \hat{Y}_{cd} + \hat{Y}_{ed}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{ed}) = 0$$

method of
derivative

Multiply both sides by $\frac{rk}{2}$, we get:

$$rk\hat{Y}_{ed} - r(\beta'_e + \hat{Y}_{ed}) - rk(T'_d + \hat{Y}_{cd} + \hat{Y}_{ed}) + G' + \hat{Y}_{cd} + \hat{Y}_{ed} = 0$$

$$(rk - r - k + 1)\hat{Y}_{ed} - rk\hat{Y}_{cd} + \hat{Y}_{cd} = r\beta'_e + kT'_d - G'$$

$$(r-1)(k-1)\hat{Y}_{ed} + (1-k)\hat{Y}_{cd} = r\beta'_e + kT'_d - G'$$

$$E\hat{Y}_{ed} + F\hat{Y}_{cd} = Z_{ed} \quad \text{--- (II)}$$

where

$$E = (r-1)(k-1)$$

$$F = 1-k$$

$$Z_{ed} = r\beta'_e + kT'_d - G'$$

To find the estimated values of $\hat{Y}_{ed}$ & $\hat{Y}_{cd}$ we solve eq (I) and eq (II) simultaneously. For this purpose multiply eq (I) by E and eq (II) by F and then subtracting (I) and (II).

$$E^2\hat{Y}_{cd} + EF\hat{Y}_{ed} = EZ_{ed}$$

$$\pm F^2\hat{Y}_{cd} \pm EF\hat{Y}_{ed} = \pm FZ_{ed}$$

$$(E^2 - F^2)\hat{Y}_{cd} = (E - F)Z_{ed} \quad EZ_{ed} - FZ_{ed}$$

$$\hat{Y}_{ed} = \frac{EZ_{ed} - FZ_{ed}}{E^2 - F^2}$$

put this value in eqr ①. we get/have:

$$E \cdot \frac{EZ_{ed} - FZ_{ed}}{E^2 - F^2} + F\hat{Y}_{ed} = Z_{ed}$$

$$\frac{E^2 Z_{ed} - EFZ_{ed}}{E^2 - F^2} + F\hat{Y}_{ed} = Z_{ed}$$

$$F\hat{Y}_{ed} = Z_{ed} - \frac{E^2 Z_{ed} - EFZ_{ed}}{E^2 - F^2}$$

$$F\hat{Y}_{ed} = \frac{E^2 Z_{ed} - F^2 Z_{ed} - E^2 Z_{ed} + EFZ_{ed}}{E^2 - F^2}$$

$$\hat{Y}_{ed} = \frac{EZ_{ed} - FZ_{ed}}{E^2 - F^2}$$

General Case:

"N" Missing Observations

Suppose the missing observations are $Y_{ed}, Y_{ef}, Y_{uv}, \dots$ (say N). Then;

$$\text{Error SS} = \text{Total SS} - \text{Block SS} - \text{Treatment SS}$$

$$\begin{aligned} ESS &= \left[\sum_i \sum_j Y_{ij}^2 + \hat{Y}_{ed}^2 + \hat{Y}_{ef}^2 + \hat{Y}_{uv}^2 + \dots \right] - C/F - \frac{1}{rk} \left\{ \sum_i B_i^2 + (B'_e + \hat{Y}_{ed})^2 + \right. \\ &\quad \left. (B'_e + \hat{Y}_{ef})^2 + (B'_u + \hat{Y}_{uv})^2 + \dots \right\} + C/F - \frac{1}{r} \left\{ \sum_j T_j^2 + (T'_d + \hat{Y}_{ed})^2 + \right. \\ &\quad \left. (T'_e + \hat{Y}_{ef})^2 + (T'_u + \hat{Y}_{uv})^2 + \dots \right\} + C.F \end{aligned}$$

$$\text{where } C.F = \frac{(G' + \hat{Y}_{ed} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots)^2}{rk}$$

To minin
method
derivative
and

$$\frac{\partial \text{Error SS}}{\partial Y_{ed}} =$$

Mult

$$= rk$$

$$= (rk - r -$$

$$(r -$$

when

Now;

$$\frac{\partial \text{Error SS}}{\partial Y_{ef}} =$$

$$= rk \hat{Y}_{ef} -$$

$$(rk - r$$

$$E$$

To minimize the error sum of square, we use the method of least square and take the partial derivative w.r.t unknown i.e $\hat{Y}_{cd}$, $\hat{Y}_{ef}$, $\hat{Y}_{uv}$, ... and equalling it to zero i.e

$$\frac{\partial \text{Error SS}}{\partial \hat{Y}_{cd}} = 2\hat{Y}_{cd} - \frac{1}{k} \{ 2(\beta'_c + \hat{Y}_{cd}) \} - \frac{1}{r} \{ (T'_d + \hat{Y}_{cd}) \} + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots) = 0$$

Multiply both sides by $\frac{rk}{2}$, we get

$$= rk\hat{Y}_{cd} - r(\beta'_c + \hat{Y}_{cd}) - k(T'_d + \hat{Y}_{cd}) + (G' + \hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots) = 0$$

$$-(rk - r - k + 1)\hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots = r\beta'_c + kT'_d - G'$$

$$(r-1)(k-1)\hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots = Z_{cd}$$

$$E\hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots = Z_{cd} \quad \text{--- (i)}$$

where

$$E = (r-1)(k-1)$$

$$Z_{cd} = r\beta'_c + kT'_d - G'$$

Now;

$\hat{Y}_{ef}$,

$$\frac{\partial \text{Error SS}}{\partial \hat{Y}_{ef}} = 2\hat{Y}_{ef} - \frac{2}{k} (\beta'_e + \hat{Y}_{ef}) - \frac{2}{l} (T'_f + \hat{Y}_{ef}) + \frac{2}{rk} (G' + \hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots) = 0$$

SS

$$+ (\beta'_c + \hat{Y}_{cd}) +$$

$$+ \hat{Y}_{cd}) +$$

Multiply both sides by $\frac{rk}{2}$, we get

$$= rk\hat{Y}_{ef} - r(\beta'_e + \hat{Y}_{cd}) - k(T'_f + \hat{Y}_{ef}) + (G' + \hat{Y}_{cd} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots) = 0$$

$$(rk - r - k + 1)\hat{Y}_{ef} + \hat{Y}_{cd} + \hat{Y}_{uv} + \dots = r\beta'_e + kT'_f - G'$$

$$E\hat{Y}_{ef} + \hat{Y}_{cd} + \hat{Y}_{uv} + \dots = Z_{ef} \quad \text{--- (ii)}$$

where, $Z_{ef} = r\beta'_e + kT'_g - G'$
 $E = (r-1)(k-1)$

let M

Now;

$$\frac{\partial \text{Error SS}}{\partial \hat{Y}_{uv}} = 2\hat{Y}_{uv} - \frac{2}{k}(\beta'_u + \hat{Y}_{uv}) - \frac{2}{r}(\tau'_v + \hat{Y}_{uv}) + \frac{2}{rk}(G' + \hat{Y}_{ed} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots) = 0$$

we

Multiply by $\frac{rk}{2}$ on both sides.

$$rk\hat{Y}_{uv} - r(\beta'_u + \hat{Y}_{uv}) - k(\tau'_v + \hat{Y}_{uv}) + (G' + \hat{Y}_{ed} + \hat{Y}_{ef} + \hat{Y}_{uv} + \dots) = 0$$

$$(rk - r - k + 1)\hat{Y}_{uv} + \hat{Y}_{ed} + \hat{Y}_{ef} + \dots = r\beta'_u + k\tau'_v - G'$$

$$E\hat{Y}_{uv} + \hat{Y}_{ef} + \hat{Y}_{ed} + \dots = Z_{uv} \quad \text{--- (iii)}$$

where

$$E = (r-1)(k-1)$$

$$Z_{uv} = r\beta'_u + k\tau'_v - G'$$

By

EP

EB

A

To find

(4) and

(5) by E

and so on, we get N equations to find the estimated values of unknown values. We solve these equations by Matrices.

Convert these equations in matrices form, we get

$$\begin{bmatrix} E & 1 & 1 & 1 & 1 \\ 1 & E & 1 & 1 & 1 \\ 1 & 1 & E & 1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & 1 & E \end{bmatrix} \begin{bmatrix} \hat{Y}_{ed} \\ \hat{Y}_{ef} \\ \hat{Y}_{uv} \\ \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} Z_{ed} \\ Z_{ef} \\ Z_{uv} \\ \vdots \\ \vdots \end{bmatrix}$$

B

B

say M $X = Z$

Thus;

$$MX = Z$$

$$X = M^{-1}Z$$

put the

$$\text{let } M^{-1} = \begin{pmatrix} A & B & B & \dots & B \\ B & A & B & \dots & B \\ B & B & A & \dots & B \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ B & B & B & \dots & A \end{pmatrix}$$

we know that

$$MM^{-1} = I \quad 1-e$$

$$\begin{pmatrix} E & 1 & 1 & \dots & 1 \\ 1 & E & 1 & \dots & 1 \\ 1 & 1 & E & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & E \end{pmatrix} \begin{pmatrix} A & B & B & \dots & B \\ B & A & B & \dots & B \\ B & B & A & \dots & B \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ B & B & B & \dots & A \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

By solving it, we get

$$EA + (N-1)B = 1 \quad (4)$$

$$EB + A + (N-2)B = 0$$

$$A + (E + N-2)B = 0 \quad (5)$$

To find the values of A and B, we solve the eq of (4) and (5) simultaneously. For this, multiply eq (5) by E and Then subtract the results from eq (4).

form,

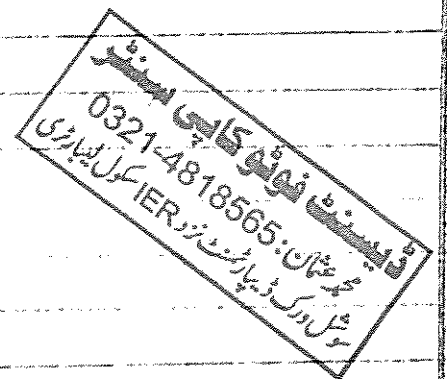
$$\begin{aligned} EA + E(E + N-2)B &= 1 \\ + EA + (N-1)B &= 0 \end{aligned}$$

$$(N-1)B - E(E + N-2)B = 1$$

$$B [(N-1) - E(E + N-2)] = 1$$

$$B = \frac{1}{(N-1) - E(E + N-2)}$$

put the value of B in eq (5), we get



$$A + (E+N-2) \left[\frac{1}{(N-1) - E(E+N-2)} \right] = 0$$

$$A = \frac{-(E+N-2)}{(N-1) - E(E+N-2)}$$

Now,

$$X = M'Z$$

$\hat{y}_{cd}$		A	B	B	...	B	Z_{cd}
$\hat{y}_{ef}$		B	A	B	...	B	Z_{ef}
$\hat{y}_{uv}$		B	B	A	...	B	Z_{uv}
$\vdots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$		B	B	B	...	A	$\vdots$

$$\hat{y}_{cd} = AZ_{cd} + B(Z_{ef} + Z_{uv} + \dots)$$

where $A = \frac{-(E+N-2)}{(N-1) - E(E+N-2)}$, $B = \frac{1}{(N-1) - E(E+N-2)}$

$$\hat{y}_{cd} = \frac{-(E+N-2)}{(N-1) - E(E+N-2)} Z_{cd} + \frac{1}{(N-1) - E(E+N-2)} (Z_{ef} + Z_{uv} + \dots)$$

$$\hat{y}_{cd} = \frac{-(E+N-2)Z_{cd} + Z_{ef} + Z_{uv} + \dots}{(N-1) - E(E+N-2)}$$

$$\hat{y}_{ef} = BZ_{cd} + AZ_{ef} + B(Z_{uv} + \dots)$$

$$\hat{y}_{ef} = AZ_{ef} + B(Z_{cd} + Z_{uv} + \dots)$$

$$\hat{y}_{ef} = \frac{-(E+N-2)Z_{ef} + Z_{cd} + Z_{uv} + \dots}{(N-1) - E(E+N-2)}$$

$$\hat{y}_{uv} =$$

$$\hat{y}_{uv} = -$$

For

$$\hat{y}_d$$

For

$$\hat{y}_c$$

$$\hat{y}_e$$

ANC

After

of various changes

degree estimates reduced

2) The

Therefore, bias and

Bias

where

When not in term ca

$$\hat{y}_{uv} = AZ_{uv} + B(Z_{cd} + Z_{ef} + \dots)$$

$$\hat{y}_{uv} = \frac{-(E+N-2)Z_{uv} + Z_{cd} + Z_{ef} + \dots}{(N-1) + E(E+N-2)} \quad \text{and so on.}$$

For $N=1$

$$\hat{y}_{cd} = \frac{Z_{cd}}{E}$$

For $N=2$

$$\hat{y}_{cd} = \frac{EZ_{cd} - Z_{ef}}{E^2 - 1}$$

$$\hat{y}_{ef} = \frac{EZ_{ef} - Z_{cd}}{E^2 - 1}$$

ANOVA AFTER MISSING OBSERVATIONS:

After the estimation of missing observations, the analysis of variance is performed as usual except the following changes.

1) The total degree of freedom and error degree of freedom is reduced by the number of estimates. e.g. if there is one missing observation, reduced the total df and error df by 1.

2) The treatment SS is biased upward i.e.

$$E(\text{Treat}^2) > \sigma^2$$

Therefore, we use a correction term to reduce this bias and bias is defined as:

$$\text{Bias} = Z = \frac{\{B'_c - (K-1)\hat{y}_{cd}\}^2}{K(K-1)}$$

where one missing observation is $\hat{y}_{cd}$.
When more than one observation is missing but not in the same blocks then the above bias term can be generalized. e.g. In case of two

missing observations.

$$\text{Bias} = Z = \frac{[B'_i - (K-1)\hat{Y}_{cd}]^2 + [B'_e - (K-1)\hat{Y}_g]^2}{k(k-1)}$$

3) When one observation is missing, the S.E for the difference b/w two means is:

$$S.E(\bar{X}_1 - \bar{X}_2) = \sqrt{MSE \left[\frac{2}{r} + \frac{k}{r(r-1)(K-1)} \right]}$$

when more than one observation is missing. We can use Taylor approximation for the S.E of the difference b/w two treatment means. is

$$S.E(\bar{X}_j - \bar{X}_h) = \sqrt{MSE \left(\frac{1}{r_j'} + \frac{1}{r_h'} \right)}$$

where r_j' and r_h' are affected number of replicates defined as:

$$r_j' = \begin{cases} 1, & \text{if both treatments are present} \\ \frac{K-2}{K-1}, & \text{if } j^{\text{th}} \text{ treatment is present and } h^{\text{th}} \text{ treatment is absent.} \\ 0, & \text{if } j^{\text{th}} \text{ treatment is absent.} \end{cases}$$

and

$$r_h' = \begin{cases} 1, & \text{if both treatments are present.} \\ \frac{r-2}{r-1}, & \text{if } h^{\text{th}} \text{ treatment is present and } j^{\text{th}} \text{ treatment is absent.} \\ 0, & \text{if } h^{\text{th}} \text{ treatment is absent.} \end{cases}$$

Efficiency

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P.F

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associated
Design
note:

Efficiency of RCB Design Relative to CR

Design:

An experimental design is said to be more efficient than others if it attains the same precision with less expenditures of time or money and its relative efficiency is usually expressed in terms of a ratio of error variances.

The efficiency of RCB Design vs CR Design can be computed as

$$R.E = \frac{\text{Estimated MSE for CR Design}}{\text{MSE for RCB Design}}$$

$$R.E = \frac{(r-1)S_b^2 + r(k-1)S_e^2}{(rk-1)S_e^2} \times 100.$$

If the error degree of freedom is less than 20, then we must multiply the ratio by a precision factor i.e.

$$P.F = \frac{(n_1+1)(n_2+3)}{(n_1+3)(n_2+1)}$$

where n_1 and n_2 are degree of freedoms associated with Mean Square Error of RCB

Design and CR Design respectively.

Note: less MSE $\rightarrow$ More Efficient.

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Example: (Efficiency).

"ANOVA TABLE"

Source of Variation	Degree of freedom	Sum of Square	Mean Square
Treatment	3	134.45	44.82
Block	4	21.46	5.365
Error	12	26.36	2.197
Total	19	182.27	.

As the error df is less than 20. Therefore;

$$R.E = \frac{(r-1)s_b^2 + r(k-1)s_e^2}{(rk-1)s_e^2} \times \frac{(n_1+1)(n_2+3)}{(n_1+3)(n_2+3)} \times 100.$$

Here;

$$r=5, k=4, n_1=12, n_2=16$$

$$s_b^2 = 5.365, s_e^2 = 2.197, rk=20$$

By putting values:

$$R.E = \frac{(4)(5.365) + (5)(3)(2.197)}{(20-1)(2.197)} \times \frac{(12+1)(16+3)}{(12+3)(16+1)} \times 100.$$

$$R.E = \frac{21.46 + 32.985}{41.743} \times \frac{247}{255} \times 100.$$

$$R.E = \frac{54.415}{41.743} \times \frac{247}{255} \times 100$$

$$R.E = 126.26\%$$

RCB Design is 26% more efficient than CR design.

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Latin Square Design:

An experimental error in RCBD is reduced by controlling the source of variation in one direction that is by grouping units in one way.

When the variation is found in two directions, it becomes necessary to remove these sources of variation simultaneously.

This is done by simultaneous blocking of experimental units in two mutually perpendicular directions known as rows and columns. Since each row and each column is a complete block. This grouping for a balanced arrangement is performed by imposing the restriction that each treatment must appear once and only once in each row and one and only once in each column.

If there are " p " treatments, the experimental area will be divided in " p " rows and " p " columns. Resulting in " p^2 " plots of experimental units.

As the experiment is laid out in a square pattern. Such a double block of experimental units and corresponding double restricted random assignment of treatment is known as Latin Square Design (LSD).

Hence a LS Design is an arrangement of ' p ' treatments in $P \times P$ squares. Where the treatments are grouped

in blocks in two directions. The directions being independent to each other and to the treatments and where the treatments appear once and only once in each direction.

Construction And layout:

Latin Square Design is always constructed by rotation e.g.

In case of four treatments A, B, C, D we get

①

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

A large no. of distinct Latin squares can be derived by interchanging rows and column of a standard Latin Square.

Standard Square:

A Latin square in which treatments in the first row and in the first column are arranged in alphabetical or numerical order.

There is only 1 Standard Square for 2x2 Latin Square Design and 3x3 LSD.

2x2:

But for are

②

A
B
C
D

Similarly,

for 5 Each square.

e.g.

and

lections
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nts
ach

2x2:

A	B
B	A

3x3:

A	B	C
B	C	A
C	A	B

B	C	A
---	---	---

C	A	B
---	---	---

But for 4x4 Latin Square Design, there are 4 possible standard squares.

ected

②	A	B	C	D	③	A	B	C	D
---	---	---	---	---	---	---	---	---	---

B	A	D	C
---	---	---	---

B	A	D	C
---	---	---	---

C	D	B	A
---	---	---	---

C	D	A	B
---	---	---	---

D	C	A	B
---	---	---	---

D	C	B	A
---	---	---	---

④	A	B	C	D
---	---	---	---	---

B	D	A	C
---	---	---	---

C	A	D	B
---	---	---	---

D	C	B	D
---	---	---	---

com
ed

Similarly,

the possible No. of standard squares for 5x5 are 56 and so on.
Each standard square of k x k latin square yields

$k! (k-1)!$ different squares.

for
3x3 LSD.

e.g. 4x4 LSD:

$$4! (4-1)! = (24)(6) = 144$$

and st. square = 4

$$\text{Total Designs} = 144 \times 4 = 576.$$

2x2 → 2 different Designs

3x3 → 12 " "

4x4 → 576 " "

5x5 → 161280 " " etc

Standard Square design is also called reduced Latin Square Design.

Mathematical Model:

In Latin Square Design, the observation $Y_{ij(k)}$ is expressed as:

$$Y_{ij(k)} = \mu + R_i + C_j + \tau_k + E_{ij(k)}$$

where;

μ = overall Mean

R_i = i th row effect.

C_j = j th column effect.

τ_k = k th treatment effect.

$E_{ij(k)}$ = Random Error $\sim NID(0, \sigma^2)$

ANOVA TABLE

S.O.V	D.O.f	S.S	M.S	F-Ratio
B/w rows	$P-1$	SS_{rows}	MSR	
B/w columns	$P-1$	$SS_{columns}$	MSC	
B/w treatments	$P-1$	$SS_{treat.}$	MST	$F = \frac{MST}{MSE}$
Error	$(P-1)(P-1)$ By Subt.	SSE	MSE	
Total	$P^2 - 1$	TSS	—	

$$H_0: \mu_A = \mu_B = \dots = \mu_P$$

Advani

- 1) A LS Control
- 2) The a relative
- 3) A LST a R

Disadv

- 1) A It is When - design
- 2) For does n to gr
- 3) Replica
- 4) In requirem in the approach diffi

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Advantages:

- 1) A LSD reduces the error variation by controlling two sources of variation.
- 2) The analysis of LSD is simple and remains relatively simple with missing observations.
- 3) A LSD is generally more efficient than a RCB Design.

on $Y_{ij}(k)$

Disadvantages:

- 1) A LSD is less flexible than RCB design. It is practical only for 5-10 treatments. When the No. of treatments exceeds 10, this design is seldom used.
- 2) For a small No. of treatments, a LSD does not provide a sufficient No. of replicates to give a valid estimate of error.
- 3) Replications in LSD is costly.
- 4) In agricultural experimentation, the land requirement is rigid. The actual layout in the field may be laborious and the approach to the central main plot becomes difficult.

atio

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A Least Square Estimator of Latin Square Design:

In Latin Square Design, the observations Y_{ijk} is expressed as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ijk}$$

Now if $\hat{\mu}$, $\hat{\alpha}_i$, $\hat{\beta}_j + \hat{\gamma}_k$ are the estimates of μ , α_i , β_j , γ_k respectively. $i = j = k = 1, 2, \dots, p$.

The sum of squares of residuals are:

$$S = \sum_j \sum_i (Y_{ijk} - \hat{Y})^2$$

$\hat{Y} = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j + \hat{\gamma}_k$

$$S = \sum_j \sum_i (Y_{ijk} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k)^2$$

To find the estimators, we differentiate the above eq. w.r.t unknown estimates and then equating it to zero.

differentiate it w.r.t $\hat{\mu}$.

$$\frac{\partial S}{\partial \hat{\mu}} = 2 \sum_j \sum_i (Y_{ijk} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k) (-1)$$

Since

$$\frac{\partial^2 S}{\partial \hat{\mu}^2} > 0$$

put $\frac{\partial S}{\partial \hat{\mu}} = 0$

For Rows

$$S = \sum_j ($$

+ ...

Now

$$\frac{\partial S}{\partial \hat{\alpha}_i} =$$

put

$\frac{\partial}{\partial}$

$$\sum_i Y_i$$

For Col

$$S = \sum_i (Y_i$$

+ ...

$$\sum_j \sum_i (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k) = 0$$

$$\sum_j \sum_i y_{ij(k)} - P\hat{\mu} - P\sum_i \hat{\alpha}_i - P\sum_j \hat{\beta}_j - P\sum_k \hat{\gamma}_k = 0$$

$$\sum_j \sum_i y_{ij(k)} = P\hat{\mu} + P\sum_i \hat{\alpha}_i + P\sum_j \hat{\beta}_j + P\sum_k \hat{\gamma}_k \quad \text{--- (1)}$$

For Rows :

$$S = \sum_j (y_{1j(k)} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta}_j - \hat{\gamma}_k)^2 + \sum_j (y_{2j(k)} - \hat{\mu} - \hat{\alpha}_2 - \hat{\beta}_j - \hat{\gamma}_k)^2$$

$$+ \dots + \sum_j (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k)^2 + \dots$$

Now differentiate

$$\frac{\partial S}{\partial \hat{\alpha}_i} = -2 \sum_j (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k)$$

Since $\frac{\partial^2 S}{\partial \hat{\alpha}_i^2} > 0$

put $\frac{\partial S}{\partial \hat{\alpha}_i} = 0$

$$\sum_j (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k) = 0$$

$$\sum_j y_{ij(k)} - P\hat{\mu} - P\hat{\alpha}_i - \sum_j \hat{\beta}_j - P\sum_k \hat{\gamma}_k = 0$$

$$\sum_j y_{ij(k)} = P\hat{\mu} + P\hat{\alpha}_i + \sum_j \hat{\beta}_j + P\sum_k \hat{\gamma}_k \quad \text{--- (2)}$$

For Columns :

$$S = \sum_i (y_{i1(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_1 - \hat{\gamma}_k)^2 + \sum_i (y_{i2(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_2 - \hat{\gamma}_k)^2$$

$$+ \dots + \sum_i (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_k)^2 + \dots$$

differentiate w.r.t $\hat{\beta}_j$.

$$\frac{\partial S}{\partial \hat{\beta}_j} = -2 \sum_i (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\tau}_k)$$

Since $\frac{\partial^2 S}{\partial \hat{\beta}_j^2} > 0$

put $\frac{\partial S}{\partial \hat{\beta}_j} = 0$

$$\sum_i (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\tau}_k) = 0$$

$$\sum_i Y_{ij(k)} - P\hat{\mu} - \sum_i \hat{\alpha}_i - P\hat{\beta}_j - \sum_k \hat{\tau}_k = 0$$

$$\sum_i Y_{ij(k)} = P\hat{\mu} + \sum_i \hat{\alpha}_i + P\hat{\beta}_j + \sum_k \hat{\tau}_k \quad (3)$$

For Treatments:

$$\frac{\partial S}{\partial \hat{\tau}_k} = -2 \sum_j (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\tau}_k)$$

put $\frac{\partial S}{\partial \hat{\tau}_k} = 0$

$$\sum_j (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\tau}_k) = 0$$

$$\sum_j Y_{ij(k)} - P\hat{\mu} - \sum_i \hat{\alpha}_i - \sum_j \hat{\beta}_j - P\hat{\tau}_k$$

$$\sum_j Y_{ij(k)} = P\hat{\mu} + \sum_i \hat{\alpha}_i + \sum_j \hat{\beta}_j + P\hat{\tau}_k \quad (4)$$

Now if we add p equations in eq (ii) & (iii) then these equations becomes the same as eq No. (1).

This is not a linear solution

$\sum_i$

Under From eq

From

$\sum_j$

$\sum_j Y_{ij}$

From

This means that these Normal equations are not linearly independent. Therefore for unique solution we apply the following conditions

$$\sum_i \hat{\alpha}_i = 0, \quad \sum_j \hat{\beta}_j = 0, \quad \sum_k \hat{\gamma}_k = 0.$$

Under these conditions:

From eq (1)

$$\sum_j \sum_i Y_{ij}(k) = P^2 \hat{\mu}$$

$$\frac{\sum_j \sum_i Y_{ij}(k)}{P^2} = \hat{\mu}$$

(3)
$$\bar{Y}_{..(.)} = \hat{\mu}$$

From eq (1):

$$\sum_j Y_{ij}(k) = P \hat{\mu} + P \hat{\alpha}_i$$

$$= P \bar{Y}_{..(.)} + P \hat{\alpha}_i$$

$$\sum_j Y_{ij}(k) = P(\bar{Y}_{..(.)} + \hat{\alpha}_i)$$

$$Y_{i..(.)} = P(\bar{Y}_{..(.)} + \hat{\alpha}_i)$$

$$\bar{Y}_{i..(.)} = \bar{Y}_{..(.)} + \hat{\alpha}_i$$

(4)
$$\hat{\alpha}_i = \bar{Y}_{i..(.)} - \bar{Y}_{..(.)}$$

eq

From eq (ii):

$$\sum_i Y_{ij}(k) = P \hat{\mu} + P \hat{\beta}_j$$

$$Y_{.j..(.)} = P(\bar{Y}_{..(.)} + \hat{\beta}_j)$$

$$\bar{Y}_{.j(i)} = \bar{Y}_{..(i)} + \hat{\beta}_j$$

$$\boxed{\hat{\beta}_j = \bar{Y}_{.j(i)} - \bar{Y}_{..(i)}}$$

From eq (iv):

$$\sum_R Y_{ij(k)} = P\hat{\mu} + P\hat{\tau}_k$$

$$Y_{..(k)} = P(\bar{Y}_{..(i)} + \hat{\tau}_k)$$

$$\bar{Y}_{..(k)} = \bar{Y}_{..(i)} + \hat{\tau}_k$$

$$\boxed{\hat{\tau}_k = \bar{Y}_{..(k)} - \bar{Y}_{..(i)}}$$

Mean Square Error in LS Design:

$$E(MSE) = \sigma^2$$

Case I:

1) Fixed Effect Model

Case II: Random Effect Model

Case III: Mixed Effect Model.

- when α_i and β_j are fixed, τ_k is random.
- when β_j and τ_k are fixed, α_i is random.
- when α_i and τ_k are fixed, β_j is random.
- τ_k is fixed, α_i, β_j are random.
- α_i is fixed, β_j, τ_k are random.
- β_j is fixed, α_i, τ_k are random.

Case II

Random

In L
express

with α_i

where

E

In random
 β_j and
with α_i

$E(\alpha_i) =$

β_j 's

$E(\beta_j)$

τ_k 's

$E(\tau_k)$

and

So;

E

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Case II :

Random Effect Model

In LS Design, the observations Y_{ijk} is expressed as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ijk}$$

with assumptions $i = j = k = 1, 2, \dots, p$

where $\epsilon_{ijk} \sim NII(0, \sigma^2)$

$$E(\epsilon_{ijk}) = 0, E(\epsilon_{ijk}^2) = \sigma^2, E(\epsilon_{ijk} \epsilon_{ghn}) = 0$$

In random effect model, let α_i , $i \neq g$
 β_j and γ_k are random $j \neq h$
with assumptions $k \neq n$

Design:

$$\alpha_i's \sim NII(0, \sigma_\alpha^2)$$

$$E(\alpha_i) = 0, E(\alpha_i^2) = \sigma_\alpha^2, E(\alpha_i \alpha_g) = 0 \quad i \neq g$$

$$\beta_j's \sim NII(0, \sigma_\beta^2)$$

$$E(\beta_j) = 0, E(\beta_j^2) = \sigma_\beta^2, E(\beta_j \beta_h) = 0 \quad j \neq h$$

$$\gamma_k's \sim NII(0, \sigma_\gamma^2)$$

$$E(\gamma_k) = 0, E(\gamma_k^2) = \sigma_\gamma^2, E(\gamma_k \gamma_n) = 0 \quad k \neq n$$

and $\alpha_i, \beta_j, \gamma_k$ are independent.
So;

$$E(\alpha_i \beta_j) = 0, E(\alpha_i \gamma_k) = 0, E(\beta_j \gamma_k) = 0$$

Collection Factor:

$$C.F. = \frac{G^2}{P^2} \quad \text{where}$$

$$G = \sum_j \sum_i (\mu + \alpha_i + \beta_j + \gamma_k) \pm E_{ij(k)}.$$

$$G = P^2 \mu + P \sum_i \alpha_i + P \sum_j \beta_j + P \sum_k \gamma_k + \sum_j \sum_i E_{ij(k)}$$

squaring on both sides.

$$G^2 = \left(P^2 \mu + P \sum_i \alpha_i + P \sum_j \beta_j + P \sum_k \gamma_k + \sum_j \sum_i E_{ij(k)} \right)^2.$$

$$G^2 = P^4 \mu^2 + P^2 \sum_i \alpha_i^2 + P^2 \sum_{i \neq g} \alpha_i \alpha_g + P^2 \sum_j \beta_j^2 + P^2 \sum_{j \neq h} \beta_j \beta_h +$$

$$P^2 \sum_k \gamma_k^2 + P^2 \sum_{k \neq n} \gamma_k \gamma_n + 2P^3 \mu \sum_j \sum_i E_{ij(k)} + \sum_{j \neq h} \sum_{i \neq g} E_{ij(k)} E_{gh(n)} +$$

$$2P^3 \mu \sum_i \alpha_i + 2P^3 \mu \sum_j \beta_j + 2P^3 \mu \sum_k \gamma_k + 2P^2 \sum_j \sum_i E_{ij(k)} +$$

$$2P^2 \sum_i \alpha_i \sum_j \beta_j + 2P^2 \sum_i \alpha_i \sum_k \gamma_k + 2P \sum_i \alpha_i \sum_j \sum_i E_{ij(k)} +$$

$$2P^2 \sum_j \beta_j \sum_k \gamma_k + 2P \sum_j \beta_j \sum_i \sum_i E_{ij(k)} + 2P \sum_k \gamma_k \sum_j \sum_i E_{ij(k)}$$

Dividing By P^2 and take expectation on both sides.

$$E\left(\frac{G^2}{P^2}\right) = P^2 \mu^2 + P \sigma_\alpha^2 + 0 + P \sigma_\beta^2 + 0 + P \sigma_\gamma^2 + 0 + \sigma^2 + \dots + 0$$

$$E(C.F.) = P^2 \mu^2 + P \sigma_\alpha^2 + P \sigma_\beta^2 + P \sigma_\gamma^2 + \sigma^2 \quad \text{--- ①}$$

Total Sum of Squares:

$$\text{Total SS} = \sum_j \sum_i Y_{ij(k)}^2 - C.F.$$

where;

$$Y_{ij(k)}^2 = \mu^2$$

$$2 \mu E_{ij(k)}$$

$$2 \gamma_k E$$

$$\sum_j \sum_i Y_{ij(k)}$$

$$2 P \mu \sum_i \alpha_i$$

$$2 \sum_i \alpha_i \sum_j$$

$$2 \sum_j \beta_j$$

Taking

$$E\left(\sum_j \sum_i Y_{ij(k)}\right)$$

By

$$E(\text{Total})$$

$$E(\text{Total})$$

where;

$$Y_{ij(k)}^2 = (\mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ij(k)})^2$$

$$Y_{ij(k)}^2 = \mu^2 + \alpha_i^2 + \beta_j^2 + \tau_k^2 + \epsilon_{ij(k)}^2 + 2\mu\alpha_i + 2\mu\beta_j + 2\mu\tau_k +$$

$$2\mu\epsilon_{ij(k)} + 2\alpha_i\beta_j + 2\alpha_i\tau_k + 2\alpha_i\epsilon_{ij(k)} + 2\beta_j\tau_k + 2\beta_j\epsilon_{ij(k)} +$$

$$\epsilon_{ij(k)}$$

$$2\tau_k\epsilon_{ij(k)}$$

2

).

$$\sum_j \sum_i Y_{ij(k)}^2 = P\mu^2 + P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + P\sum_k \tau_k^2 + \sum_j \sum_i \epsilon_{ij(k)}^2 +$$

+

$$2P\mu\sum_i \alpha_i + 2P\mu\sum_j \beta_j + 2P\mu\sum_k \tau_k + 2\mu\sum_j \sum_i \epsilon_{ij(k)} +$$

$$\epsilon_{ij(k)}\epsilon_{ij(k)} +$$

$$2\sum_i \alpha_i \sum_j \beta_j + 2\sum_i \alpha_i \sum_k \tau_k + 2\sum_i \alpha_i \sum_j \epsilon_{ij(k)} + 2\sum_j \beta_j \sum_k \tau_k +$$

$$\epsilon_{ij(k)} +$$

$$2\sum_j \beta_j \sum_i \epsilon_{ij(k)} + 2\sum_k \tau_k \sum_i \epsilon_{ij(k)}.$$

Taking Expectation on both sides.

).

$$E\left(\sum_j \sum_i Y_{ij(k)}^2\right) = P\mu^2 + P\sigma_\alpha^2 + P\sigma_\beta^2 + P\sigma_\tau^2 + P\sigma^2 \quad \text{--- (2)}$$

By eq (2) - eq (1)

$$+ 0 + \sigma^2$$

$$E(\text{Total SS}) = P\mu^2 + P\sigma_\alpha^2 + P\sigma_\beta^2 + P\sigma_\tau^2 + P\sigma^2 - P\mu^2 - P\sigma_\alpha^2 - P\sigma_\beta^2 - P\sigma_\tau^2 - \sigma^2$$

).

$$E(\text{Total SS}) = P(P-1)\sigma_\alpha^2 + P(P-1)\sigma_\beta^2 + P(P-1)\sigma_\tau^2 + (P^2-1)\sigma^2 \quad \text{--- (III)}$$

Rows Sums of Squares:

$$RSS = \sum_i \frac{R_i^2}{P} - C.F$$

$$R_i = \sum_j Y_{ij}(K)$$

$$R_i = \sum_j (\mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ij}(K))$$

$$R_i^2 = \left(P\mu + P\alpha_i + \sum_j \beta_j + \sum_k \gamma_k + \sum_j \epsilon_{ij}(K) \right)^2$$

$$R_i^2 = P^2\mu^2 + P^2\alpha_i^2 + \sum_j \beta_j^2 + \sum_{j \neq h} \beta_j \beta_h + \sum_k \gamma_k^2 + \sum_{k \neq n} \gamma_k \gamma_n +$$

$$\sum_j \epsilon_{ij}(K)^2 + \sum_{j \neq h} (\epsilon_{ij}(K) \epsilon_{ih}(K)) + 2P\mu\alpha_i + 2P\mu \sum_j \beta_j +$$

$$2P\mu \sum_k \gamma_k + 2P\mu \sum_j \epsilon_{ij}(K) + 2P\alpha_i \sum_j \beta_j + 2P\alpha_i \sum_k \gamma_k +$$

$$2P\alpha_i \sum_j \epsilon_{ij}(K) + 2 \sum_j \beta_j \sum_k \gamma_k + 2 \sum_j \beta_j \sum_j \epsilon_{ij}(K) + 2 \sum_k \gamma_k \sum_j \epsilon_{ij}(K)$$

Applying $\sum_i$ and $\div$ by 'P':

$$\frac{\sum_i R_i^2}{P} = P\mu^2 + P \sum_i \alpha_i^2 + \frac{1}{P} P \sum_j \beta_j^2 + 0 + \frac{1}{P} P \sum_k \gamma_k^2 + 0 +$$

$$\frac{1}{P} \sum_j \sum_j \epsilon_{ij}(K)^2 + 0 + \dots + 0$$

By taking Expectation on both side

$$E\left[\frac{\sum_i R_i^2}{P}\right] = P\mu^2 + P\sigma_\alpha^2 + P\sigma_\beta^2 + P\sigma_\gamma^2 + P\sigma^2 \quad \text{--- (iv)}$$

By eq (iv) - eq (i)

$$E(\text{Rows SS}) = P\mu^2 + P\sigma_\alpha^2 + P\sigma_\beta^2 + P\sigma_\gamma^2 + P\sigma^2 - \cancel{P\mu^2} - \cancel{P\sigma_\alpha^2} - \cancel{P\sigma_\beta^2} - \cancel{P\sigma_\gamma^2} - \sigma^2$$

E(Rows

Column

C:

where;

$C_j =$

$C_j^2 =$

$C_j^2 = P\mu^2 +$

$\sum_i \epsilon_{ij}(K)^2$

$2P\mu \sum_k \gamma_k$

$2 \sum_i \alpha_i$

By Ap on

$$E\left[\frac{\sum_i C_i^2}{P}\right] =$$

$$E(CSS) = P\sigma$$

$$E(\text{Rows SS}) = P(P-1)\sigma_\alpha^2 + P(P-1)\sigma_\beta^2 + (P-1)\sigma^2 \quad \text{--- (v)}$$

Columns Sums Of Squares:

$$CSS = \sum_j \frac{C_j^2}{P} - C.F$$

where;

$$C_j = \sum_i Y_{ij}(x)$$

$$C_j = \sum_i (\mu + \alpha_i + \beta_j + r_k + \epsilon_{ij}(x))$$

$$C_j^2 = \left(P\mu + \sum_i \alpha_i + P\beta_j + \sum_k r_k + \sum_i \epsilon_{ij}(x) \right)^2$$

$$C_j^2 = P^2\mu^2 + \sum_i \alpha_i^2 + \sum_{i \neq g} (\alpha_i \alpha_g) + P^2\beta_j^2 + \sum_k r_k^2 + \sum_{k \neq n} r_k r_n +$$

$$\sum_i \epsilon_{ij}(x)^2 + \sum_{i \neq g} \epsilon_{ij}(x) \epsilon_{gj}(x) + 2P\mu \sum_i \alpha_i + 2P\mu P\beta_j +$$

$$2P\mu \sum_k r_k + 2P\mu \sum_i \epsilon_{ij}(x) + 2P\beta_j \sum_i \alpha_i + 2\sum_i \alpha_i \sum_k r_k +$$

$$2\sum_i \alpha_i \sum_i \epsilon_{ij}(x) + 2P\beta_j \sum_k r_k + 2P\beta_j \sum_i \epsilon_{ij}(x) + 2\sum_k r_k \sum_i \epsilon_{ij}(x)$$

By Applying $\sum_j$ and taking Expectation on both sides and divided by 'P'

$$E\left[\frac{\sum_j C_j^2}{P}\right] = P^2\mu^2 + P\sigma_\alpha^2 + 0 + P^2\sigma_\beta^2 + P\sigma_r^2 + 0 + P\sigma^2 + 0 + 0$$

$$E(CSS) = \frac{P^2\mu^2 + P\sigma_\alpha^2 + P^2\sigma_\beta^2 + P\sigma_r^2 + P\sigma^2 - P^2\mu^2 - P\sigma_\alpha^2 - P\sigma_\beta^2 - P\sigma_r^2 - P\sigma^2}{P} =$$

$$= P(P-1)\sigma_\beta^2 + (P-1)\sigma^2 \quad \text{--- (vi)}$$

Treatment Sum of Squares:

$$Tr SS = \sum_k \frac{T_k^2}{p} - G.F$$

where

$$T_k = \sum_{j \text{ or } i} (Y_{ij(k)})$$

$$T_k = \sum_{j \text{ or } i} (\mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ij(k)})$$

$$T_k^2 = \left(p\mu + \sum_i \alpha_i + \sum_j \beta_j + p\tau_k + \sum_i \varepsilon_{ij(k)} \right)^2$$

$$T_k^2 = p^2 \mu^2 + \sum_i \alpha_i^2 + \sum_{i \neq g} \alpha_i \alpha_g + \sum_j \beta_j^2 + \sum_{j \neq h} \beta_j \beta_h + p^2 \tau_k^2 +$$

$$\sum_i \varepsilon_{ij(k)}^2 + \sum_{i \neq g} \varepsilon_{ij(k)} \varepsilon_{ig(k)} + 2p\mu \sum_i \alpha_i + 2p\mu \sum_j \beta_j + 2p\mu p\tau_k +$$

$$2p\mu \sum_i \varepsilon_{ij(k)} + 2 \sum_i \alpha_i \sum_j \beta_j + 2 \sum_i \alpha_i p\tau_k + 2 \sum_i \alpha_i \sum_j \varepsilon_{ij(k)} +$$

$$2 \sum_j \beta_j p\tau_k + 2 \sum_j \beta_j \sum_i \varepsilon_{ij(k)} + 2p\tau_k \sum_i \varepsilon_{ij(k)}$$

Dividing by 'p', applying $\sum_k$ and

taking expectation

$$E\left(\sum_k \frac{T_k^2}{p}\right) = p\mu^2 + p\sigma_\alpha^2 + p\sigma_\beta^2 + p\sigma_\tau^2 + p\sigma$$

$$E(TrSS) = \cancel{p\mu^2} + \cancel{p\sigma_\alpha^2} + \cancel{p\sigma_\beta^2} + \cancel{p\sigma_\tau^2} + p\sigma - \cancel{p\mu^2} -$$

$$\cancel{p\sigma_\alpha^2} - \cancel{p\sigma_\beta^2} - \cancel{p\sigma_\tau^2} - \sigma^2$$

$$= p(p-1)\sigma_\tau^2 + (p-1)\sigma^2 \quad \text{--- (iii)}$$

Error

E(Error)

By

E(Error)

E(Error SS)

E(M)

Case

E(M)

In
expos

with

i.e

Error Sums of Square:

$$E(\text{Error SS}) = E(\text{Total SS}) - E(\text{Tr. SS}) - E(\text{CSS}) - E(\text{RSS})$$

By eq (3) - eq (vi) - eq (vi) - eq (v)

$$E(\text{Error SS}) = (P^2 - 1 - P + 1 - P + 1 - P + 1) \sigma^2$$

$$= (P^2 - 3P + 2) \sigma^2$$

$$= (P^2 - 2P - P + 2) \sigma^2$$

$$= [P(P-2) - P(P-2)] \sigma^2$$

$$E(\text{Error SS}) = (P-2)(P-1) \sigma^2$$

Dividing by its degree of freedom

$$E(\text{MSE}) = \sigma^2$$

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Case I:

Fixed Effect Model

$$E(\text{MSE}) = \sigma^2$$

In LS Design, the observations $Y_{ijk(k)}$ is expressed as

$$Y_{ijk(k)} = \mu + \alpha_i + \beta_j + \gamma_k + E_{ijk(k)}$$

with assumptions

$$i = j = k = 1, 2, \dots, P$$

$$E_{ijk(k)} \sim NII(0, \sigma^2)$$

i.e

$$E(E_{ijk(k)}) = 0, \quad E(E_{ijk(k)}^2) = \sigma^2, \quad E(E_{ijk(k)} E_{qmn(m)}) = 0$$

$$\begin{aligned} i &\neq q \\ j &\neq h \\ k &\neq n \end{aligned}$$

In fixed effect Model, $\alpha_i, \beta_j, \gamma_k$ are fixed with assumptions

$$\sum_i \alpha_i = 0, \sum_j \beta_j = 0, \sum_k \gamma_k = 0$$

also the rows, columns and treatment effects are independently distributed.

Correction Factor:

$$C.F. = \frac{G^2}{P^2}$$

where

$$G = \sum_j \sum_i (Y_{ij(k)}) = \sum_j \sum_i (\mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ij(k)})$$

$$G^2 = \left(P\mu + P \sum_i \alpha_i + P \sum_j \beta_j + P \sum_k \gamma_k + \sum_j \sum_i \epsilon_{ij(k)} \right)^2$$

$$G^2 = \left(P\mu + \sum_j \sum_i \epsilon_{ij(k)} \right)^2$$

$$G^2 = P^2 \mu^2 + \sum_j \sum_i \epsilon_{ij(k)}^2 + \sum_{\substack{j \neq h \\ k \neq n}} \sum_i \epsilon_{ij(k)} \epsilon_{ih(n)} + 2 P \mu \sum_j \sum_i \epsilon_{ij(k)}$$

Dividing both sides by P^2 and apply expectation we get.

$$E\left(\frac{G^2}{P^2}\right) = P\mu^2 + \frac{1}{P^2} P \sigma^2 + 0 + 0$$

$$E(C.F.) = P\mu^2 + \sigma^2 \quad \text{--- (1)}$$

are

Total Sum of Squares:

$$TSS = \sum_j \sum_i Y_{ij(k)}^2 - C.F$$

and

$$Y_{ij(k)}^2 = (\mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ij(k)})^2$$

effects

$$= \mu^2 + \alpha_i^2 + \beta_j^2 + \tau_k^2 + \epsilon_{ij(k)}^2 + 2\mu\alpha_i + 2\mu\beta_j + 2\mu\tau_k + 2\mu\epsilon_{ij(k)} + 2\alpha_i\beta_j + 2\alpha_i\tau_k + 2\alpha_i\epsilon_{ij(k)} + 2\beta_j\tau_k + 2\beta_j\epsilon_{ij(k)} + 2\tau_k\epsilon_{ij(k)}$$

$$\sum_j \sum_i Y_{ij(k)}^2 = P\mu^2 + P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + P\sum_k \tau_k^2 + \sum_j \sum_i \epsilon_{ij}^2 + 2\mu P\sum_i \alpha_i$$

+ $\epsilon_{ij(k)}$

$$+ 2\mu P\sum_j \beta_j + 2\mu P\sum_k \tau_k + 2\mu \sum_j \sum_i \epsilon_{ij(k)} + 2\sum_i \alpha_i \sum_j \beta_j +$$

$$2\sum_i \alpha_i \sum_k \tau_k + 2\sum_i \alpha_i \sum_j \epsilon_{ij(k)} + 2\sum_j \beta_j \sum_k \tau_k + 2\sum_j \beta_j \sum_i \epsilon_{ij(k)} +$$

$$2\sum_k \tau_k \sum_i \epsilon_{ij(k)}$$

By Applying expectation on Both sides:

$$E\left[\sum_j \sum_i Y_{ij(k)}^2\right] = P\mu^2 + P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + P\sum_k \tau_k^2 + P\sigma^2 \quad \text{--- (I)}$$

$$E(TSS) = \text{eq (I)} - \text{eq (I)}$$

ply

$$E(TSS) = P\mu^2 + P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + P\sum_k \tau_k^2 + P\sigma^2 - P\mu^2 - \sigma^2$$

$$E(TSS) = P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + P\sum_k \tau_k^2 + (P-1)\sigma^2 \quad \text{--- (II)}$$

Row Sum of Squares:

$$RSS = \sum_i \frac{R_i^2}{P} - C.F$$

$$R_i^2 = \left(\sum_j Y_{ij(k)}\right)^2 = \sum_j (\mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ij(k)})$$

$$R_i = \mu + \alpha_i + \sum_j \beta_j + \sum_k \gamma_k + \sum_j \epsilon_{ij(k)}$$

$$R_i^2 = \left(\mu + \alpha_i + \sum_j \epsilon_{ij(k)} \right)^2$$

$$R_i^2 = \mu^2 + \alpha_i^2 + \sum_j \epsilon_{ij(k)}^2 + \sum_{\substack{j \neq h \\ k \neq n}} \epsilon_{ij(k)} \epsilon_{ih(n)} +$$

$$2\mu\alpha_i + 2\mu \sum_j \epsilon_{ij(k)} + 2\alpha_i \sum_j \epsilon_{ij(k)}$$

Sum it over "i" and dividing by "P" and then applying expectation on both sides we get

$$E\left[\frac{\sum_i R_i^2}{P}\right] = \mu^2 + P \sum_i \alpha_i^2 + \frac{1}{P} P \sigma^2 + 0 + 0 + 0 + 0$$

$$= \mu^2 + P \sum_i \alpha_i^2 + P \sigma^2 \quad \text{--- (iv)}$$

$$\text{eq (iv)} \stackrel{!}{=} \text{eq (i)}$$

$$E(RSS) = P\mu^2 + P \sum_i \alpha_i^2 + P\sigma^2 - \frac{P^2 \mu^2}{P} - \sigma^2$$

$$E(RSS) = P \sum_i \alpha_i^2 + (P-1)\sigma^2 \quad \text{--- (v)}$$

Columns Sums of Squares:

$$CSS = \sum_j \frac{C_j^2}{P} - C.F$$

$$\text{where } C_j = \sum_i Y_{ij(k)} = \sum_i (\mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ij(k)})$$

$$C_j = \mu + \sum_i \alpha_i + P\beta_j + \sum_k \gamma_k + \sum_i \epsilon_{ij(k)}$$

$$C_j^2 = \left(\mu + P\beta_j + \sum_i \epsilon_{ij(k)} \right)^2$$

$$\sum_i \alpha_i = 0$$

$$\sum_k \gamma_k = 0$$

$$C_j^2 = P\mu^2$$

$$2\mu \sum_i \epsilon_{ij(k)}$$

Sum it
applying

$$E\left[\frac{\sum_j C_j^2}{P}\right]$$

$$E(CSS)$$

$$E(CSS)$$

$$E$$

Theater

Tr

where;

$$T_k =$$

$$T_k^2 =$$

$$T_k^2 = P^2$$

$$2\mu \sum_i \epsilon_{ij(k)}$$

$$C_j^2 = P\mu^2 + P\beta_j^2 + \sum_i E_{ij(k)}^2 + \sum_{\substack{i \neq j \\ k \neq h}} E_{ij(k)} E_{jh(k)} + 2P\mu\beta_j + 2P\mu \sum_i E_{ij(k)} + 2P\beta_j \sum_i E_{ij(k)}$$

Sum it over "j" values, divided by "P" and applying expectation on both sides. we get.

$$E\left[\frac{\sum_j C_j^2}{P}\right] = P\mu^2 + P \sum_j \beta_j^2 + \frac{1}{P} P\sigma^2 + 0 \dots + 0$$

$$= P\mu^2 + P \sum_j \beta_j^2 + P\sigma^2 \quad (6)$$

$$E(CSS) = \text{eq (6)} - \text{eq (1)}$$

$$E(CSS) = \cancel{P\mu^2} + P \sum_j \beta_j^2 + P\sigma^2 - \cancel{P\mu^2} - \sigma^2$$

$$E(CSS) = P \sum_j \beta_j^2 + (P-1)\sigma^2 \quad (7)$$

Treatment Sum of Square:

$$Tr. SS = \sum_k \frac{T_k^2}{P} - C.F$$

where;

$$T_k = \sum_{i \in j} (\mu + \alpha_i + \beta_j + \tau_k + E_{ij(k)})$$

$$T_k = P\mu + \sum_i \alpha_i + \sum_j \beta_j + P\tau_k + \sum_i E_{ij(k)}$$

$$T_k^2 = (P\mu + P\tau_k + \sum_i E_{ij(k)})^2$$

$$\sum_i \alpha_i = \sum_j \beta_j = 0$$

$$T_k^2 = P^2\mu^2 + P^2\tau_k^2 + \sum_i E_{ij(k)}^2 + \sum_{\substack{i \neq j \\ k \neq h}} E_{ij(k)} E_{jh(k)} + 2P\mu P\tau_k + 2P\mu \sum_i E_{ij(k)} + 2P\tau_k \sum_i E_{ij(k)}$$

Sum it over "ik", dividing by "P", and apply expectation on both sides.

$$E\left[\frac{\sum_k T_k^2}{P}\right] = P\mu^2 + P\sum_k \bar{r}_k^2 + \frac{1}{P} P\sigma^2 + 0 \dots + 0$$

$$= P\mu^2 + P\sum_k \bar{r}_k^2 + P\sigma^2 \quad (8)$$

$$E(T_r SS) = \text{eq (8)} - \text{eq (1)}$$

$$E(T_r SS) = P\mu^2 + P\sum_k \bar{r}_k^2 + P\sigma^2 - P\mu^2 - \sigma^2$$

$$= P\sum_k \bar{r}_k^2 + (P-1)\sigma^2 \quad (9)$$

Error Sums of Square:

$$E(\text{Error SS}) = E(\text{Total SS}) - E(\text{Rows SS}) - E(\text{Columns SS}) - E(\text{Treatment SS})$$

$$\text{eq (3)} - \text{eq (5)} - \text{eq (7)} - \text{eq (9)}$$

$$= P\sum_i \bar{d}_i^2 + P\sum_j \bar{\beta}_j^2 + P\sum_k \bar{r}_k^2 + (P^2-1)\sigma^2 - P\sum_i \bar{d}_i^2 - (P-1)\sigma^2$$

$$- P\sum_j \bar{\beta}_j^2 - (P-1)\sigma^2 - P\sum_k \bar{r}_k^2 - (P-1)\sigma^2$$

$$E(ESS) = (P^2 - 1 - P + 1 - P + 1 - P + 1)\sigma^2$$

$$= (P^2 - 3P + 2)\sigma^2$$

$$= (P-1)(P-2)\sigma^2$$

Dividing by all df $(P-1)(P-2)$ on both sides

$$E(MSE) = \sigma^2$$

Mixed Effect Model:

f) When β_j is constant/fixed and α_i, r_k are random.

$$E(MSE) = \sigma^2$$

In LS Design, the observation y_{ijk} is written as

$$y_{ijk} = \mu + \alpha_i + \beta_j + r_k + \epsilon_{ijk} \quad i, j, k = 1, 2, \dots, p$$

with assumptions

$$\epsilon_{ijk} \sim NII(0, \sigma^2)$$

i.e.

$$E(\epsilon_{ijk}) = 0, E(\epsilon_{ijk}^2) = \sigma^2, E(\epsilon_{ijk}\epsilon_{lmn}) = 0$$

α_i 's and r_k 's are random

with assumptions

$$\alpha_i \sim NII(0, \sigma_\alpha^2)$$

i.e.

$$E(\alpha_i) = 0, E(\alpha_i^2) = \sigma_\alpha^2, E(\alpha_i \alpha_j) = 0 \quad i \neq j$$

$$r_k \sim NII(0, \sigma_r^2)$$

i.e.

$$E(r_k) = 0, E(r_k^2) = \sigma_r^2, E(r_k r_n) = 0 \quad k \neq n$$

and the columns effects are fixed such that

$$\sum_j \beta_j = 0$$

and Rows, Columns, treatments effects are treated independently.

Now;

C.F:

$$C.F = \frac{G^2}{p^2}$$

where $G = \sum_j \sum_i y_{ij(k)}$

$$G = \sum_j \sum_i (\mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ij}(k))$$

$$G = P^2 \mu + P \sum_i \alpha_i + P \sum_j \beta_j + P \sum_k \gamma_k + \sum_j \sum_i \epsilon_{ij}(k)$$

$$\because \sum_j \beta_j = 0$$

$$G^2 = \left(P^2 \mu + P \sum_i \alpha_i + P \sum_k \gamma_k + \sum_j \sum_i \epsilon_{ij}(k) \right)^2$$

By Equating on both sides

$$G^2 = P^2 \mu^2 + P^2 \sum_i \alpha_i^2 + P^2 \sum_{i \neq j} (\alpha_i \alpha_j) + P^2 \sum_k \gamma_k^2 + P^2 \sum_{k \neq n} (\gamma_k \gamma_n) +$$

$$\sum_j \sum_i \epsilon_{ij}(k)^2 + \sum_{\substack{i \neq j \\ k \neq n}} \sum_j \sum_i \epsilon_{ij}(k) \epsilon_{jn}(n) + 2P^2 \mu \sum_i \alpha_i + 2P^2 \mu \sum_k \gamma_k +$$

$$2P^2 \mu \sum_j \sum_i \epsilon_{ij}(k) + 2P^2 \sum_i \alpha_i \sum_k \gamma_k + 2P^2 \sum_i \alpha_i \sum_j \epsilon_{ij}(k) +$$

$$2P^2 \sum_k \gamma_k \sum_j \epsilon_{ij}(k)$$

Dividing both sides by P^2 and applying expectation we get

$$E\left[\frac{G^2}{P^2}\right] = P^2 \mu^2 + P^2 \sigma_\alpha^2 + 0 + P^2 \sigma_\gamma^2 + 0 + \frac{1}{P^2} P^2 \sigma^2 + 0 + 0.$$

$$E(C.F) = P^2 \mu^2 + P^2 \sigma_\alpha^2 + P^2 \sigma_\gamma^2 + \sigma^2 \quad \text{--- (1)}$$

$$TSS = \sum_j \sum_i y_{ij}(k)^2 - C.F.$$

$$y_{ij}(k)^2 = (\mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ij}(k))^2$$

$$y_{ij}(k)^2 = \mu^2 + \alpha_i^2 + \beta_j^2 + \gamma_k^2 + \epsilon_{ij}(k)^2 + 2\mu\alpha_i + 2\mu\beta_j + 2\mu\gamma_k + 2\mu\epsilon_{ij}(k) + 2\alpha_i\beta_j + 2\alpha_i\gamma_k + 2\alpha_i\epsilon_{ij}(k) + 2\beta_j\gamma_k + 2\beta_j\epsilon_{ij}(k) + 2\gamma_k\epsilon_{ij}(k).$$

$$\sum_j \sum_i y_{ij}(k)^2$$

$$+ 2P\mu$$

$$+ 2\sigma^2$$

By A

$$E\left(\sum_j \sum_i y_{ij}(k)^2\right)$$

$$= P^2 \mu^2$$

$$E(TSS) = P$$

Row

$$R_i =$$

$$R_i =$$

$$R_i^2 = (P)$$

$$R_i^2 = P^2 \mu^2 + P$$

$$2P\mu\alpha_i +$$

$$2\epsilon_{ij}(k) \sum_j$$

$$\begin{aligned} \sum_i \sum_j Y_{ij(k)}^2 &= P\mu^2 + P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + P\sum_k r_k^2 + \sum_i \sum_j \epsilon_{ij(k)}^2 + 2P\mu \sum_i \alpha_i \\ &+ 2P\mu \sum_j \beta_j + 2P\mu \sum_k r_k + 2\mu \sum_i \sum_j \epsilon_{ij(k)} + 2\sum_i \alpha_i \sum_j \beta_j + 2\sum_i \alpha_i \sum_k r_k \\ &+ 2\sum_i \alpha_i \sum_j \epsilon_{ij(k)} + 2\sum_j \beta_j \sum_k r_k + 2\sum_j \beta_j \sum_i \epsilon_{ij(k)} + 2\sum_k r_k \sum_i \epsilon_{ij(k)} \end{aligned}$$

By Applying expectation on both sides.

$$E\left(\sum_i \sum_j Y_{ij(k)}^2\right) = P\mu^2 + P\sum_i \sigma_\alpha^2 + P\sum_j \sigma_\beta^2 + P\sum_k r_k^2 + P\sigma^2 \quad (2)$$

$$\text{eq(2)} - \text{eq(1)}$$

$$= P\mu^2 + P\sum_i \sigma_\alpha^2 + P\sum_j \sigma_\beta^2 + P\sum_k r_k^2 + P\sigma^2 - P\mu^2 - P\sigma_\alpha^2 - P\sigma_\beta^2 - P\sigma^2$$

$$E(TSS) = P(P-1)\sigma_\alpha^2 + P\sum_j \sigma_\beta^2 + P(P-1)r_k^2 + P(P-1)\sigma^2 \quad (3)$$

$$\text{Rows SS} = \frac{\sum_i R_i^2}{P} - C.F.$$

$$R_i = \sum_j Y_{ij(k)} = \sum_j (\mu + \alpha_i + \beta_j + r_k + \epsilon_{ij(k)})$$

$$R_i = P\mu + P\alpha_i + \sum_j \beta_j + \sum_k r_k + \sum_j \epsilon_{ij(k)}$$

$$\sum_j \beta_j = 0$$

$$R_i^2 = \left(P\mu + P\alpha_i + \sum_k r_k + \sum_j \epsilon_{ij(k)} \right)^2$$

$$R_i^2 = P\mu^2 + P\alpha_i^2 + \sum_k r_k^2 + \sum_{k \neq n} r_k r_n + \sum_i \epsilon_{ij(k)}^2 + \sum_{i \neq h} \epsilon_{ij(k)} \epsilon_{ih(n)} +$$

$$2P\mu\alpha_i + 2P\mu \sum_k r_k + 2P\mu \sum_j \epsilon_{ij(k)} + 2P\alpha_i \sum_k r_k + 2P\alpha_i \sum_j \epsilon_{ij(k)} +$$

$$2\sum_k r_k \sum_j \epsilon_{ij(k)}$$

Sum it over "i" values, divided by "P" and apply expectation on both sides. we get.

$$E\left[\frac{\sum R_i^2}{P}\right] = \frac{P^2 \mu^2}{P} + \frac{P^2 \sigma_a^2}{P} + \frac{P^2 \sigma_k^2}{P} + \frac{P^2 \sigma^2}{P} + \dots + 0$$

$$= P\mu^2 + P\sigma_a^2 + P\sigma_k^2 + P\sigma^2 \quad (4)$$

$$E(RSS) = \cancel{P\mu^2} + P\sigma_a^2 + \cancel{P\sigma_k^2} + P\sigma^2 - \cancel{P\mu^2} - \cancel{P\sigma_a^2} - \cancel{P\sigma_k^2} - \sigma^2$$

$$E(RSS) = P(P-1)\sigma_a^2 + (P-1)\sigma^2 \quad (5)$$

$$CSS = \frac{\sum C_j^2}{P} - C.F.$$

$$C_j = \sum_i Y_{ij}(x) = \sum_i (\mu + \alpha_i + \beta_j + r_k + \epsilon_{ij}(x))$$

$$C_j^2 = \left(P\mu + \sum_i \alpha_i + P\beta_j + \sum_k r_k + \sum_i \epsilon_{ij}(x) \right)^2$$

By solving it we get

$$E\left[\frac{\sum C_j^2}{P}\right] = \frac{P \sum \beta_j^2}{P} + (P-1)\sigma^2 \quad (6)$$

Treatment SS:

$$Tr. SS = \frac{\sum T_k^2}{P} - C.F.$$

where

$$T_k = \sum_{i \alpha j} Y_{ij}(x) = \sum_{i \alpha j} (\mu + \alpha_i + \beta_j + r_k + \epsilon_{ij}(x))$$

$$T_k^2 = \left(P\mu + \sum_i \alpha_i + \sum_j \beta_j + PR_k + \sum_j \epsilon_{ij}(x) \right)^2$$

$$\therefore \sum_j \beta_j = 0$$

By

$$E\left[\frac{\sum R_i^2}{P}\right]$$

Ex

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ply By solving it, we get.

$$E\left[\frac{\sum_k T_k^2}{P}\right] = E(C \cdot F) = P(P-1)\sigma_r^2 + (P-1)\sigma^2 \quad (7)$$

Error SS :

$$E(ESS) = E(\text{Total SS}) - E(\text{Rows SS}) - E(\text{Columns SS}) - E(\text{Tr. SS}).$$

$$E(ESS) = (P^2 - 1 - P + 1 - P + 1 - P + 1)\sigma^2$$

$$E(ESS) = (P^2 - 3P + 2)\sigma^2$$

dividing by its df

$$E(MSE) = \sigma^2.$$

Q. 1) α_i is fixed, β_j and r_k are random:

In LS Design, Y_{ijk} is expressed as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + r_k + \epsilon_{ijk} \quad i=j=k=1, 2, \dots, P$$

with assumptions

$$\epsilon_{ijk} \sim NII(0, \sigma^2)$$

i.e.

$$E(\epsilon_{ij}) = 0, \quad E(\epsilon_{ij}^2) = \sigma^2, \quad E(\epsilon_{ijk} \epsilon_{ghn}) = 0 \quad \begin{matrix} i \neq g \\ j \neq h \\ k \neq n \end{matrix}$$

In this case

The rows effect is fixed i.e.

$$\sum \alpha_i = 0$$

and column and treatment effects are random with assumptions.

$$\beta_j's \sim NII(0, \sigma_\beta^2)$$

$$\gamma_k's \sim NII(0, \sigma_\gamma^2)$$

$$E(\beta_j) = 0$$

$$E(\gamma_k) = 0$$

$$E(\beta_j^2) = \sigma_\beta^2$$

$$E(\gamma_k^2) = \sigma_\gamma^2$$

$$E(\beta_j \beta_h) = 0, j \neq h$$

$$E(\gamma_k \gamma_n) = 0, k \neq n$$

$$E(C.F) = P\mu^2 + P\sigma_\beta^2 + P\sigma_\gamma^2 + \sigma^2 \quad \text{--- (i)}$$

$$E(TSS) = P \sum_i \alpha_i^2 + P(P-1)\sigma_\beta^2 + P(P-1)\sigma_\gamma^2 + (P-1)\sigma^2 \quad \text{--- (ii)}$$

$$E(RSS) = P \sum_i \alpha_i^2 + (P-1)\sigma^2 \quad \text{--- (iii)}$$

$$E(CSS) = P(P-1)\sigma_\beta^2 + (P-1)\sigma^2 \quad \text{--- (iv)}$$

$$E(Tv.SS) = P(P-1)\sigma_\gamma^2 + (P-1)\sigma^2 \quad \text{--- (v)}$$

$$E(ESS) = \text{eq (ii)} - \text{eq (iii)} - \text{eq (iv)} - \text{eq (v)}$$

$$E(ESS) = (P^2 - 1 - P + 1 - P + 1 - P + 1)\sigma^2$$

$$E(ESS) = (P^2 - 3P + 2)\sigma^2$$

$$E(ESS) = (P-1)(P-2)\sigma^2$$

$$\boxed{E(MSE) = \sigma^2}$$

Case d:

γ_k is fixed, α_i, β_j are random.

In LS Design, the obs $y_{ij(k)}$ is expressed as

$$y_{ij(k)} = \mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ij(k)} \quad i=j=k=1, 2, \dots, P$$

with assumptions.

$$\epsilon_{ij(k)}$$

i.e.

In this α_i

$$E(\alpha_i) =$$

$$\beta_j$$

$$E(\beta_j) =$$

and

$$E(C.F)$$

$$E(TSS)$$

$$E(RSS) =$$

$$E(CSS) =$$

$$E(Tv.SS) =$$

$$E(ESS) =$$

$$E(ESS) =$$

$$E(ESS) =$$

σ^2

$$E_{ij(n)} \sim NII(0, \sigma^2)$$

i.e. $E(E_{ij(n)}) = 0, E(E_{ij(n)}^2) = \sigma^2, E(E_{ij(n)} E_{gh(n)}) = 0$

 $k+n$

In this case α_i, β_j are random with assumptions
 $\alpha_i's \sim NII(0, \sigma_\alpha^2)$ i.e.

$$E(\alpha_i) = 0, E(\alpha_i^2) = \sigma_\alpha^2, E(\alpha_i \alpha_g) = 0 \quad i \neq g$$

 σ^2 (II)

$$\beta_j's \sim NII(0, \sigma_\beta^2) \quad \text{i.e.}$$

$$E(\beta_j) = 0, E(\beta_j^2) = \sigma_\beta^2, E(\beta_j \beta_h) = 0 \quad j \neq h$$

and r_k is fixed such that $\sum_k r_k = 0$.

$$E(C.F) = P\mu^2 + P\sigma_\alpha^2 + P\sigma_\beta^2 + \sigma^2 \quad \text{--- (I)}$$

$$E(TSS) = P\mu^2 + P(P-1)\sigma_\alpha^2 + P(P-1)\sigma_\beta^2 + P\sum_k r_k^2 + (P^2-1)\sigma^2 \quad \text{--- (II)}$$

$$E(RSS) = P(P-1)\sigma_\alpha^2 + (P-1)\sigma^2$$

$$E(CSS) = P(P-1)\sigma_\beta^2 + (P-1)\sigma^2$$

$$E(Tv.SS) = P\sum_k r_k^2 + (P-1)\sigma^2$$

$$E(ESS) = \text{eq (I)} - \text{eq (II)} - \text{eq (IV)} - \text{eq (V)}$$

$$E(ESS) = (P^2 - 1 + 1 - P - P + 1 - P + 1)\sigma^2$$

$$= (P^2 - 3P + 2)\sigma^2$$

$$E(\text{error SS}) = (P-1)(P-2)\sigma^2$$

$$E(MSE) = \sigma^2$$

 $-b_2 = P$

Case a:

α_i, β_j are fixed, r_k is random.

In LS Design, the obs. y_{ijk} is expressed as

$$y_{ijk} = \mu + \alpha_i + \beta_j + r_k + \epsilon_{ijk} \quad i=1, \dots, P \quad j=1, \dots, P \quad k=1, \dots, P$$

with assumptions

$\epsilon_{ijk} \sim NII(0, \sigma^2)$ i.e.

$$E(\epsilon_{ijk}) = 0, \quad E(\epsilon_{ijk}^2) = \sigma^2, \quad E(\epsilon_{ijk} \epsilon_{lmn}) = 0 \quad \begin{matrix} i \neq l \\ j \neq m \\ k \neq n \end{matrix}$$

In this case, α_i and β_j are fixed i.e.

$$\sum_i \alpha_i = 0, \quad \sum_j \beta_j = 0$$

and r_k is random i.e.

$r_k \sim NII(0, \sigma_r^2)$ i.e.

$$E(r_k) = 0, \quad E(r_k^2) = \sigma_r^2, \quad E(r_k r_n) = 0 \quad k \neq n$$

$$E(C.F) = P\mu^2 + P\alpha_i^2 + \sigma^2 \quad \text{--- (i)}$$

$$E(TSS) = P(P-1)\sigma_r^2 + P\sum_i \alpha_i^2 + P\sum_j \beta_j^2 + (P^2-1)\sigma^2 \quad \text{--- (ii)}$$

$$E(RSS) = P\sum_i \alpha_i^2 + (P-1)\sigma^2 \quad \text{--- (iii)}$$

$$E(CSS) = P\sum_j \beta_j^2 + (P-1)\sigma^2 \quad \text{--- (iv)}$$

$$E(Tr.SS) = P(P-1)\sigma_r^2 + (P-1)\sigma^2 \quad \text{--- (v)}$$

$$E(ESS) = \text{eq (ii)} - \text{eq (iii)} - \text{eq (iv)} - \text{eq (v)}$$

$E(ES$

E

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In LS

with y_{ijk}

$E(\epsilon_{ijk})$

α_i

and β

$E(\beta_j) = 0$

$E(C.F)$

$E(TSS) =$

$E(RSS) =$

$E(Tr.SS)$

$E(CSS) =$

$$E(ESS) = \frac{(P^2 - 1 - P + 1 - P + 1 - P + 1) \sigma^2}{(P-1)(P-2)} = \sigma^2$$

as

$$E(MSE) = \sigma^2$$

P

5) α_i, τ_k are fixed, β_j is random.

In LS Design, the obs y_{ijk} is expressed as

$$y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ijk} \quad i, j, k = 1, 2, \dots, P$$

with assumptions:

$$\epsilon_{ijk} \sim NII(0, \sigma^2) \quad \text{i.e.}$$

$$E(\epsilon_{ijk}) = 0, \quad E(\epsilon_{ijk}^2) = \sigma^2, \quad E(\epsilon_{ijk} \epsilon_{ghn}) = 0 \quad \begin{matrix} i \neq g \\ j \neq h \\ k \neq n \end{matrix}$$

α_i and τ_k are fixed i.e.

$k \neq n$

$$\sum_i \alpha_i = 0, \quad \sum_k \tau_k = 0$$

$k \neq n$

and β_j is random. with assumptions

$$\beta_j \sim NII(0, \sigma_\beta^2) \quad \text{i.e.}$$

$$E(\beta_j) = 0, \quad E(\beta_j \beta_h) = 0, \quad E(\beta_j^2) = \sigma_\beta^2$$

$$E(C.F) = P\mu^2 + P\sigma_\beta^2 + \sigma^2 \quad \text{--- (I)}$$

$$E(TSS) = P(P-1)\sigma_\beta^2 + P\sum_i \alpha_i^2 + P\sum_k \tau_k^2 + (P-1)\sigma^2 \quad \text{--- (II)}$$

$$E(RSS) = P\sum_i \alpha_i^2 + (P-1)\sigma^2 \quad \text{--- (III)}$$

$$E(Tr.SS) = P\sum_k \tau_k^2 + (P-1)\sigma^2 \quad \text{--- (IV)}$$

$$E(CSS) = P(P-1)\sigma_\beta^2 + (P-1)\sigma^2 \quad \text{--- (V)}$$

$$E(ESS) = \text{eq (2)} - \text{eq (3)} - \text{eq (4)} - \text{eq (5)}$$

$$= (P^2 - 1 - P + 1 - P + 1 - P + 1) \sigma^2$$

$$= (P^2 - 3P + 2) \sigma^2$$

$$= (P-1)(P-2) \sigma^2$$

$$E(MSE) = \sigma^2$$

$$E(C.F.) =$$

$$E(\text{Total SS}) =$$

$$E(RSS) =$$

$$E(CSS) =$$

Case b:

β_j, τ_k are fixed, α_i is random.

$$E(\text{Tr. SS}) =$$

$$E(ESS) =$$

In LS Design, the obs Y_{ijk} is expressed as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ijk} \quad i=j=k=1, 2, \dots, P$$

with assumptions

$$E(\epsilon_{ijk}) = 0$$

$$\epsilon_{ijk} \sim NII(0, \sigma^2) \quad \text{i.e.}$$

$$E(\epsilon_{ijk}) = 0, \quad E(\epsilon_{ijk}^2) = \sigma^2, \quad E(\epsilon_{ijk} \epsilon_{ghn}) = 0$$

$$i \neq g$$

$$j \neq h$$

$$k \neq n$$

In this case, column and treatment effects are fixed such that

$$\sum_j \beta_j = 0, \quad \sum_k \tau_k = 0$$

and rows effect is random such that

$$\alpha_i \sim NII(0, \sigma_\alpha^2) \quad \text{i.e.}$$

$$E(\alpha_i) = 0$$

$$E(\alpha_i^2) = \sigma_\alpha^2$$

$$E(\alpha_i \alpha_g) = 0 \quad i \neq g$$

ڈیسٹنٹ فوٹو کاپی سینٹر

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وشل ورک ڈیپارٹمنٹ IER کولمبیا روڈ

$$E(C.F) = P\mu^2 + P\sigma_\alpha^2 + \sigma^2 \quad \text{--- (1)}$$

$$E(\text{Total SS}) = P(P-1)\sigma_\alpha^2 + P \sum_j B_j^2 + P \sum_k r_k^2 + (P^2-1)\sigma^2 \quad \text{--- (2)}$$

$$E(RSS) = P(P-1)\sigma_\alpha^2 + (P-1)\sigma^2 \quad \text{--- (3)}$$

$$E(CSS) = P \sum_j B_j^2 + (P-1)\sigma^2 \quad \text{--- (4)}$$

$$E(Ty.SS) = P \sum_k r_k^2 + (P-1)\sigma^2 \quad \text{--- (5)}$$

$$E(ESS) = \text{eq (2)} - \text{eq (3)} - \text{eq (4)} - \text{eq (5)}$$

$$= (P^2-1-P+1-P+1-P+1)\sigma^2$$

$$= (P-1)(P-2)\sigma^2$$

$$E(MSE) = \sigma^2$$

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Estimation Of Missing Observation in LS Design:

a) One Missing observation:

In LS Design, the observation Y_{ijk} is expressed as
 $Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ijk}$; $i, j, k = 1, 2, \dots, P$

Let the missing observation be $\hat{Y}_{cd}(t)$ is the missing observation is in the d^{th} column, c^{th} row and t^{th} treatment. Then the expected total are

$$\begin{aligned} R_c &= R'_c + \hat{Y}_{cd}(t) \\ C_d &= C'_d + \hat{Y}_{cd}(t) \\ T_t &= T'_t + \hat{Y}_{cd}(t) \\ G &= G' + \hat{Y}_{cd}(t) \end{aligned}$$

Now the error sums of square

$$\begin{aligned} \text{Error SS} &= \sum_i \sum_j Y_{ij}^2 + \hat{Y}_{cd}^2 - C \cdot F - \frac{1}{P} \left\{ \sum_i R_i^2 + (R'_c + \hat{Y}_{cd}(t))^2 \right\} \\ &\quad + C \cdot F - \frac{1}{P} \left\{ \sum_j C_j^2 + (C'_d + \hat{Y}_{cd}(t))^2 \right\} + C \cdot F - \frac{1}{P} \left\{ \sum_k T_k^2 + \right. \\ &\quad \left. (T'_t + \hat{Y}_{cd}(t))^2 \right\} + C \cdot F \end{aligned}$$

$$\therefore G \cdot F = \frac{(G' + \hat{Y}_{cd}(t))^2}{P^2}$$

Differentiate w.r.t $\hat{Y}_{cd}(t)$
 and then equating it to zero we get.

$$\begin{aligned} \frac{\partial S}{\partial \hat{Y}_{cd}(t)} &= 2\hat{Y}_{cd}(t) - \frac{2}{P} (R'_c + \hat{Y}_{cd}(t)) - \frac{2}{P} (C'_d + \hat{Y}_{cd}(t)) - \\ &\quad \frac{2}{P} (T'_t + \hat{Y}_{cd}(t)) + 2 \left\{ \frac{2}{P^2} (G' + \hat{Y}_{cd}(t)) \right\} = 0 \end{aligned}$$

$$\frac{\partial S}{\partial \hat{Y}_{cd}(t)} =$$

$$(P^2 - 1)$$

$$(P^2 - 1)$$

where;

b) To

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2) S

3)

4)

Case I:

In L

Y_{ijk}

Multiply both sides by $\frac{P^2}{2}$, we get

$$\frac{\partial S}{\partial \hat{Y}_{cd(t)}} = P \hat{Y}_{cd(t)} - P(R'_c + \hat{Y}_{cd(t)}) - P(C'_d + \hat{Y}_{cd(t)}) - P(T'_k + \hat{Y}_{cd(t)}) + 2(G'_c + \hat{Y}_{cd(t)}) = 0$$

$$(P^2 - P - P - P + 2) \hat{Y}_{cd(t)} = PR'_c + PC'_d + PT'_k - 2G'_c$$

$$(P^2 - 3P + 2) \hat{Y}_{cd(t)} = Z_{cd(t)}$$

$$B \hat{Y}_{cd(t)} = Z_{cd(t)}$$

$$\hat{Y}_{cd(t)} = \frac{Z_{cd(t)}}{B}$$

where;

$$B = (P-1)(P-2)$$

$$Z_{cd(t)} = PR'_c + PC'_d + PT'_k - 2G'_c$$

b) Two Missing Observations:

There are four cases.

- 1) Different rows, different columns, different treatment.
- 2) Same row, " " " "
- 3) Same column, diff rows, diff treatment
- 4) Same treatment, diff rows, diff columns.

Case I:

Different rows, columns, treatments.

In LS Design, the obs Y_{ijk} is expressed as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ijk} \quad i=j=k=1, 2, \dots, P.$$

Let the two missing obs are $\hat{Y}_{cd(t)}$, $\hat{Y}_{ef(u)}$ i.e.
 in the c^{th} row, d^{th} column, T^{th} treatment and
 e^{th} row, f^{th} column and u^{th} treatment.
 Then the effected totals are

$$\begin{aligned} B_c &= B'_c + \hat{Y}_{cd(t)} & B_e &= B'_e + \hat{Y}_{ef(u)} \\ C_d &= C'_d + \hat{Y}_{cd(t)} & C_f &= C'_f + \hat{Y}_{ef(u)} \\ T_t &= T'_t + \hat{Y}_{cd(t)} & T_u &= T'_u + \hat{Y}_{ef(u)} \end{aligned}$$

$$G = G' + \hat{Y}_{cd(t)} + \hat{Y}_{ef(u)}$$

Now

Error Sums of Square is:

$$\text{Error SS} = \text{Total SS} - \text{Row SS} - \text{Column SS} - \text{Treatment SS}$$

$$\begin{aligned} &= \sum_j \sum_i Y_{ij(t)}^2 + \hat{Y}_{cd(t)}^2 + \hat{Y}_{ef(u)}^2 - C \cdot F - \frac{1}{P} \left\{ \sum_i R_i^2 + (R'_c + \hat{Y}_{cd(t)})^2 + \right. \\ &\quad \left. (R'_e + \hat{Y}_{ef(u)})^2 \right\} + C \cdot F - \frac{1}{P} \left\{ \sum_j C_j^2 + (C'_d + \hat{Y}_{cd(t)})^2 + (C'_f + \hat{Y}_{ef(u)})^2 \right\} \\ &\quad + C \cdot F - \frac{1}{P} \left\{ \sum_T T_k^2 + (T'_t + \hat{Y}_{cd(t)})^2 + (T'_u + \hat{Y}_{ef(u)})^2 \right\} + C \cdot F \end{aligned}$$

diff w.r.t $\hat{Y}_{cd(t)}$ and equating it to zero.

$$\begin{aligned} \frac{\partial S}{\partial \hat{Y}_{cd(t)}} &= 2\hat{Y}_{cd(t)} - \frac{2}{P} (B'_c + \hat{Y}_{cd(t)}) - \frac{2}{P} (C'_d + \hat{Y}_{cd(t)}) - \\ &\quad \frac{2}{P} (T'_t + \hat{Y}_{cd(t)}) + \frac{4}{P^2} (G' + \hat{Y}_{cd(t)} + \hat{Y}_{ef(u)}) = 0 \end{aligned}$$

Multiply by $\frac{P^2}{2}$ on both sides we have

$$\begin{aligned} P^2 \hat{Y}_{cd(t)} - P(B'_c + \hat{Y}_{cd(t)}) - P(C'_d + \hat{Y}_{cd(t)}) - P(T'_t + \hat{Y}_{cd(t)}) + \\ 2(G' + \hat{Y}_{cd(t)} + \hat{Y}_{ef(u)}) = 0 \end{aligned}$$

$$(P^2 - P - P - P + 2) \hat{Y}_{cd(t)} + 2\hat{Y}_{ef(u)} = PR'_c + PE'_d + PT'_t - 2G'$$

(P-1)

B)

Now

diff

we

$$\frac{\partial S}{\partial \hat{Y}_{ef(u)}} = 2\hat{Y}_{ef(u)} +$$

Multi

$$P^2 \hat{Y}_{ef(u)}$$

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t.

$$(P-1)(P-2)\hat{Y}_{cd}(t) + 2\hat{Y}_{ef}(u) = Z_{cd}(t)$$

$$B\hat{Y}_{cd}(t) + 2\hat{Y}_{ef}(u) = Z_{cd}(t) \quad \text{--- eq (1)}$$

Now

diff w.r.t $\hat{Y}_{ef}(u)$ and equating it to zero.
we get.

$$\frac{\partial S}{\partial \hat{Y}_{ef}(u)} = 2\hat{Y}_{ef}(u) - \frac{2}{P}(B_e' + \hat{Y}_{ef}(u)) - \frac{2}{P}(C_f' + \hat{Y}_{ef}(u)) - \frac{2}{P}(T_u' + \hat{Y}_{ef}(u)) + \frac{4}{P^2}(G' + \hat{Y}_{cd}(t) + \hat{Y}_{ef}(u)) = 0$$

Multiply both sides by $\frac{P^2}{2}$, we get.

nt ss.

$$P^2\hat{Y}_{ef}(u) - P(R_e' + \hat{Y}_{ef}(u)) - P(C_f' + \hat{Y}_{ef}(u)) - P(T_u' + \hat{Y}_{ef}(u)) + 2(G' + \hat{Y}_{cd}(t) + \hat{Y}_{ef}(u)) = 0$$

$(\hat{Y}_{cd}(t))^2$

$(\hat{Y}_{ef}(u))^2$

$$(P^2 - 3P + 2)\hat{Y}_{ef}(u) + 2\hat{Y}_{cd}(t) = P(R_e' + C_f' + T_u') - 2G'$$

- C.F.

$$B\hat{Y}_{ef}(u) + 2\hat{Y}_{cd}(t) = Z_{ef}(u) \quad \text{--- (2)}$$

zero.

Multiply eq (1) by 'B', eq (2) by '2' and then subtract the results. we get.

$$\begin{aligned} B^2\hat{Y}_{cd}(t) + 2B\hat{Y}_{ef}(u) &= BZ_{cd}(t) \\ \pm 2\hat{Y}_{cd}(t) \pm 2B\hat{Y}_{ef}(u) &= \pm 2Z_{ef}(u) \end{aligned}$$

$$(B^2 - 4)\hat{Y}_{cd}(t) = (BZ_{cd}(t) - 2Z_{ef}(u))$$

) +

$$\hat{Y}_{cd}(t) = \frac{BZ_{cd}(t) - 2Z_{ef}(u)}{B^2 - 4}$$

2G'

Substituting this result in eq (1), we get.

$$\hat{Y}_{ef(u)} = \frac{B Y_{ef(u)} - 2 Y_{cd(t)}}{B^2 - 4}$$

Case II:

Same Rows, Different Columns, treatments:

In LS Design, the Y_{ijkr} is expressed as:

$$Y_{ijkr} = \mu + \alpha_i + \beta_j + \gamma_r + \epsilon_{ijkr} \quad i=j=k=1, 2, \dots, P$$

Let the two missing obs are $\hat{Y}_{cd(t)}$, $\hat{Y}_{ef(u)}$
i.e. in the c^{th} row, d^{th} column and t^{th} treatment
and c^{th} row, f^{th} column and u^{th} treatment.

Then the effected totals are:

$$\begin{aligned} B_c &= B_c' + \hat{Y}_{cd(t)} + \hat{Y}_{cf(u)} \\ C_d' &= C_d' + \hat{Y}_{cd(t)} & T_t &= T_t' + \hat{Y}_{cd(t)} \\ C_f &= C_f' + \hat{Y}_{cf(u)} & T_u &= T_u' + \hat{Y}_{cf(u)} \end{aligned}$$

$$G = G' + \hat{Y}_{cd(t)} + \hat{Y}_{cf(u)}$$

Now, the error SS is:

$$\text{Error SS} = \text{Total SS} - \text{Row SS} - \text{Column SS} - \text{Treatment SS}$$

$$\begin{aligned} ESS &= \sum_i \sum_j Y_{ijkr}^2 + \hat{Y}_{cd(t)}^2 + \hat{Y}_{cf(u)}^2 - C \cdot F - \frac{1}{P} \left\{ \sum_i R_i^2 + (R_c' + \hat{Y}_{cd(t)} + \hat{Y}_{cf(u)})^2 \right\} \\ &\quad + C \cdot F - \frac{1}{P} \left\{ \sum_j C_j^2 + (C_d' + \hat{Y}_{cd(t)})^2 + (C_f' + \hat{Y}_{cf(u)})^2 \right\} + C \cdot F \\ &\quad - \frac{1}{P} \left\{ \sum_k T_k^2 + (T_t' + \hat{Y}_{cd(t)})^2 + (T_u' + \hat{Y}_{cf(u)})^2 \right\} + C \cdot F \end{aligned}$$

Diff it w.r.t $\hat{Y}_{cd}(t)$, equating it to zero.

$$\frac{\partial S}{\partial \hat{Y}_{cd}(t)} = 2\hat{Y}_{cd}(t) - \frac{2}{P}(R'_c + \hat{Y}_{cd}(t) + \hat{Y}_{cf}(u)) - \frac{2}{P}(C'_d + \hat{Y}_{cd}(t)) \\ - \frac{2}{P}(T'_t + \hat{Y}_{cd}(t)) + \frac{4}{P^2}(G' + \hat{Y}_{cd}(t) + \hat{Y}_{cf}(u)) = 0$$

Multiply by $\frac{P^2}{2}$ we have:

$$P^2 \hat{Y}_{cd}(t) - P(R'_c + \hat{Y}_{cd}(t) + \hat{Y}_{cf}(u)) - P(C'_d + \hat{Y}_{cd}(t)) -$$

$$P(T'_t + \hat{Y}_{cd}(t)) + 2(G' + \hat{Y}_{cd}(t) + \hat{Y}_{cf}(u)) = 0$$

$$(P^2 - 3P + 2)\hat{Y}_{cd}(t) - (P - 2)\hat{Y}_{cf}(u) = P(R'_c + C'_d + T'_t) - 2G'$$

$$B\hat{Y}_{cd}(t) - W\hat{Y}_{cf}(u) = Z_{cd}(t) \quad \text{--- (1)}$$

Similarly, when differentiating w.r.t $\hat{Y}_{cf}(u)$ and equating it to zero then multiply by $\frac{P^2}{2}$, we get

$$B\hat{Y}_{cf}(u) - W\hat{Y}_{cd}(t) = Z_{cf}(u) \quad \text{--- (2)}$$

* eq (1) by B, eq (2) by W, then by adding, we have.

$$B^2 \hat{Y}_{cd}(t) - BW \hat{Y}_{cf}(u) = BZ_{cd}(t)$$

$$-W^2 \hat{Y}_{cd}(t) + BW \hat{Y}_{cf}(u) = W(Z_{cf}(u))$$

$$(B^2 - W^2)\hat{Y}_{cd}(t) = BZ_{cd}(t) + WZ_{cf}(u)$$

$$\hat{Y}_{cd}(t) = \frac{BZ_{cd}(t) + WZ_{cf}(u)}{B^2 - W^2} \quad \text{--- (3)}$$

put it in eq (1)

$$\hat{Y}_{cd}(u) = \frac{B Z_{cf}(u) + W Z_{cd}(t)}{B^2 - W^2} \quad \text{--- (4)}$$

eq (3) and eq (4) are the required results where;

$$B = (P-1)(P-2)$$

$$W = P-2$$

$$Z_{cd}(t) = P(R'_c + C'_d + T'_t) - 2G'$$

$$Z_{cf}(u) = P(R'_c + C'_f + T'_u) - 2G'$$

Case III :

Same Columns, Different rows, treatment.

In LS Design, the Y_{ijk} can be written as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + G_{ijk} \quad i=j=k=1, 2, 3, \dots, P$$

Let the two missing obs are $\hat{Y}_{cd}(t)$, $\hat{Y}_{ed}(u)$ i.e. in the c^{th} row, d^{th} column, T^{th} treatment and in the e^{th} row, d^{th} column, u^{th} treatment. Then the affected totals are:

$$R'_c = R'_c + \hat{Y}_{cd}(t), \quad R'_e = R'_e + \hat{Y}_{ed}(t)$$

$$C'_d = C'_d + \hat{Y}_{cd}(t) + \hat{Y}_{ed}(t)$$

$$T'_t = T'_t + \hat{Y}_{cd}(t), \quad T'_u = T'_u + \hat{Y}_{ed}(u)$$

$$G' = G' + \hat{Y}_{cd}(t) + \hat{Y}_{ed}(t)$$

Now

$$ESS = \sum_i$$

$$(R'_e + Y)$$

$$= \frac{1}{P} \sum_k$$

By Differ

$$(P-1)$$

and

Multipl

B

- W

$$(B^2 -$$

put

Now

$$ESS = \sum_i \sum_j \hat{Y}_{ij}(t)^2 + \hat{Y}_{cd}(t)^2 + \hat{Y}_{cd}(t)^2 - C \cdot F - \frac{1}{P} \left\{ \sum_i R_i^2 + (R_c' + \hat{Y}_{cd}(t))^2 + (R_c' + \hat{Y}_{cd}(t))^2 \right\} + C \cdot F - \frac{1}{P} \left\{ \sum_i C_i^2 + (C_d' + \hat{Y}_{cd}(t) + \hat{Y}_{cd}(t))^2 \right\} + C \cdot F - \frac{1}{P} \left\{ \sum_k T_k^2 + (T_c' + \hat{Y}_{cd}(t))^2 + (T_u' + \hat{Y}_{cd}(t))^2 \right\} + C \cdot F$$

$$\therefore C \cdot F = \frac{(G' + \hat{Y}_{cd}(t) + \hat{Y}_{cd}(t))^2}{P^2}$$

By Differentiating w.r.t. $\hat{Y}_{cd}(t)$, $\hat{Y}_{cd}(u)$ and equating it to zero then Multiply by $\frac{P^2}{2}$, we get

$$(P-1)(P-2)\hat{Y}_{cd}(t) - (P-2)\hat{Y}_{cd}(u) = Z_{cd}(t)$$

$$B\hat{Y}_{cd}(t) - W\hat{Y}_{cd}(u) = Z_{cd}(t) \quad (1)$$

$$B\hat{Y}_{cd}(u) - W\hat{Y}_{cd}(t) = Z_{cd}(u) \quad (2)$$

Multiply (1) by B and eq(2) by W, then adding

$$B^2\hat{Y}_{cd}(t) - BW\hat{Y}_{cd}(u) = BZ_{cd}(t)$$

$$-W^2\hat{Y}_{cd}(t) + BW\hat{Y}_{cd}(u) = WZ_{cd}(u)$$

$$(B^2 - W^2)\hat{Y}_{cd}(t) = BZ_{cd}(t) + WZ_{cd}(u)$$

$$\hat{Y}_{cd}(t) = \frac{BZ_{cd}(t) + WZ_{cd}(u)}{B^2 - W^2}$$

put it in (1)

$$\hat{Y}_{cd(u)} = \frac{B\hat{Y}_{cd(u)} + W\hat{Y}_{cd(t)}}{B^2 - W^2}$$

where,

$$B = (P-1)(P-2)$$

$$W = P-2$$

$$Z_{cd(t)} = P(R'_c + C'_d + T'_t) - 2G'$$

$$Z_{cd(u)} = P(R'_c + C'_d + T'_t) - 2G'$$

Case IV:

Same Treatment, Different rows, columns.

In LS Design, the obs $Y_{ij}(k)$ is expressed as

$$Y_{ij}(k) = \mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ij}(k) \quad i=j=k=1,2,3,\dots,P.$$

Let the two missing obs are $\hat{Y}_{cd(t)}$, $\hat{Y}_{ef(t)}$ i.e. in the c^{th} row, d^{th} column, t^{th} treatment and e^{th} row, f^{th} column and t^{th} treatment. Then the expected totals are:

$$B_c = B'_c + \hat{Y}_{cd(t)} \quad , \quad B_e = B'_e + \hat{Y}_{ef(t)}$$

$$C_d = C'_d + \hat{Y}_{cd(t)} \quad , \quad C_f = C'_f + \hat{Y}_{ef(t)}$$

$$T_t = T'_t + \hat{Y}_{cd(t)} + \hat{Y}_{ef(t)}$$

$$G = G' + \hat{Y}_{cd(t)} + \hat{Y}_{ef(t)}$$

Now,

$$ESS = \text{Total SS} - \text{Row SS} - \text{Column SS} - \text{Treatment SS}.$$

$$ESS = \sum_{j=1}^P \sum_{k=1}^P Y_{ij}(k)^2 + \hat{Y}_{cd(t)}^2 + \hat{Y}_{ef(t)}^2 - C.F. - \frac{1}{P} \left\{ \sum_i R_i^2 + (R'_c + \hat{Y}_{cd(t)})^2 + \right.$$

$$(R'_e + \hat{Y}_{ef(t)})^2 \left. \right\} + C.F. - \frac{1}{P} \left\{ \sum_j C_j^2 + (C'_d + \hat{Y}_{cd(t)})^2 + (C'_f + \hat{Y}_{ef(t)})^2 \right\}$$

$$+ C.F. - \frac{1}{P} \left\{ \sum_k T_k^2 + (T'_t + \hat{Y}_{cd(t)} + \hat{Y}_{ef(t)})^2 \right\} + C.F.$$

$$\therefore C.F = \frac{(G' + \hat{Y}_{cd}(t) + \hat{Y}_{ef}(t))^2}{P^2}$$

By differentiating w.r.t $\hat{Y}_{cd}(t)$ and $\hat{Y}_{ef}(t)$ and equating it to zero and then multiply by $\frac{P^2}{2}$ we get

$$(P-1)(P-2)\hat{Y}_{cd}(t) - (P-2)\hat{Y}_{ef}(t) = P(R_c' + C_d' + T_c') - 2G'$$

$$B\hat{Y}_{cd}(t) - W\hat{Y}_{ef}(t) = Z_{cd}(t) \quad \text{--- (1)}$$

and

$$B\hat{Y}_{ef}(t) - W\hat{Y}_{cd}(t) = Z_{ef}(t) \quad \text{--- (2)}$$

Multiply eq (1) by B and eq (2) by W and then adding.

$$\begin{aligned} B^2\hat{Y}_{cd}(t) - BW\hat{Y}_{ef}(t) &= BZ_{cd}(t) \\ -W^2\hat{Y}_{cd}(t) + BW\hat{Y}_{ef}(t) &= WZ_{ef}(t) \end{aligned}$$

in the

$$(B^2 - W^2)\hat{Y}_{cd}(t) = BZ_{cd}(t) + WZ_{ef}(t)$$

$$\hat{Y}_{cd}(t) = \frac{BZ_{cd}(t) + WZ_{ef}(t)}{B^2 - F^2} \quad \text{put it in eq (1) we get}$$

$$\hat{Y}_{ef}(t) = \frac{BZ_{ef}(t) + WZ_{cd}(t)}{B^2 - F^2}$$

where;

$$B = (P-1)(P-2)$$

$$W = P-2$$

$$Z_{cd}(t) = P(R_c' + C_d' + T_c') - 2G'$$

$$Z_{ef}(t) = P(R_e' + C_f' + T_e') - 2G'$$

Least Square Estimator in CR Design:

$$\frac{\partial S}{\partial \hat{\alpha}_j} =$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

$$\frac{\partial S}{\partial \hat{\alpha}_j}$$

The observation Y_{ij} is written as

$$Y_{ij} = \mu + \alpha_j + \epsilon_{ij} \quad \text{for } i = 1, 2, \dots, r, \\ j = 1, 2, \dots, k.$$

Let $\hat{\alpha}_j, \hat{\mu}$ are estimators of α_j and μ .
The sum of squares of error is

$$S = \sum_j \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j)^2$$

Diff w.r.t " $\hat{\mu}$ ".

$$\frac{\partial S}{\partial \hat{\mu}} = 2 \sum_j \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j)(-1)$$

$$\therefore \frac{\partial S}{\partial \hat{\mu}^2} > 0$$

so put $\frac{\partial S}{\partial \hat{\mu}} = 0$

Now $\hat{\mu}$ w

$$\sum_j \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j) = 0$$

$$\sum_j \sum_i Y_{ij} - r k \hat{\mu} - r \sum_j \hat{\alpha}_j = 0$$

$$\sum_j \sum_i Y_{ij} = r k \hat{\mu} + r \sum_j \hat{\alpha}_j \quad \text{--- (1)}$$

Now;

For Treatments i

And from

$$S = \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1)^2 + \sum_i (Y_{i2} - \hat{\mu} - \hat{\alpha}_2)^2 + \dots + \sum_i (Y_{ik} - \hat{\mu} - \hat{\alpha}_k)^2$$

Diff w.r.t $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_k$.

Design:

$$\frac{\partial S}{\partial \hat{\alpha}_1} = -2 \sum_i (Y_{ij} - \hat{\alpha} - \alpha_1)$$

$$\frac{\partial S}{\partial \hat{\alpha}_2} = -2 \sum_i (Y_{ij} - \hat{\alpha} - \alpha_2)$$

$$\frac{\partial S}{\partial \hat{\alpha}_j} = -2 \sum_i (Y_{ij} - \hat{\alpha} - \alpha_j)$$

put $\frac{\partial S}{\partial \hat{\alpha}_j} = 0$

$$\sum_i (Y_{ij} - \hat{\alpha} - \hat{\alpha}_j) = 0$$

$$\sum_i Y_{ij} - r\hat{\alpha} - r\hat{\alpha}_j = 0$$

$$\sum_i Y_{ij} = r\hat{\alpha} + r\hat{\alpha}_j \quad \text{--- (2)}$$

$$Y_{.j} = r\hat{\alpha} + r\hat{\alpha}_j$$

Now if we add "k" equations in eq (1) it will become as same as eq (1)

This means that the Normal equations are normal not linearly independent. Therefore, in order to get unique solution we have to apply conditions. i.e.

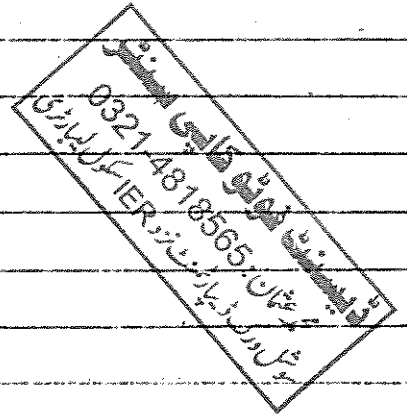
$$\sum \alpha_j = 0$$

Under the condition from eq (1)

$$\sum_j \sum_i Y_{ij} = rk\hat{\alpha} + 0$$

$$\hat{\alpha} = \frac{\sum_j \sum_i Y_{ij}}{rk} = \frac{G}{rk} = \frac{Y_{..}}{rk} = \bar{Y}_{..}$$

$$\boxed{\hat{\alpha} = \bar{Y}_{..}}$$



from eq (ii)

$$y_{.j} = \frac{r(V_{..})}{r} + r\hat{\alpha}_j$$

$$\frac{y_{.j}}{r} = \bar{y}_{..} + \hat{\alpha}_j$$

$$\bar{y}_{.j} - \bar{y}_{..} = \hat{\alpha}_j$$

$$\boxed{\hat{\alpha}_j = \bar{y}_{.j} - \bar{y}_{..}}$$

Least Square Estimate of RCB Design:

In RCB Design, the obs y_{ij} is expressed as

$$y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij} \quad \text{for } i=1, 2, \dots, r, j=1, 2, \dots, k$$

Let $\hat{\mu}$, $\hat{\alpha}_j$, $\hat{\beta}_i$ are the estimators of μ , α_j , β_i .
Then the sum of squares of errors is

$$S = \sum_j \sum_i (y_{ij} - \hat{y})^2$$

$$S = \sum_j \sum_i (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)^2$$

To find the estimates of μ , α_j , β_i , we differentiate w.r.t unknown estimators and equating it to zero.

$$\frac{\partial S}{\partial \hat{\mu}} = -2 \sum_j \sum_i (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)$$

For Row

$$S = \sum_j ($$

diff

$$\frac{\partial S}{\partial \hat{\beta}_i} =$$

$$\frac{\partial S}{\partial \hat{\beta}_i} =$$

$$\frac{\partial S}{\partial \hat{\beta}_i} =$$

put

$$\frac{\partial^2 S}{\partial \hat{\mu}^2} > 0$$

put $\frac{\partial S}{\partial \hat{\mu}} = 0$

$$\sum_j \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \beta_i) = 0$$

$$\sum_j \sum_i Y_{ij} - nk\hat{\mu} - r \sum_j \hat{\alpha}_j - k \sum_i \beta_i = 0$$

$$\sum_j \sum_i Y_{ij} = nk\hat{\mu} + r \sum_j \hat{\alpha}_j + k \sum_i \beta_i \quad \text{--- (1)}$$

For Rows/Blocks:

Design:

$$S = \sum_j (Y_{j1} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_1)^2 + \sum_j (Y_{j2} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_2)^2 + \dots + \sum_j (Y_{jr} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_r)^2$$

diff w.r.t $\beta_1, \beta_2, \dots, \beta_r$

$$\frac{\partial S}{\partial \hat{\beta}_1} = -2 \sum_j (Y_{j1} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_1)$$

$$\frac{\partial S}{\partial \hat{\beta}_2} = -2 \sum_j (Y_{j2} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_2)$$

$$\frac{\partial S}{\partial \hat{\beta}_i} = -2 \sum_j (Y_{ji} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)$$

Potential
ing

$$\frac{\partial^2 S}{\partial \hat{\beta}_i^2} > 0$$

put $\frac{\partial S}{\partial \hat{\beta}_i} = 0$

$$\sum_j (Y_{ji} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i) = 0$$

$$\sum_j Y_{ij} = k\hat{\mu} + \sum_j \hat{\alpha}_j + k\hat{\beta}_i \quad (2)$$

for treatments:

$$S = \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta}_i)^2 + \sum_i (Y_{i2} - \hat{\mu} - \hat{\alpha}_2 - \hat{\beta}_i)^2 + \dots + \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)^2 + \dots + \sum_i (Y_{ik} - \hat{\mu} - \hat{\alpha}_k - \hat{\beta}_i)^2$$

diff w.r.t $\alpha_1, \alpha_2, \dots, \alpha_k$

$$\frac{\partial S}{\partial \hat{\alpha}_1} = -2 \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta}_i)$$

$$\frac{\partial S}{\partial \hat{\alpha}_2} = -2 \sum_i (Y_{i2} - \hat{\mu} - \hat{\alpha}_2 - \hat{\beta}_i)$$

$$\frac{\partial S}{\partial \hat{\alpha}_j} = -2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)$$

$$\therefore \frac{\partial S}{\partial \hat{\alpha}_j} > 0$$

so put

$$\frac{\partial S}{\partial \hat{\alpha}_j} = 0$$

$$\sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i) = 0$$

$$\sum_j Y_{ij} = r\hat{\mu} + r\hat{\alpha}_j + \sum_i \hat{\beta}_i \quad (3)$$

If we add 'k' equations in eq (3) and 'r' equations in eq (2) then it will become the same as eq (1).

This is linearly unique i.e.

Uncl. From eq

From eq

From eq

This means that these normal equations are not linearly independent. Therefore, in order to find unique solutions, we have to apply conditions i.e. $\sum_j d_j = 0$, $\sum_i \beta_i = 0$

Under this condition.

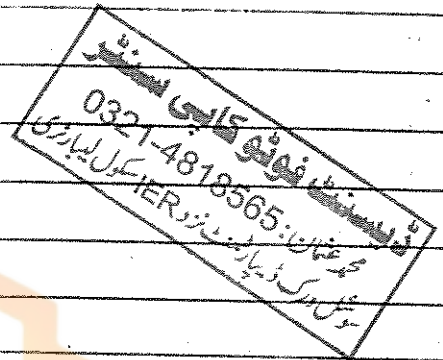
From eq (i)

$$\sum_i \sum_j Y_{ij} = nk\hat{u} + 0 + 0$$

$$Y_{..} = nk\hat{u}$$

$$\frac{Y_{..}}{nk} = \hat{u}$$

$$\hat{u} = \bar{Y}_{..}$$



From eq (ii)

$$\sum_j Y_{ij} = k\hat{u} + 0 + k\hat{\beta}_i$$

$$\frac{\sum_j Y_{ij}}{k} = (\hat{u} + \hat{\beta}_i)$$

$$\bar{Y}_{i.} = \bar{Y}_{..} + \hat{\beta}_i$$

$$\because \hat{u} = \bar{Y}_{..}$$

$$\hat{\beta}_i = \bar{Y}_{i.} - \bar{Y}_{..}$$

From eq (iii)

$$\sum_i Y_{ij} = r\hat{u} + 0 + r\hat{\alpha}_j$$

$$\sum_i Y_{ij} = r(\hat{u} + \hat{\alpha}_j)$$

$$\frac{Y_{.j}}{r} = \hat{u} + \hat{\alpha}_j$$

$$\because \hat{u} = \bar{Y}_{..}$$

$$\bar{Y}_{.j} = \bar{Y}_{..} + \hat{\alpha}_j$$

$$\hat{\alpha}_j = \bar{Y}_{.j} - \bar{Y}_{..}$$

In Latin Square Design

"N" MISSING OBSERVATIONS:

In LSD, the observation y_{ijk} is written as:

$$y_{ijk} = \mu + \alpha_j + \beta_i + \tau_k + \epsilon_{ijk}$$

$$i=j=k=1, 2, \dots, P$$

Suppose the missing obs are in different rows, columns and treatments, and let

The missing obs are $y_{12(A)}, y_{34(B)}, y_{56(C)} \dots$ (say N), then

$$SSE = \left[\sum_j \sum_i y_{ijk}^2 + \hat{y}_{12(A)}^2 + \hat{y}_{34(B)}^2 + \hat{y}_{56(C)}^2 + \dots \right] - \frac{1}{P} \left\{ \sum_i R_i^2 + \right.$$

$$\left. (R'_1 + \hat{y}_{12(A)})^2 + (R'_3 + \hat{y}_{34(B)})^2 + (R'_5 + \hat{y}_{56(C)})^2 + \dots \right\} - \frac{1}{P}$$

$$\left\{ \sum_j C_j^2 + (C'_2 + \hat{y}_{12(A)})^2 + (C'_4 + \hat{y}_{34(B)})^2 + (C'_6 + \hat{y}_{56(C)})^2 + \dots \right\} -$$

$$\frac{1}{P} \left\{ \sum_k T_k^2 + (T'_A + \hat{y}_{12(A)})^2 + (T'_B + \hat{y}_{34(B)})^2 + (T'_C + \hat{y}_{56(C)})^2 + \dots \right\}$$

$$+ \frac{2}{P^2} (G' + \hat{y}_{12(A)} + \hat{y}_{34(B)} + \hat{y}_{56(C)} + \dots)^2$$

Differentiate it w.r.t unknowns i.e.

$\hat{y}_{12(A)}, \hat{y}_{34(B)}, \hat{y}_{56(C)}, \dots$ and equate them to zero, we have.

$$\frac{\partial SSE}{\partial \hat{y}_{12(A)}} = 2\hat{y}_{12(A)} - \frac{2}{P} (R'_1 + \hat{y}_{12(A)}) - \frac{2}{P} (R'_2 + \hat{y}_{12(A)}) -$$

$$\frac{2}{P} (T'_A + \hat{y}_{12(A)}) + \frac{4}{P^2} (G' + \hat{y}_{12(A)} + \hat{y}_{34(B)} + \hat{y}_{56(C)} + \dots) = 0$$

Multiply both sides by $\frac{P^2}{2}$, we get.

$$= P^2 \hat{y}_{12(A)} + 2$$

$$(P^2 - 3P)$$

$$B$$

$$N_{obs}$$

$$\frac{\partial SSE}{\partial \hat{y}_{34(B)}} =$$

$$M_u$$

$$= P^2 \hat{y}_{34(B)} + 2$$

$$(P^2 - 3P)$$

Similarly

$$B \hat{y}_{56(C)}$$

To find

we

this

form

$$= P^2 \hat{Y}_{12(A)} - P(R'_1 + \hat{Y}_{12(A)}) - P(C'_2 + \hat{Y}_{12(A)}) - P(T'_A + \hat{Y}_{12(A)}) + 2(G'_1 + \hat{Y}_{12(A)} + \hat{Y}_{34(B)} + \hat{Y}_{56(C)} + \dots) = 0$$

$$(P^2 - 3P + 2) \hat{Y}_{12(A)} + 2\hat{Y}_{34(B)} + 2\hat{Y}_{56(C)} + \dots = P(R'_1 + C'_2 + T'_A) - 2G'_1$$

then as:

$$B \hat{Y}_{12(A)} + A \hat{Y}_{34(B)} + A \hat{Y}_{56(C)} + \dots = Z_{12(A)} \quad (1)$$

let

Now diff w.r.t $\hat{Y}_{34(B)}$ and equate it to zero.

$$\frac{\partial SSE}{\partial \hat{Y}_{34(B)}} = 2\hat{Y}_{34(B)} - \frac{2(R'_3 + \hat{Y}_{34(B)})}{P} - \frac{2(C'_4 + \hat{Y}_{34(B)})}{P} - \frac{2(T'_B + \hat{Y}_{34(B)})}{P} + \frac{2}{P^2} (G'_1 + \hat{Y}_{12(A)} + \hat{Y}_{34(B)} + \hat{Y}_{56(C)} + \dots) = 0$$

Multiply both sides by $\frac{P^2}{2}$, we get.

$$= P^2 \hat{Y}_{34(B)} - P(R'_3 + \hat{Y}_{34(B)}) - P(C'_4 + \hat{Y}_{34(B)}) - P(T'_B + \hat{Y}_{34(B)}) + 2(G'_1 + \hat{Y}_{12(A)} + \hat{Y}_{34(B)} + \hat{Y}_{56(C)} + \dots) = 0$$

$$(P^2 - 3P + 2) \hat{Y}_{34(B)} + 2\hat{Y}_{12(A)} + 2(\hat{Y}_{56(C)} + \dots) = P(R'_3 + C'_4 + T'_B) - 2G'_1$$

$$B \hat{Y}_{34(B)} + A \hat{Y}_{12(A)} + A \hat{Y}_{56(C)} + \dots = Z_{34(B)} \quad (2)$$

Similarly;

$$B \hat{Y}_{56(C)} + A \hat{Y}_{12(A)} + A \hat{Y}_{34(B)} + \dots = Z_{56(C)} \quad (3)$$

and so on.

To find the estimated values of unknowns we solve eq (1) (ii) (iii) by matrices. For this we convert these eq in matrix form i.e

$$X(Y) + \dots = 0$$

B	A	A	A	$\hat{y}_{12}(A)$	$Z_{12}(A)$
A	B	A	A	$\hat{y}_{31}(B)$	$Z_{31}(B)$
A	A	B	A	$\hat{y}_{56}(C)$	$Z_{56}(C)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
A	A	A	B	$\vdots$	$\vdots$

To solve
and
results

2f

± 2)

$$M\hat{y} = Z$$

We knew that

$$\hat{y} = M^{-1}Z$$

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$$MM^{-1} = I$$

Let

$$M^{-1} = \begin{bmatrix} W & V & V & V \\ V & W & V & V \\ V & V & W & V \\ \vdots & \vdots & \vdots & \vdots \\ V & V & V & W \end{bmatrix}$$

put

BW +

$$MM^{-1} = I$$

BW -

$$\begin{bmatrix} B & A & A & A \\ A & B & A & A \\ A & A & B & A \\ \vdots & \vdots & \vdots & \vdots \\ A & A & A & B \end{bmatrix} \begin{bmatrix} W & V & V & V \\ V & W & V & V \\ V & V & W & V \\ \vdots & \vdots & \vdots & \vdots \\ V & V & V & W \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

BW =

W -

W and V can be obtained as:

$$BW + (N-1)AV = 1 \quad (4)$$

$$AW + BV + (N-2)AV = 0$$

W =

N-1

$$AW + (B + (N-2)A)V = 0 \quad (5)$$

To solve eq (4), (5), multiply eq (4) by 'A' and eq (5) by 'B' and then subtract the results we have.

$$\begin{array}{r} 2ABW + A^2(N-1)V = A \\ \underline{+ 2ABW + B^2V + AB(N-2)V = 0} \end{array}$$

$$V[A^2(N-1) - B^2 - AB(N-2)] = A$$

$$V = \frac{A}{A^2(N-1) - B^2(B+N-2)}$$

$$V = \frac{1}{A(N-1) - B(B+N-2)}$$

put This value in eq (4), we get

$$BW + (N-1)A \left[\frac{1}{A(N-1) - B(B+N-2)} \right] = 1$$

$$BW = \frac{1}{A(N-1) - B(B+N-2)} - A(N-1)$$

$$BW = \frac{A(N-1) - B(B+N-2) - A(N-1)}{A(N-1) - B(B+N-2)}$$

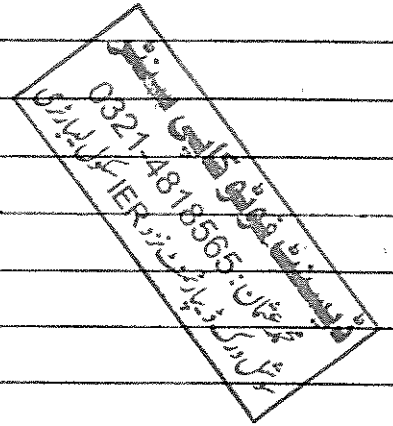
$$W = \frac{-B(B+N-2)}{AB(N-1) - B^2(B+N-2)}$$

OR

$$W = \frac{-(B+(N-2)A)}{A^2(N-1) - AB(B+N-2)}$$

Now let

$$B = A^2(N-1) - AB(B+N-2)$$



$$\hat{Y} = \begin{bmatrix} \hat{Y}_{12(A)} \\ \hat{Y}_{34(B)} \\ \hat{Y}_{56(C)} \\ \vdots \end{bmatrix} = \frac{1}{S} \begin{bmatrix} -(B+(N-2)A) & A & A & \dots & A \\ A & -(B+(N-2)A) & A & \dots & A \\ A & A & -(B+(N-2)A) & \dots & A \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A & A & A & \dots & -(B+(N-2)A) \end{bmatrix} \begin{bmatrix} Z_{12(A)} \\ Z_{34(B)} \\ Z_{56(C)} \\ \vdots \end{bmatrix}$$

$$\hat{Y}_{12(A)} =$$

$$=$$

$$\hat{Y}_{56(C)} =$$

where

$$\hat{Y}_{12(A)} = \frac{-(B+(N-2)A)Z_{12(A)} + AZ_{34(B)} + AZ_{56(C)} + \dots}{A^2(N-1) - AB(B+(N-2))}$$

$$A^2(N-1) - AB(B+(N-2))$$

$$\hat{Y}_{34(B)} = \frac{-(B+(N-2)A)Z_{34(B)} + AZ_{12(A)} + AZ_{56(C)} + \dots}{A^2(N-1) - AB(B+(N-2))}$$

$$A^2(N-1) - AB(B+(N-2))$$

$$\hat{Y}_{56(C)} = \frac{-(B+(N-2)A)Z_{56(C)} + AZ_{12(A)} + AZ_{34(B)} + \dots}{A^2(N-1) - AB(B+(N-2))}$$

$$A^2(N-1) - AB(B+(N-2))$$

put $N = 1$

ANO

$$\hat{Y}_{12(A)} = \frac{-(B+(1-2)A)Z_{12(A)}}{A^2(1-1) - AB(B+(1-2))}$$

The

$$= \frac{-(B-A)Z_{12(A)}}{-AB(B-1)}$$

$$\hat{Y}_{12(A)} = \frac{Z_{12(A)}}{B}$$

put $N = 2$, we get

1) Reduc

2) of es

so

A

Z =

$$2) A) \begin{bmatrix} Z_{12(A)} \\ Z_{34(B)} \\ Z_{56(C)} \\ \vdots \end{bmatrix}$$

$$\begin{aligned} \hat{Y}_{12(A)} &= \frac{-[B + (2-2)A]Z_{12(A)} + AZ_{34(B)}}{(2-1)A^2 - B(B + (2-2)A)} \\ &= \frac{BZ_{12(A)} - AZ_{34(B)}}{B^2 - A^2} \end{aligned}$$

$$\hat{Y}_{56(C)} = \frac{BZ_{56(C)} - AZ_{12(A)}}{B^2 - A^2}$$

where;

$$B = (P-1)(P-2)$$

$$A = 2$$

$$Z_{12(A)} = P(R'_1 + G'_2 + T'_A) - 2G'_1$$

$$Z_{34(B)} = P(R'_3 + G'_4 + T'_B) - 2G'_3$$

$$Z_{56(C)} = P(G'_6 + R'_5 + T'_C) - 2G'_6$$

ANOVA AFTER ESTIMATION OF

MISSING OBSERVATION:

The following changes are made in the analysis.

- 1) Reduce the total degree of freedom and error degree of freedom by the number of estimates.
- 2) Treatment sum of squares is upward biased with estimates of missing observations so we used the following correction

$$Z = \text{Factor Bias} = \frac{(G'_i - R'_i - (P-1)T'_i)^2}{(P-1)^2(P-2)^2} \quad \text{For 1 Missing obs.}$$

This selection term can be generalized for more than one missing obs but with diff treatment.

If the observations are missing in the same treatments then we can't apply it.

If $\hat{Y}_{d(i)}$, $\hat{Y}_{e(i)}$ are M.Obs then

$$Z = \text{Bias} = \frac{(G' - R'_i - C'_i - (P-1)T'_i)^2 + (G' - R'_i - C'_i - (P-1)T'_u)^2}{(P-1)^2 (P-2)^2}$$

$$r'_A = \begin{cases} 1, \\ 2/3, \\ 1/3, \\ 0, \end{cases}$$

- 3) The standard error of the difference b/w the treatment mean A with the missing unit and a diff b/w treatment B with all units presents is

$$S.E(\bar{Y}_A - \bar{Y}_B) = \sqrt{MSE \left\{ \frac{2}{P} + \frac{1}{(P-1)(P-2)} \right\}}$$

- 4) S.E for the diff b/w two treatment means each with missing obs can be obtained by Yates approx.

$$S.E(\bar{Y}_A - \bar{Y}_B) = \sqrt{MSE \left\{ \frac{1}{r'_A} + \frac{1}{r'_B} \right\}}$$

where

r'_A and r'_B are affected NO. of replicates obtained as:

Exam

r'_A

r'_B

$$r'_A = \begin{cases} 1, & \text{if the other treatment B is present in the} \\ & \text{corresponding row and column.} \\ \frac{2}{3}, & \text{if " " is missing in either row or column but} \\ & \text{not both.} \\ \frac{1}{3}, & \text{if " " both row and column} \\ 0, & \text{when the treatment in question (i.e. A)} \\ & \text{is missing.} \end{cases}$$

Example:

Consider 5x5 LSD:

B	(A)	C	E	D
D	B	E	A	(C)
E	D	A	C	B
C	E	B	D	A
A	C	D	(B)	E

$$r'_A = \frac{2}{3} + 0 + 1 + 2 + 1 = \frac{10}{3}$$

$$r'_B = \frac{2}{3} + \frac{2}{3} + 1 + 0 + 1 = \frac{10}{3}$$

Efficiency Of Latin Square Design

Now

a) With RCB Design:

Let $s_r^2, s_c^2, s_t^2, s_e^2$ denotes the mean squares for rows, columns and treatments and error in a $P \times P$ latin square design.

and we factor

Then the efficiency of a LS Design relative to a RCB design is estimated as

1) Taking rows as blocks, then MSE of RCB design can be estimated as

for

$$\begin{aligned} \text{MSE(RCB)} &= \frac{n_c s_c^2 + (n_t + n_e) s_e^2}{n_c + n_t + n_e} \\ &= \frac{1}{P} \left[\frac{s_c^2 + (P-1) s_e^2}{s_e^2} \right] \end{aligned}$$

MSE

MS

where

n_c = column df.

n_e = Error df.

n_t = Treatment df.

PUACP

and

ii) Taking columns as blocks, then MSE for RCB Design can be estimated as

and then equat

$$\text{MSE(RCB)} = \frac{n_r s_r^2 + (n_t + n_e) s_e^2}{n_r + n_t + n_e}$$

where

n_r = rows df.

$$= \frac{1}{P} \left[\frac{s_r^2 + (P-1) s_e^2}{s_e^2} \right]$$

wher

Design

Now Efficiency (LS to RCB).

$$= \frac{\text{MSE(RCB)}}{\text{MSE(LSD)}} \times 100$$

Squares
error

and if the error df is less than 20
we must multiply by the precision
factor i.e

n relative
is

$$\frac{(n_1+1)(n_2+3)}{(n_1+3)(n_2+1)} \text{ where } n_1, n_2 \text{ are } ^{\text{error}} \text{ df}$$

for LSD and RCB respectively

of

b) With CR Design:

MSE for CR Design can be estimated as:

$$\text{MSE(CRD)} = \frac{s_r^2 + s_e^2 + (P-1)s_e^2}{(P-1)}$$

and Relative efficiency (LSD to CRD):

$$= \frac{\text{MSE(CRD)}}{\text{MSE(LSD)}} \times 100.$$

ISE
red

and if the error df is less than 20
then we must multiply the above
equation by the precision factor i.e

$$\frac{(n_1+1)(n_2+3)}{(n_1+3)(n_2+1)}$$

where n_1 and n_2 are error df for
LSD and CRD respectively.

Example:

Suppose we have a latin square design

S.V	df	S.S	M.S
Rows	$3 = n_r$	1.95	$0.65 = s_r^2$
columns	$3 = n_c$	6.80	$2.27 = s_c^2$
Treatments	$3 = n_t$	78.93	$26.31 = s_t^2$
Error	$6 = n_e$	2.72	$0.45 = s_e^2$
Total	$15 =$	90.40	

Efficien.

a) With RCBD:

1) Rows as Blocks:

LSD

$$MSE(RCBD) = \frac{n_c s_c^2 + (n_t + n_e) s_e^2}{n_c + n_t + n_e}$$

$$= \frac{(3)(2.27) + (3+6)(0.45)}{3+3+6}$$

$$= 0.91$$

Efficiency (LSD to RCBD):

$$= \frac{MSE(RCB)}{MSE(LSD)} \times 100$$

$$= \frac{0.91}{0.45} \times 100 \times \frac{(6+1)(9+3)}{(6+3)(9+1)}$$

$$= 189\%$$

The LS Design is 89% more efficient than RCB Design.

ii)

MSE(

b) With

M.S

Efficien

ii) Column and blocks:

$$MSE(RCBD) = \frac{(3)(0.65) + (3+6)(0.45)}{3+3+6}$$

$$= 0.50$$

Efficiency:

$$= \frac{0.50}{0.45} \times 100 \times \frac{(6+1)(9+3)}{(6+3)(9+1)}$$

$$= 104\%$$

LSD is 4% more efficient than RCBD.

b) With CR Design:

$$MSE(CRD) = \frac{sr^2 + sc^2 + (P-1)se^2}{(P-1)}$$

$$= \frac{0.65 + 2.27 + 3(0.45)}{3-1}$$

$$= 1.423$$

Efficiency:

$$= \frac{MSE(CR)}{MSE(LS)} \times 100 \times \frac{(\eta_1+1)(\eta_2+3)}{(\eta_1+3)(\eta_2+1)}$$

$$= \frac{1.423}{0.45} \times 100 \times \frac{(6+1)(12+3)}{(12+1)(6+3)}$$

$$= 311\%$$

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How To Perform Duncan's Multiple Range Test:

To perform the test, following steps should be performed/followed.

1) Arrange the 'k' treatment means in order from low to high/ minimum to maximum

2) Enter the ANOVA table and take the error mean square with its d.f

3) Obtain the standard error of the mean for each treatment i.e

$$S_{\bar{y}.j} = \sqrt{\frac{MSE}{n}}$$

4) For unequal sample size, replace 'r' by the harmonic mean r_h of all r_i values i.e

$$r_h = \frac{k}{\sum (1/r_i)}$$

5) From the Duncan's Table of significant ranges obtain the values of

$q_{\alpha}(p, v)$ for $p = 2, 3, \dots, k$.
where α = significant level and v = d.f of error.

6) Multiply these ranges by $S_{\bar{y}.j}$ to form a group of $(k-1)$ least significant ranges i.e

$$R_p = q_{\alpha}(p, v) S_{\bar{y}.j} \text{ for } p = 2, 3, 4, 5, \dots, k$$

7) Test statistic

is compared

$p = k$

with

and

Conf

limits

7) Test the observed ranges b/w means starting with the largest vs smallest which is compared with least significant range for $p=k$; then take largest vs 2nd smallest with the least significant range for $p=k-1$ and so on.

Continue this until $\frac{k(k-1)}{2}$ possible pairs have been tested.



Factorial Design:

OR

Factorial Experiment:

"An experiment in which the treatments consists of all possible combinations of several levels of several factors"

When each treatment combination is used the same number of times, the factorial experiment is known as complete factorial experiment.

A factorial experiment, therefore is concerned with examining simultaneously a number of basic treatments or factors, each of which takes a number of possible forms or levels.

"A particular combination of the levels of the factors determines a 'treatment'."

The term 'factor' is the characteristic or feature that is under the control of the experimenter can be varied from trial to trial.

Example:

In comparing weight increases animals kept on different diets, animals of both sexes from several breeds are fed by different methods may be used in factorial design. By using each method of feeding of each breed, it is possible to determine the best method

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of diet and breeds. It is also possible to study whether the best method of feeding from diet to diet or whether the optimum method and diet depends upon the sex and breed of animal.

Consequently, by using a factorial design, we can study the manner in which the effects may vary with changes in other experimental factors, i.e. the interaction of the experimental factor. The factor design by using each combination of series of treatment and experimental conditions allows the average effect and more important their interaction one another to be estimated simultaneously.

By this approach, consistency of each effect may test and where it is shown to be dependent on the other treatments any inter relation may be study.

If the set of factors don't interact, all the observations may be used in making treatment comparison. If, however, they do interact, then attention may have to be restricted to particular combination. The existence of interaction can only be verified by the use of factorial experiment and the simultaneous determination of significant interaction greatly facilitated.

The recognition of which interaction are relevant, then allows attention to be focused on these. e.g. if we find that the best method of feeding depends upon the diet but not upon the breed or sex of animals. We may consider the different methods of feeding for each diet separately. But averaged over both sexes and all breeds.

E_{ijk}
STATIST

a) 1
b) 1
c) 1

Factor A

STATISTICAL MODEL:

Let us assume that there are two factors 'A' and 'B'. There are 'a' levels of factor 'A' and 'b' levels of factor 'B'. Then the observation y_{ijk} is the k^{th} observation for the i^{th} level of factor 'A' and j^{th} level of factor 'B' and can be expressed as:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + E_{ijk}$$

where;

μ = overall Mean

α_i = effect of i^{th} level of factor 'A'.

β_j = effect of j^{th} level of factor 'B'.

$i = 1, 2, \dots, a$

$j = 1, 2, \dots, b$

$k = 1, 2, \dots, r$

$(\alpha\beta)_{ij}$ = interaction for the i^{th} level of factor 'A' and j^{th} level of factor 'B'

Total $B_1 =$

a $y_{\text{all}} =$

1 y_{11}

2 y_{21}

i y_{i1}

a y_{a1}

$\epsilon_{ijk} = \text{Random Error} \sim NII(0, \sigma^2)$

Statistical Analysis:

- Fixed Effect Model
- Random Effect Model / Variance of Component model.
- Mixed effect Model.

Factor A	Factor B			Total.
	1	j	b	
1	$y_{111} + y_{112} + \dots + y_{11r}$ $= y_{11\cdot}$	$y_{1j1} + \dots + y_{1jr}$ $= y_{1j\cdot}$	$y_{1b1} + \dots + y_{1br}$ $= y_{1b\cdot}$	$A_1 = y_{1\cdot\cdot} = \sum_k y_{1jk}$
2	$y_{211} + y_{212} + \dots + y_{21r}$ $= y_{21\cdot}$	$y_{2j1} + \dots + y_{2jr}$ $= y_{2j\cdot}$	$y_{2b1} + \dots + y_{2br}$ $= y_{2b\cdot}$	$A_2 = y_{2\cdot\cdot} = \sum_k y_{2jk}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	$y_{i11} + y_{i12} + \dots + y_{i1r}$ $= y_{i1\cdot}$	$y_{ij1} + \dots + y_{ijr}$ $= y_{ij\cdot}$	$y_{ib1} + \dots + y_{ibr}$ $= y_{ib\cdot}$	$A_i = y_{i\cdot\cdot} = \sum_k y_{ijk}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
a	$y_{a11} + y_{a12} + \dots + y_{a1r}$ $= y_{a1\cdot}$	$y_{aj1} + \dots + y_{ajr}$ $= y_{aj\cdot}$	$y_{ab1} + \dots + y_{abr}$ $= y_{ab\cdot}$	$A_a = y_{a\cdot\cdot} = \sum_k y_{ajk}$
Total	$B_1 = y_{\cdot 1\cdot}$	$B_j = y_{\cdot j\cdot}$	$B_b = y_{\cdot b\cdot}$	$y_{\cdot\cdot\cdot} = T_{\cdot\cdot\cdot} = \sum_i \sum_j \sum_k y_{ijk}$

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1) Statistical Model for Fixed Effect Model (Model I):

In factorial designs, the observation y_{ijk} is expressed as:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

$$i=1, 2, \dots, a$$

$$j=1, 2, \dots, b$$

$$k=1, 2, \dots, r$$

with assumptions that different levels of factor A and different levels of factor B are chosen by the experimenter i.e. they are fixed.

$$\sum_{i=1}^a \alpha_i = 0, \quad \sum_{j=1}^b \beta_j = 0, \quad \sum_{i,j=1}^a \sum_{k=1}^r (\alpha\beta)_{ij} = 0$$

$$\sum_{i=1}^a (\alpha\beta)_{ij} = 0$$

and

$$\epsilon_{ijk} \sim NII(0, \sigma^2)$$

Let $\bar{y}_{i..}$, $\bar{y}_{.j.}$, $\bar{y}_{...}$ are the rows, columns and grand mean i.e.

$$\bar{y}_{i..} = \frac{y_{i..}}{br} = \frac{\sum_{j,k} y_{ijk}}{br} \quad \text{For } i=1, 2, \dots, a$$

$$\bar{y}_{.j.} = \frac{y_{.j.}}{ar}$$

$$\text{For } j=1, 2, \dots, b$$

$$\bar{y}_{...} = \frac{y_{...}}{abr}$$

$$\text{For } k=1, 2, \dots, r$$

The

$$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r y_{ijk} = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r y_{ijk}$$

$$= \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r y_{ijk}$$

By Applying

$$= br \sum_{i=1}^a$$

So, we

$$TSS =$$

where:

$$TSS = \sum_{i,j,k} y_{ijk}^2$$

$$SS_A = br \sum_{i=1}^a$$

$$SS_B = ar \sum_{j=1}^b$$

$$SS_{AB} =$$

$$SS_{AB} = \sum_{i,j} y_{ij.}^2$$

$$=$$

fect

The total sums of square can be written as:

$$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{...})^2 = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{i..} + \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{.j.} - \bar{Y}_{ij.} + \bar{Y}_{ij.} - \bar{Y}_{...} + \bar{Y}_{...})^2$$

$$= \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r \left[(\bar{Y}_{i..} - \bar{Y}_{...}) + (\bar{Y}_{.j.} - \bar{Y}_{...}) + (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}) + (Y_{ijk} - \bar{Y}_{ij.}) \right]^2$$

ation

...a

...b

...r

By Applying Summation:

$$= br \sum_{i=1}^a (\bar{Y}_{i..} - \bar{Y}_{...})^2 + ar \sum_{j=1}^b (\bar{Y}_{.j.} - \bar{Y}_{...})^2 + r \sum_{i=1}^a \sum_{j=1}^b (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$$

are

$$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{ij.})^2 + 6 \text{ cross products}$$

∴ These cross product term vanishes (zero).

So, we have.

$$TSS = SS_A + SS_B + SS_{AB} + SS_E$$

where:

$$TSS = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{...})^2 = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r Y_{ijk}^2 - \frac{(Y_{...})^2}{abr}$$

$$SS_A = br \sum_{i=1}^a (\bar{Y}_{i..} - \bar{Y}_{...})^2 = \frac{\sum_{i=1}^a Y_{i..}^2}{br} - \frac{(Y_{...})^2}{abr}$$

$$SS_B = ar \sum_{j=1}^b (\bar{Y}_{.j.} - \bar{Y}_{...})^2 = \frac{\sum_{j=1}^b Y_{.j.}^2}{ar} - \frac{(Y_{...})^2}{abr}$$

a

$$SS_{AB} = ar \sum_{i=1}^a \sum_{j=1}^b (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$$

$$SS_{AB} = \left\{ \sum_{i=1}^a \sum_{j=1}^b \frac{Y_{ij.}^2}{r} - \frac{(Y_{...})^2}{abr} \right\} - \left\{ \frac{\sum_{i=1}^a Y_{i..}^2}{br} - \frac{(Y_{...})^2}{abr} \right\} - \left\{ \frac{\sum_{j=1}^b Y_{.j.}^2}{ar} - \frac{(Y_{...})^2}{abr} \right\}$$

$$= TSS_{\text{subtable AB}} - SS_A - SS_B$$

$$SS_{\text{Error}} = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{ij})^2$$

$$= \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r Y_{ijk}^2 - \frac{\sum_{i=1}^a \sum_{j=1}^b Y_{ij}^2}{r}$$

OR

$$SS_E = TSS - SS_A - SS_B - SS_{AB}$$

$$\therefore SS_{AB} = TSS_{AB} - SS_A - SS_B$$

$$SS_E = TSS - \cancel{SS_A} - \cancel{SS_B} - TSS_{AB} + \cancel{SS_A} + \cancel{SS_B}$$

$$SS_E = TSS - TSS_{AB}$$

Remarks:

1) The TSS has been partitioned into sums of squares due to rows, sum of squares due to columns, sum of squares due to interaction b/w the factors A and B and sum of squares of error.

2) we must have atleast two replicates i.e. ($r \geq 2$) to obtain the error SS.

3) The t.d.f, df_A , df_B , df_{AB} and df_E are as follows:

$$\text{d.f for } TSS = abr - 1$$

$$\text{d.f for } SS_A = a - 1$$

$$\text{d.f for } SS_B = b - 1$$

$$\text{d.f for } SS_{AB} = (a-1)(b-1)$$

$$\text{d.f for } SS_E = ab(r-1).$$

When each sum of squares is divided by the respective d.f and we take expectation on it, we get;

$$E(MSA) = E\left(\frac{SS_A}{a-1}\right) = \sigma^2 + \frac{br}{a-1} \sum_{i=1}^a \alpha_i^2$$

$$E(MSB) = E\left(\frac{SS_B}{b-1}\right) = \sigma^2 + \frac{ar}{b-1} \sum_{j=1}^b \beta_j^2$$

$$E(MS_{AB}) = E\left(\frac{SS_{AB}}{(a-1)(b-1)}\right) = \sigma^2 + \frac{r \sum_{i,j}^a \sum_{j=1}^b (\alpha\beta)_{ij}}{(a-1)(b-1)}$$

and

$$E(MSE) = E\left(\frac{SSE}{ab(r-1)}\right) = \sigma^2$$

The F-Statistic for Factor A, Factor B and for the interaction is called out as:

$$F\text{-Statistic for A} = \frac{MSA}{MSE}$$

$$F\text{-Statistic for B} = \frac{MSB}{MSE}$$

$$F\text{-Statistic AB} = \frac{MS_{AB}}{MSE}$$

The Analysis of variance table is carried out as below:

ANOVA TABLE

Source of Variation	Degree of freedom	Sum of Squares	Mean Squares	F-Ratio
Factor A	$a-1$	$br \sum_i (\bar{Y}_{i.} - \bar{Y}_{..})^2$	MS_A	$F_A = MS_A / MSE$
Factor B	$b-1$	$ar \sum_j (\bar{Y}_{.j} - \bar{Y}_{..})^2$	MS_B	$F_B = MS_B / MSE$
Interaction AB	$(a-1)(b-1)$	$r \sum_{ij} (\bar{Y}_{ij.} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2$	MS_{AB}	$F_{AB} = MS_{AB} / MSE$
Error	$ab(r-1)$	$\sum_{ijk} (Y_{ijk} - \bar{Y}_{ij.})^2$	MSE	
Total	$abr-1$	$\sum_{ijk} (Y_{ijk} - \bar{Y}_{...})^2$		

Null Hypotheses:

- i) $H_0' : \alpha_i = 0$
- ii) $H_0'' : \beta_j = 0$
- iii) $H_0''' : (\alpha\beta)_{ij} = 0$

2) Random Effect Model or Component of Variance Model:

In factorial design, the obs Y_{ijk} is expressed as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

$$i=1, 2, \dots, a$$

$$j=1, 2, \dots, b$$

$$k=1, 2, \dots, r$$

Ratio

In this model, different levels of factor A and factor B are chosen at random from larger population. This means that α_i , β_j and $(\alpha\beta)_{ij}$ are distributed randomly with assumptions

MSA/MSE

$$\alpha_i \sim NII(0, \sigma_\alpha^2)$$

MSB/MSE

$$\beta_j \sim NII(0, \sigma_\beta^2)$$

MSAB/MSE

$$(\alpha\beta)_{ij} \sim NII(0, \sigma_{\alpha\beta}^2)$$

$$\varepsilon_{ijk} \sim NII(0, \sigma^2)$$

and the variance of any obs is:

$$\text{Var}(Y_{ijk}) = \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_{\alpha\beta}^2 + \sigma^2$$

where σ_α^2 , σ_β^2 , $\sigma_{\alpha\beta}^2$ are called variance components and various hypothesis are:

$$i) H_0' : \sigma_\alpha^2 = 0$$

$$ii) H_0'' : \sigma_\beta^2 = 0$$

$$iii) H_0''' : \sigma_{\alpha\beta}^2 = 0$$

The basic ANOVA remains unchanged i.e. SS_A , SS_B , SS_{AB} , SS_E , TSS are all calculated as in the fixed effect Model/Case.

However, to perform the test statistic we must examine the expected mean square i.e.

$$E(MSA) = \sigma^2 + r\sigma_{\alpha\beta}^2 + b\sigma_\alpha^2$$

$$E(MSB) = \sigma^2 + r\sigma_{\alpha\beta}^2 + a\sigma_\beta^2$$

$$E(MS_{AB}) = \sigma^2 + r\sigma_{\alpha\beta}^2$$

$$E(MSE) = \sigma^2$$

And F-Statistic is calculated as:

$$F_A = \frac{MS_A}{MSE}, F_B = \frac{MS_B}{MSE}, F_{AB} = \frac{MS_{AB}}{MSE}$$

when

Mixed Effect Model:

Case I:

Different levels of factor A are selected randomly and that of factor B are chosen by experimenter. It means that effect of factor A is random and effect of factor B is fixed i.e.

$$\alpha_i's \sim NII(0, \sigma_\alpha^2)$$

$$\sum_j \beta_j = 0 \quad \sum_j (\alpha\beta)_{ij} = 0$$

$$(\alpha\beta)_{ij} \sim NII(0, \sigma_{\alpha\beta}^2)$$

Null Hypothesis are:

$$H_0': \sigma_\alpha^2 = 0$$

$$H_0'': \beta_j = 0$$

$$H_0''': \sigma_{\alpha\beta}^2 = 0$$

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Case II:

Different levels of factor A are chosen by experimenter and that of factor B are selected randomly. i.e. α_i is fixed and β_j is random.

With assumptions

$$\sum_{i=1}^a \alpha_i = 0$$

$$\sum_j (\alpha\beta)_{ij} = 0$$

$$\beta_j's \sim NII(0, \sigma_{\beta}^2)$$

$$(\alpha\beta)_{ij} \sim NII(0, \sigma_{\alpha\beta}^2)$$

Null Hypothesis Are:

i) $H_0': \alpha_i = 0$

ii) $H_0'': \sigma_{\beta}^2 = 0$

iii) $H_0''': \sigma_{\alpha\beta}^2 = 0$

Example:

A soft drink bottler is studying the effect of percent carbonation (A) operating pressure in the filter (B) and line speed (C) on the volume of carbonation soft drink on each bottle. Three levels of carbonation, 2 levels of pressure and 2 levels of speed are selected and a factorial design with two replicates is conducted. The order in which 24 observations are taken is determined

randomly. The coded data that results from this experiment is shown in the table below. Analysis the data.

A	B ₁ 25%		B ₂ 30%		Total
	C ₁	C ₂	C ₁	C ₂	
	200	250	200	250	
10	-3	-1	-1	1	10
	-1	0	0	1	
12	0	2	2	6	12
	1	1	3	5	
14	5	7	7	10	14
	4	6	9	11	
					Total
					Co

Solution:

Testing Procedure:

1) Null and Alternative Hypothesis are:

TSS

H₀: The effect of A is zero

H₀': The effect of B is zero

H₀": The effect of C is zero

H₀''': There is no interaction effect of AB

H₀'''': " " AC

H₀''': " " BC

H₀''': " " ABC

SSA

2) level of significance

$$\alpha = 0.05$$

3) Test-Statistic

$$F = \frac{\text{MS for the effect}}{\text{MS for error}}$$

SSB

4) Calculation.

from
table

	B ₁		B ₂		
A	C ₁	C ₂	C ₁	C ₂	Total
10	-3 $\boxed{4}$	-1 $\boxed{4}$	-1	1	-4
	-1	0	0	1	
12	0	2	2	6	20
	1	1	3	5	
14	5	7	7	10	59
	4	6	9	11	
Total	6	15	20	34	75

Correction Factor:

$$C.F. = \frac{(\sum \sum \sum V_{ijk})^2}{abcr}$$

a = 3

b = 2

c = 2

r = 2

$$C.F. = \frac{(75)^2}{24} = 234.375$$

Ans:

$$TSS = \sum \sum \sum V_{ijk}^2 - C.F.$$

$$= \{(-3)^2 + (-1)^2 + \dots + (11)^2\} - 234.375$$

$$TSS = 336.625$$

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$$SS_A = \frac{\sum V_{i..}^2}{bcr} - C.F.$$

$$= \frac{(-4)^2 + (20)^2 + (59)^2}{(2)(2)(2)} - 234.375$$

$$SS_A = 252.75$$

$$SS_B = \frac{\sum V_{.j.}^2}{acr} - C.F.$$

$$= \frac{(20)^2 + (34)^2}{12} - 234.375$$

$$SS_B = 45.375$$

$$SS_C = \frac{\sum \frac{V_{..j}^2}{abr} - C.F}{12}$$

$$= \frac{(26)^2 + (49)^2}{12} - 234.375$$

$$SS_C = 22.042$$

Now for the interaction AB:

The subtable of interaction of AB is:

A \ B	25%	30%	Total
10	-5	1	-4
12	4	16	20
14	22	37	59
Total	21	54	75

$$TSS_{AB} = \sum_{i,j} \frac{V_{ij}^2}{cr} - C.F$$

$$= \frac{\{(-5)^2 + (4)^2 + \dots + (16)^2 + (37)^2\}}{(2)(2)} - 234.375$$

$$TSS_{AB} = 303.375$$

$$SS_{AB} = TSS_{AB} - SS_A - SS_B$$

$$SS_{AB} = 5.28$$

The subtable for interaction AC is:

A \ C	10	12	14	Total
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B \ C

25%

30%

Total

T

A \ C	200	250	Total
10	-5	1	-4
12	6	4	20
14	25	34	59
Total	26	49	75

is:

$$TSS_{AC} = \frac{\sum_{i,j} \frac{Y_{ij}^2}{br}}{br} - C.F$$

$$= \frac{\{(-5)^2 + \dots + (34)^2\}}{(2)(2)} - 234.375$$

$$TSS_{AC} = 275.375$$

$$SS_{AC} = TSS_{AC} - SS_A - SS_C$$

$$SS_{AC} = 0.583$$

The Subtable for the interaction BC is:

B \ C	200	250	Total
25%	6	15	21
30%	20	34	54
Total	26	49	75

$$TSS_{BC} = \frac{\sum_{i,j} \frac{Y_{ij}^2}{br}}{br} - C.F$$

$$= \frac{\{(6)^2 + \dots + (34)^2\}}{6} - 234.375$$

$$= 68.458.$$

$$SS_{BC} = TSS_{BC} - SS_B - SS_C$$

$$SS_{BC} = 1.041$$

$$\begin{aligned} TSS_{ABC} &= \sum_{i=1}^n \sum_{j=1}^2 \frac{Y_{ij}^2}{r} - C.F \\ &= \frac{(-4)^2 + (1)^2 + \dots + (21)^2}{2} - 234.375 \\ &= 328.125 \end{aligned}$$

$$SS_{ABC} = TSS_{ABC} - SS_A - SS_B - SS_C - SS_{AB} - SS_{BC} - SS_{AC}$$

$$SS_{ABC} = 1.084$$

$$ESS = TSS - TSS_{ABC}$$

$$ESS = 8.50$$

ANOVA TABLE

S.O.V	Df	Sum of S	M.S	F-Ratio	F-Table
A	2	252.75	126.375	178.50	$> F_{0.05(2,12)} = 3.88$
B	1	45.375	45.375	64.089	> 4.75
C	1	22.042	22.042	31.133	> 4.75
AB	2	5.25	2.625	3.708	< 3.88
AC	2	0.583	0.2915	0.412	< 4.75
BC	1	1.041	1.041	1.470	< 3.88
ABC	2	1.084	0.542	0.766	< 4.75
Error	12	8.50	0.708		
Total	23	336.625	—		

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6) Conclusion:

The effects of A, B, C is ^{not} zero As their calculated values of F falls in Rejection Region. and The effects of AB, BC, CA, ABC is zero as their cal values of F lies in Acceptance Region.

F-Table

$$F_{0.05(3,12)} = 3.88$$

$$> 4.75$$

$$> 4.75$$

$$3.88$$

$$4.75$$

$$3.88$$

$$4.75$$

2^k Factorial Design:

Factorial designs are widely used in experiments involving several factors where it is necessary to study the joint effect of these factors on a response. Uptil now we have presented a general method for the analysis of factorial design. However, there are special cases of the general factorial design that are important because they are widely used in research work and also because they form the basis of other designs of considerable practical value.

The first of these special cases is that of ' k ' factors each had two levels only. These levels may be quantitative or qualitative. A complete replicate of such design required $2 \times 2 \times 2 \dots 2$ (k factor) = 2^k observations and is called 2^k factorial design.

The second special case is that of k factors each had 3 levels and is called 3^k factorial design.

Assumptions:

- i) The factors are fixed.
- ii) The design is completely randomized (CR)
- iii) The usual normality.

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A Low, B Low
A high, B Low
A Low, B High
A high, B High

2^2 Factorial Design:

The simplest case to consider is in which two factors are of interest and each factor is set at just two levels. This type of design is called 2^2 factorial design.

The levels of factors say A and B may arbitrarily be called 'low' and 'high' or 'absent' or 'present'.

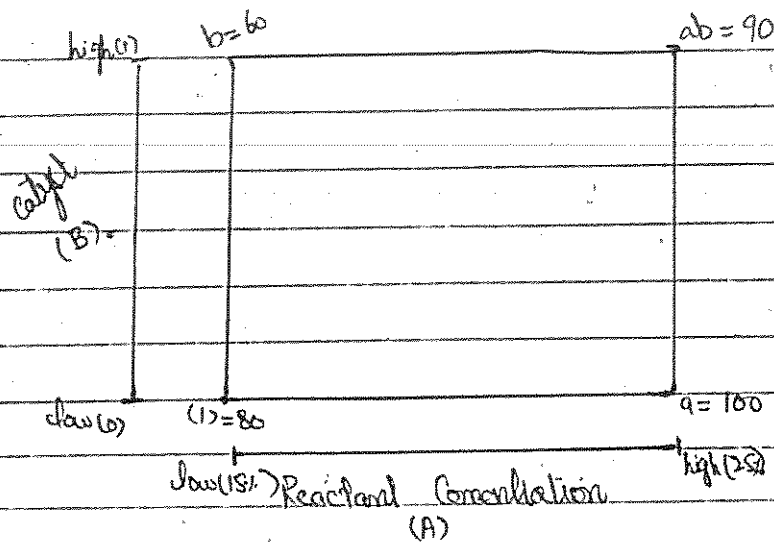
Example:

Consider an investigation into the effect of concentration of reactant and presence of catalyst on the reaction time of a chemical process. Let the concentration of reactants be known as factor 'A' and the two levels of interest be 15% and 25%. Catalyst being factor B the high level denoting the presence of catalyst and the low level denoting the absence of catalyst.

Assuming three replicates, the data are given below

Replicates

	I	II	III		
1 (CR)	A Low, B Low	28	25	27	80 = (1)
	A high, B Low	36	32	32	100 (a)
	A Low, B high	18	19	23	60 = b
	A high, B high	31	30	29	90 = ab



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By convention, we denote the effect of a factor by a capital letter. Thus 'A' refers to the effect of factor 'A' and 'B' refers to the effect of factor 'B' and 'AB' refers to the interaction 'AB'.

The treatment combinations are usually represented by a lower case letter.

Conventionally, 'a' represents the treatment combination of factor 'A' at high level and factor B at low level. 'b' represents the t.c. of factor B at high level and factor A at low level and '1' represents both factors at low levels.

Define the average effect of a factor as the change in response produced by a change in the levels of that factor averaged over the levels of other factors.

Remarks:

The Jewel case letters "1", "a", "b" and "ab" also represented the total of 'n' replicates taken at the treatment combination.

Now the effect A at the low level of B is $\frac{[a - (1)]}{2n}$ and the

effect of A at high level of B is $\frac{[ab - b]}{2n}$. Averaging these two quantities n yields:

$$A = \frac{1}{2n} [(ab - b) + (a - (1))]$$

$$A = \frac{1}{2n} [ab - b + a - (1)]$$

$$A = \frac{1}{2n} [a(b+1) - 1(b+1)]$$

$$A = \frac{1}{2n} [(a-1)(b+1)]$$

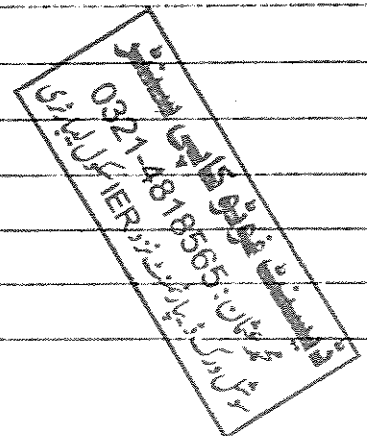
Similarly;

The average effect of B is found as

$$B = \frac{1}{2n} [(b - (1)) + (ab - a)]$$

$$B = \frac{1}{2n} [b - (1) + ab - a]$$

$$B = \frac{1}{2n} (b-1)(a+1)$$



Define the interaction effect AB as average difference b/w at the high level of B and high effect of A at low level of B.

$$AB = \frac{1}{2n} [(ab - b) - (a - 1)]$$

$$AB = \frac{1}{2n} (a-1)(b-1)$$

And the usual mean represent by 'M' is computed as:

$$M = \frac{1}{4n} \{ (1) + a + b + ab \}$$

$$M = \frac{1}{4n} (a+1)(b+1)$$

These effects can be conveniently found/written in a table of "+" and "-" sign as below i.e.

Effects	(1)	T.C	a	b	ab	Divisor
M	+	+	+	+	+	4n
A	-	+	+	-	-	2n
B	-	-	+	+	-	2n
AB	+	-	-	+	+	2n

where n = No. of replicates

Note that effect A, B and AB are three mutually orthogonal contrast each based on one degree of freedom. The contrast coefficients for estimating the interaction effect are the first product of the corresponding coefficients for the two main effects. A table of "+" and "-" (minus) signs can be used to determine the proper sign for each treatment combinations.

Using the above sample data, the solution is as follows:

Average Effects:

$$A = \frac{1}{2n} [ab + a - b - (1)].$$

$$A = \frac{1}{2(3)} [90 + 100 - 60 - 80]$$

$$A = 8.33$$

$$B = \frac{1}{2n} [ab + b - a - (1)].$$

$$B = \frac{1}{6} [90 + 60 - 100 - 80]$$

$$B = -5.00.$$

$$AB = \frac{1}{2n} [(1) - a - b + ab].$$

$$= \frac{1}{6} [80 - 100 - 60 + 90]$$

$$AB = 1.67.$$

[A]; Total of A and A Average effect

$$SS_A = \frac{[A]^2}{2^k \cdot n} = \frac{(50)^2}{4(3)} = 208.33$$

$$SS_B = \frac{[B]^2}{2^k \cdot n} = \frac{(-30)^2}{4(3)} = 75$$

$$SS_{AB} = \frac{[AB]^2}{2^k \cdot n} = \frac{(10)^2}{12} = 8.33$$

$$TSS = \sum \sum \sum \frac{1}{n} y_{ijk}^2 - \frac{(V_{..})^2}{2^k \cdot n}$$

$$= 9398.00 - 9075.00$$

$$TSS = 323.00$$

$$SSE = TSS - SS_A - SS_B - SS_{AB}$$

$$SSE = 31.34$$

ANOVA TABLE

S.O.V	Df	S.S	M.S	F-Ratio	Fo-Table
A	2-1=1	208.33	208.33	53.15 >	4.84
B	1	75.00	75.00	19.13 >	4.84
AB	1	8.33	8.33	2.13 <	4.84
Error	8	31.34	3.92		
Total	11	323.00	—		

Conclusion:

The effects of A and B are significant and effect of AB is insignificant.

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Single Replicate of 2^k Design:

From even a moderate number of factors, the total number of t.c. in a 2^k factorial design is very large. For example, a 2^5 has 32 treatment combinations, a 2^6 has 64 t.c., or 2^7 has 128 t.c. and so on.

Since the resources are usually limited, the number of replicates that the experimenter can employ may be restricted.

Frequently available resources only allow a single replicate of the design to be run unless the experimenter is willing to omit some of the original factors. We will only a single replicate of 2^k design.

It is impossible to compute an estimate of experimental error i.e. Mean Square Error (MSE). Hence it seems

that hypothesis concerning main effect and interactions cannot be tested. In order to perform these tests, we have to estimate the MSE either from some external sources or by pooling some of the higher order interactions that may be considered insignificant.

For Table

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Yates Technique For Computing Contrast:

For large, there is a difficulty in writing out the table of sign. A systematic method for computing the contrast in a 2^k factorial experiment developed by Yates Method is given below

- 1) Write down the treatment combinations and observations in standard form.
- 2) Add the treatment combinations/observations in pairs and write down in column mark I. This fills up the top half of the column I.

For the bottom half write down the differences of pairs by subtracting the upper from the lower treatment combination. This completes column I.

- 3) Obtain column mark II in exactly the same way using the results in column I i.e. operating on the results of preceding column.

4) Continue this operation until the k th column for a 2^k experiment is reached. The column k corresponds to the effect. The first value will be the grand total of the entries experiment each remaining value will be a contrast in a t.c

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In that case, the sum of square for the main effect and the interaction effect are calculated by squaring each of the value give C_k and divide the results by " $2^k \cdot n$ " where n is the No. of the replicates.

2^2 Design:

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Treatment Combinations	Response	Column I	Column II
(1)		(1) + a	(1) + a + b + ab $\rightarrow$ grand total
a		b + ab	a - (1) + ab - b $\rightarrow$ Total of effect A
b		a - (1)	-(1) + a + b + ab $\rightarrow$ " B
ab		ab - b	ab - b - a + (1) $\rightarrow$ " AB

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Example:

$$(1) = 80, \quad a = 100, \quad b = 60, \quad ab = 90$$

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By Yates Method:

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T.C	Response	C_1	C_2	$SS = (\text{Contrast})^2 / 2^k \cdot n = 12$
(1)	80	180	330	9075
a	100	150	50	20833
b	60	20	-30	75
ab	90	30	10	8.33

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Rest of the procedure is same.

2^3 Factorial Design:

In this case, consider 3 factors each at two levels. This design is known as 2^3 factorial design and the 8 treatment combination can be displayed graphically as a cube.

Extending the previous notation, we write the t.c in standard order as (1), a, b, ab, c, ac, bc, abc. The 7 d.f. w/ the 8 t.c are associated with the main effect A, B, C; the two interaction effect AB, BC, AC; and the three factor effect ABC.

Consider estimating the main effect A, the effect of A when B and C are at low level is $(a - (1))$, the effect of A when B is at high n level and C is at low level is $(ab - b)$, the effect of A when B is at n low level and C is at high level is $(ac - c)$.

Finally, the effect of A when both factors A and B are at high level is $(abc - bc)$.

The average effect A is just the average of these 4:

$$A = \frac{1}{4n} [(a - (1)) + (ab - b) + (ac - c) + (abc - bc)]$$

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$$AB = \frac{1}{4n}$$

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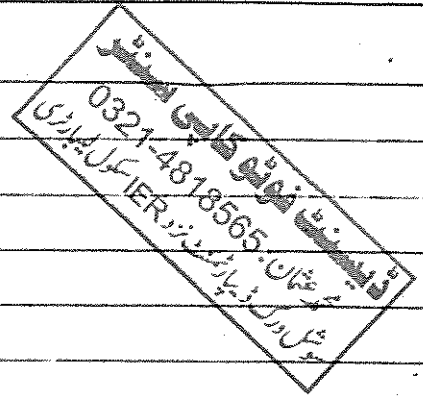
AC

$$A = \frac{1}{4n} [(a-1)(b+1)(c+1)]$$

Similarly;

$$C = \frac{1}{4n} [(c-1)(b+1)(a+1)]$$

$$B = \frac{1}{4n} [(b-1)(a+1)(c+1)]$$



The effect of interaction AB when C is at low level is the average difference b/w the effect of A at high level of B and the effect of A at low level of B.

i.e. $AB = \frac{1}{2n} [(ab-b) - (a-c)]$.

But

When C is at high level, the average effect of AB is the average difference b/w the effect of A at high level of B and at low level of B.

i.e.

$$AB = \frac{1}{2n} [(abc-bc) - (ac-c)]$$

factors is

Thus; the average effect of AB is the average of the above 2 eqs:

$$AB = \frac{1}{4n} [(a-1)(b-1)(c+1)]$$

Similarly;

$$AC = \frac{1}{4n} [(a-1)(c-1)(b+1)]$$

-bc)]

$$BC = \frac{1}{4n} [(a+1)(b-1)(c-1)]$$

$$ABC = \frac{1}{4n} [abc - bc - ac + c - ab + b + a - (1)]$$

$$ABC = \frac{1}{4n} [(a-1)(b-1)(c-1)]$$

The sign table for 2^3 factorial design is

Effects		T. C						
	(1)	a	b	ab	c	ac	bc	abc
M	+	+	+	+	+	+	+	+
A	-	+	-	+	-	+	-	+
B	-	-	+	+	-	-	+	+
C	-	-	-	-	+	+	+	+
AB	+	-	-	+	+	-	-	+
AC	+	-	+	-	-	+	-	+
BC	+	+	-	-	-	-	+	+
ABC	-	+	+	-	+	-	-	+

$$\text{Average effect} = \frac{[A] \rightarrow \text{effect}}{2^{k-1} \cdot n}$$

$$SS \text{ of effect} = \frac{(\text{Effect})^2}{2^{k-1} \cdot n}$$

Properties of Sign Table:

The sign table has the following properties.

- 1) Except Column I, every column has an equal No. of (+) & (-) signs.

2) The

3) By with a remain identically

4) The row in

Question

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B₀
C₀

D₀ 45 (1)
D₁ 43 (d) 7

Analyze
i) Yates
ii) By 6

2) The sum of the product of the sign in any two columns is zero.

3) By multiplying the signs of Column Row "M" with another row, the next row will remain unchanged i.e. the column is an identity column.

4) The product of any two rows yields a row in a table i.e.

$$A \times B = AB$$

and

$$AB \times B = AB^2 = A$$

Question :

A chemical product is produced in a factorial design was carried out to study the four factors A, B, C & D. Each factor is present at two levels and data obtained from single replicate are shown below:

	A ₀				A ₁			
	B ₀		B ₁		B ₀		B ₁	
	C ₀	C ₁	C ₀	C ₁	C ₀	C ₁	C ₀	C ₁
D ₁	45 ^(a)	68 ^(c)	48 ^(b)	80 ^(d)	71 ^a	60 ^{ac}	65 ^{ab}	65 ^{abc}
D ₂	43 ^(d)	75 ^(cd)	45 ^(bd)	79 ^(bcd)	100 ^{ad}	86 ^{acd}	104 ^{abd}	96 ^{abcd}

Analyze the data by using

i) Yates Method

ii) By Sign Table Method.

Solution:

172

1) Yates Method:

T.C	Response	C ₁	C ₂	C ₃	C ₄	SS = (Contrast) ² / 2 ^k · n	
(1)	45	116	229	502	1115	77701.5625	BC +
a	71	113	273	613	167	1743.0625	BD +
b	48	128	292	20	31	60.0625	CD +
ab	65	145	321	147	7	3.0625	ABC -
c	68	143	43	14	73	333.0625	ABD -
ac	60	149	-23	17	-151	1425.0625	BCD -
bc	80	155	116	-16	25	39.0625	ABCD +
abc	65	166	31	23	21	27.5625	
d	43	26	-3	44	11	770.0625	
ad	100	17	17	29	127	1008.0625	
bd	45	-8	6	-66	3	0.5625	
abd	104	-15	11	-85	39	95.0625	
cd	75	57	-9	20	-15	14.0625	
acd	80	59	-7	5	-19	22.5625	
bcd	70	5	2	2	-15	14.0625	
abcd	96	26	21	19	17	18.0625	

$$SS_A = \frac{1}{2^k}$$

$$SS_A = \frac{1}{16}$$

$$SS_B = \frac{1}{2^k}$$

By Using Sign table:

Effects	T. C															
	(1)	a	b	ab	c	ac	bc	abc	d	ad	bd	abd	cd	acd	bcd	abcd
M	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
A	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+
B	-	-	+	+	-	-	+	+	-	-	+	+	-	-	+	+
C	-	-	-	+	+	+	+	+	-	-	-	+	+	+	+	+
D	-	-	-	-	-	-	-	-	+	+	+	+	+	+	+	+
AB	+	-	-	+	+	-	-	+	+	-	-	+	+	-	-	+
AC	+	-	+	-	-	+	-	+	+	-	+	-	-	+	-	+
AD	+	-	+	-	+	-	+	-	-	+	+	-	-	+	-	+

$$SS_C =$$

$$SS_{AC} =$$

BC	+	+	-	-	-	-	+	+	+	+	-	-	-	-	+	+
BD	+	+	-	-	+	+	-	-	+	+	+	+	-	-	+	+
CD	+	+	+	+	-	-	-	-	-	-	-	-	+	+	+	+
ABC	-	+	+	-	+	-	-	+	-	+	+	-	+	-	-	+
ABD	-	+	+	-	-	+	+	-	+	-	-	+	+	-	-	+
BCD	-	-	+	+	+	+	-	-	+	+	-	-	-	-	+	+
ABCD	+	-	-	+	+	+	-	-	+	+	-	-	+	-	-	+

$$SS_A = \frac{1}{2^k \cdot n} \left[(-1) + a - b + ab - c + ac - bc + abc - d + ad - bd + abd - cd + acd - bcd + abcd \right]^2$$

$$SS_A = \frac{1}{16} \left[-45 + 71 - 48 + 65 - 68 + 60 - 80 + 65 - 43 + 100 - 45 + 104 - 75 + 80 - 70 + 96 \right]^2$$

$$SS_A = 1743 \cdot 0625$$

$$SS_B = \frac{1}{2^k \cdot n} \left[(a+1)(b-1)(c+1)(d+1) \right]^2$$

$$= \frac{1}{16} (31)^2 = 60 \cdot 0625$$

$$SS_{AB} = \frac{1}{2^k \cdot n} \left[(a-1)(b-1)(c+1)(d+1) \right]^2$$

$$= \frac{1}{16} (-7)^2 = 3 \cdot 0625$$

$$SS_C = \frac{1}{2^k \cdot n} \left[(a+1)(b+1)(c-1)(d+1) \right]^2$$

$$= \frac{1}{16} (73)^2 = 333 \cdot 0625$$

$$SS_{AC} = \frac{1}{2^k \cdot n} \left[(a-1)(c-1)(b+1)(d+1) \right]^2$$

$$= \frac{(-151)^2}{16} = 1425 \cdot 0625$$

$$SS_{BC} = \frac{1}{2^k \cdot n} [(b-1)(c-1)(a+1)(d+1)]^2$$

$$= \frac{1}{16} (25)^2 = 39.0625$$

$$SS_{ABC} = \frac{1}{2^k \cdot n} [(a-1)(b-1)(c-1)(d+1)]^2$$

$$= \frac{1}{16} (21)^2 = 27.5625$$

$$SS_D = \frac{1}{2^k \cdot n} [(a+1)(b+1)(c+1)(d-1)]^2$$

$$= \frac{(111)^2}{16} = 770.0625$$

$$SS_{AD} = \frac{(127)^2}{16} = 1008.0625$$

$$SS_{BD} = \frac{(3)^2}{16} = 0.5625$$

$$SS_{ABD} = \frac{(39)^2}{16} = 95.0625$$

$$SS_{CD} = \frac{(-15)^2}{16} = 14.0625$$

$$SS_{ACD} = \frac{(-19)^2}{16} = 22.5625$$

$$SS_{BCD} = \frac{(-15)^2}{16} = 14.0625$$

$$SS_{ABCD} = \frac{(17)^2}{16} = 18.0625$$

S.O.V	df
A	1
B	1
C	1
D	1
AB	1
AC	1
AD	1
BC	1
BD	1
CD	1
ABC	1
ABD	1
ACD	1
BCD	1
ABCD	1
Total	(16-1)=

Conclude

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ANOVA TABLE

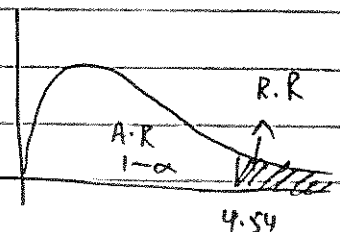
S.O.V	df	S.S	M.S	F-Ratio	F-Table
A	1	1743.0625	1743.0625	9.83	> 6.61
B	1	60.0625	60.0625	0.338	< 6.61
C	1	333.0625	333.0625	1.8784	< 6.61
D	1	770.0625	770.0625	4.343	< 6.61
AB	1	3.0625	3.0625	0.0172	< 6.61
AC	1	1425.0625	1425.0625	8.0370	> 6.61
AD	1	1008.0625	1008.0625	5.6852	< 6.61
BC	1	39.0625	39.0625	0.2203	< 6.61
BD	1	0.5625	0.5625	0.003172	< 6.61
CD	1	14.0625	14.0625	0.07931	< 6.61
ABC	1	27.5625	27.5625		
ABD	1	95.0625	95.0625		
ACD	1	22.5625	22.5625		
BCD	1	14.0625	14.0625		
ABCD	1	18.0625	18.0625		
Total	16-1=15	5573.4375			

$$F_{0.05}(1,5) = 6.61$$

Conclusion:

The effects of A, AC, are significant because their calculated values of F lies in Rejection Region.

And the remaining effects are insignificant because their calculated values of F falls in Acceptance Region.



Confounding:

In a factorial experiment, to reduce the size of the blocks and obtain a greater error control, the device confounding is introduced.

The confounding involves reflection of certain comparisons seems relatively unimportant and arranging the treatment combinations in such a way that these unimportant comparisons can't be separate from the block effect. The block being the subdivision of the replicate. The comparison entangled (mixed up) with block effect are said to be confounded with blocks.

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Confounding in 2^k Factorial Design:

The simplest confounding scheme are those for the 2^k factorial experiment and these designs can be run in two blocks, four blocks, 8 blocks and so on.

2^k Factorial Design In Two Blocks:

Suppose that we wish to run a single replicate of the 2^k design.

F

B

Each of $2^2 = 4$ treatment combinations (T.C.) required the quantity of raw material.

For example:

Each batch of raw material is only large enough for two T.C. to be tested. Thus, two batches of raw material are required. If batches of raw material considered as blocks then we must assign 2 of the 4 T.C. to each block. Conduct the design shown below.

Block I

Block II

(1)

a

ab

b

Note that block I contains the T.C. (1) and ab and that block II contains 'a' and 'b'. Of course, the order in which the T.C.s are run within the blocks is randomly determined and we would also run randomly decide which run will first.

Suppose, we estimate the main effect A and B just as if no blocking has occurred then

$$A = \frac{1}{2} (ab + a - b - (1)).$$

$$B = \frac{1}{2} (ab + b - a - (1)).$$

Note that both A and B are unaffected by blocking. Since in each estimate there is one (+) and one (-) T.C. from each block i.e. any difference b/w Block I and Block II will cancel out.

Consider the interaction AB:

$$AB = \frac{1}{2} [(a-1)(b-1)] = \frac{1}{2} [ab+1-a-b]$$

Since the two T.C's with '+' sign are in block I and two T.C's with '-' sign are in block II.

Hence the block effect and the interaction effect are identical i.e. AB is confounded with blocks.

The reason for this is apparent from the table of (+) and (-) sign for 2^2 factorial design.

	M	A	B	AB
(1)	+	-	-	+
a	+	+	-	-
b	+	-	+	-
ab	+	+	+	+

From table we see that all T.C's that have a plus (+) sign on AB are assigned to block I and all T.C's that have a (-) sign on AB are assigned to block II.

These confound with 'ab' as 'a' then confound confound

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This approach can be used to confound any effect i.e. A, B and AB with blocks. For example, if '12' and '3' are assigned to block I and '4' and '5' are assigned to block II then the main effect will be confounded with blocks (A will be confounded with blocks).

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The normal practice is to confound highest order interaction with blocks. This scheme can be used to confound any 2^k factorial design in two blocks.

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Sign Table Method:

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Consider a 2^3 design runs in two blocks. Suppose we wish to confound the 3 factor interaction ABC with blocks then from the table of '+' and '-' signs shown below.

Effects

T.C	M	A	B	AB	C	AC	BC	ABC
(1)	+	-	-	+	-	+	+	-
a	+	+	-	-	-	-	+	+
b	+	-	+	-	-	+	-	+
ab	+	+	+	+	-	-	-	-
c	+	-	-	+	+	-	-	+
ac	+	+	-	-	+	+	-	-
bc	+	-	+	-	+	-	+	-
abc	+	+	+	+	+	+	+	+

have for i.e. linear L:

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we assign treatment combinations that (-) on ABC to block I and (+) on ABC to block II.

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The resulting design is shown below:

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Block I (-)

Block II (+)

(1)

a

ab

b

ac

c

bc

abc

the

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It is also noted that the t.c.s within a block are run in random order.

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Kempthorn Hicks And Adason Mclean Method / Defining Contrast Method:

Kempthorn Hicks and Adason Mclean have described another method for constructing the blocks of t.c's i.e. confounded. Their method uses the linear combination.

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k \quad \text{--- (A)}$$

where x_i is the level of the i th factor appearing in a particular T.C. and α_i is the exponent appearing on the i th factor in the effect to be confounded.

For the 2^k design, we have α_i either equal to zero ($\alpha_i = 0$) or equal to 1 ($\alpha_i = 1$) and either ($x_i = 0$) low or equal to 1 ($x_i = 1$) high.

The equation (A) is called the defining contrast.

Treatment Combination that produced the same value of linear combination $L \pmod{2}$ will be placed in the same block. Since the only possible of $L \pmod{2}$ are '0' and '1'.

This will assign 2^k t.c's to exactly two blocks. To illustrate this approach,

Construct a 2^3 design with ABC confounded with blocks.

Here α_1 corresponds to factor A, α_2 corresponds to factor B, α_3 corresponds to factor C.

$$\text{and } \alpha_1 = \alpha_2 = \alpha_3 = 1$$

Thus the defining contrast corresponding to ABC is

$$L = \alpha_1 + \alpha_2 + \alpha_3$$

T.C is written in '0' and '1' notations as

(1)	$L = (1)(0) + (1)(0) + (1)(0) = 0$	$0 \pmod{2} = 0$
a	$L = (1)(1) + (1)(0) + (1)(0) = 1$	$1 \pmod{2} = 1$
b	$L = (1)(0) + (1)(1) + (1)(0) = 1$	$1 \pmod{2} = 1$
ab	$L = (1)(1) + (1)(1) + (1)(0) = 2$	$2 \pmod{2} = 0$
c	$L = (1)(0) + (1)(0) + (1)(1) = 1$	$1 \pmod{2} = 1$
ac	$L = (1)(1) + (1)(0) + (1)(1) = 2$	$2 \pmod{2} = 0$
bc	$L = (1)(0) + (1)(1) + (1)(1) = 2$	$2 \pmod{2} = 0$
abc	$L = (1)(1) + (1)(1) + (1)(1) = 3$	$3 \pmod{2} = 1$

Thus, (1), ab, ac and abc are run in Block I and a, b, c and abc are run in block II, i.e.

Block I

Block II

(1)

a

ab

b

ac

c

bc

abc

Note $\left\{ \begin{array}{l} \alpha_i = \text{Represents the factors} \\ \alpha_i = \text{represents the co-efficients of factor} \end{array} \right.$

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This is the same design as shown previously that was generated from the table of '+' and '-' sign.

Principle Block Method:

Another method may be used to construct the confounded design. The block containing (1) is called the principle block. The t.c's in this block have a useful group theoretical property namely, they form a group with respect to multiplication (Mod 2).

This implies any treatment/element except (1) in the principle block may be generated by multiplying the two other elements in the principle block (Mod 2).

For example, consider the principle block of 2^3 design with ABC confounded as shown below:

$$ab \times ac = a^2bc; \quad a^2bc \pmod{2} = bc$$

$$ab \times bc = ab^2c; \quad ab^2c \pmod{2} = ac$$

Treatment combinations in other block or blocks may be generated by multiplying one element in the new block by each element in the principle block.

For 2^3 design, the principle block is (1), ab, ac, bc, and we know that 'b' is not in that block.

Thus, the elements are another block is

$$\begin{aligned} 1 \cdot b &= b; & b(\text{Mod } 2) &= b \\ ab \cdot b &= ab^2; & ab^2(\text{Mod } 2) &= a \\ ac \cdot b &= abc; & abc(\text{Mod } 2) &= abc \\ bc \cdot b &= cb^2; & cb^2(\text{Mod } 2) &= c \end{aligned}$$

This agrees the result obtained previously.

Unless the experimenter has an independent estimate of the experimental error or certain interactions can be assumed negligible. Replication of experiment will be necessary. To conduct the formal ANOVA and to test the hypothesis.

Complete Confounding:

Suppose that a 2^3 factorial design must be run in two incomplete blocks with ABC confounded and the experimenter decides to replicate four times. The resulting design appearing in the following table and the effect ABC is confounded for each replicate.

Replicate

B,

(1)

ac

bc

ab

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S.O.V

Replicate

Block or ABC

Error, For ABC (R

A

B

C

AB

AC

BC

Error (Rep EF

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ABC Confounded.

Replicate I

 R_2 R_3 R_4 B_1 B_2 B_1 B_2 B_1 B_2 B_1 B_2

(1)

a

ac

c

bc

abc

(1)

a

ac

b

bc

b

ac

a

ab

c

bc

c

ab

a

ab

b

ac

b

ab

abc

(1)

abc

(1)

c

bc

abc

ANOVA TABLE:

S.O.V

df

SS

MS

F-Ratio

Replicate

 $4-1=3$ SS_{Rep} MS_{Rep} MS_{Rep} / MSE_1

Block or ABC

 $2-1=1$ SS_{Blocks} MS_B MS_B / MSE_1 Error₁ For ABC (RxB) $7-4=3$ SS_{E_1} MS_{E_1}

—

A

 $2-1=1$ SS_A MS_A MS_A / MSE_2

B

1

 SS_B MS_B MS_B / MSE_2

C

1

 SS_C MS_C MS_C / MSE_2

AB

1

 SS_{AB} MS_{AB} MS_{AB} / MSE_2

AC

1

 SS_{AC} MS_{AC} MS_{AC} / MSE_2

BC

1

 SS_{BC} MS_{BC} MS_{BC} / MSE_2 Error₂ (Residual Effect)

18

 SS_{Error} MSE_2

—

Total

 $32-1=31$

—

There are 32 observations and 31 degree of freedom. Furthermore, since there are 8 blocks 7 degree of freedom must be associated with these blocks. The breakdown of the 7 degree of freedom as suggested by the error SS actually consists of two factor interactions b/w replicates and each of the effects A, B, C, AB, AC and

Solution

BC. Main effects and two factor interactions are tested against MS_{Error_2} . Replicates and blocks are tested against the higher MSEI.

If the same effect is confounded in all the replicates then at the time of analysis, that particular effect is completely lost/mixed up with the blocks and we can't estimate the confounded effect separately.

Such confounding scheme is known as "complete confounding".

Question:

Suppose there are four factors Temperature (A), Pressure (B), Concentration of reactant (C) and stirring rate (D).

Suppose now that $2^4 = 16$ t.e can't be run on the same day. The experimenter can run 8 t.e in one day. So, a 2^4 design confounding in two blocks seem appropriate.

$$(1) = 45, ab = 65, ac = 60, bc = 80, ad = 100$$

$$bd = 45, a = 71, b = 48, c = 68, cd = 75$$

$$d = 43, abcd = 96, abc = 65, acd = 86$$

$$bcd = 70, abd = 104.$$

Analysis the data.

$$(1); L =$$

$$a; L =$$

$$b; L =$$

$$c; L =$$

$$d; L =$$

$$ab; L =$$

$$ac; L =$$

$$ad; L =$$

$$bc; L =$$

$$bd; L =$$

$$cd; L =$$

$$abc; L =$$

$$abcd; L =$$

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Solution:

It is logical to confound the highest order interaction ABCD

By Defining Contrast Method:

The defining contrast is:

$$L = X_1 + X_2 + X_3 + X_4$$

because $d_i = 1$ for $i = 1, 2, 3, 4$

To assign the t.c in blocks we proceed as follow.

$$(1); L = (1)(0) + 1(0) + 1(0) + 1(0) = 0, \quad 0(\text{Mod } 2) = 0$$

$$a; L = (1)(1) + 0 + 0 + 0 = 1, \quad 1(\text{Mod } 2) = 1$$

$$b; L = 0 + 1 + 0 + 0 = 1, \quad 1(\text{Mod } 2) = 1$$

$$c; L = 0 + 0 + 1 + 0 = 1, \quad 1(\text{Mod } 2) = 1$$

$$d; L = 0 + 0 + 0 + 1 = 1, \quad 1(\text{Mod } 2) = 1$$

$$ab; L = 1 + 1 + 0 + 0 = 2, \quad 2(\text{Mod } 2) = 0$$

$$ac; L = 1 + 0 + 1 + 0 = 2, \quad 2(\text{Mod } 2) = 0$$

$$ad; L = 1 + 0 + 0 + 1 = 2, \quad 2(\text{Mod } 2) = 0$$

$$bc; L = 0 + 1 + 1 + 0 = 2, \quad 2(\text{Mod } 2) = 0$$

$$bd; L = 0 + 1 + 0 + 1 = 2, \quad 2(\text{Mod } 2) = 0$$

$$cd; L = 0 + 0 + 1 + 1 = 2, \quad 2(\text{Mod } 2) = 0$$

$$abc = abd = bcd; L = 1$$

$$abcd; L = 4, \quad 4(\text{Mod } 2) = 0$$

Hence the assignment of 16 treatments in two blocks with ABCD confounding is

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Block I (o)

Block II (1)

$(D) = 45$
 $ab = 65$
 $ac = 60$
 $ad = 100$
 $abcd = 96$
 $bd = 45$
 $bc = 80$
 $cd = 75$
 $Total = 566$

$a = 71$
 $b = 48$
 $c = 68$
 $d = 43$
 $abc = 65$
 $abd = 104$
 $bcd = 70$
 $acd = 86$
 $Total = 555$

bd
 abd
 cd
 acd
 bcd
 $abcd$

S.O.V

Hence the effect of block is

$$SSB = \frac{B_1^2 + B_2^2}{8} - C.F$$

$$= \frac{78547.6250 - 78540.0625}{8}$$

$$SS_B = 7.5625$$

Now we can compute the other SS by Yates Method.

T.C	Response	C ₁	C ₂	C ₃	C ₄	SS	
(D)	45	116	229	502	1121	78540.0625	(ABC
a	71	113	273	619	173	1870.5625	ABD
b	48	128	292	20	25	39.0625	ACD
ab	65	145	327	153	1	0.0625	BCD
c	68	143	43	14	79	390.0625	Block or ABCD
ac	60	149	-23	11	-145	1314.0625	
bc	80	161	116	-16	19	22.0625	
abc	65	166	37	17	15	14.0625	
d	43	26	-3	44	117	855.5625	
ad	100	17	17	35	133	1105.5625	

bd	45	-8	6	-66	-3	0.5625
abd	104	-15	5	-79	33	68.0625
cd	75	57	-9	20	-9	5.0625
acd	86	59	-7	-1	-13	10.5625
bcd	70	11	2	2	-21	27.5625
abcd	96	26	15	13	11	7.5625
						$T = 5730.9375$

ANOVA TABLE

S.O.V	df	SS	MS	F-Ratio	F-Table
A	1	1870.5625	"	62.222	> 7.71
B	1	39.0625	"	1.299	< 7.71
C	1	390.0625	"	12.975	> 7.71
D	1	855.5625	"	28.489	> 7.71
AB	1	0.0625	"	0.00267	< 7.71
AC	1	1314.0625	"	43.7116	> 7.71
AD	1	1165.5625	"	36.775	> 7.71
BC	1	22.5625	"	0.7505	< 7.71
BD	1	0.5625	"	0.0187	< 7.71
CD	1	5.0625	"	0.1684	< 7.71
ABC	1	14.0625	$120.25/4 = 30.0625$	0.46778	< 7.71
ABD	1	68.0625			
ACD	1	10.5625			
BCD	1	27.5625			
Block or ABCD	1	7.5625		0.25155	< 7.71
		5730.9375			

$$F_p \geq F_{\alpha}(v_1, v_2)$$

$$F_c \geq F_{0.05}(1, 4) = 7.71$$

Conclusion:

The effect of A, C, D, AC, AD are significant because their calculated value of F falls in Rejection Region. Other are insignificant because calculated values of F falls in acceptance Region.

2^k Factorial Design In 4 Blocks:

It is possible to construct a 2^k factorial design confounded in 4 blocks of 2^{k-2} observation each. These designs are particularly useful in situation where the number of factors is moderately large and the block size is relatively small.

Example:

Consider a 2^5 design if each block can only hold 8 treatment combination then 4 blocks must be used. The construction of this design is relatively straight forward.

Select 2 effects to be confounded with blocks say ADE and BEF. These effects have two defining contrast associated with them

$$L_1 = X_1 + X_4 + X_5$$

$$L_2 = X_2 + X_3 + X_5$$

Now
pair
i-1
(L₁, L₂)
T₁
(L₁, L₂)
val
Now;

(1) L₁
L₂
a L₁
L₂
b L₁
L₂
ab L₁
L₂
c L₁
L₂
ac L₁
L₂
bc L₁
L₂
abc L₁
L₂
d L₁
L₂
ad L₁
L₂
bd L₁
L₂
abd L₁
L₂

Now every T.C will yield a particular pair of values $L_1(\text{Mod } 2)$ and $L_2(\text{Mod } 2)$ i.e. either

$$(L_1, L_2) = (0, 0), (1, 0), (0, 1), (1, 1).$$

T.C yielding same values of (L_1, L_2) are assigned to the same values blocks.

Now;

Blocks:

$$(1) \quad L_1 = 0+0+0 = 0(\text{Mod } 2) = 0$$

$$L_2 = 0+0+0 = 0(\text{Mod } 2) = 0$$

$$a \quad L_1 = 1+0+0 = 1(11) = 1$$

$$L_2 = 0+0+0 = 0(11) = 0$$

$$b \quad L_1 = 0+0+0 = 0(\text{Mod } 2) = 0$$

$$L_2 = 1+0+0 = 1(\text{Mod } 2) = 1$$

$$ab \quad L_1 = 1+0+0 = 1(11) = 1$$

$$L_2 = 0+1+0 = 1(11) = 1$$

$$c \quad L_1 = 0+0+0 = 0(11) = 0$$

$$L_2 = 0+0+0 = 0(11) = 0$$

$$ac \quad L_1 = 0+0+0 = 0(11) = 0$$

$$L_2 = 0+1+0 = 1(11) = 1$$

$$bc \quad L_1 = 0+0+0 = 0(\text{Mod } 2) = 0$$

$$L_2 = 1+1+0 = 2(\text{Mod } 2) = 0$$

$$abc \quad L_1 = 1+0+0 = 1, = 1$$

$$L_2 = 0+1+1 = 2, = 0$$

$$d \quad L_1 = 0+1+0 = 1, = 1$$

$$L_2 = 0+0+0 = 0, = 0$$

$$ad \quad L_1 = 1+1+0 = 2, = 0$$

$$L_2 = 0+0+0 = 0, = 0$$

$$bd \quad L_1 = 0+1+0 = 1, = 1$$

$$L_2 = 1+0+0 = 1, = 1$$

$$abd \quad L_1 = 1+1+0 = 2, = 0$$

$$L_2 = 1+0+0 = 1, = 1$$

cd	$L_1 = 0+1+0 = 1, 1$ $L_2 = 0+1+0 = 1, 1$	cde	$L_1 = 0+1+1 = 2, 0$ $L_2 = 0+1+1 = 2, 0$
acd	$L_1 = 1+1+0 = 2, 0$ $L_2 = 0+1+0 = 1, 1$	acde	$L_1 = 1+1+1 = 3, 1$ $L_2 = 0+1+1 = 2, 0$
bcd	$L_1 = 0+1+0 = 1, 1$ $L_2 = 1+1+0 = 2, 0$	bde	$L_1 = 0+1+1 = 2, 0$ $L_2 = 1+1+1 = 3, 1$
abcd	$L_1 = 1+1+0 = 2, 0$ $L_2 = 1+1+0 = 2, 0$	abcde	$L_1 = 1+1+1 = 3, 1$ $L_2 = 1+1+1 = 3, 1$
e	$L_1 = 0+0+1 = 1, 1$ $L_2 = 0+0+1 = 1, 1$		
ae	$L_1 = 1+0+1 = 2, 0$ $L_2 = 0+0+1 = 1, 1$		
be	$L_1 = 0+0+1 = 1, 1$ $L_2 = 1+0+1 = 2, 0$		
abe	$L_1 = 1+0+1 = 2, 0$ $L_2 = 1+0+1 = 2, 0$		
ce	$L_1 = 0+0+1 = 1, 1$ $L_2 = 0+1+1 = 2, 0$		
ace	$L_1 = 1+0+1 = 2, 0$ $L_2 = 0+1+1 = 2, 0$		
bce	$L_1 = 0+0+1 = 1, 1$ $L_2 = 1+1+1 = 3, 1$		
abce	$L_1 = 1+0+1 = 2, 0$ $L_2 = 1+1+1 = 3, 1$		
de	$L_1 = 0+1+1 = 2, 0$ $L_2 = 0+0+1 = 1, 1$		
ade	$L_1 = 1+1+1 = 3, 1$ $L_2 = 0+0+1 = 1, 1$		
bde	$L_1 = 0+1+1 = 2, 0$ $L_2 = 1+0+1 = 2, 0$		
abde	$L_1 = 1+1+1 = 3, 1$ $L_2 = 1+0+1 = 2, 0$		

Block I

(0,0)

(1)

ad

bc

abcd

abe

ace

bde

cde

This
effects
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$$+1 = 2, 0$$

$$+1 = 2, 0$$

$$-1 = 3, 1$$

$$-1 = 2, 0$$

$$+1 = 2, 0$$

$$-1 = 3, 1$$

$$+1 = 3, 1$$

$$-1 = 3, 1$$

Block I

(0,0)

11

ad

bc

abcd

abe

ace

bde

cde

Block II

(0,1)

b

c

abd

acd

ae

abce

de

bcde

Block III

(1,0)

a

abe

d

bcd

be

ce

abde

acde

Block IV

(1,1)

e

ab

ac

bd

cd

bce

ade

abcde

This is 2^5 design in 4 blocks with effects ADE, BCE, ABDC confounded with blocks.

With a little reflection, we realize that another effect in addition to ADE and BCE must be confounded with blocks. Since there are 4 blocks with 3 degree of freedom b/w them and since ADE and BCE have only one degree of freedom each.

Clearly an additional effect with '1' df must be confounded. The effect is the generalized interaction of ADE and BCE defined as the product of these two effects (Mod 2). In our example the generalized interaction of these two effects is $ADE \times BCE = ABCD$. is also confounded with blocks.

It is easy to verify this by referring to the table of 4th and signs for this design. Inspection of such table reveals that the T.C are assigned to the blocks as follow.

T.C	Sign on ADE	Sign on BCE	Sign on ABCD
Block I	-	-	+
Block II	+	-	-
Block III	-	+	-
Block IV	+	+	+

Notice that the product of sign for any two effects/signs for the particular block yield the sign of other block. In this case, this is the sign of ABCD.

Hence ADE, BCE and ABCDE are confounded with blocks.

However, the allocation of T.C to the 4 blocks can be facilitated by using group theoretical property of Principal Block. For example, the product of two T.C in the Principal block yields another element of the same block. In such a manner the other elements of other blocks can be produced.

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In practice the principle block can be obtained using the defining contrast and using group theoretical property of principle block and the remaining block determined by the multiplication of principle block with the T.C. that is not present in that block.

Example:

From defining contrast, the three elements of Principle Block are

Block I : (1), b, ad, abd, abe, ace, bde, cde

To form block I, multiply Block I by b,

Block II : b, c, adb, acd, ae, abce, de, bde

To form block III, multiply Block I by a.

Block III : a, abe, d, bcd, be, ce, abde, acde

To form block IV, multiply Block I by e.

Block IV : e, be, ade, abde, ab, ac, bd, cd.

The general practice for constructing a 2^k design confounded in 4 blocks is to choose 2 effect to generate the blocks automatically confounding a third effect that is generalized interaction of the first two. Then the design is constructed by using the two defining contrast and the group theoretical property of Principal Block.

In selecting the effects to

be confounded with blocks care must be exercised to obtain a design that does not confound effect that be any potential interest

2^k Factorial Design In 2^p Blocks:

The method described above may be extended to the construction of a 2^k factorial design confounded in 2^p blocks where $p < k$ where each block contain exactly two runs.

We select 'P' independent effects to be confounded where independent means that no effect chosen is the generalized interaction of the other
e.g. (CE is for ABDE and ABD)

The blocks may be generated by the use of 'P' defining contrast $L_1, L_2, \dots, L_p$ associated with these effects. In addition exactly $(2^p - p - 1)$ other effects will be confounded with blocks, these being the generalized interactions of those 'p' independent effects, initially chosen. Care should be exercised in selecting effects to be confounded so that the information on the effects that may be of any interest is not sacrifice.

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blc

No. of
factors No
Blc

3 2
4

4 2
4
8

The statistical analysis of these designs is straight forward. Sum of squares of all effects are computed as if no blocking had occurred. The Block SS is found by adding the sum of square for all effects with confounding with blocks.

No. of factors	No. of Blocks	Block Size	Effects chosen to generate blocks	Interaction confounded with blocks.
3	2	4	ABC	ABC
	4	2	AB, AC	AB, BC, AC
4	2	8	ABCD	ABCD
	4	4	ABC, ACD	ABC, ACD, BD
	8	2	AB, BC, CD	AB, BC, CD, AC, ABCD, BD, AD.

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Partial Confounding:

Since the procedure adopted up till now is open to some criticism, in that we have completely sacrificed one or more effect. It is of interest to inquire into another confounding procedure which enables us to distribute confounding over more than one effect by confounding a different effect in each replicate. This is known as partial confounding.

Partial confounding in 2^3 experiment can be obtained by confounding in AB in replicate I, AC in 2nd replicate, BC in III replicate and ABC in 4th replicate.

Rep I	(1)	ab	c	abc	a	b	ac	bc	AB Confounded
Rep II	(1)	ac	b	abc	a	c	ab	bc	AC "
Rep III	(1)	bc	a	abc	b	c	ab	ac	BC "
Rep IV	(1)	ab	ac	bc	a	b	c	abc	ABC "

In this schema we can't speak of an interaction such as AB being particular confounded because it is confounded in only one replicate out of 4. The analysis of this design can be outlined as follows.

S.O.V

Replicate
Block within
A
B
C
AB (I, II)
AC (I, II)
BC (I, II)
ABC (I, II)
Error
Total

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ANOVA TABLE

S.O.V df

Replicate 3

Block within Replicates 4

A 1

B 1

C 1

AB (I, II, III, IV) 1

AC (I, II, III, IV) 1

BC (I, II, III, IV) 1

ABC (I, II, III, IV) 1

Error 17

Total 31

The precision of the interaction will be 3/4 because they are tested in 3/4 of the replicates. This method of analysis is determined sum of squares of blocks and main effects in the usual way.

The interaction sum of square from these blocks in which they are not confounded.

Example:

Suppose the study is performed to determine the effect of 3 factors. Suppose that each batch of serum is only large enough to test only the 4 t.c. Thus each replicate of 2^3 design must be run in 2 blocks. 2 replicates are run with ABC confounding.

in replicate I and AB confounding in replicate II. The data is as follow.
Analyze the data.

Replicate I
ABC Confounded

Block	I	II
(1)	-3	a = 0
ab	-2	b = -1
ac	-2	c = -1
bc	1	abc = 6

Replicate II
AB Confounded

Block	I	II
(1)	-1	a = 4
c	0	b = 0
ab	3	ac = 1
abc	5	bc = 1

$$SS_{ABC(II)} = \frac{1}{2^3} = 1$$

$$SS_{ABC} \text{ Simultaneous}$$

$$SS_{ABC(I)} = \frac{1}{2}$$

Solution:

To find the sum of squares for A, B, C, AB, AC, BC. We use Yates Method.

$$SS_{AB}$$

$$SS_{Blocks}$$

$$SS_{AB(II)}$$

T.C	Response	C ₁	C ₂	C ₃	$SS = \frac{(\text{contrast})^2}{2^3 \cdot n}$
(1)	-4]	-3]	1]	16	16
a	1]	4]	15]	24	36
b	-1]	2]	11]	18	20.25
ab	5]	13]	13]	6	2.25
c	-1]	5]	7]	14	12.25
ac	3]	6]	11]	2	0.25
bc	2]	4]	1]	4	1
abc	11]	9]	5]	4	1

$$SS_{ABC}$$

$$SS_{Block}$$

$$SS_{Rep}$$

As ABC is confounded in Replicate I, so sum of square for ABC is computed from replicate II.

$$SS_{Rep}$$

in

$$SS_{ABC(II)} = \frac{1}{2^k \cdot n} [a + b + c + abc - ab - ac - bc - (1)]^2$$

$$= \frac{1}{8} [1 + 0 + 0 + 5 - 3 - 1 - 1 + 1]^2$$

$$SS_{ABC} = 0.50$$

Similarly AB is confounded in replicate II
So its effect is computed from replicate I

$$SS_{AB(I)} = \frac{1}{2^k \cdot n} [(1) + c + ab + abc - a - b - ac - bc]^2$$

$$= \frac{1}{8} [-3 - 1 + 2 + 6 - 0 + 1 - 2 - 1]^2$$

$$SS_{AB} = 0.50$$

$$SS_{Blocks} = SS_{ABC(I)} + SS_{ABC(II)}$$

$$SS_{AB(II)} = \frac{1}{8} [-1 + 0 + 3 + 5 - 1 - 0 - 1 - 1]^2$$

$$= 2$$

$$SS_{ABC(I)} = \frac{1}{8} [0 + (-1) - 1 + 6 - 2 - 2 - 1 + 3]^2$$

$$= 0.5$$

$$SS_{Blocks} = 0.5 + 2 = 2.5$$

$$SS_{Rep} = \frac{\sum P_i^2}{8} - C.F$$

$$= \frac{(6)^2 + (10)^2}{8} - 16$$

$$SS_{Rep} = 1$$

Yates
Yates

$= \frac{(\text{contrast})^2}{2^k \cdot n}$

16

36

26.25

2.25

12.25

0.25

1

1

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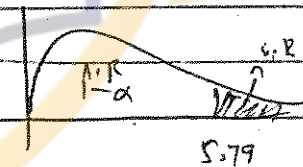
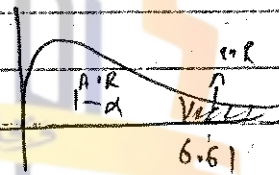
ANOVA Table

S.O.V	df	SS	M.S	F-Ratio	F-Table
Replicate	2-1-1	1.00	1.00	$F = 1.33$	< 6.61
B with R	2	2.50	1.25	$F = 1.667$	< 5.79
A	1	36	36	$F = 48$	> 6.61
B	1	20.25	20.25	$F = 27$	> 6.61
C	1	12.25	12.25	$F = 16.33$	> 6.61
AB(I)	1	0.50	0.50	$F = 0.667$	< 6.61
AC	1	0.25	0.25	$F = 0.333$	< 6.61
BC	1	1.00	1.00	$F = 1.333$	< 6.61
ABC(II)	1	0.50	0.50	$F = 0.667$	< 6.61
Error	5	3.75	0.75	—	—
Total	15	78	—	—	—

$$F > F_{\alpha}(1,5)$$

$$F > F_{0.05}(1,5) \Rightarrow F > 6.61$$

$$F > F_{0.05}(2,5) \Rightarrow F > 5.79$$



Conclusion:

The effects of A, B and C are significant and remaining all effects are significant because their calculated values of F falls in Acceptance Region.

Design And Analysis Of Experiments

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Incomplete Block Design:

In some randomized block design, it may not be possible to apply all the treatments in every block.

For example, if there were 6 brands of tires to test, only 4 could be test on a given car. Such a block would be incomplete having only 4 out of 6 on it.

Take the problem of determining the effect on current flow of 4 treatment applied to the coils of T.V. tube filament. As each treatment application requires some time. It is not possible to run the several observations of these treatments in one day. If days were taken as blocks, all treatments must be run in random order on each of several days in order to have a RCB design. After checking it was found that all 4 treatments can't be run in one day. Three are the most that can be

(Not all treatments are tested in each block. There is an incomplete set of treatments in each block. Designs are complex and are usually looked up in tables)

run. The question is then arises is that which treatment would be run first and which would be later.

If information is desired on all treatment. For such problems it is possible to run RCB design in which every treatment is not present in every block. These designs are known as "Randomized Incomplete block design".

Types of Incomplete block design:

There are two types of these designs.

- 1) Balanced Incomplete Block Design (BIBD)
- 2) Partially Balanced Incomplete Block Design (PBIBD)

1) BIB Design:

"A BIB design is an

incomplete block design in which every pair of treatment occurs same number of times in an experiment."

The number of blocks necessary for balancing will depend upon the number of treatments that can be run in a single block.

Suppose that there are ' a ' treatments and each block can hold ' k ' ($k < a$) treatments. A BIB design can be constructed by taking aC_k blocks and assigning a different combination of treatment to each block. Frequently, however, balance can be obtained with fewer than aC_k blocks.

Tables of BIB designs are given in Fisher's and Yates Table, Cochran and Cox Table etc.

Statistical Analysis:

If there are ' a ' treatments and ' b ' blocks we assume that each block contains ' k ' treatments and

and each treatment is repeated 'r' times in the design.

Thus there are:

$$N = ar = bk = \text{Total No. of observations.}$$

Furthermore, the No. of times each pair of treatment appears in the same block is

$$\lambda = \frac{r(k-1)}{a-1}$$

For BIB design, the parameter λ should be an integer.

Example:

Treatments	Blocks			
	I	II	III	IV
1	✓	✓	✓	—
2	—	✓	✓	✓
3	✓	—	✓	✓
4	✓	✓	—	✓

In this example

$$a = 4$$

$$b = 4$$

$$k = 3$$

$$r = 3$$

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$$\lambda = \frac{r(k-1)}{a-1}$$

$$\lambda = \frac{3(3-1)}{4-1} = \frac{6}{3} = 2$$

Treatment pairs in each block

Blocks	I	II	III	IV
	(1,3)	(1,2)	(1,2)	(2,3)
	(1,4)	(1,4)	(1,3)	(2,4)
	(3,4)	(2,4)	(2,3)	(3,4)

Statistical Model:

The statistical model for BIB design is

$$Y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij} \quad \begin{matrix} i=1, 2, \dots, a \\ j=1, 2, \dots, b \end{matrix}$$

where

μ = Overall mean

α_i = Effect of i th treatment.

β_j = Effect of j th block.

G_{ij} = Random Error $\sim NII(0, \sigma^2)$

$$C.F = \frac{G^2}{N}$$

$$TSS = \sum_i \sum_j Y_{ij}^2 - C.F.$$

(6)

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$$TSS = SS_{\text{blocks}} + SS_{Tr(\text{adj})} + SSE$$

where the sum of squares of treatment is adjusted to separate the block effect and treatment effect. This adjustment is necessary because every treatment is represented in a different set of 'r' blocks. i.e. Difference b/w unadjusted treatment total are also affected by the difference b/w blocks.

$$\text{Block SS} = \frac{1}{k} \sum_j T_{.j}^2 - C.F$$

$$\text{Treatment SS} = \frac{1}{r} \sum_i T_i^2 - C.F$$

$$\text{Adjusted Treatment SS} = \frac{k}{ra} \sum_i Q_i^2$$

where

$$Q_i = Y_i - \frac{1}{k} \sum_j n_{ij} Y_{.j}$$

with

$$n_{ij} = \begin{cases} 1, & \text{if the } i^{\text{th}} \text{ treatment appears in block } j^{\text{th}} \\ 0, & \text{otherwise.} \end{cases}$$

Error SS = TSS - SS_{blocks} - SS_{tr(adj)}
 with df $N - a - b + 1$

ANOVA TABLE

S.O.V	Degree of freedom	SS	MS	F-Ratio
Treatments (Adjusted)	$a - 1$	Tr.SS	MSTr.	$F_1 = \frac{MSTr}{MSE}$
Blocks	$b - 1$	BSS	MSB	$F_2 = \frac{MSB}{MSE}$
Error	$N - a - b + 1$	E.SS	MSE	
Total	$N - 1$	TSS	—	

Note :

If No. of block = No. of treatments then this design is known as symmetric and we can also find

Adjusted block SS = $\frac{r}{ab} \sum_j Q_j^2$ with df $b - 1$

where

$Q_j = y_j - \frac{1}{r} \sum_i n_{ij} y_i$

with

$n_{ij} = \begin{cases} 1, & \text{if the block } j^{\text{th}} \text{ is present in } i^{\text{th}} \text{ tr} \\ 0, & \text{otherwise} \end{cases}$

Question: (2011) Q# 7(b)

An engineer is studying the mileage performance characteristics of 5 types of gasoline additives in the road test. He wishes to use cars as blocks. However, because of time constrained, he must use incomplete block design. He runs BIB design with 5 blocks. Analyze the data.

Treatments	Blocks					y_{ij}
	1	2	3	4	5	
1	—	17	14	13	12	56
2	14	14	—	13	10	51
3	12	—	13	12	9	46
4	13	11	11	12	—	47
5	11	12	10	—	8	41
$y_{.j}$	50	54	48	50	39	$241 = G$

Here

$$a = 5, b = 5$$

$$k = 4, r = 4, N = ar = bk = 20.$$

$$\lambda = \frac{r(k-1)}{a-1} = \frac{4(4-1)}{5-1} = \frac{12}{4} = 3$$

which is an integer.

$$C.F = \frac{G^2}{N} = \frac{(241)^2}{20} = 2904.05$$

$$TSS = \sum_{i=1}^k \sum_{j=1}^4 y_{ij}^2 - C.F$$

$$= 2981 - 2904.05$$

$$TSS = 76.95$$

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$$SS_{blocks} = \frac{1}{k} \sum_i y_{i.}^2 - C.F$$

$$= \frac{(50)^2 + \dots + (39)^2}{4} - 2904.05$$

$$= 31.2$$

$$\text{Treatment SS (Adjusted)} = \frac{k}{n} \sum_i Q_i^2$$

where

$$Q_i = y_{i.} - \frac{1}{k} \sum_j n_{ij} y_{.j}$$

$$Q_1 = 56 - \frac{1}{4} (54 + 48 + 50 + 39) = 8.25$$

$$Q_2 = 51 - \frac{1}{4} (50 + 54 + 50 + 39) = 2.75$$

$$Q_3 = 46 - \frac{1}{4} (50 + 48 + 50 + 39) = -0.75$$

$$Q_4 = 47 - \frac{1}{4} (50 + 54 + 48 + 50) = -3.5$$

$$Q_5 = 41 - \frac{1}{4} (50 + 54 + 48 + 39) = -6.75$$

$$Tr. SS_{adj} = \frac{(4)}{(3)(5)} \left[(8.25)^2 + \dots + (-6.75)^2 \right]$$

$$= \frac{(4)(134)}{15}$$

$$= 35.73$$

$$ESS = TSS - (SSB - SS_{Tr. (adj)})$$

$$= 76.95 - 35.73 - 31.2$$

$$ESS = 10.02$$

ANOVA TABLE

Source of variation	Degree of freedom	Sums of Square	Mean Square	F-Ratio
Treatment (adj)	5-1 = 4	35.73	8.9325	$F_1 = 9.81$
Blocks	5-1 = 4	31.2	7.8	$F_2 = 8.56$
Error	11	10.02	0.9109	$F_3 = 9.20$
Total	20-1 = 19	76.25	—	

Critical Regions:

$$F_1 \geq F_{\alpha(a-1, N-a-b+1)}$$

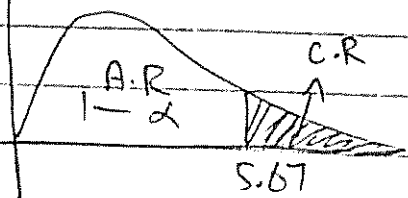
$$F_1 \geq F_{0.05(4, 11)}$$

$$F_1 \geq 5.67$$

$$F_2 \geq F_{\alpha(b-1, N-a-b+1)}$$

$$F_2 \geq F_{0.05(4, 11)}$$

$$F_2 \geq 5.67$$



Conclusion:

As both calculated values of F falls in rejection region. So, we reject H_0 at 5% level of significance. i.e. the effect of treatment means and the effect of block means are significant.

Youden Square:

These are incomplete Latin Square designs in which number of rows does not equal to the number of columns and number of treatments.

For example, consider the design shown in the table below with 5 treatments A, B, C, D and E.

Rows	Columns			
	1	2	3	4
1	A	B	C	D
2	B	C	D	E
3	C	D	E	A
4	D	E	A	B
5	E	A	B	C

This design is known as Youden Square design.

Now, if we append a column (E, A, B, C, D) to the design then the result is a 5×5 Latin Square design.

These designs are developed by W. J. Youden, therefore these are known as Youden Square design.

Although, a Youden Square is always a Latin Square from which atleast one column, or row or diagonal is missing.

It is not necessarily true that every Latin Square with more than one column or row or diagonal deleted is a Youden Square.

In general, Youden Square is a symmetric balanced incomplete block design in which rows corresponds to blocks and each Treatment occurs exactly once in each column or position of the block.

Statistical Model:

The statistical model for Youden Square is

$$Y_{ijk} = \mu + \alpha_i + \pi_j + \beta_k + \epsilon_{ijk}$$

where

$$i = 1, 2, \dots, a$$

$$j = 1, 2, \dots, b$$

$$k = 1, 2, 3, \dots, h$$

$\mu =$ overall mean

$\alpha_i =$ i^{th} block effect

$\pi_j =$ j^{th} treatment effect.

$\beta_k =$ k^{th} column/position effect.

The analysis of Youden Square is similar to the analysis of BIB design except that the sum of squares b/w the position totals may also be calculated.

$$C.F = \frac{(Y_{...})^2}{N}$$

$$TSS = \sum \sum \sum y_{ijk}^2 - C.F$$

$$\text{Block/Row SS} = \frac{1}{K} \sum_i y_{i..}^2 - C.F$$

$$\text{Column SS} = \frac{1}{b} \sum_k y_{..k}^2 - C.F$$

$$\text{Adjusted Tr. SS} = \frac{K \cdot r}{a \cdot b} \sum Q_i^2$$

$$Q_i = y_{i.} - \frac{1}{K} \sum_j n_j y_{ij}$$

$$\text{Error SS} = TSS - BSS - CSS - TrSS.$$

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ANOVA TABLE

Source of Variation	df	SS	MS	F-Ratio
Treatments	a-1	SS _{tr}	MST _r	$F = \frac{MST_r}{MSE}$
Rows	b-1	SS _R	MS _R	
Columns	h-1	SS _C	MS _C	
Error	By sub	SS _E	MSE	
Total	N-1	TSS	—	

Example:

An industrial engineer is studying the effect of 5 elimination levels on the occurrence of defects in an assembly operation, because time may be a factor in the experiment. She has decided to run the exp. in 5 blocks where block is the day of the week however the department in which the experiment is conducted has only 4 work stations and these stations represents a potential source of variability. The engineer decides to run a Youden Square with 5 rows (days/Blocks)

4 columns (work stations) and 5 treatments (illumination). The coded data are given in the table below. Analyze the data.

Columns

Days (Blocks)	1	2	3	4	Totals
1	A = 3	B = 1	C = -2	D = 0	2
2	B = 0	C = 0	D = -1	E = 7	6
3	C = -1	D = 0	E = 5	A = 3	7
4	D = -1	E = 6	A = 4	B = 0	9
5	E = 5	A = 2	B = 1	C = -1	7
Totals	6	9	7	9	31

Here

$$a = 5, \quad b = 5$$

$$k = 4, \quad r = 4$$

$$\lambda = \frac{r(k-1)}{a-1}$$

$$\lambda = \frac{4(4-1)}{5-1} = \frac{12}{4} = 3$$

C.F.

$$C.F. = \frac{G^2}{N} = \frac{(31)^2}{20} = 48.05$$

$$TSS = \sum \sum Y_{ij}^2 - C.F.$$

$$= 188 - 48.05$$

$$TSS = 139.95$$

$$\begin{aligned}
 SS_{\text{blocks/rows}} &= \frac{1}{K} \sum y_{i..}^2 - C.F \\
 &= \frac{1}{4} [(2)^2 + \dots + (7)^2] - 48.05 \\
 &= 6.70
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{columns}} &= \frac{1}{b} \sum y_{..k}^2 - C.F \\
 &= \frac{1}{5} [(6)^2 + \dots + (9)^2] - 48.05 \\
 &= 4.35
 \end{aligned}$$

Now for adjusted treatment SS:

Blocks	Treatments					Totals
	A	B	C	D	E	
1	3	1	-2	0	—	2
2	—	0	0	-1	7	6
3	3	—	-1	0	5	7
4	4	0	—	-1	6	9
5	2	1	-1	—	5	7
Totals	12	2	-4	-2	23	31

$$SS_{\text{Treatments(adj)}} = \frac{K}{a\lambda} \sum Q_i^2$$

where

$$Q_i = y_i - \frac{1}{K} \sum_j n_{ij} y_{.j}$$

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$$Q_1 = 12 - \frac{1}{4} (2+7+9+7) = \frac{23}{4}$$

$$Q_2 = 2 - \frac{1}{4} (2+6+9+7) = \frac{-16}{4}$$

$$Q_3 = -4 - \frac{1}{4} (2+6+7+7) = \frac{-38}{4}$$

$$Q_4 = -2 - \frac{1}{4} (2+6+7+9) = \frac{-32}{4}$$

$$Q_5 = 23 - \frac{1}{4} (6+7+9+7) = \frac{63}{4}$$

$$SS_{tr(adj)} = \frac{(4)}{(5)(3)} \left[\left(\frac{23}{4} \right)^2 + \dots + \left(\frac{63}{4} \right)^2 \right]$$

$$= 120.37$$

$$SSE = TSS - SS_{blocks} - SS_{columns} - SS_{tr(adj)}$$

$$= 134.95 - 6.70 - 1.35 - 120.37$$

$$SSE = 6.53$$

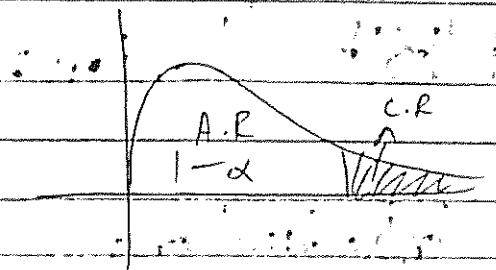
ANOVA TABLE

S.O.V	df	SS	MS	F-Ratio
Treatment	5-1=4	120.37	30.0925	F = 36.8667.
Rows	5-1=4	6.70	1.675	
Columns	4-1=3	1.35	0.45	
Error	8	6.53	0.81625	
Total	20-1=19	134.95	—	

Critical Region:

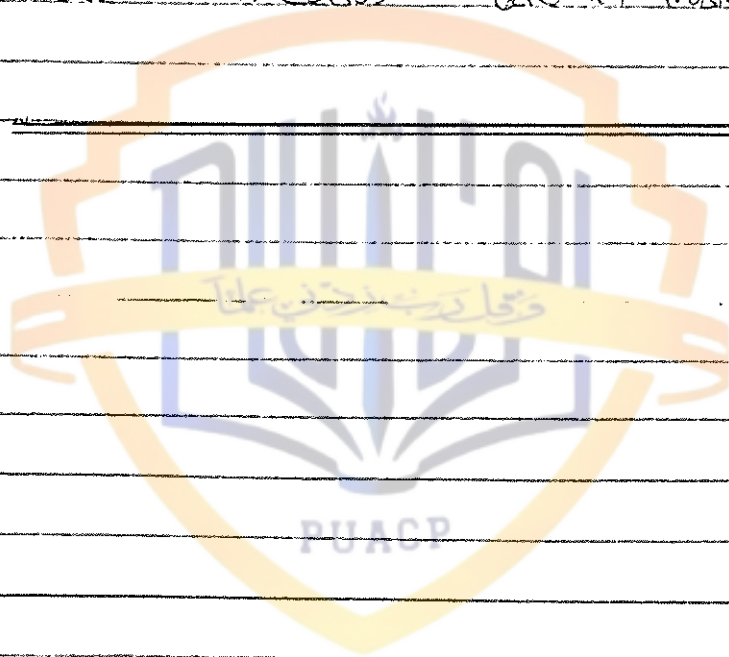
$$F_{\gamma}, F_{0.05(4,8)}$$

$$F_{\gamma} = 3.84$$



Conclusion:

Since $F_c = 36.867$ falls in Rejection Region. So, we reject H_0 at 5% level of significance i.e. All treatment means are not insignificant.



PBIB Design:

Introduction:

BIB designs, though may have optimal properties, do not fit well to many experimental situations as these designs require a large number of replications. Moreover, these designs are not available for all number of treatments and block size. Since the constraint that n to be an integer can force either the number of blocks or block size to be excessively large. e.g. if there are 8 treatments and block size is 3 then for n to be an integer the smallest number of replications r is 21. This leads to the design of 36 blocks which is too large for practical problems.

To overcome these difficulties a class of binary, equireplicate and proper design that are called PBIB design were introduced.

"In these designs the variance of every estimated elementary contrast among treatment effects is not the

Some. In this design, some pair of treatments occurs λ_1 times, some appears λ_2 and so on the remaining pairs appear λ_m times. The definition of PBIB design is based on association scheme.

Association Scheme:

Given 'a' treatments symbols $(1, 2, \dots, a)$, a relation satisfying the following conditions is called m-class association scheme. ($m \geq 2$).

- 1) Any two symbols are either 1st, 2nd, or m th associates; the relation of association being symmetric i.e. if the symbol α is the i th associate of β then β is i th associate of α .
- 2) Each symbol α has n_i i th associate. The number n_i being independent of α .
- 3) If any two symbols α and β are i th associates then the number of symbols that are j th associate of α and k th associate of β is p_{jk}^i and is independent of the pair of i th associates α and β .

The quantities a, n_i, p_{jk}^i ($i, j, k = 1, 2, \dots, m$) are called the parameters of association scheme.

PBIB Design:

Given an association scheme $\mathcal{A}$ with m classes ($m \geq 2$), we have a PBIB design with m associate classes based on the association scheme. If the ' a ' treatments can be arranged into ' b ' blocks, such that

- Every symbol occurs at most once in a block
- Every symbol occurs in exactly ' r ' blocks.
- If the two symbols are i th associates, then they occur together in λ_i blocks. The No. λ_i being independent of the particular pair of i th associates α and β .

The numbers $a, b, r, k, \lambda_i, n_i$ are called parameters of 1st kind. It can be easily seen that

$$\sum_{i=1}^{m-1} n_i \lambda_i = r(k-1).$$

Two-class association scheme and the two

associates PBIB design have been extensively studied in the literature and are simple to use.

Example:

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Consider 12 treatments symbols denoted by numbers 1 to 12. Let us form 3 groups of 4 symbols each as follows

(1, 2, 3, 4)

(5, 6, 7, 8)

(9, 10, 11, 12)

we now define i) any two treatment symbols are first associates if they belong to the same group. ii) any two symbols are 2nd associates if they belong to different groups. Here $v = 12$
 $n_1 = 3$, $n_2 = 8$

$$P_{11}^1 = 2, \quad P_{12}^1 = 0, \quad P_{21}^1 = 0, \quad P_{22}^1 = 8$$

$$P_{11}^2 = 0, \quad P_{12}^2 = 3, \quad P_{21}^2 = 3, \quad P_{22}^2 = 4$$

It can be verified that these values of P_{jk}^i ($i, j, k = 1, 2$) remains unchanged for any choice of two first or 2nd associates. These parameters are

usually written in the form of the following matrices.

$$P^1 = [P_{ij}^1] = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}$$

$$P^2 = [P_{ij}^2] = \begin{bmatrix} 0 & 3 \\ 3 & 4 \end{bmatrix}$$

$P_{11}^1, P_{12}^1, P_{21}^1, P_{22}^1, P_{11}^2, P_{12}^2, P_{21}^2, P_{22}^2$ are called parameters of 2nd kind.

Question: (2005) Q # 7(b):

Consider the following PBIB design

Blocks	Treatments		
	1	2	3
1	1	2	3
2	3	4	5
3	2	5	6
4	1	2	4
5	3	4	6
6	1	5	6

find the parameters of 1st kind and 2nd kind of PBIB design.

ing and also verify the following relationship among the parameters.

$$1) n_1 + n_2 = a - 1$$

$$2) P_{11}^1 + P_{12}^1 = n_1$$

$$3) P_{11}^1 + P_{12}^1 = n_1 - 1$$

$$4) n_1 P_{11}^1 = n_2 P_{11}^2$$

$$5) n_1 \lambda_1 + n_2 \lambda_2 = r(k-1)$$

$$6) P_{21}^1 + P_{22}^1 = n_2$$

$$7) P_{21}^2 + P_{22}^2 = n_2 - 1$$

$$8) n_1 P_{22}^1 = n_2 P_{12}^2$$

ii) Analyze the data, test the significance difference b/w the treatments by computing adjusted treatment SS

Solution:

Here

$$a = 6$$

$$b = 6$$

$$k = 3$$

$$r = 3$$

$$\lambda = \frac{r(k-1)}{a-1} = \frac{3(3-1)}{6-1} = \frac{6}{5}$$

which is not an integral therefore, it is PBIB design. it means that pair of treatments do not occur the same number of times.

Blocks	Treatment			Treatment pairs			Tr.
1	1	2	3	(1,2)	(1,3)	(2,3)	Tr. 1
2	3	4	5	(3,4)	(3,5)	(4,5)	Tr. 2
3	2	5	6	(2,5)	(2,6)	(5,6)	Tr. 3
4	1	2	4	(1,2)	(1,4)	(2,4)	Tr. 4
5	3	4	6	(3,4)	(3,6)	(4,6)	Tr. 5
6	1	5	6	(1,5)	(1,6)	(5,6)	Tr. 6

The pairs (1,2), (3,4) and (5,6) occurs two times in the design

$$\lambda_1 = 2$$

and the remaining pairs occur one time in the design so $\lambda_2 = 1$

It is two associates PBIB design.

Treatment 1 is a 1st associate with treatment 2

" 2 " " " 1

" 3 " " " 4

" 4 " " " 3

" 5 " " " 6

" 6 " " " 5

Now

associates for $i = 1$

let us take (1,2)

Tr. 1 is a 1st associate with tr. 2 } No common point
 Tr. 2 is a 1st associate with tr. 1 } $P_{11}^1 = 0$

i
 P_{ij}

Tr. 1 is a 1st associate with Tr. 2 } No common point
 Tr. 2 is a 2nd associate with Tr. 3, 4, 5, 6 } $P_{12}^1 = 0$

Tr. 1 is a 2nd associate with Tr. 3, 4, 5, 6 } No common point
 Tr. 2 is a 1st associate with Tr. 1 } $P_{21}^1 = 0$

Tr. 1 is a 2nd associate with Tr. 3, 4, 5, 6 } 4 Common points
 Tr. 2 is a 2nd associate with Tr. 3, 4, 5, 6 } $P_{22}^1 = 4$

Now Associates for $i = 2$
 let us take (1, 3)

Tr. 1 is a 1st associate with Tr. 2 } No common point
 Tr. 3 is a 1st associate with Tr. 4 } $P_{11}^2 = 0$

Tr. 1 is a 1st associate with Tr. 2 } 1 Common point
 Tr. 3 is a 2nd associate with Tr. 1, 2, 5, 6 } $P_{12}^2 = 1$

Tr. 1 is a 2nd associate with Tr. 3, 4, 5, 6 } 1 Common point
 Tr. 3 is a 1st associate with Tr. 4 } $P_{21}^2 = 1$

Tr. 1 is a 2nd associate with Tr. 3, 4, 5, 6 } 2 Common points
 Tr. 3 is a 2nd associate with Tr. 1, 2, 5, 6 } $P_{22}^2 = 2$

The No. of treatments that appeared as the 1st associate is denoted by n_1 and the No. of treatments that appeared as the 2nd associate is denoted by n_2 .

Thus

$$\text{Here } n_1 = 1$$

$$n_2 = 4$$

The quantities $a, b, r, k, \lambda_1, \lambda_2, n_1$ and n_2 are called parameters of 1st kind.

$$a=6, b=6, r=3, k=3, \lambda_1=2, \lambda_2=1$$

$$n_1=1, n_2=4$$

The quantities $P'_{11}, P'_{12}, P'_{21}, P'_{22},$

$P''_{12}, P''_{21}, P''_{12}$ and P''_{22} are called parameters of 2nd kind.

For BIB design, the parameters of 2nd kind may also be written as in Matrix form as

$$P'_{jk} = \begin{bmatrix} P'_{11} & P'_{12} \\ P'_{21} & P'_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$$

$$P''_{jk} = \begin{bmatrix} P''_{11} & P''_{12} \\ P''_{21} & P''_{22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

Relationships:

$$1) \quad n_1 + n_2 = a - 1$$

$$1 + 4 = 6 - 1$$

$$5 = 5$$

$$2) \quad P'_{11} + P'_{12} = n_1$$

$$0 + 1 = 1$$

$$1 = 1$$

$$3) \quad P'_{11} + P'_{12} = n_1 - 1$$

$$0 + 0 = 1 - 1$$

$$0 = 0$$

$$4) \quad n_1 P'_{11} = n_2 P'_{12}$$

$$(1)(0) = (4)(0)$$

$$0 = 0$$

$$5) \quad n_1 r_1 + n_2 r_2 = r(k-1)$$

$$(1)(2) + (4)(1) = 3(3-1)$$

$$6 = 6$$

$$6) \quad P'_{21} + P'_{22} = n_2$$

$$0 + 4 = 4$$

$$4 = 4$$

$$7) \quad P'_{21} + P'_{22} = n_2 - 1$$

$$1 + 4 = 4 - 1$$

$$3 = 3$$

$$8) n_1 P_{22}^1 = n_2 P_{12}^2$$

$$(1)(4) = (4)(1)$$

$$4 = 4$$

(Related to Theory)

A PBIB design with 2 associate classes is described by the following parameters.

- 1) There are 'a' treatments arranged in 'b' blocks. Each block contains 'k' runs and each treatment appears 'r' times (or in 'r' blocks).
- 2) 2 treatments which are i^{th} associate appear together in n_i blocks. $i = 1, 2$
- 3) Each treatment has exactly n_i i^{th} associate. $i = 1, 2$
- 4) If two treatments are i^{th} associates then the No. of treatments that are j^{th} associate of one treatment of the other treatment and k^{th} associate of one treatment is P_{jk}^i ($i, j, k = 1, 2$).

Question:

Write down the advantages and disadvantages of RCB design and incomplete block design.

a) Incomplete Block Design:

Advantages:

Following are the advantages of incomplete block design.

- 1) Small blocks are more homogeneous than larger blocks, so experimental error is lower.
- 2) It is used when there is variability within larger blocks.
- 3) It is used to increase precision.

Disadvantages:

1) These designs can require a fixed number of treatments, a fixed number of replicates or both.

2) More replicates are needed for it.

3) The complexity of the analysis is increased.

4) There is unequal precision for certain comparisons of treatment means.

Uses:

1) It is used to reduce the block size, in single factor experiments where there is large number of treatments.

2) All comparisons have equal precision i.e. variance is constant.

3) Treatments and blocks are not orthogonal.
 * These designs are not available for all No. of treatments. The No. of rep must be large and does not always suit the convenience of the experimenter.

RCB Design:

Advantages:

Following are the advantages of RCB design.

- 1) The source of extraneous variation is controlled by grouping the experimental material and hence the estimate of the experimental error is decreased.
- 2) The design is flexible i.e. any number (but not ^{less} ~~more~~ than 2) of replications may be run and any number of treatments may be tested.
- 3) The experiment can be set up easily.
- 4) The statistical analysis is simple and straight forward.
- 5) It is easy to adjust ^{for} the missing observation.

Disadvantages:

1) It controls variability only in one direction

2) It is not suitable when the treatments is very large or when the blocks are not homogeneous.

PBIB Design:

(Define PBIB)

An incomplete block design is said to be partially balanced if it satisfies the following conditions

1) The experimental material is divided into ' b ' blocks of ' k ' units each different treatments being applied to the units in the same block

2) There are ' a ' treatments each of which occurs in ' r ' blocks

3) There can be established a relation of association b/w any two treatments satisfying the following requirements

- a) Two treatments are either 1st, 2nd, ..., m^{th} associates.
- b) Each treatment has exactly n_i i^{th} associates ($i=1, 2, \dots, m$).
- c) Given any two treatments which are i^{th} associates, the number of treatments common to the j^{th} associate of the first and the k^{th} associate of the 2nd is p_{jk}^i ($i, j, k=1, 2, \dots, m$) and is independent of the pair of treatment with which we start.
- 4) Two treatments which are i^{th} associates occur together in λ_i blocks.
- 5) The quantities $a, b, r, k, \lambda_i, n_i$ are called parameters of 1st kind and p_{jk}^i ($i, j, k=1, \dots, m$) are called parameters of 2nd kind.

Youden Square:

Youden Square is an incomplete block design by means of which two sources of variations can be eliminated. These are basically symmetrical balanced incomplete block designs by which the block to block variation can be eliminated.

The k -units in each block can be thought of occupying k -different positions. With the help of Youden Square designs, the effects of such positions can also be eliminated.

These designs were used for the first time by Youden (1940) for green house studies.

(These are available for all No. of treatments) The No. of replications of t are flexible in these design. It is a useful design for balancing out the effects of treatments order in a repeated measure sequence.

Split Plot Design:

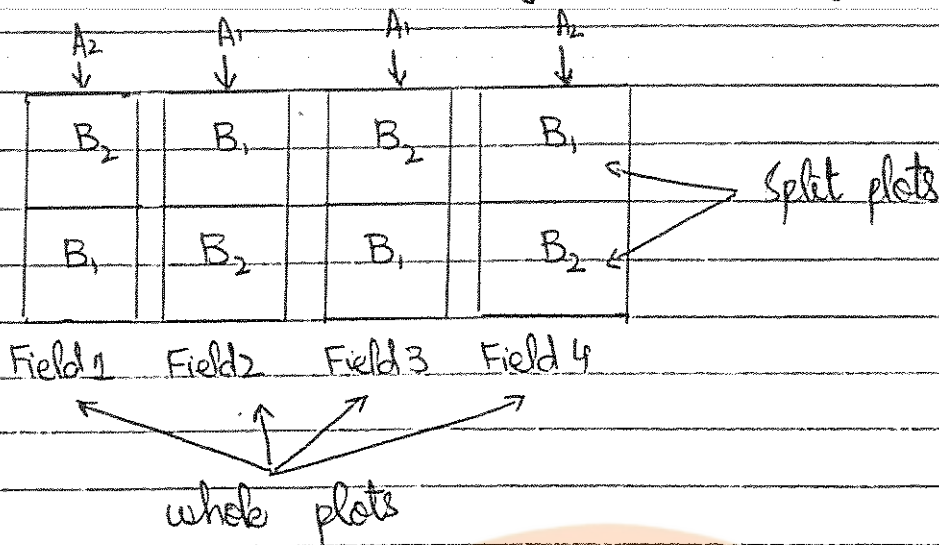
A split plot experiment is a blocked experiment where the blocks themselves serve as experimental units for a subset of the factors.

Thus, there are two levels of experimental units. The blocks are referred to as whole plots while the experimental units within each block are called split plots, split units or sub plots.

Corresponding the two levels of experimental units are two levels of randomization. One randomization is conducted to determine the assignment of block-level treatments to whole plots. Then, as always in a blocked experiment, a randomization of treatments to split plot experimental units occurs within each block or whole plot.

Split plot designs were originally developed by Fisher (1925) for use in agricultural experiments.

Split-Plot: agricultural layout:



(Factor A is the whole plot factor and factor B is the split plot factor.)

Example:

A study of the effects of two irrigation methods (factor A) and two fertilizers (factor B) on yield of a crop, using 4 available fields as experimental units.

In this investigation, it is not possible to apply different irrigation method in areas smaller than a field, although different fertilizer types could be applied in relatively small areas. For example, if we

subdivide each whole plot (field) into two split plots, each of the two fertilizer types can be applied once within each whole plot.

In this split plot design, a first randomization assigns the two irrigation types to the four fields (whole plots); then within each field, a separate randomization is conducted to assign the two fertilizer types to the two split plots within each field.

In industrial experiments, factors are often differentiated with respect to the ease with which they can be changed from exp run to experimental run. This may be due to the fact that a particular treatment is expensive or time consuming to change, or it may be due to the fact that the experiment is to be run in large batches and the batches can be subdivided later for additional treatments.

The most important factor is placed in the smallest subplot and the least important factor is

placed in the biggest main plot.

Example 2:

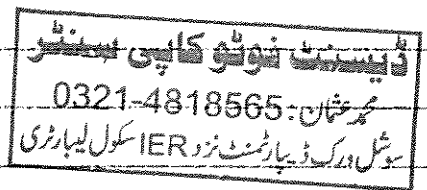
An experiment involving four prototypes combustion engine designs and two grades of gasoline. The engineers preferred to make up the four motor designs and test both of the gasoline types while a particular motor was on a test stand. Here hard to change factor is motor type and the easy to change factor is gasoline type.

Description about analysis of Split Plot Design:

The analysis of split plot design is more complex than that for a completely randomized design due to the presence of both whole plot and split plot errors.

Split plot errors might arise due to lack of repeatability of the measurement system. we will assume that whole plot errors are $NII(0, \sigma_w^2)$ and split plot errors are $NII(0, \sigma^2)$ and they are mutually independent.

Variations among whole plots is much larger than split plots. Factors that have larger relative effect (compared to the S.D) should be placed at the main plot level and factors that have smaller relative effect should be placed at the sub plot level.



Advantages:

1) Experimental units which are larger by necessity or design may be utilized to compare subsidiary treatments.

Design:

2) Increased precision over RCB design is attained on the sub plot treatments and the interaction between sub plot and main plot treatments.

3) The overall precision of the split plot design relative to RCB design may be increased by designing the main plot treatments in a Latin Square design or in an incomplete Latin Square design.

4) The cost of running a set of treatments is less than the cost of the same experiment when completely randomized.

5) These are more efficient than RCB designs.

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Disadvantages:

1) The main plot treatments are measured with less precision than they are in a RCB design.

2) When missing data occur, the analysis is more complex than for a RCB design with missing data.

3) Different treatment comparisons have different basic error variances which make the analysis more complex than with the RCB design especially if some unusual type of comparison is made.

Uses of Split Plot Design:

1) Split plot designs are used in situations when practical limit for plot size is much larger for one factor compared with the other. For example, in an

experiment to compare irrigation treatments and population densities; irrigation treatments require large plots and should, therefore, be assigned to the main plots while population densities should be assigned to the subplots.

2) It is used when greater precision is desired in one factor relative to the other. For example, if several varieties are being compared at different fertilizer levels and the factor of primary interest is the varieties, then it should be assigned to the subplots and fertilizer levels assigned to the main plots because main plot treatments are less precise than sub plot treatments.

3) A common usage of split plot design occurs when 'Time' is a second factor. e.g. in many experiments, the first factor is randomized to individual experimental units, and then these same units are measured at several time points. This is called a split plot in time.

Greco-Latin Square Design:

Considers a $P \times P$ Latin Square design and superimpose on it a second $P \times P$ Latin Square design in which treatments are denoted by Greek letters. If the two squares when superimposed have the property that each Greek letter appears once and only once with each Latin letter. The two Latin squares are said to be orthogonal and independent and the design is said to be Greco-Latin Square Design.

The GLSD can be used to control three sources of extraneous variation i.e., we investigate four factors (Rows, Columns, Latin Letters, Greek Letters) each at P -levels with P^2 observations.

GLSD exists for all $P \geq 3$ except when $P=6$.

The layout of 5×5 GLSD is given below:

Rows	Columns				
	1	2	3	4	5
1	A α	B γ	C θ	D β	E δ
2	B β	C δ	D α	E γ	A θ
3	C γ	D θ	E β	A δ	B α
4	D δ	E α	A γ	B θ	C β
5	E θ	A β	B δ	C α	D γ

Statistical Model:

The observation Y_{ij} in GLSD is expressed as

$$Y_{ij(k,m)} = \mu + \beta_i + \alpha_j + \gamma_k + \delta_m + \epsilon_{ij(k,m)}$$

where

$i, j, k, m = 1, 2, \dots, P$

μ = overall mean

β_i = i^{th} row effect

α_j = j^{th} column effect

γ_k = k^{th} Latin letter effect

δ_m = m^{th} Greek letter effect.

$$\epsilon_{ij(k,m)} \sim NII(0, \sigma^2)$$

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$$C.F = \frac{G^2}{P^2} \quad \text{where } G = \sum Y_{...} = T_{...}$$

$$TSS = \sum \sum Y_{ij(k,m)}^2 - C.F$$

$$SSR = \frac{\sum R_i^2}{P} - C.F$$

$$SSC = \frac{\sum C_j^2}{P} - C.F$$

$$LLSS = \frac{\sum L_k^2}{P} - C.F$$

$$GLSS = \frac{\sum G_m^2}{P} - C.F$$

$$ESS = TSS - SSR - SSC - LLSS - RLSS$$

ANOVA TABLE

S.O.V	df	SS	MS	F-Ratio
Rows	P-1	SSR	MSR	
Columns	P-1	SSC	MSC	
Greek letters	P-1	SSGL	MSGI	$F_1 = \frac{MSGI}{MSE}$
Latin Letters	P-1	SSL	MSL	$F_2 = \frac{MSL}{MSE}$
Error	(P-1)(P-3)	SSE	MSE	
Total	P ² -1	TSS		

C.R

$$F_1 \geq F_{\alpha(p-1, (p-1)(p-3))}$$

$$F_2 \geq F_{\alpha(p-1, (p-1)(p-2))}$$

Conclusion:

Depending upon the calculated values of F and tableted values of F , we conclude that either the Greek letter means and Latin letter means are equal or not.

Example:

The yield of a chemical process measured by 5 batches of raw material by using 5 acid concentration, 5 standing times (A, B, C, D, E) and 5 catalyst concentration ($\alpha, \beta, \gamma, \delta, \epsilon$). The GLSD shown below was used. Analyse the data and draw conclusion.

Solution:

- 1) H_0 : All row means are equal
 H_1 : " " " unequal
- H_0' : All column means are equal
 H_1' : " " " unequal
- H_0'' : All Latin letter means are equal
 H_1'' : " " " unequal
- H_0''' : All Greek letter means are equal
 H_1''' : " " " unequal

$$2) \alpha = 0.05$$

$$3) F = \frac{\text{MS for effect}}{\text{MSE error}}$$

4) Calculation

Batches	Acid Concentration					Total
	1	2	3	4	5	
1	Ad=26	BB=16	CX=19	DS=16	EE=13	90
2	BH=18	CS=21	DE=18	EX=11	AB=21	89
3	CE=20	DA=12	EB=16	AF=25	BS=13	86
4	DB=15	EX=15	AS=22	BE=14	CA=17	83
5	ES=10	AE=24	BA=17	CB=17	DX=14	82
Total	89	88	92	83	78	430

$$C.F = \frac{(430)^2}{25} = 7396$$

$$TSS = 7832 - 7396 = 436$$

$$RSS = \frac{\sum R_i^2}{P} - C.F$$

$$= \frac{(90)^2 + \dots + (82)^2}{5} - C.F$$

$$= 10$$

$$CSS = \frac{(89)^2 + \dots + (78)^2}{5} - C.F$$

$$= 2404$$

For Latin letters:

A	B	C	D	E	Total
118	78	94	75	65	430

$$SSL = \frac{\sum Y_{ij}^2}{p} - C.F$$

$$SSL = \frac{(118)^2 + \dots + (65)^2}{5} - C.F$$

$$= 342.8$$

For Greek letters:

α	β	γ	δ	ϵ	Total
83	85	91	82	89	430

$$SSB = \frac{(83)^2 + \dots + (89)^2}{5} - C.F$$

$$= 12$$

$$ESS = 46.8$$

(PTG)

ANOVA TABLE

S.O.V	df	SS	MS	F-Ratio
Rows	5-1=4	10	2.5	
Columns	4	24.4	6.1	
Latin letters	4	342.8	85.7	$F_1 = 14.65 > 3.84$
Greek letters	4	12	3	$F_2 = 0.51 < 3.84$
Error	8	46.8	5.85	
Total	24	436	—	

C.R

$F_1 \geq 3.84$

$F_2 \geq 3.84$

Conclusion:

All Latin letters means are unequal and all Greek letters means are equal.

PUACP

Fractional Replication:

As the number of factors in 2^k factorial design increases, the number of treatment combination increases rapidly. This can be seen with a 2^k factorial exp when $k=5$, requires 32 T.Cs for one replicate, for $k=6$ requires 64 and for $k=7$ requires 128 and so on.

Along with this increase in the amount of experimentation comes an increase in the number of higher order interactions.

Some of these interactions may be used as error. Second order would be difficult to explain if found to be significant.

"When information is obtained by running a fraction of a complete factorial experiment, then these designs are called fractional factorial designs."

These are among the most widely used designs for product and process designs and for process improvement.

A major use of f.f.s is in screening experiments. These are experiments in which many factors are considered with the purpose of identifying those factors that have large effect.

→ The successful use of fractional factorial designs is based on three key ideas.

1) The sparsity of effects principle:

When there are several variables the system or process is likely to be driven primarily by some of the main effects and lower order interactions.

2) The Projection Property:

These designs can be projected into stronger (larger) designs in the subset of significant factors.

3) Sequential experimentation:

It is possible to combine the runs of two (or more) fractional factorials to assemble sequentially a larger design to estimate the factor effects and interactions of interest.

Advantages: The two main reasons for employing 2^k factorial replications are:

1) In the 2^7 experiment, a full replicated design would involve 128 treatment combinations for a single replicate to be made up. This may be sometimes impracticable or too costly. In this case, fractional replication is very useful. The use of fractional design means that only a fraction of 128 T.C would be run.

2) For practical implications, an experimenter or investigator may prefer to carry out his exp as a series of small experiments.

He may be able to discover quickly which are significant factors.

Alternatively, he may wish to carry out his experiment in a sequential manner. Planning of his new exp according to the results of the current one and building up a whole set of experimental results.

Disadvantages:

- 1) The accuracy of the estimates of the effects may be reduced.
- 2) Outcomes of a fractional experiment is open to mis-interpretation as compared to fully replicated experiment.

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The One-half fraction of 2^k design:

Consider a situation in which 3 factors, each at two levels, are of interest. But the experiment can't afford to run all $2^3 = 8$ T.C's.

This can however afford for 4 runs. So, this suggests a one half fraction of 2^3 design because the design contains a one half fraction of the 2^3 design.

"One half fraction of 2^3 design is often called a 2^{3-1} design."

The table of '+' and '-' signs is given below:

T.C	I=M	A	B	C	AB	AC	BC	ABC
a	+	+	-	-	-	-	+	+
b	+	-	+	-	+	+	-	+
c	+	-	-	+	+	-	-	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	-	+	-	-	-
ac	+	+	-	+	-	+	-	-
bc	+	-	+	+	-	-	+	-
1 (abc)	+	-	-	-	+	+	+	-

Suppose we select the first four t.c's a, b, c, abc as a half fraction.

Notice that the 2^{3-1} design is formed by selecting only those t.c that yield a '+' sign on ABC effect. Hence ABC is called **generator** of this particular fraction.

Furthermore, the identity element (1) is always +ve so we call

$$I = ABC$$

the defining relation of the design.

The t.c's in 2^{3-1} design yields 3 of which are used to estimate the main effects.

Linear combination of observations used to estimate the main effects are

$$l_A = \frac{1}{2} [a - b - c + abc]$$

$$l_B = \frac{1}{2} [-a + b - c + abc]$$

$$l_C = \frac{1}{2} [-a - b + c + abc]$$

It is also easy to verify that the linear combination of observation to estimate the two factor interactions are:

$$J_{BC} = \frac{1}{2} [a - b - c + abc]$$

$$J_{AC} = \frac{1}{2} [-a + b - c + abc]$$

$$J_{AB} = \frac{1}{2} [-a - b + c + abc]$$

Hence

$$J_A = J_{BC}$$

$$J_B = J_{AC}$$

$$J_C = J_{AB}$$

It is impossible to differentiate b/w A and BC, B and AC, C and AB. In fact, when we estimate A, B and C, we are really estimating the $A+BC$, $B+AC$, $C+AB$.

"Two or more effects that have this property (having same linear combination or effect) are called aliases."

In our example

A and BC are aliases

B and AC are aliases

C and AB are aliases.

we indicate this by notation as

$$\phi_A \rightarrow A + BC$$

$$\phi_B \rightarrow B + AC$$

$$\phi_C \rightarrow C + AB$$

A quick way to find aliases of an effect is to multiply the effect by the terms in the defining contrast (Mod 2)

In our example:

$$I = ABC$$

$$\text{aliases of } A = (A)(ABC) = A^2BC \text{ (Mod 2)} = BC$$

$$\text{" } B = (B)(ABC) = AB^2C \text{ (Mod 2)} = AC$$

$$\text{" } C = (C)(ABC) = ABC^2 \text{ (Mod 2)} = AB$$

ANOVA Table

S.o.V	df	SS	MS	F Ratio
A(BC)	1	SSA	MSA	
B(AC)	1	SSB	MSB	
C(AB)	1	SSC	MSE	

Now suppose we have chosen the other half fraction i.e. the treatment in the table of (+) and (-) signs associated with (-) on ABC. Then the defining relation for this design is

$$I = -ABC$$

Then A has the value $[-(1) + ab + ac - bc]$ and BC has the value $[(1) - ab - ac + bc]$

which gives the same total except for the signs. The sum of squares due to A and BC are same, again they are called aliases.

Here

$$A = -BC$$

$$B = -AC$$

$$C = -AB$$

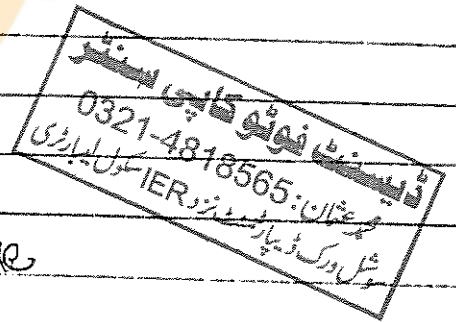
The linear combinations are

$$JA' \rightarrow A - BC$$

$$JB' \rightarrow B - AC$$

$$JC' \rightarrow C - AB$$

Suppose that after running the half fraction of the 2^3 design, the other half has also run. Thus, all t.c. associated with



full replicate of 2^3 design are now available
 we can now obtain de-aliases
 of 8-run as a full replicate of 2^3
 design in two blocks of 4 runs
 each. This could also be done
 by adding or subtracting the three
 combination of effects from the two
 individual fractions

e.g. consider

$$J_A \rightarrow A + BC$$

$$J_{A'} \rightarrow A - BC$$

$$\frac{1}{2} [J_A + J_{A'}] = \frac{1}{2} [A + BC + A - BC] \rightarrow A$$

$$\frac{1}{2} [J_A - J_{A'}] = \frac{1}{2} [A + BC - A + BC] \rightarrow BC$$



Analysis of Co-Variance:

Occasionally, when a study is being made of the effect of one or more factors on some response variable (say Y), there is another variable which may vary along Y . It is often not possible to control this variable at some constant level throughout the experiment but this variable can be measured along with response variable. This variable is called **Concomitant Variable** or co-variate and is denoted by X . The response/main variable is represented by Y . The data thus consists of pair of values of the variables X and Y , one pair for each unit.

"The statistical technique applied to analyze the effect of classification or treatment on the main variable Y , after removing the effect of the concomitant variable by regression method, is called analysis of co-variance."

Hence, the analysis of co-variance is a mixture of regression analysis and analysis of variance.

Like ANOVA, this technique partitions the total covariation of a bivariate sample, measured by the sum of product of the deviation of the variables from their means, into components associated with specific source of variation.

The covariance effects are sorted out and a test of existence of regression is made. Finding evidence of any regression, its effect are removed before tests are made on the significance of the treatments.

R.A Fisher (1890-1962) has expressed that the ANCOVA "combines the advantages and reconciles the requirement of the two widely applicable procedures known as regression and analysis of variance."

To illustrate how the ANOVA is adjusted to provide for the removal of the effects of concomitant variable X .

Let us consider the simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

Then the estimated regression line is

$$\hat{Y} = a + bX \quad (1)$$

where a and b are the least square estimates of α and β respectively

As the regression line goes through the means we have

$$\bar{Y} = a + b\bar{X} \quad (2)$$

By subtracting (1) and (2), we get

$$\hat{Y} - \bar{Y} = b(X - \bar{X})$$

Let

$$\hat{Y} - \bar{Y} = \hat{y}$$

$$X - \bar{X} = x$$

$$\hat{y} = bx$$

This implies that

$$e = Y - \hat{Y} = Y - bx$$

Now the sum of squares of the estimate is

$$\Sigma e^2 = \Sigma (\hat{y} - bx)^2$$

$$\Sigma e^2 = \Sigma [y^2 + b^2 x^2 - 2bxy]$$

$$\Sigma e^2 = \Sigma y^2 + b^2 \Sigma x^2 - 2b \Sigma xy$$

$$\therefore b = \frac{\Sigma xy}{\Sigma x^2}$$

$$= \Sigma y^2 + \left(\frac{\Sigma xy}{\Sigma x^2} \right)^2 \Sigma x^2 - 2 \left(\frac{\Sigma xy}{\Sigma x^2} \right) \Sigma xy$$

$$= \Sigma y^2 + \frac{(\Sigma xy)^2}{\Sigma x^2} - 2 \frac{(\Sigma xy)^2}{\Sigma x^2}$$

$$\Sigma e^2 = \Sigma y^2 - \frac{(\Sigma xy)^2}{\Sigma x^2}$$

The term $\frac{(\Sigma xy)^2}{\Sigma x^2}$, which is actually

the sum of squares due to the regression with '1' d.f., is the amount of reduction in the sum of squares of Y due to its linear regression on X.

Thus, to remove the effect of the regression, the term $\frac{(\Sigma xy)^2}{\Sigma x^2}$ is to be subtracted from the Σx^2 sums of square for the Y and the result is the corrected and adjusted

Sum of Squares of Y. The procedure after correction for the regression is that of ANOVA.

Statistical Analysis:

Suppose that a random sample of 'n' pair of values of two variables 'X' and 'Y' is taken from the bivariate normal population from that the paired observations are classified into 'k' groups.

These groups presented some factors of classification or treatment.

Let X_{ij} and Y_{ij} be the i th observation in j th block/group/treatment.

Groups / Treatments

$i \backslash j$	1	...	j	...	k	
1	X_{11}	Y_{11}	X_{1j}	Y_{1j}	X_{1k}	Y_{1k}
2	X_{21}	Y_{21}	X_{2j}	Y_{2j}	X_{2k}	Y_{2k}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	X_{i1}	Y_{i1}	X_{ij}	Y_{ij}	X_{ik}	Y_{ik}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	X_{n1}	Y_{n1}	X_{nj}	Y_{nj}	X_{nk}	Y_{nk}

Total	$X_{.1}$	$Y_{.1}$	$X_{.j}$	$Y_{.j}$	$X_{.k}$	$Y_{.k}$	$T_{..}$	$T_{..}$
Mean	$\bar{X}_{.1}$	$\bar{Y}_{.1}$	$\bar{X}_{.j}$	$\bar{Y}_{.j}$	$\bar{X}_{.k}$	$\bar{Y}_{.k}$		

$$i = 1, 2, \dots, n$$

$$j = 1, 2, \dots, k$$

$$\begin{aligned}
 S_{xx} &= T_{xx} + E_{xx} \\
 S_{yy} &= T_{yy} + E_{yy} \\
 S_{xy} &= T_{xy} + E_{xy}
 \end{aligned}$$

where

$$S_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{..})^2$$

$$T_{xx} = \sum_j n_j (\bar{x}_{.j} - \bar{x}_{..})^2$$

$$E_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{.j})^2$$

$$S_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{..})^2, \quad T_{yy} = \sum_j n_j (\bar{y}_{.j} - \bar{y}_{..})^2$$

$$E_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{.j})^2$$

$$S_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{..})(y_{ij} - \bar{y}_{..})$$

$$T_{xy} = \sum_j n_j (\bar{x}_{.j} - \bar{x}_{..})(\bar{y}_{.j} - \bar{y}_{..})$$

$$E_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{.j})(y_{ij} - \bar{y}_{.j})$$

S.O.V	df	Sum of squares and products			Coefficient of regression
		$\sum x^2$	$\sum y^2$	$\sum xy$	
Treatments	K-1	T_{xx}	T_{yy}	T_{xy}	$b = E_{xy}/E_{xx}$ $b' = S_{xy}/S_{xx}$
Error	N-K	E_{xx}	E_{yy}	E_{xy}	
Total	N-1	S_{xx}	S_{yy}	S_{xy}	

We may calculate the estimate of regression co-efficient, a regression line and partition the ~~tot~~ sum of squares due to error and total.

For example; the error sum of squares is partitioned into sum of squares due to regression of $\bar{E}_{xy}$ with 1 df and a sum of E_{xx} squares due to deviations about the regression line of

$$E_{yy} - \frac{E_{xy}^2}{E_{xx}} \text{ with } (n-k-1) \text{ df}$$

The corresponding sum of squares for total SS terms are S_{xy} with 1 df and $S_{yy} - \frac{S_{xy}^2}{S_{xx}}$ with $(n-2)$ df S_{xx}

The next problem is to find out if there is evidence of any regression b/w X and Y. For this purpose, we compare $\frac{E_{xy}^2}{E_{xx}}$ with 1 df with the

term $E_{yy} - \frac{E_{xy}^2}{E_{xx}}$ with $(n-k-1)$ df by

applying F-statistic. If the regression proves to be significant then we adjust the treatments. The procedure is presented in another ANOVA Table as follows:

S.o.V	Regression	Adjusted SS	MS	F-Ratio
	df SS	df SS		
Total	1 S_{xy}^2/S_{xx}	$n-2$ $S_{yy}' = S_{yy} - S_{xy}^2/S_{xx}$		
Error	1 E_{xy}/E_{xx}	$n-k-1$ $E_{yy} = E_{yy} - E_{xy}/E_{xx}$	$se^2 = E_{yy}/(n-k-1)$	$F = \frac{st^2}{se^2}$
Treatment (By Sub)		$k-1$ $T_{yy}' = S_{yy}' - E_{yy}$	$st^2 = T_{yy}'/k-1$	

The quantity se^2 is an unbiased estimate of the variance of y ($V(y)$) in the population after correction for regression. Similarly;

st^2 is an unbiased estimate of $V(\bar{y})$. Hence, the statistic F has F -dist with $(V_1 = k-1, V_2 = n-k-1)$ degree of freedom.

Assumptions:

Following are the assumptions of ANCOVA

- 1) The populations are normally distributed with homogeneous variances.
- 2) The samples are random and independent. (PTO)

- 3) The regression is linear and slope is not equal to zero.
- 3) The treatment, block and regression effects are additive.
- 4) The concomitant variable (X) is fixed, implying that it is not being affected by the treatment.
- 5) The residuals e_{ij} are normally and independently distributed with zero mean and common variance.

Use of Covariance Analysis:

Uses of ANCOVA are:

- 1) It increases the precision of the experiment by removing certain environmental effects that can't be controlled by experimental design.
- 2) The covariance adjustments remove the bias due to regression. The nature of the bias is the term $\beta(\bar{X}_{ij} - \bar{X}_{..})$

3) As it partitions the total covariance or the sum of cross products into components parts, it is therefore used for testing a regression co-efficient or the homogeneity of k linear regression co-efficients.

4) It assists in analysing the treatment effects properly when a concomitant variable is playing its part in producing the treatment effects.

5) It can be profitably used to estimate missing observations by setting $Y=0$ for each missing value and introducing dummy covariate X in such a way that $X=1$ corresponds to the missing value and $X=0$ to all others.

Question:

If A, B, C are three methods of teachings in X and Y stands for original spelling performance and later spelling performance. For each of the four students allocated to the system as given below. Perform the analysis of variance.

Solution:

A			B			C			
X	Y	XY	X	Y	XY	X	Y	XY	
3	10	30	4	12	48	1	6	6	
2	8	16	3	12	36	2	5	10	
1	8	8	3	10	30	3	8	24	
2	11	22	5	13	65	1	7	7	
8	37	76	15	47	179	7	26	47	$T_{..} = 30 \quad T'_{..} = 110$

For X:

$$C.F = \frac{(T_{..})^2}{12} - \frac{(30)^2}{12} = 75$$

$$S_{xx} = TSS = \sum X_{ij}^2 - C.F = 92 - 75 = 17$$

$$T_{xx} = \text{Treatment SS} = \frac{\sum T_{.j}^2}{4} - C.F = 9.5$$

$$ESS = E_{xx} = S_{xx} - T_{xx} = 7.5$$

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For Y:

$$C.F = \frac{(T_{.}')^2}{12} = \frac{(110)^2}{12} = 1008.33$$

$$S_{yy} = \sum Y_{ij}^2 - C.F = 1080 - 1008.33 = 71.67.$$

$$T_{yy} = \frac{\sum T_{ij}^2}{4} - C.F = \frac{4254}{4} - C.F = 55.17$$

$$E_{yy} = S_{yy} - T_{yy} = 16.5$$

For XY:

$$C.F = \frac{(T_{.})(T_{.}')}{12} = \frac{(30)(110)}{12} = 275$$

$$S_{xy} = \sum X_{ij}Y_{ij} - C.F = 302 - 275 = 27.$$

$$T_{xy} = \frac{\sum T_{ij}T_{ij}'}{4} - C.F = \frac{1183}{4} - C.F = 20.75$$

$$E_{xy} = S_{xy} - T_{xy} = 6.25.$$

ANO

S.O.V

 $\sum X^2$ $\sum Y^2$ $\sum XY$

C.F

75

1008.33

275

Total SS

17

71.67

27

Treatment SS

9.50

55.17

20.75

Error SS

7.5

16.5

6.25

ANOVA Table

S.O.V	df	Σx^2	Σy^2	Σxy	Regression Co-efficient
Treatment	3-1=2	9.5	55.7	20.75	$b = E_{xy}/E_{xx} = 0.833$
Error	12-3=9	7.5	16.5	6.25	
Total	12-1=11	17	71.67	27	

To test the significance of regression co-efficient, we use the following test statistic

$$H_0: \beta = 0$$

$$H_1: \beta \neq 0$$

$$F = \frac{E_{xy}^2/E_{xx}}{E_{yy} - \frac{E_{xy}^2}{E_{xx}}} \times (N-K-1)$$

$$F = \frac{(6.25)^2/7.5}{16.5 - \frac{(6.25)^2}{7.5}} \times (12-3-1)$$

$$F = \frac{5.2083}{11.29167} \times 8$$

$$F = 3.69$$

C.R. : Reject H_0 if

$$F_c > F_{\alpha}(1, N-K-1)$$

$$F_c > F_{0.05}(1, 8)$$

$$F_c > 5.318$$

Conclusion: As $F_{\alpha} = 3.69$ falls in acceptance region so we accept H_0 at 5% level of significance i.e. there is no impact of X on Y. This means that there is no need to consider original spelling performance in order to compare the later spelling performance.

These ANOVA Table is

S.O.V	df	SS	MS	F-Ratio	F-Table
Treatment	2	55.17	27.585	$F = 15.04$	$F_{(2,9)} = 4.25$
Error	9	16.5	1.833		
Total	11	71.67	6.5154		

Reject H_0 i.e. the average teaching methods are not same.

Note: If $\beta \neq 0$, then totals for total SS and Error SS must be adjusted

Adjusted SS for Error:

$$E_{yy}' = E_{yy} - \frac{E_{xy}^2}{E_{xx}} \quad \text{with } n-k-1 \text{ df}$$

Adjusted SS for total:

$$S_{yy}' = S_{yy} - \frac{S_{xy}^2}{S_{xx}} \quad \sim n-2 \text{ df}$$

Adj SS for treatment:

$$T_{yy}' = S_{yy}' - E_{yy}' \quad \sim k-1 \text{ df}$$

Two Way Analysis of Covariance: (with No Interaction) (RCB Design):

Let a random sample of n pair of observations (X_{ij}, Y_{ij}) from a homogeneous population, be classified according to two factors of classifications say treatments and blocks. Assuming one pair of obs per cell. We may arrange the data in a rectangular array having c columns and r rows so that $n = rc$. Let the column represents treatment and row block represents block. Then the pair (X_{ij}, Y_{ij}) belonging to the i th block and j th treatment will appear in the i th row and the j th column.

For partitioning the sum of squares for the two variables X and Y we have the identities

$$(X_{ij} - \bar{X}_{..}) = X_{ij} - \bar{X}_{i.} + \bar{X}_{i.} + \bar{X}_{.j} - \bar{X}_{.j} - \bar{X}_{..} + \bar{X}_{..} - \bar{X}_{..}$$

$$= (\bar{X}_{i.} - \bar{X}_{..}) + (\bar{X}_{.j} - \bar{X}_{..}) + (X_{ij} - \bar{X}_{i.} - \bar{X}_{.j} + \bar{X}_{..})$$

and

$$(Y_{ij} - \bar{Y}_{..}) = (\bar{Y}_{i.} - \bar{Y}_{..}) + (\bar{Y}_{.j} - \bar{Y}_{..}) + (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})$$

Squaring on both sides and summing over all values, we have.

$$\sum_i^r \sum_j^c (X_{ij} - \bar{X}_{..})^2 = c \sum_i (\bar{X}_{i.} - \bar{X}_{..})^2 + r \sum_j (\bar{X}_{.j} - \bar{X}_{..})^2 + \sum_i \sum_j (X_{ij} - \bar{X}_{i.} - \bar{X}_{.j} + \bar{X}_{..})^2$$

$$S_{xx} = B_{xx} + T_{xx} + E_{xx}$$

$$\sum_i^r \sum_j^c (Y_{ij} - \bar{Y}_{..})^2 = c \sum_i (\bar{Y}_{i.} - \bar{Y}_{..})^2 + r \sum_j (\bar{Y}_{.j} - \bar{Y}_{..})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2$$

$$S_{yy} = B_{yy} + T_{yy} + E_{yy}$$

Partitioning the sum of product is:

$$\sum_i^r \sum_j^c (X_{ij} - \bar{X}_{..})(Y_{ij} - \bar{Y}_{..}) = c \sum_i (\bar{X}_{i.} - \bar{X}_{..})(\bar{Y}_{i.} - \bar{Y}_{..}) +$$

$$r \sum_j (\bar{X}_{.j} - \bar{X}_{..})(\bar{Y}_{.j} - \bar{Y}_{..}) + \sum_i \sum_j (X_{ij} - \bar{X}_{i.} - \bar{X}_{.j} + \bar{X}_{..})(Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})$$

$$S_{xy} = B_{xy} + T_{xy} + E_{xy}$$

(PT6).

S.O.V	df	Sum of Squares and Sx ² product Sy ²			Regression Co-efficient	mi
Treatments	c-1	Txx	Tyy	Txy		S.O.
Blocks	r-1	Bxx	Byy	Bxy		Total (T+E)
Error	(r-1)(c-1)	Exx	Eyy	Exy	$b = \frac{E_{xy}}{E_{xx}}$	Error
Total	rc-1	Sxx	Syy	Sxy		Treat

Firstly, we must find out if there is evidence of any regression b/w X and Y.

For this purpose, b is obtained by $\frac{E_{xy}}{E_{xx}}$.

To test the significance of B

$$H_0: B = 0$$

$$H_1: B \neq 0$$

$$\alpha = 0.05$$

$$F = \frac{E_{xy}^2 / E_{xx}}{E_{yy} - \frac{E_{xy}^2}{E_{xx}}} * (rc - r - c)$$

$$F_{\alpha(1, rc-r-c)}$$

$$F_c \geq F_{\alpha(1, rc-r-c)}$$

If the regression proves to be significant then sum of squares

must be corrected

S.O.V	df	Regression	Adjusted Sum of square
Total (T+E)	1	$\frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}}$	$r(c-1)-1$ $(T+E)'_{yy} = (T_{yy} + E_{yy}) - \frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}}$
Error	1	$\frac{(E_{xy})^2}{E_{xx}}$	$(r-1)(c-1)-1$ $E'_{yy} = E_{yy} - \frac{E_{xy}^2}{E_{xx}}$
Treatment By sub			$c-1$ $T_{yy} = (T+E)'_{yy} - E'_{yy}$

$$s_e^2 = \frac{E'_{yy}}{(r-1)(c-1)-1}$$

$$s_t^2 = \frac{T_{yy}}{c-1}$$

$$F = \frac{s_t^2}{s_e^2}$$

with $v_1 = c-1$ and $v_2 = (r-1)(c-1)-1$

Questions Related to ANCOVA:

(2012) Q # 3(b):

Solution:

$$c = 3, r = 3$$

$$n = cr = 9$$

S.O.V	df	XY	XX	YY	Reg Co-efficient
Blocks	$r-1$ $3-1=2$	-72	126	42	
Treatments	$c-1$ $3-1=2$	-15	2	186	
Error	4	18	32	12	$b = \frac{E_{xy}}{E_{xx}} = \frac{18}{32} = 0.5625$
Total	$q-1=8$				

a) Is the regression of Y on X significant at 5% d.o.f.

i) $H_0: \beta = 0$

$H_1: \beta \neq 0$

ii) $\alpha = 0.05$

iii) $F = \frac{E_{xy}^2 / E_{xx} \times (rc - r - c)}{E_{yy} - \frac{E_{xy}^2}{E_{xx}}}$

$$F = \frac{(18)^2 / 32 \times (9 - 3 - 3)}{12 - \frac{(18)^2}{32}}$$

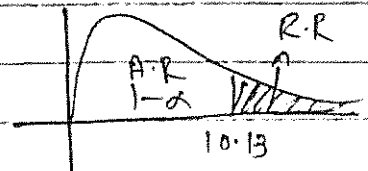
$$iv) F = \frac{10.125}{1.875} (3) = 16.2$$

v) C.R.

Reject H_0 if

$$F_c \geq F_{\alpha, 0.5}(1, 3)$$

$$F_c \geq 10.13$$



vi) Reject H_0 at $\alpha = 0.05$ i.e. the regression of Y on X is significant.

b) Construct ANOVA and write inference. First we have to find corrected SS for total and error. For this purpose.

S.O.V	df	XY	YY	XX
Blocks	2	-72	42	126
Treatments	2	-15	186	2
Error	4	18	12	32
(T+E)	6	3	198	34

Adjusted SS for error:

$$\begin{aligned} \text{Adjusted } \Sigma y^2 &= \Sigma y^2 - \frac{(\Sigma xy)^2}{\Sigma x^2} \\ &= 12 - \frac{(18)^2}{32} = 1.875 \end{aligned}$$

Adjusted SS for treatment plus error:

$$\text{Adjusted } \sum y^2 = \sum y^2 - \frac{(\sum xy)^2}{\sum x^2}$$

$$= 198 - \frac{(3)^2}{34}$$

$$= 197.7353$$

ANOVA Table for adjusted SS is:

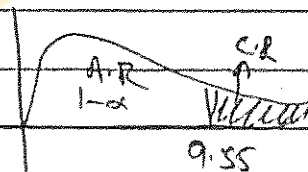
S.O.V	df	$\sum y^2$	MS	F-Ratio
(T+E)	$3(3-1)-1=5$	197.7353	—	—
Error	$(r-1)(c-1)-1=3$	1.875	$s_e^2 = 0.625$	—
Treatment	$3-1=2$	195.8603	$s_t^2 = 97.9301$	$F = 156.69$

C.R.:

Reject H_0 if

$$F_c \geq F_{\alpha, (2, 3)}$$

$$F_c \geq 9.55$$



Conclusion:

Reject H_0 i.e. treatment means are not equal.

c) Use the LSD test at 5% level of significance to analyze the corrected treatment means.

Adjusted Mean: $\bar{Y}_{ij} - b(\bar{X}_{.j} - \bar{X}_{..})$

For this
we have to find

$$b = \frac{\sum xy}{\sum x^2} = \frac{18}{32} = 0.5625$$

$$\bar{X}_{..} =$$

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PUACP



(2010) Q # 3(b):

S.O.V	df	XY	YY	XX
Blocks	2	1.2667	0.933	1.9111
Treatments	14	19.4000	68.1333	10.3111
Error	28	7.4000	13.7333	10.7556
Total	44	28.0667	82.8000	22.9778

$b = 0.6880$

First, we need to check that is there any evidence of regression b/w X and Y. For this purpose

i) $H_0: \beta = 0$

$H_1: \beta \neq 0$

ii) $\alpha = 0.05$

iii) $F = \frac{E^2_{xy} / E_{xx}}{E_{yy} - \frac{E^2_{xy}}{E_{xx}}} \times (rc - r - c)$

$r = 3$

$c = 15$

$rc = 45$

iv) $F = \frac{(7.4)^2 / 10.7556}{13.7333 - \frac{(7.4)^2}{10.7556}} \times (45 - 3 - 15)$

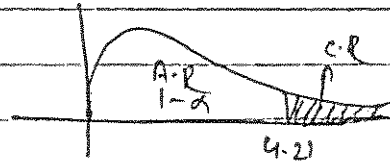
$F = \frac{5.09130 \times 27}{8.642}$

$F = 3.713411$

v) C.R.Reject H_0 if

$$F_c \geq F_{0.05(1,27)}$$

$$F_c \geq 4.21$$



vi) As $F_c = 3.7134$ lies in acceptance region, so we accept H_0 at $\alpha = 0.05$ i.e. there is no relationship b/w X and Y or the regression is insignificant of Y on X .
Hence Adjustment for sum of squares will not performed.

S.O.V	df	SS	MS	F-Ratio
Blocks	2	0.933	—	
Treatment	14	68.1333	4.8667	$F = 9.9224$
Error	28	13.7333	0.4905	
Total	44	82.8000	—	

C.R. :

$$F_c \geq F_{0.05(14,28)}$$

$$F_c \geq 2.085$$

Conclusion

Reject H_0 . Treatment means are not equal.

(20)

a)

iv)

v)

vi)

b)

S.O.V

Blocks

Treatment

error

(T+E)

(2003) Q # 3(b):

a) As the regression of Y on X is significant at $\alpha = 5\%$

i) $H_0: B = 0$

$H_1: B \neq 0$

$r = 6$

$c = 7$

$rc = 42$

ii) F_{α}
iii)

$\alpha = 0.05$

$F = \frac{E_{xy}^2 / E_{xx}}{E_{yy} - \frac{E_{xy}^2}{E_{xx}}} * (rc - r - c)$

$E_{yy} - \frac{E_{xy}^2}{E_{xx}}$

iv) $F = \frac{(682.20)^2 / 28665.1}{23.23 - (682.20)^2 / 28665.1} * (42 - 6 - 7)$

$F = \frac{16.23566}{6.9943} * (29)$

$F = 67.3168$

v)

Reject H_0 if

$F_c \geq F_{\alpha=0.05}(1, 29)$

$F_c \geq 4.18$

vi)

Reject H_0 at $\alpha = 0.05$ i.e. $B \neq 0$

b) First, we have to find adjusted SS

S.O.V	df	XX	XY	YY
Blocks	6-1=5	7472.6	-111.65	6.31
Treatments	7-1=6	116020.3	3598.05	112.86
error	30	28665.1	682.20	23.23
(T+E)	36	144685.4	4280.25	136.09

adjusted SS for error : $\therefore$

(20

$$\text{adjusted } \Sigma y^2 = \Sigma y^2 - \frac{(\Sigma xy)^2}{\Sigma x^2}$$

$$= 23.23 - \frac{(682.20)^2}{28665.1}$$

First
reg

$$= 6.9943$$

For treatment + Error:

$$\text{adj } \Sigma y^2 = \Sigma y^2 - \frac{(\Sigma xy)^2}{\Sigma x^2}$$

$$= 136.09 - \frac{(4280.25)^2}{144685.4}$$

$$= 9.46671$$

iv

S.O.V	df	Σy^2	MS	F-Ratio
(T+E)	35	9.46671	0.27047	
Error	29	6.9943	0.241182 = $\frac{S^2}{2}$	$F = \frac{M^2}{S^2} = 1.07085$
Treatment	6	2.47241	0.412068 = $\frac{M^2}{2}$	

v)

v) C.F. : Reject H_0 if

$$F_c > F_{\alpha, (6, 29)}$$

$$F_c > 2.43$$

vi)

reg

vi) Accept H_0 : treatment means are insignificant.

Now,

S.O.V

Blocks

Treatments

Error

Total

(2002) Q# 3(b):

First, we have to check the significance of regression of Y on X.

$$r = 4$$

$$c = 3$$

$$rc = 12$$

i) $H_0: \beta = 0$

$H_1: \beta \neq 0$

ii) $\alpha = 0.05$

iii)
$$F = \frac{E_{reg} / E_{xx}}{E_{reg} - \frac{E_{xy}^2}{E_{xx}}} \times (rc - r - c)$$

iv)
$$F = \frac{(46)^2 / 86}{48 - \frac{(46)^2}{86}} \times (12 - 4 - 3)$$

$$F = \frac{24.6046}{23.3953} \times (5)$$

$$F = 5.2584$$

v) C.R

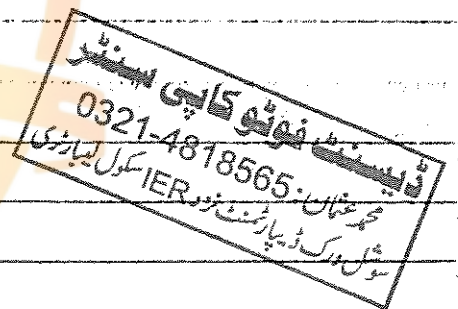
$$F_c > F_{obs}(1, 5)$$

$$F_c > 6.61$$

vi) Accept H_0 . There is insignificant regression b/w X and Y.

Now;

S.O.V	df	YY	MS	F-Ratio	F-Table	Conclusion
Blocks	3	252	84			
Treatments	2	24	12	$F = 1.5$	< 5.14	Insignificant
Error	6	48	8			
Total	11	324	—			



ANCOVA In CR Design:

In CR design, of analysis of covariance, the observation y_{ij} is expressed as

$$y_{ij} = \mu + \alpha_j + \beta x_{ij} + \epsilon_{ij}$$

$i=1, 2, \dots, n$
 $j=1, 2, \dots, k$

let $\hat{\mu}$, $\hat{\alpha}_j$ and $\hat{\beta}$ are the estimates of μ , α_j and β respectively. So, the sum of squares of error is

$$S = \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} x_{ij})^2$$

differentiating it w.r.t unknown values and equating it to zero, we get

$$\frac{\partial S}{\partial \hat{\mu}} = 2 \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} x_{ij}) (-1) = 0$$

$$\sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} x_{ij}) = 0$$

$$\sum_i \sum_j y_{ij} - nk \hat{\mu} - n \sum_j \hat{\alpha}_j - \hat{\beta} \sum_i \sum_j x_{ij} = 0$$

$$\sum_i \sum_j y_{ij} = nk \hat{\mu} + n \sum_j \hat{\alpha}_j + \hat{\beta} \sum_i \sum_j x_{ij} \quad \text{--- (1)}$$

For treatment:

$$S = \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta} X_{i1})^2 + \sum_i (Y_{i2} - \hat{\mu} - \hat{\alpha}_2 - \hat{\beta} X_{i2})^2 + \\ \sum_i (Y_{i3} - \hat{\mu} - \hat{\alpha}_3 - \hat{\beta} X_{i3})^2 + \dots + \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} X_{ij})^2 + \\ \dots + \sum_i (Y_{ik} - \hat{\mu} - \hat{\alpha}_k - \hat{\beta} X_{ik})^2$$

differentiating it w.r.t $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_j, \hat{\alpha}_k$
and equating them to zero

$$\frac{\partial S}{\partial \hat{\alpha}_1} = 2 \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta} X_{i1}) (-1) = 0$$

$$\frac{\partial S}{\partial \hat{\alpha}_j} = 2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} X_{ij}) (-1) = 0$$

$$\frac{\partial S}{\partial \hat{\alpha}_k} = 2 \sum_i (Y_{ik} - \hat{\mu} - \hat{\alpha}_k - \hat{\beta} X_{ik}) (-1) = 0$$

Taking only j^{th} term

$$2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} X_{ij}) = 0$$

$$\sum_i Y_{ij} - n\hat{\mu} - n\hat{\alpha}_j - \hat{\beta} \sum_i X_{ij} = 0$$

$$\sum_i Y_{ij} = n\hat{\mu} + n\hat{\alpha}_j + \hat{\beta} \sum_i X_{ij}$$

$$Y_{.j} = n\hat{\mu} + n\hat{\alpha}_j + \hat{\beta} X_{.j} \quad \text{--- (1)}$$

For $\hat{\beta}$:

$$\frac{\partial S}{\partial \hat{\beta}} = 2 \sum_i \sum_j^k (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} x_{ij}) x_{ij} = 0$$

$$\sum_i \sum_j^k (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta} x_{ij}) x_{ij} = 0$$

$$\sum \sum x_{ij} y_{ij} - \hat{\mu} \sum \sum x_{ij} - \sum \sum \hat{\alpha}_j x_{ij} - \hat{\beta} \sum_i \sum_j^k x_{ij}^2 = 0$$

$$\sum \sum x_{ij} y_{ij} = \hat{\mu} \sum \sum x_{ij} + \sum_j \hat{\alpha}_j \sum_i x_{ij} + \hat{\beta} \sum_i \sum_j^k x_{ij}^2 \quad (3)$$

Now, if we add 'k' equations in equation (2), it will become same as eq (1). Hence eq (1) and (11) are not linearly independent.

For unique solution, we put the following restriction

$$\sum_j \hat{\alpha}_j = 0$$

From eq (1)

$$Y_{..} = nk\hat{\mu} + 0 + \hat{\beta} X_{..}$$

$$nk\hat{\mu} = Y_{..} - \hat{\beta} X_{..}$$

$$\hat{\mu} = \frac{Y_{..}}{nk} - \frac{\hat{\beta} X_{..}}{nk}$$

$$\boxed{\hat{\mu} = \bar{Y} - \hat{\beta} \bar{X}}$$

From eq (II):

$$Y_{.j} = n(\bar{Y}_{..} - \hat{\beta}\bar{X}_{..}) + n\hat{\alpha}_{.j} + \hat{\beta}X_{.j}$$

$$Y_{.j} = n\bar{Y}_{..} - n\hat{\beta}\bar{X}_{..} + n\hat{\alpha}_{.j} + \hat{\beta}X_{.j}$$

$$n\hat{\alpha}_{.j} = Y_{.j} - n\bar{Y}_{..} + n\hat{\beta}\bar{X}_{..} - \hat{\beta}X_{.j}$$

$$\hat{\alpha}_{.j} = \frac{Y_{.j}}{n} - \bar{Y}_{..} + \hat{\beta}\bar{X}_{..} - \hat{\beta}\frac{X_{.j}}{n}$$

$$\hat{\alpha}_{.j} = (\bar{Y}_{.j} - \bar{Y}_{..}) + \hat{\beta}\bar{X}_{..} - \hat{\beta}\bar{X}_{.j}$$

$$\boxed{\hat{\alpha}_{.j} = (\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta}(\bar{X}_{.j} - \bar{X}_{..})}$$

From Equation (III):

$$X_{..}Y_{..} = (\bar{Y}_{..} - \hat{\beta}\bar{X}_{..})X_{..} + \sum_j [(\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta}(\bar{X}_{.j} - \bar{X}_{..})]X_{.j} + \hat{\beta} \sum_j \sum_i X_{ij}^2$$

$$X_{..}Y_{..} = \bar{Y}_{..}X_{..} - \hat{\beta}X_{..}\bar{X}_{..} + \sum_j X_{.j}(\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta} \sum_j X_{.j}(\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} \sum \sum X_{ij}^2$$

$$\therefore \bar{Y}_{..} = \frac{Y_{..}}{nK}$$

$$\therefore \sum_j X_{.j}(\bar{Y}_{.j} - \bar{Y}_{..}) = n \sum_j (\bar{X}_{.j} - \bar{X}_{..})(\bar{Y}_{.j} - \bar{Y}_{..})$$

$$\therefore \sum_j X_{.j}(\bar{X}_{.j} - \bar{X}_{..}) = n \sum_j (\bar{X}_{.j} - \bar{X}_{..})(\bar{X}_{.j} - \bar{X}_{..}) = n \sum_j (\bar{X}_{.j} - \bar{X}_{..})^2$$

$$\sum_{i=1}^{n-k} \sum_j X_{ij} Y_{ij} = \frac{X_{..} Y_{..}}{nk} - \hat{\beta} \frac{X_{..} X_{..}}{nk} + n \sum_j (\bar{X}_{.j} - \bar{X}_{..}) (\bar{Y}_{.j} - \bar{Y}_{..})$$

$$- \hat{\beta} \sum_j (\bar{X}_{.j} - \bar{X}_{..})^2 + \hat{\beta} \sum_{i,j} X_{ij}^2$$

Now

$$\sum \sum X_{ij} Y_{ij} = \frac{X_{..} Y_{..}}{nk} = \hat{\beta} \left[\sum \sum X_{ij}^2 - \frac{(X_{..})^2}{nk} \right] + n \sum_j (\bar{X}_{.j} - \bar{X}_{..}) (\bar{Y}_{.j} - \bar{Y}_{..})$$

$$- \hat{\beta} \sum_j (\bar{X}_{.j} - \bar{X}_{..})^2$$

So

abs

regl

$$\therefore \sum \sum X_{ij} Y_{ij} - \frac{X_{..} Y_{..}}{nk} = S_{xy}$$

$$\sum \sum X_{ij}^2 - \frac{(X_{..})^2}{nk} = S_{xx}$$

E

u

$$n \sum_j (\bar{X}_{.j} - \bar{X}_{..}) (\bar{Y}_{.j} - \bar{Y}_{..}) = T_{xy}$$

reg(u

$$n \sum_j (\bar{X}_{.j} - \bar{X}_{..})^2 = T_{xx}$$

By using these results:

$$S_{xy} = \hat{\beta} S_{xx} + T_{xy} - \hat{\beta} T_{xx}$$

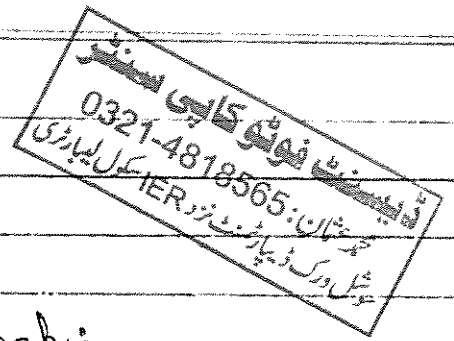
$$S_{xy} - T_{xy} = \hat{\beta} (S_{xx} - T_{xx})$$

$$\therefore E_{xy} = S_{xy} - T_{xy}$$

$$E_{xx} = S_{xx} - T_{xx}$$

$$E_{xy} = \hat{\beta} E_{xx}$$

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}$$



Now by regression approach:

Sum of Squares due to fitting the above full model is

$$\text{reg}(\hat{\mu}, \hat{\alpha}_j, \hat{\beta}) = \hat{\mu} Y_{..} + \sum_j \hat{\alpha}_j Y_{.j} + \hat{\beta} \sum_j \sum_i X_{ij} Y_{ij}$$

By putting the values of $\hat{\mu}$ and $\hat{\alpha}_j$ we get

$$\text{reg}(\mu, \alpha_j, \beta) = (\bar{Y}_{..} - \hat{\beta} \bar{X}_{..}) Y_{..} + \sum_j [(\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta} (\bar{X}_{.j} - \bar{X}_{..})] Y_{.j} + \hat{\beta} \sum_j \sum_i X_{ij} Y_{ij}$$

$$= Y_{..} \bar{Y}_{..} - \hat{\beta} \bar{X}_{..} Y_{..} + \sum_j Y_{.j} (\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta} \sum_j Y_{.j} (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} \sum_j \sum_i X_{ij} Y_{ij}$$

$$= \frac{(Y_{..})^2}{nk} - \hat{\beta} \left(\frac{X_{..} Y_{..}}{nk} \right) + n \sum_j (\bar{Y}_{.j} - \bar{Y}_{..})^2 - \hat{\beta} n \sum_j (\bar{Y}_{.j} - \bar{Y}_{..}) (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} \sum_j \sum_i X_{ij} Y_{ij}$$

$$= n \sum_j (\bar{Y}_{.j} - \bar{Y}_{..})^2 + \frac{(Y_{..})^2}{nk} - n \hat{\beta} \sum_j (\bar{Y}_{.j} - \bar{Y}_{..}) (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} \left[\sum_j \sum_i X_{ij} Y_{ij} - \frac{X_{..} Y_{..}}{nk} \right]$$

$$= T_{yy} + \frac{(Y_{..})^2}{nk} - \hat{\beta} T_{xy} + \beta S_{xy}$$

$$= T_{yy} + \frac{(Y_{..})^2}{nk} + \hat{\beta} (S_{xy} - T_{xy})$$

$$= T_{yy} + \frac{(Y_{..})^2}{nk} + \hat{\beta} E_{xy}$$

$$\because \hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

$$\text{reg}(y_i, x_i, \beta) = T_{yy} + \frac{(Y_{..})^2}{nk} + \frac{E_{xy}^2}{E_{xx}} \quad \text{--- (A)}$$

Next we assume that the treatment effects are insignificant, then full model reduces to

$$Y_{ij} = \mu + \beta' X_{ij} + E_{ij}$$

The Normal equations are

$$\sum \sum Y_{ij} = nk\hat{\mu} + \hat{\beta}' \sum \sum X_{ij} \quad \text{--- (4)}$$

$$\sum \sum X_{ij} Y_{ij} = \hat{\mu} \sum \sum X_{ij} + \hat{\beta}' \sum \sum X_{ij}^2 \quad \text{--- (5)}$$

From eq (4):

$$Y_{..} = nk\hat{\mu} + \hat{\beta}' X_{..}$$

$$\boxed{\hat{\mu} = \bar{Y}_{..} - \hat{\beta}' \bar{X}_{..}}$$

From eq (5):

$$\begin{aligned}\sum \sum X_{ij} Y_{ij} &= (\bar{Y}_{..} - \hat{\beta}' \bar{X}_{..}) \sum \sum X_{ij} + \hat{\beta}' \sum \sum X_{ij}^2 \\ &= \bar{Y}_{..} X_{..} - \hat{\beta}' \bar{X}_{..} X_{..} + \hat{\beta}' \sum \sum X_{ij}^2\end{aligned}$$

$$= \frac{X_{..} Y_{..}}{nk} - \hat{\beta}' \frac{(X_{..})^2}{nk} + \hat{\beta}' \sum \sum X_{ij}^2$$

$$\sum \sum X_{ij} Y_{ij} - \frac{X_{..} Y_{..}}{nk} = \hat{\beta}' \left[\sum \sum X_{ij}^2 - \frac{(X_{..})^2}{nk} \right]$$

$$S_{xy} = \hat{\beta}' S_{xx}$$

$$\boxed{\hat{\beta}' = \frac{S_{xy}}{S_{xx}}}$$

Now the variation due to fitting the reduced regression model is

$$\text{Reg}(w', \beta') = \hat{u}' Y_{..} + \hat{\beta}' \sum \sum X_{ij} Y_{ij}$$

$$= (\bar{Y}_{..} - \hat{\beta}' \bar{X}_{..}) Y_{..} + \hat{\beta}' \sum \sum X_{ij} Y_{ij}$$

$$= \frac{(Y_{..})^2}{nk} - \hat{\beta}' \left(\frac{X_{..} Y_{..}}{nk} \right) + \hat{\beta}' \sum \sum X_{ij} Y_{ij}$$

$$= \frac{(Y_{..})^2}{nk} + \hat{\beta}' S_{xy}$$

$$\text{reg}(u', \beta') = \frac{(Y_{..})^2}{nk} + \frac{\sum_{i,j} \bar{y}_{ij}^2}{S_{xx}} \quad \text{(B)}$$

Difference of eq (A) and eq (B), gives

the variation due to treatment effects.

$$\text{eq (A)} - \text{eq (B)}$$

$$SS_{\text{treatments}} = SS_{\text{reg}}(\text{full model}) - SS_{\text{reg}}(\text{reduced model})$$

$$= \frac{(Y_{..})^2}{nk} + T_{yy} + \frac{E_{yy}^2}{E_{xx}} - \frac{(Y_{..})^2}{nk} - \frac{S_{xy}^2}{S_{xx}}$$

$$\therefore T_{yy} = R_{yy} - E_{yy}$$

=

$$= S_{yy} - E_{yy} + \frac{E_{yy}^2}{E_{xx}} - \frac{S_{xy}^2}{S_{xx}}$$

$$= \left(S_{yy} - \frac{S_{xy}^2}{S_{xx}} \right) - \left(E_{yy} - \frac{E_{xy}^2}{E_{xx}} \right)$$

$$T_{yy} = S_{yy} - E_{yy}$$

where S_{yy} and E_{yy} are total SS and Error SS for covariate X . Then

$$\text{Error SS} = E_{yy} = \sum \sum y_{ij}^2 - \text{reg}(u, a_j, \beta)$$

$$y_{ij} =$$

where

$$y_{ij} = \frac{1}{n} \sum$$

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$$= \sum \sum y_{ij}^2 - \frac{(y_{..})^2}{nk} = T_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

$$= S_{yy} - T_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

$$E_{yy} = E_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

The F-Statistic is:

$$F = \frac{T_{yy} / k - 1}{E_{yy} / (N - k - 1)}$$

$$E_{yy} / (N - k - 1) \quad (N = k - 1)$$

is distributed with F-dist with
 $v_1 = k - 1$ and $v_2 = N - k - 1$ degree
 of freedom.

ANCOVA In RCB Model:

Mathematical Model:

$$y_{ij} = \mu + \alpha_i + \rho_j + \beta(x_{ij} - \bar{x}_{..}) + \epsilon_{ij}$$

where;

$i = 1, 2, \dots, r$

$j = 1, 2, \dots, k$

y_{ij} = obs for i th treatment and
 j th block.

μ = overall mean.

α_i = i^{th} treatment effect

ρ_j = j^{th} block effect

β = linear regression co-efficient indicating the dependency of X_{ij} and Y_{ij} .

$\bar{X}_{..}$ = grand mean of X_{ij} values.

$E_{ij} \sim NII(0, \sigma^2)$

Least Square Estimates and ANCOVA

By Regression Approach:

In RCB design of ANCOVA, the obs Y_{ij} can be expressed as

$$Y_{ij} = \mu + \alpha_i + \rho_j + \beta(X_{ij} - \bar{X}_{..}) + E_{ij}$$

$$i = 1, 2, \dots, r$$

$$j = 1, 2, \dots, k$$

Let $\hat{\mu}$, $\hat{\alpha}_i$, $\hat{\rho}_j$ and $\hat{\beta}$ are the estimates of μ , α_i , ρ_j and β resp

Then sum of squares of error is

$$S = \sum_{i=1}^r \sum_{j=1}^k (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\rho}_j - \hat{\beta}(X_{ij} - \bar{X}_{..}))^2$$

To find the values of estimates we differentiate it w.r.t unknown values and equating them to zero

for $\hat{\mu}$:

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$$\frac{\partial S}{\partial \hat{\mu}} = 2 \sum_i \sum_j^k (y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(x_{ij} - \bar{x}_{..}))(-1) = 0$$

$$\frac{\partial^2 S}{\partial \hat{\mu}^2} > 0$$

$$= \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(x_{ij} - \bar{x}_{..})) = 0$$

$$\sum_i \sum_j y_{ij} - r.k.\hat{\mu} - k \sum_i \hat{\alpha}_i - r \sum_j \hat{e}_j - \hat{\beta} \sum_i \sum_j (x_{ij} - \bar{x}_{..}) = 0$$

$$\therefore \sum_i \sum_j (x_{ij} - \bar{x}_{..}) = r.k.\bar{x}_{..} - r.k.\bar{x}_{..} = 0$$

$$\sum_i \sum_j y_{ij} = r.k.\hat{\mu} + k \sum_i \hat{\alpha}_i + r \sum_j \hat{e}_j + \hat{\beta} \sum_i \sum_j (x_{ij} - \bar{x}_{..}) \quad (1)$$

For treatment:

$$S = \sum_j^k (y_{1j} - \hat{\mu} - \hat{\alpha}_1 - \hat{e}_j - \hat{\beta}(x_{1j} - \bar{x}_{..}))^2 + \sum_j (y_{2j} - \hat{\mu} - \hat{\alpha}_2 - \hat{e}_j - \hat{\beta}(x_{2j} - \bar{x}_{..}))^2 + \dots + \sum_j^k (y_{rj} - \hat{\mu} - \hat{\alpha}_r - \hat{e}_j - \hat{\beta}(x_{rj} - \bar{x}_{..}))^2$$

$$\frac{\partial S}{\partial \hat{\alpha}_1} = -2 \sum_j (y_{1j} - \hat{\mu} - \hat{\alpha}_1 - \hat{e}_j - \hat{\beta}(x_{1j} - \bar{x}_{..})) = 0$$

$$\frac{\partial S}{\partial \hat{\alpha}_2} = -2 \sum_j (y_{2j} - \hat{\mu} - \hat{\alpha}_2 - \hat{e}_j - \hat{\beta}(x_{2j} - \bar{x}_{..})) = 0$$

$$\frac{\partial S}{\partial \hat{\alpha}_i} = -2 \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(x_{ij} - \bar{x}_{..})) = 0$$

$$\sum_j y_{ij} - k.\hat{\mu} - k.\hat{\alpha}_i - \sum_j \hat{e}_j - \hat{\beta} \sum_j (x_{ij} - \bar{x}_{..}) = 0$$

$$\sum_j Y_{ij} = r\hat{\mu} + r\hat{\alpha}_i + \sum_j \hat{e}_j + \hat{\beta} \sum_j (X_{ij} - \bar{x}_{..})$$

for

$$Y_{ij} = r\hat{\mu} + r\hat{\alpha}_i + \sum_j \hat{e}_j + r\hat{\beta} (\bar{x}_{ij} - \bar{x}_{..}) \quad (2)$$

 $\frac{\partial S}{\partial \hat{\beta}}$ for $\hat{\alpha}_i$:

This is normal eq for $\hat{\alpha}_i$:
For $\hat{e}_j$:

$$S = \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(X_{ij} - \bar{x}_{..}))^2 + \sum_i (Y_{2j} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(X_{2j} - \bar{x}_{..}))^2 + \dots + \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(X_{ij} - \bar{x}_{..}))^2$$

$$\frac{\partial S}{\partial \hat{\beta}} = -2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(X_{ij} - \bar{x}_{..})) = 0$$

 $\sum \sum$

$$\frac{\partial S}{\partial \hat{e}_2} = -2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_2 - \hat{\beta}(X_{2j} - \bar{x}_{..})) = 0$$

$$\frac{\partial S}{\partial \hat{e}_j} = -2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(X_{ij} - \bar{x}_{..})) = 0$$

r k
 $\sum \sum$

$$\sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{e}_j - \hat{\beta}(X_{ij} - \bar{x}_{..})) = 0$$

$$\sum_i Y_{ij} - r\hat{\mu} - \sum_i \hat{\alpha}_i - r\hat{e}_j - \hat{\beta} \sum_i (X_{ij} - \bar{x}_{..}) = 0$$

$$\sum_i Y_{ij} = r\hat{\mu} + \sum_i \hat{\alpha}_i + r\hat{e}_j + \hat{\beta} \sum_i (X_{ij} - \bar{x}_{..})$$

$$Y_{ij} = r\hat{\mu} + \sum_i \hat{\alpha}_i + r\hat{e}_j + r\hat{\beta} (\bar{x}_{ij} - \bar{x}_{..}) \quad (3)$$

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for $\hat{\beta}$:

$$\frac{\partial S}{\partial \hat{\beta}} = -2 \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\epsilon}_j - \hat{\beta}(x_{ij} - \bar{x}_{..}))(x_{ij} - \bar{x}_{..}) = 0$$

$\frac{\partial S}{\partial \hat{\beta}} = 0$

$$= \sum_i \sum_j y_{ij}(x_{ij} - \bar{x}_{..}) - \hat{\mu} \sum_i \sum_j (x_{ij} - \bar{x}_{..}) - \sum_i \sum_j \hat{\alpha}_i (x_{ij} - \bar{x}_{..}) \\ - \sum_i \sum_j \hat{\epsilon}_j (x_{ij} - \bar{x}_{..}) - \hat{\beta} \sum_i \sum_j (x_{ij} - \bar{x}_{..})(x_{ij} - \bar{x}_{..}) = 0$$

$$\sum_i \sum_j y_{ij}(x_{ij} - \bar{x}_{..}) = \hat{\mu} \sum_i \sum_j (x_{ij} - \bar{x}_{..}) + \sum_i \sum_j \hat{\alpha}_i (x_{ij} - \bar{x}_{..}) \\ + \sum_i \sum_j \hat{\epsilon}_j (x_{ij} - \bar{x}_{..}) + \hat{\beta} \sum_i \sum_j (x_{ij} - \bar{x}_{..})(x_{ij} - \bar{x}_{..})$$

$$\sum_i^r \sum_j^k (y_{ij} - \bar{y}_{..})(x_{ij} - \bar{x}_{..}) = 0 + k \sum_i \hat{\alpha}_i (\bar{x}_{i.} - \bar{x}_{..}) + r \sum_j \hat{\epsilon}_j (\bar{x}_{.j} - \bar{x}_{..}) \\ + \hat{\beta} \sum_i \sum_j (x_{ij} - \bar{x}_{..})^2$$

$$S_{xy} = k \sum_i \hat{\alpha}_i (\bar{x}_{i.} - \bar{x}_{..}) + r \sum_j \hat{\epsilon}_j (\bar{x}_{.j} - \bar{x}_{..}) + \hat{\beta} S_{xx} \quad (4)$$

Now if we add 'k' equations in eq (2) and 'r' equations in eq (3) then they will become same as eq (1). This means that eq (1), (2), (3) are not linearly independent.

So, For unique solution, we put the following restrictions.

$$i) \sum_i \hat{\alpha}_i = 0$$

$$ii) \sum_j \hat{\rho}_j = 0$$

From eq (i):

$$\sum_i \sum_j y_{ij} = rk\hat{\alpha} + 0 + 0.$$

$$Y_{..} = rk\hat{\alpha}$$

$$\hat{\alpha} = \frac{Y_{..}}{rk}$$

$$\boxed{\hat{\alpha} = \bar{y}_{..}}$$

From eq (ii):

$$y_{ij} = k\bar{y}_{..} + k\hat{\alpha}_i + 0 + k\hat{\beta}(\bar{x}_i - \bar{x}_{..})$$

$$k\hat{\alpha}_i = y_{ij} - k\bar{y}_{..} - k\hat{\beta}(\bar{x}_i - \bar{x}_{..})$$

$$\hat{\alpha}_i = \frac{y_{ij}}{k} - \bar{y}_{..} - \hat{\beta}(\bar{x}_i - \bar{x}_{..})$$

$$\boxed{\hat{\alpha}_i = (\bar{y}_{i.} - \bar{y}_{..}) - \hat{\beta}(\bar{x}_i - \bar{x}_{..})}$$

From eq (iii):

$$y_{ij} = r \bar{y}_{..} + 0 + r \hat{e}_j + r \hat{\beta} (\bar{x}_{.j} - \bar{x}_{..})$$

$$r \hat{e}_j = y_{.j} - r \bar{y}_{..} - r \hat{\beta} (\bar{x}_{.j} - \bar{x}_{..})$$

$$\hat{e}_j = \frac{y_{.j}}{r} - \bar{y}_{..} - \hat{\beta} (\bar{x}_{.j} - \bar{x}_{..})$$

$$\hat{e}_j = (\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta} (\bar{x}_{.j} - \bar{x}_{..})$$

From eq (4):

$$S_{xy} = k \sum_i [(\bar{y}_{i.} - \bar{y}_{..}) - \hat{\beta} (\bar{x}_{i.} - \bar{x}_{..})]^2 + \sum_j r \sum_j [(\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta} (\bar{x}_{.j} - \bar{x}_{..})] * (\bar{x}_{.j} - \bar{x}_{..}) + \hat{\beta} S_{xx}$$

$$S_{xy} = k \sum_i (\bar{y}_{i.} - \bar{y}_{..})(\bar{x}_{i.} - \bar{x}_{..}) - k \hat{\beta} \sum_i (\bar{x}_{i.} - \bar{x}_{..})^2 + r \sum_j (\bar{y}_{.j} - \bar{y}_{..})(\bar{x}_{.j} - \bar{x}_{..}) - r \hat{\beta} \sum_j (\bar{x}_{.j} - \bar{x}_{..})^2 + \hat{\beta} S_{xx}$$

$$S_{xy} = T_{xy} - \hat{\beta} T_{xx} + B_{xy} - \hat{\beta} B_{xx} + \hat{\beta} S_{xx}$$

$$S_{xy} = T_{xy} + B_{xy} + \hat{\beta} (S_{xx} - T_{xx} - B_{xx})$$

$$S_{xy} - T_{xy} - B_{xy} = \hat{\beta} (S_{xx} - T_{xx} - B_{xx})$$

$$E_{xy} = \hat{\beta} (E_{xx})$$

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

By Regression Approach:

Now, the variation due to fitting the full regression model is

$$reg(u, \alpha_i, \beta_j, \hat{\beta}) = \hat{\mu} Y_{..} + \sum_i \alpha_i Y_{i.} + \sum_j \beta_j Y_{.j} + \hat{\beta} S_{xy}$$

$$= (\bar{Y}_{..}) Y_{..} + \sum_i [(\bar{Y}_{i.} - \bar{Y}_{..}) - \hat{\beta} (\bar{X}_{i.} - \bar{X}_{..})] Y_{i.} +$$

$$\sum_j [(\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta} (\bar{X}_{.j} - \bar{X}_{..})] Y_{.j} + \hat{\beta} S_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + \sum_i Y_{i.} (\bar{Y}_{i.} - \bar{Y}_{..}) - \hat{\beta} \sum_i Y_{i.} (\bar{X}_{i.} - \bar{X}_{..}) +$$

$$\sum_j Y_{.j} (\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta} \sum_j Y_{.j} (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} S_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + k \sum_i (\bar{Y}_{i.} - \bar{Y}_{..})^2 - k \hat{\beta} \sum_i (\bar{Y}_{i.} - \bar{Y}_{..}) (\bar{X}_{i.} - \bar{X}_{..}) +$$

$$r \sum_j (\bar{Y}_{.j} - \bar{Y}_{..})^2 - r \hat{\beta} \sum_j (\bar{Y}_{.j} - \bar{Y}_{..}) (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} S_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + T_{yy} - \hat{\beta} T_{xy} + B_{yy} - \hat{\beta} B_{xy} + \hat{\beta} S_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + \hat{\beta} (S_{xy} - T_{xy} - B_{xy}) + T_{yy} + B_{yy}$$

$$= \frac{(Y_{..})^2}{rk} + \frac{E_{xy} (E_{xy})}{E_{xy}} + T_{yy} + B_{yy}$$

$$= \frac{(Y_{..})^2}{rk} + \frac{E_{xy}^2}{E_{xy}} + T_{yy} + B_{yy} \quad \text{--- (A)}$$

Now if we assume that the treatment effects are insignificant i.e. $H_0: \alpha_i = 0$, under this hypothesis the full regression model reduces to

$$Y_{ij} = \mu' + \beta' (X_{ij} - \bar{X}_{..}) + \epsilon_{ij}$$

The Normal equations are:

$$\sum Y_{ij} = r.k.\hat{\mu}' + r \sum_j \hat{\epsilon}_j' + \hat{\beta}' \sum \sum (X_{ij} - \bar{X}_{..})$$

$$\therefore \sum \sum (X_{ij} - \bar{X}_{..}) = 0$$

$$Y_{..} = r.k.\hat{\mu}' + r \sum_j \hat{\epsilon}_j' \quad \text{--- (5)}$$

$$\sum_i Y_{ij} = r.\hat{\mu}' + r\hat{\epsilon}_j' + \hat{\beta}' \sum_i (X_{ij} - \bar{X}_{..})$$

$$Y_{.j} = r.\hat{\mu}' + r\hat{\epsilon}_j' + r\hat{\beta}' (\bar{X}_{.j} - \bar{X}_{..}) \quad \text{--- (6)}$$

$$S_{xy} = r \sum_j \hat{\epsilon}_j' (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta}' S_{xy} \quad \text{--- (7)}$$

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Now, if we add 'k' equations in eq (6) we get the same as eq (5),

For unique solution

$$\sum_{j=1}^n \hat{e}_j' = 0$$

From eq (5):

$$y_{..} = r k \hat{u}' + 0$$

$$\hat{u}' = \bar{y}_{..}$$

From eq (6):

$$y_{.j} = r(\bar{y}_{..}) + r \hat{e}_j' + r \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..})$$

$$r \hat{e}_j' = y_{.j} - r \bar{y}_{..} - r \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..})$$

$$\hat{e}_j' = \frac{y_{.j}}{r} - \bar{y}_{..} - \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..})$$

$$\hat{e}_j' = (\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..})$$

From eq (7):

$$S_{xy} = r \sum_j \left[(\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..}) \right] (\bar{x}_{.j} - \bar{x}_{..}) + \hat{\beta}' S_{xx}$$

$$S_{xy} = r \sum_j (\bar{y}_{.j} - \bar{y}_{..}) (\bar{x}_{.j} - \bar{x}_{..}) - r \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..}) (\bar{x}_{.j} - \bar{x}_{..}) + \hat{\beta}' S_{xx}$$

$$V(6) \quad S_{xy} = B_{xy} - \hat{\beta} B_{xx} + \hat{\beta} S_{xx}$$

$$S_{xy} = B_{xy} + \hat{\beta} (S_{xx} - B_{xx})$$

$$S_{xy} - B_{xy} = \hat{\beta} (S_{xx} - B_{xx})$$

$$\hat{\beta} = \frac{S_{xy} - B_{xy}}{S_{xx} - B_{xx}}$$

OR

$$\hat{\beta} = \frac{T_{xy} + E_{xy}}{T_{xx} + E_{xx}}$$

$$\begin{aligned} \because S_{xx} &= T_{xx} - B_{xx} = E_{xx} \\ S_{xx} - B_{xx} &= E_{xx} + T_{xx} \\ \therefore S_{xy} - B_{xy} &= E_{xy} + T_{xy} \end{aligned}$$

Now, the variation due to fitting the above reduced model is

$$\text{reg}(u_i', \ell_j', \beta') = \hat{\alpha}' Y_{..} + \sum_j \hat{\ell}_j' Y_{.j} + \hat{\beta}' S_{xy}$$

$$= (\bar{Y}_{..}) Y_{..} + \sum_j [(\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta}' (\bar{X}_{.j} - \bar{X}_{..})] Y_{.j} + \hat{\beta}' S_{xy}$$

$$= \frac{(Y_{..})^2}{rK} + \sum_j Y_{.j} (\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta}' \sum_j Y_{.j} (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta}' S_{xy}$$

$$= \frac{(Y_{..})^2}{rK} + r \sum_j (\bar{Y}_{.j} - \bar{Y}_{..})^2 - r \hat{\beta}' \sum_j (\bar{Y}_{.j} - \bar{Y}_{..}) (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta}' S_{xy}$$

$$= \frac{(N_{..})^2}{rK} + B_{yy} - \hat{\beta}' B_{xy} + \hat{\beta}' S_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + B_{yy} - \hat{\beta}' B_{xy} + \hat{\beta}' S_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + B_{yy} + \hat{\beta}' (S_{xy} - B_{xy})$$

$$\hat{\beta}' = \frac{T_{xy} + E_{xy}}{T_{xx} + E_{xx}} \quad \text{and} \quad S_{xy} - B_{xy} = T_{xy} + E_{xy}$$

$$= \frac{(Y_{..})^2}{rk} + B_{yy} + \frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}} \quad \text{--- (B)}$$

Difference of eq (A) and eq (B) gives the variation due to treatment effects which can be used to test the hypothesis that treatment effects are insignificant.

$$SS_{\text{treatment}} = \text{reg}(u, x_i, \beta) - \text{reg}(u, \beta)$$

$$= \frac{(Y_{..})^2}{rk} + T_{yy} + B_{yy} + \frac{E_{xy}^2}{E_{xx}} - \frac{(Y_{..})^2}{rk} - B_{yy} - \frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}}$$

$$= \frac{T_{yy} + E_{xy}^2}{E_{xx}} - \frac{(T_{xy} + E_{xy})^2}{E_{xx} + T_{xx}}$$

By adding and subtracting E_{yy} , we get

$$= (T_{yy} + E_{yy}) - \frac{(T_{xy} + E_{xy})^2}{E_{xx} + T_{xx}} + \frac{E_{xy}^2}{E_{xx}} - E_{yy}$$

$$SS_{\text{treatment}} = \left\{ (T_{yy} + E_{yy}) - \frac{(T_{xy} + E_{xy})^2}{E_{xx} + T_{xx}} \right\} - \left\{ E_{yy} - \frac{E_{xy}^2}{E_{xx}} \right\}$$

$$= (T+E)_{yy} - E_{yy}$$

Now

adjusted error SS is:

$$E_{yy} = \sum \sum y_{ij}^2 - \text{reg}(\mu, d_i, l_j, \beta)$$

$$= \sum \sum y_{ij}^2 - \frac{(Y_{..})^2}{rk} - T_{yy} - B_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

$$= S_{yy} - T_{yy} - B_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

$$E_{yy} = E_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

Hence, the F-statistic is

$$F = \frac{[(T+E)_{yy} - E_{yy}] / (r-1)(k-1)}{E_{yy} / (r-1)(k-1)-1}$$

having an F-dist with $v_1 = r-1$
and $v_2 = (r-1)(k-1)-1$

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Latin Square Design In ANCOVA:

Mathematical Model:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + W_k + B(X_{ijk} - \bar{X}_{..}) + \epsilon_{ijk}$$

where

$$i, j, k = 1, 2, \dots, p$$

μ = overall mean

α_i = i th row effect

β_j = j th column effect

W_k = k th treatment effect

B = linear reg co-efficient indicating the dependency of X_{ijk} and Y_{ijk} .

Y_{ijk} = obs in i th row, j th column and k th treatment

$\epsilon_{ijk} \sim N(0, \sigma^2)$

Least Square Estimates and ANCOVA

By Regression Approach:

In LS Design in ANCOVA, the obs Y_{ijk} is expressed as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + W_k + B(X_{ijk} - \bar{X}_{..}) + \epsilon_{ij}$$

$i, j, k = 1, 2, \dots, p$

A: let $\hat{\mu}$, $\hat{\alpha}_i$, $\hat{\rho}_j$, $\hat{w}_k$ and $\hat{\beta}$ are the estimates of μ , α_i , ρ_j , w_k and β respectively.

To find the values of these estimates, The sum of squares of error is:

$$S = \sum_i^P \sum_j^P (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\rho}_j - \hat{w}_k - \hat{\beta}(x_{ij(k)} - \bar{x}_{..}))^2$$

differentiating it w.r.t unknown values and equating them to zero i.e.

For $\hat{\mu}$:-

$$\frac{\partial S}{\partial \hat{\mu}} = -2 \sum_i^P \sum_j^P (y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\rho}_j - \hat{w}_k - \hat{\beta}(x_{ij(k)} - \bar{x}_{..})) (-1) = 0$$

$\therefore \frac{\partial^2 S}{\partial \hat{\mu}^2} > 0$

$$\sum_i^P \sum_j^P y_{ij(k)} - P^2 \hat{\mu} - P \sum_i \hat{\alpha}_i - P \sum_j \hat{\rho}_j - P \sum_k \hat{w}_k - \hat{\beta} \sum_i \sum_j (x_{ij(k)} - \bar{x}_{..}) = 0$$

$$\sum_i^P \sum_j^P y_{ij(k)} = P^2 \hat{\mu} + P \sum_i \hat{\alpha}_i + P \sum_j \hat{\rho}_j + P \sum_k \hat{w}_k + \hat{\beta} \sum_i \sum_j (x_{ij(k)} - \bar{x}_{..})$$

$$\because \sum_i \sum_j (x_{ij(k)} - \bar{x}_{..}) = 0$$

$$\sum_i \sum_j y_{ij(k)} = P^2 \hat{\mu} + P \sum_i \hat{\alpha}_i + P \sum_j \hat{\rho}_j + P \sum_k \hat{w}_k \quad \text{--- (1)}$$

For $\hat{\alpha}_i$:

$$S = \sum_j (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\ell}_j - \hat{\omega}_k - \hat{\beta}(X_{ij(k)} - \bar{X}_{..}))^2 + \dots +$$

$$\sum_j (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\ell}_j - \hat{\omega}_k - \hat{\beta}(X_{ij(k)} - \bar{X}_{..}))^2$$

$$\frac{\partial S}{\partial \hat{\alpha}_i} = -2 \sum_j (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\ell}_j - \hat{\omega}_k - \hat{\beta}(X_{ij(k)} - \bar{X}_{..})) = 0$$

$$\frac{\partial S}{\partial \hat{\alpha}_i} = -2 \sum_j (Y_{ij(k)} - \hat{\mu} - \hat{\alpha}_i - \hat{\ell}_j - \hat{\omega}_k - \hat{\beta}(X_{ij(k)} - \bar{X}_{..})) = 0$$

$$\therefore \frac{\partial^2 S}{\partial \hat{\alpha}_i^2} > 0$$

$$= -\sum_j Y_{ij(k)} = P\hat{\mu} - P\hat{\alpha}_i - P\hat{\ell}_j -$$

$$= \sum_j Y_{ij(k)} - P\hat{\mu} - P\hat{\alpha}_i - \sum_j \hat{\ell}_j - \sum_k \hat{\omega}_k - \hat{\beta} \sum_j (X_{ij(k)} - \bar{X}_{..}) = 0$$

$$\sum_j Y_{ij(k)} - P\hat{\mu} + P\hat{\alpha}_i + \sum_j \hat{\ell}_j + \sum_k \hat{\omega}_k + \hat{\beta} \sum_j (X_{ij(k)} - \bar{X}_{..})$$

$$Y_{i..} - R_i - P\hat{\mu} + P\hat{\alpha}_i + \sum_j \hat{\ell}_j + \sum_k \hat{\omega}_k + P\hat{\beta}(\bar{X}_{i.} - \bar{X}_{..}) \quad (2)$$

For $\hat{\ell}_j$:

$$S = \sum_i (Y_{i..} - \hat{\mu} - \hat{\alpha}_i - \hat{\ell}_j - \hat{\omega}_k - \hat{\beta}(X_{i..} - \bar{X}_{..}))^2 + \dots +$$

$$\sum_i (Y_{i..} - \hat{\mu} - \hat{\alpha}_i - \hat{\ell}_j - \hat{\omega}_k - \hat{\beta}(X_{i..} - \bar{X}_{..}))^2$$

$$\frac{\partial S}{\partial \hat{\mu}} = -2 \sum_i (Y_{i1}(k) - \hat{\mu} - \hat{\alpha}_i - \hat{\epsilon}_1 - \hat{\omega}_k - \hat{\beta} (X_{i1}(k) - \bar{X})) = 0$$

$$\frac{\partial S}{\partial \hat{\epsilon}_j} = -2 \sum_i (Y_{ij}(k) - \hat{\mu} - \hat{\alpha}_i - \hat{\epsilon}_j - \hat{\omega}_k - \hat{\beta} (X_{ij}(k) - \bar{X})) = 0$$

$$\therefore \frac{\partial S}{\partial \hat{\epsilon}_j} > 0$$

$$\sum_i Y_{ij}(k) - P \hat{\mu} - \sum_i \hat{\alpha}_i - P \hat{\epsilon}_j - \sum_k \hat{\omega}_k - \hat{\beta} \sum_i (X_{ij}(k) - \bar{X}) = 0$$

$$Y_{.j}(k) = C_j = P \hat{\mu} + \sum_i \hat{\alpha}_i + P \hat{\epsilon}_j + \sum_k \hat{\omega}_k + P \hat{\beta} (\bar{X}_{.j}(k) - \bar{X}) \quad (3)$$

Similarly:

$$Y_{.k}(k) = T_k = P \hat{\mu} + \sum_i \hat{\alpha}_i + \sum_j \hat{\epsilon}_j + P \hat{\omega}_k + P \hat{\beta} (\bar{X}_k - \bar{X}) \quad (4)$$

Now if we add 'p' equations in eq (i), (iii), (iv) we get the same eq as eq (i). This means that these normal equations are not linearly independent. For unique solution, we put the following restrictions

$$\sum_i \hat{\alpha}_i = 0, \quad \sum_j \hat{\epsilon}_j = 0, \quad \sum_k \hat{\omega}_k = 0$$

From eq (1):

$$G = Y_{...} = P^2 \hat{\mu} + 0 + 0 + 0$$

$$\hat{\mu} = \frac{G}{P^2} = \bar{Y}_{..} = \bar{\bar{Y}}$$

$$\boxed{\hat{\mu} = \bar{\bar{Y}}}$$

From eq (ii):

$$y_i = P(\bar{y}) + P\hat{\alpha}_i + 0 + 0 + P\hat{\beta}(\bar{x}_i - \bar{x}_{..})$$

$$P\hat{\alpha}_i = y_i - \bar{y}P - P\hat{\beta}(\bar{x}_i - \bar{x}_{..})$$

$$\hat{\alpha}_i = \frac{y_i}{P} - \bar{y} - \hat{\beta}(\bar{x}_i - \bar{x}_{..})$$

$$\boxed{\hat{\alpha}_i = (\bar{y}_i - \bar{y}_{..}) - \hat{\beta}(\bar{x}_i - \bar{x}_{..})}$$

From eq (iii):

$$y_{ij} = P\hat{\alpha}_i + 0 + P\hat{\epsilon}_j + 0 + P\hat{\beta}(\bar{x}_{ij} - \bar{x}_{..})$$

$$\hat{\epsilon}_j = \frac{y_{ij}}{P} - \frac{P\bar{y}_{..}}{P} - \frac{P\hat{\beta}}{P}(\bar{x}_{ij} - \bar{x}_{..})$$

$$\boxed{\hat{\epsilon}_j = (\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta}(\bar{x}_{.j} - \bar{x}_{..})}$$

From eq (iv)

$$y_k = P\bar{y}_{..} + 0 + 0 + P\hat{\omega}_k + P\hat{\beta}(\bar{x}_k - \bar{x}_{..})$$

$$\boxed{\hat{\omega}_k = (\bar{y}_k - \bar{y}_{..}) - \hat{\beta}(\bar{x}_k - \bar{x}_{..})}$$

* (Now by putting the values of $\hat{\alpha}_i$, $\hat{\epsilon}_j$, $\hat{\omega}_k$ in eq (5))

For $\hat{\beta}$:

$$\frac{\partial S}{\partial \hat{\beta}} = -2 \sum_i^P \sum_j^P (Y_{ijk}) - \hat{\mu} - \hat{\alpha}_i - \hat{\epsilon}_j - \hat{\omega}_k - \hat{\beta} (X_{ijk} - \bar{X}_{..}) (X_{ijk} - \bar{X}_{..}) = 0$$

$$\begin{aligned} & \sum_i^P \sum_j^P Y_{ijk} (X_{ijk} - \bar{X}_{..}) - \hat{\mu} \sum_i^P \sum_j^P (X_{ijk} - \bar{X}_{..}) - P \sum_i^P \sum_j^P \hat{\alpha}_i (X_{ijk} - \bar{X}_{..}) \\ & - \sum_i^P \sum_j^P \hat{\epsilon}_j (X_{ijk} - \bar{X}_{..}) - \sum_i^P \sum_j^P \hat{\omega}_k (X_{ijk} - \bar{X}_{..}) - \hat{\beta} \sum_i^P \sum_j^P (X_{ijk} - \bar{X}_{..})^2 \end{aligned}$$

$$\bullet \sum_i^P \sum_j^P (Y_{ijk} - \bar{Y}_{..}) (X_{ijk} - \bar{X}_{..}) = 0 + P \sum_i^P \hat{\alpha}_i (\bar{X}_{i.} - \bar{X}_{..}) +$$

$$\sum_j^P \hat{\epsilon}_j (\bar{X}_{.j} - \bar{X}_{..}) + P \sum_k^P \hat{\omega}_k (\bar{X}_{k.} - \bar{X}_{..}) + \hat{\beta} S_{XX}$$

$$S_{XY} = P \sum_i^P \hat{\alpha}_i (\bar{X}_{i.} - \bar{X}_{..}) + P \sum_j^P \hat{\epsilon}_j (\bar{X}_{.j} - \bar{X}_{..}) +$$

$$P \sum_k^P \hat{\omega}_k (\bar{X}_{k.} - \bar{X}_{..}) + \hat{\beta} S_{XX} \quad \text{--- (5)}$$

Putting the values of $\hat{\mu}$, $\hat{\alpha}_i$, $\hat{\epsilon}_j$ and $\hat{\omega}_k$ in eq (5), we get

$$S_{XY} = P \sum_i^P [(\bar{Y}_{i.} - \bar{Y}_{..}) - \hat{\beta} (\bar{X}_{i.} - \bar{X}_{..})] (\bar{X}_{i.} - \bar{X}_{..}) +$$

$$P \sum_j^P [(\bar{Y}_{.j} - \bar{Y}_{..}) - \hat{\beta} (\bar{X}_{.j} - \bar{X}_{..})] (\bar{X}_{.j} - \bar{X}_{..}) + P \sum_k^P [(\bar{Y}_{k.} - \bar{Y}_{..}) - \hat{\beta} (\bar{X}_{k.} - \bar{X}_{..})]$$

$$(\bar{X}_{k.} - \bar{X}_{..}) + \hat{\beta} S_{XX}$$

$$S_{XY} = P \sum_i^P (\bar{Y}_{i.} - \bar{Y}_{..}) (\bar{X}_{i.} - \bar{X}_{..}) - P \hat{\beta} \sum_i^P (\bar{X}_{i.} - \bar{X}_{..})^2 +$$

$$P \sum_j^P (\bar{Y}_{.j} - \bar{Y}_{..}) (\bar{X}_{.j} - \bar{X}_{..}) - \hat{\beta} P \sum_j^P (\bar{X}_{.j} - \bar{X}_{..}) (\bar{X}_{.j} - \bar{X}_{..}) +$$

$$P \sum_k^P (\bar{Y}_{k.} - \bar{Y}_{..}) (\bar{X}_{k.} - \bar{X}_{..}) - \hat{\beta} P \sum_k^P (\bar{X}_{k.} - \bar{X}_{..})^2 + \hat{\beta} S_{XX}$$

$$S_{xy} = R_{xy} - \hat{\beta} R_{xx} + C_{xy} - \hat{\beta} C_{xx} + T_{xy} - \hat{\beta} T_{xx} + \hat{\beta} S_{xx}$$

$$S_{xy} = R_{xy} - C_{xy} - T_{xy} - \hat{\beta} (S_{xx} - R_{xx} - C_{xx} - T_{xx})$$

$$E_{xy} = \hat{\beta} E_{xx}$$

$$\boxed{\hat{\beta} = \frac{E_{xy}}{E_{xx}}}$$

By Regression Approach:

Sum of squares (variations) due to fitting the full regression model is obtained as

$$\begin{aligned} \text{reg}(\mu, \alpha_i, \beta_j, \omega_k, \beta) &= \hat{\mu} G + \sum_i \hat{\alpha}_i R_i + \sum_j \hat{\beta}_j C_j + \sum_k \hat{\omega}_k T_k + \hat{\beta} S_{xy} \\ &= \bar{Y} G + \sum_i [(\bar{Y}_i - \bar{Y}_{..}) - \hat{\beta}(\bar{X}_i - \bar{X}_{..})] P \bar{Y}_i + \sum_j [(\bar{Y}_j - \bar{Y}_{..}) - \hat{\beta}(\bar{X}_j - \bar{X}_{..})] P \bar{Y}_j \\ &\quad + \sum_k [(\bar{Y}_k - \bar{Y}_{..}) - \hat{\beta}(\bar{X}_k - \bar{X}_{..})] P \bar{Y}_k + \hat{\beta} S_{xy} \end{aligned}$$

$$= \bar{Y} G + P \sum_i (\bar{Y}_i - \bar{Y}_{..})^2 - \hat{\beta} P \sum_i (\bar{Y}_i - \bar{Y}_{..})(\bar{X}_i - \bar{X}_{..}) +$$

$$P \sum_j (\bar{Y}_j - \bar{Y}_{..})^2 - \hat{\beta} P \sum_j (\bar{Y}_j - \bar{Y}_{..})(\bar{X}_j - \bar{X}_{..}) +$$

$$P \sum_k (\bar{Y}_k - \bar{Y}_{..})^2 - \hat{\beta} P \sum_k (\bar{Y}_k - \bar{Y}_{..})(\bar{X}_k - \bar{X}_{..}) + \hat{\beta} S_{xy}$$

$$\begin{aligned}
 &= \bar{Y}_G + R_{yy} - \beta R_{xy} + C_{yy} - \hat{\beta} C_{xy} + T_{yy} - \hat{\beta} T_{xy} + \hat{\beta} S_{xy} \\
 &= \bar{Y}_G + (R_{yy} + C_{yy} + T_{yy}) + \hat{\beta} (C_{xy} - T_{xy} - R_{xy} - C_{xy}) \\
 &= \bar{Y}_G + (R_{yy} + C_{yy} + T_{yy}) + \frac{E_{xy}}{E_{xx}} (E_{xy}) \\
 &= \bar{Y}_G + (R_{yy} + C_{yy} + T_{yy}) + \frac{E_{xy}^2}{E_{xx}} \quad \text{--- (A)}
 \end{aligned}$$

Now if we assume that the treatment effects are insignificant, the full regression model reduces to

$$Y_{ij}(k) = \mu' + \alpha_i' + \rho_j' + \beta'(X_{ij}(k) - \bar{X}_{..}) + \epsilon_{ij}(k)$$

$i, j, k = 1, \dots, P$

Again the normal eqs are obtained by minimizing the sum of squares of error w.r.t unknown and these eqs are:

$$G = P^2 \hat{\mu}' + P \sum_i \hat{\alpha}_i' + P \sum_j \hat{\rho}_j' \quad \text{--- (6)}$$

$$Y_{i.} = P \hat{\mu}' + P \hat{\alpha}_i' + \sum_j \hat{\rho}_j' + P \hat{\beta}' (\bar{X}_{i.} - \bar{X}_{..}) \quad \text{--- (7)}$$

$$Y_{.j} = P \hat{\mu}' + P \hat{\rho}_j' + \sum_i \hat{\alpha}_i' + P \hat{\beta}' (\bar{X}_{.j} - \bar{X}_{..}) \quad \text{--- (8)}$$

$$S_{xy} = P \sum_i \hat{\alpha}_i' (\bar{X}_{i.} - \bar{X}_{..}) + P \sum_j \hat{\rho}_j' (\bar{X}_{.j} - \bar{X}_{..}) + \hat{\beta} S_{xx} \quad \text{--- (9)}$$

If we add 'p' equations in eq (7), (8) it will become the same as eq (6).
It means that they are not

linearly independent. So, we put the restriction
i.e. $\sum \hat{\alpha}_i = 0$, $\sum \hat{\beta}_j = 0$

From eq (6):

$$G = P^2 \hat{u}'$$

$$\hat{u} = \frac{G}{P^2} = \bar{y}_{..} - \bar{y}$$

From eq (7):

$$y_{i.} = P \hat{u}' + P \hat{\alpha}_i' + P \hat{\beta}' (\bar{x}_{i.} - \bar{x}_{..})$$

$$\hat{\alpha}_i' = \frac{y_{i.}}{P} - \frac{P \hat{u}'}{P} - \frac{P}{P} \hat{\beta}' (\bar{x}_{i.} - \bar{x}_{..})$$

$$\hat{\alpha}_i' = (\bar{y}_{i.} - \bar{y}_{..}) - \hat{\beta}' (\bar{x}_{i.} - \bar{x}_{..})$$

From eq (8):

$$\hat{\beta}_j' = (\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..})$$

From eq (9):

$$S_{xy} = P \sum_i \left[(\bar{y}_{i.} - \bar{y}_{..}) - \hat{\beta}' (\bar{x}_{i.} - \bar{x}_{..}) \right] (\bar{x}_{i.} - \bar{x}_{..}) + \\ P \sum_j \left[(\bar{y}_{.j} - \bar{y}_{..}) - \hat{\beta}' (\bar{x}_{.j} - \bar{x}_{..}) \right] (\bar{x}_{.j} - \bar{x}_{..}) + \hat{\beta}' S_{xx}$$

$$S_{xy} = P \sum_i (\bar{y}_i - \bar{y}_{..})(\bar{x}_i - \bar{x}_{..}) - \hat{\beta} P \sum_i (\bar{x}_i - \bar{x}_{..})^2 + \\ P \sum_j (\bar{x}_j - \bar{x}_{..})(\bar{y}_j - \bar{y}_{..}) - \hat{\beta} P \sum_j (\bar{x}_j - \bar{x}_{..})^2 + \hat{\beta} S_{xx}$$

$$S_{xy} = R_{xy} - \hat{\beta} R_{xx} + C_{xy} - \hat{\beta} C_{xx} + \hat{\beta} S_{xx}$$

$$S_{xy} - R_{xy} - C_{xy} = \hat{\beta} (S_{xx} - R_{xx} - C_{xx})$$

$$\therefore S_{xy} - R_{xy} - C_{xy} = T_{xy} + E_{xy}$$

$$T_{xy} + E_{xy} = \hat{\beta}' (T_{xx} + E_{xx})$$

$$\hat{\beta}' = \frac{T_{xy} + E_{xy}}{T_{xx} + E_{xx}}$$

Now, the variation due to fitting the reduced model is

$$\text{reg}(\mu_i', \alpha_i', \epsilon_j', \hat{\beta}) = \hat{\mu}' G + \sum_i \hat{\alpha}_i' \bar{x}_i + \sum_j \hat{\epsilon}_j' \bar{y}_j + \hat{\beta} S_{xy} \\ = \bar{y} G + \sum_i [(\bar{y}_i - \bar{y}_{..}) - \hat{\beta}'(\bar{x}_i - \bar{x}_{..})] \bar{x}_i + \sum_j [(\bar{y}_j - \bar{y}_{..}) - \hat{\beta}'(\bar{x}_j - \bar{x}_{..})] \bar{y}_j \\ + \hat{\beta}' S_{xy}$$

$$= \bar{y} G + P \sum_i (\bar{y}_i - \bar{y}_{..})^2 - \hat{\beta}' P \sum_i (\bar{x}_i - \bar{x}_{..})(\bar{y}_i - \bar{y}_{..}) +$$

$$P \sum_j (\bar{y}_j - \bar{y}_{..})^2 - \hat{\beta}' P \sum_j (\bar{y}_j - \bar{y}_{..})(\bar{x}_j - \bar{x}_{..}) + \hat{\beta} S_{xy}$$

$$= \bar{y} G + R_{yy} - \hat{\beta} R_{xy} + C_{yy} - \hat{\beta} C_{xy} + \hat{\beta} S_{xy}$$

$$= \bar{Y}_G + R_{yy} + C_{yy} + \cancel{T_{yy}} + \beta'(S_{xy} - R_{xy} - C_{xy})$$

E_y

$$= \bar{Y}_G + (R_{yy} + C_{yy} + \cancel{T_{yy}}) + \frac{T_{xy} + E_{xy}}{T_{xx} + E_{xx}} (T_{xy} + E_{xy})$$

Hence

$$= \bar{Y}_G + (R_{yy} + C_{yy} + \cancel{T_{yy}}) + \frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}} \quad \text{--- (B)}$$

Difference of eq (A) and eq (B) gives the variation due to treatment effect i.e.

$$SS_{\text{treatment}} = \text{eq (A)} - \text{eq (B)}$$

$$= \cancel{\bar{Y}_G} + \cancel{R_{yy}} + \cancel{C_{yy}} + \cancel{T_{yy}} + \frac{E_{xy}^2}{E_{xx}} - \cancel{\bar{Y}_G} - \cancel{R_{yy}} - \cancel{C_{yy}} - \frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}}$$

$$= \left\{ (T_{yy} + E_{yy}) - \frac{(T_{xy} + E_{xy})^2}{T_{xx} + E_{xx}} \right\} - \left\{ E_{yy} - \frac{E_{xy}^2}{E_{xx}} \right\}$$

$$= (T+E)_{yy} - E_{yy}$$

Now

$$SSE(\text{adj}) = \sum \sum y_{ij}^2 - \text{reg}(\mu, \alpha_i, \beta_j, w_i, B)$$

$$= \sum_{i=1}^p \sum_{j=1}^p y_{ij}^2 - \bar{Y}_G - R_{yy} - C_{yy} - T_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

$$= S_{yy} - R_{yy} - C_{yy} - T_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

$$E_{yy} = E_{yy} - \frac{E_{yx}^2}{E_{xx}}$$

Hence the F-statistic is

$$F = \frac{SS_{\text{treatment}} / p-1}{SSE / (p-1)(p-2) - 1}$$

$$F = \frac{\{(T+E)_{yy} - E_{yx}^2\} / p-1}{E_{yy} / (p-1)(p-2) - 1}$$

having an F-dist with $v_1 = p-1$
and $v_2 = (p-1)(p-2) - 1$ degree of
freedom.

Estimation Of Missing Observation

by Covariance:

The procedure which gives an unbiased estimate of a missing obs is as follows:

Step 1 :

Set $Y=0$ for missing obs.

Step 2 :

Define a covariate as $X=0$ for an observed Y and $X=+1$ or $X=-1$ for $Y=0$

Step 3 :

Carry out the analysis of covariance.

Step 4 :

Compute $b = \frac{E_{xy}}{E_{xx}}$ and

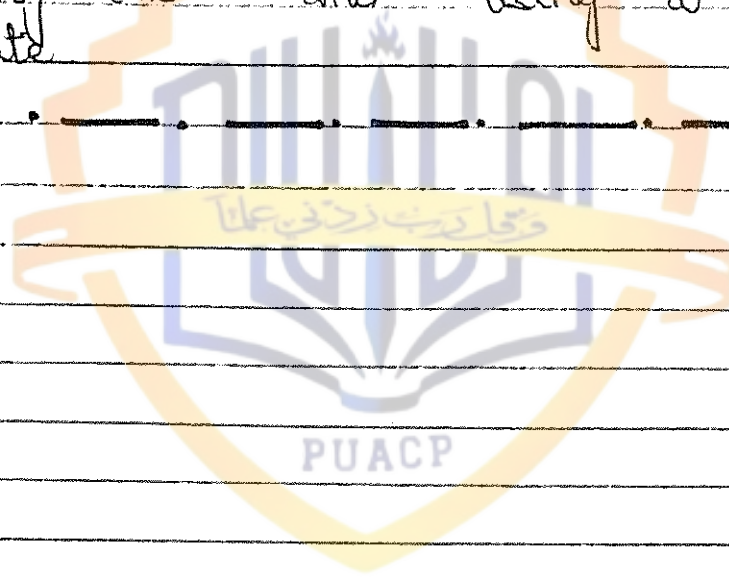
change sign to estimate the missing value i.e.

$$\text{Missing value} = -b = -\frac{E_{xy}}{E_{xx}}$$

The missing plot (value), $-b$ is essentially an adjustment to the so called obs $Y=0$ to give an

estimate of the Y that could have...
been obtained, if X had been 0
instead of 1.

On applications, the covariance method
for missing obs is convenient and
simple. The technique can be
extended to handle several missing
plots (values) by introducing new
independent variate for each
missing obs and using a multiple
variate.



Theorem:Proof:

In RCB design, the obs y_{ij} is expressed as

$$y_{ij} = \mu + \alpha_j + \beta_i + \epsilon_{ij}$$

where,

$\alpha_j = j^{\text{th}}$ treatment effect

$\beta_i = i^{\text{th}}$ block effect

$\mu = \text{overall mean}$

$y_{ij} = \text{obs of } i^{\text{th}} \text{ block and } j^{\text{th}} \text{ treatment}$

$\epsilon_{ij} \sim N(0, \sigma^2)$

let $\hat{\mu}$, $\hat{\alpha}_j$, $\hat{\beta}_i$ are the estimates of μ , α_j , β_i respectively. Then the error sum of squares is

$$S = \sum_{i=1}^r \sum_{j=1}^k (y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)^2$$

To find the values of estimates, differentiate it w.r.t unknown values and equating them to zero.

For $\hat{\mu}$:

$$\frac{\partial S}{\partial \hat{\mu}} = +2 \sum_i^r \sum_j^k (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i) (-1) = 0$$

$$\sum_i^r \sum_j^k Y_{ij} - r k \hat{\mu} - r \sum_j^k \hat{\alpha}_j - k \sum_i^r \hat{\beta}_i = 0 \quad \because \frac{\partial^2 S}{\partial \hat{\mu}^2} > 0$$

$$\sum_i^r \sum_j^k Y_{ij} = r k \hat{\mu} + r \sum_j^k \hat{\alpha}_j + k \sum_i^r \hat{\beta}_i$$

$$V_{..} = r k \hat{\mu} + r \sum_j^k \hat{\alpha}_j + k \sum_i^r \hat{\beta}_i \quad \text{--- (1)}$$

For $\hat{\alpha}_j$:

$$S = \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta}_i)^2 + \sum_i (Y_{i2} - \hat{\mu} - \hat{\alpha}_2 - \hat{\beta}_i)^2 + \dots +$$

$$+ \dots + \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i)^2$$

$$\frac{\partial S}{\partial \hat{\alpha}_1} = +2 \sum_i (Y_{i1} - \hat{\mu} - \hat{\alpha}_1 - \hat{\beta}_i) (-1) = 0$$

$$\frac{\partial S}{\partial \hat{\alpha}_2} = +2 \sum_i (Y_{i2} - \hat{\mu} - \hat{\alpha}_2 - \hat{\beta}_i) (-1) = 0$$

⋮

$$\frac{\partial S}{\partial \hat{\alpha}_j} = +2 \sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i) (-1) = 0$$

$$\because \frac{\partial^2 S}{\partial \hat{\alpha}_j^2} > 0$$

$$\sum_i (Y_{ij} - \hat{\mu} - \hat{\alpha}_j - \hat{\beta}_i) = 0$$

$$\sum_i Y_{ij} - r \hat{\mu} - r \hat{\alpha}_j - \sum_i \hat{\beta}_i = 0$$

$$\sum_i Y_{ij} = r \hat{\mu} + r \hat{\alpha}_j + \sum_i \hat{\beta}_i$$

$$\text{For } \hat{\beta}_i: y_{ij} = x_i \hat{\alpha} + x_j \hat{\alpha}_j + \sum_i \hat{\beta}_i \quad \text{--- (11)}$$

$$S = \sum_j (y_{ij} - \hat{\alpha} - \hat{\alpha}_j - \hat{\beta}_i)^2 + \dots + \sum_j (y_{ij} - \hat{\alpha} - \hat{\alpha}_j - \hat{\beta}_i)^2 + \dots$$

$$\frac{\partial S}{\partial \hat{\beta}_i} = -2 \sum_j (y_{ij} - \hat{\alpha} - \hat{\alpha}_j - \hat{\beta}_i) = 0$$

$$\frac{\partial S}{\partial \hat{\beta}_i} = -2 \sum_j (y_{ij} - \hat{\alpha} - \hat{\alpha}_j - \hat{\beta}_i) = 0$$

$$\sum_j (y_{ij} - \hat{\alpha} - \hat{\alpha}_j - \hat{\beta}_i) = 0$$

$$\sum_j y_{ij} = k \hat{\alpha} + \sum_j \hat{\alpha}_j + k \hat{\beta}_i$$

$$y_{ij} = k \hat{\alpha} + \sum_j \hat{\alpha}_j + k \hat{\beta}_i \quad \text{--- (11)}$$

Now, if we add 'k' equations in eq (11) and 'r' equations in eq (11), then we get the equations same as eq (1). This means that (1), (11), (11) are not linearly independent.

For unique solution, put the restriction as follows:

$$\sum_j \hat{\alpha}_j = 0$$

$$\sum_i \hat{\beta}_i = 0$$

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From eq (I):

$$Y_{..} = r_k \hat{u}$$

$$\hat{u} = \frac{Y_{..}}{r_k} = \bar{Y}_{..}$$

From eq (II):

$$Y_{.j} = r_k \hat{u} + r_j \hat{\alpha}_j$$

$$r_j \hat{\alpha}_j = Y_{.j} - r_k \hat{u}$$

$$\hat{\alpha}_j = \frac{Y_{.j} - r_k \hat{u}}{r_j} = \bar{Y}_{.j} - \bar{Y}_{..}$$

From eq (III):

$$Y_i = r_k \hat{u} + r_i \hat{\beta}_i$$

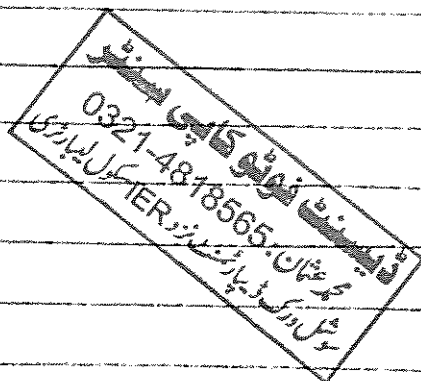
$$\hat{\beta}_i = \bar{Y}_i - \hat{u} = \bar{Y}_i - \bar{Y}_{..}$$

(Now, the variation due to fitting the full model)

To obtain regression SS, due to this model, consider error SS and substitute the best estimates

$$S = \sum_i \sum_j e_{ij}^2$$

$$S = \sum_i \sum_j \left(Y_{ij} - (\hat{u} + \hat{\alpha}_j + \hat{\beta}_i) \right)^2$$



$$S = \sum_i \sum_j y_{ij}^2 + \sum_i \sum_j (\hat{\alpha}_i + \hat{\alpha}_j + \hat{\beta}_i)^2 - 2 \sum_i \sum_j y_{ij} (\hat{\alpha}_i + \hat{\alpha}_j + \hat{\beta}_i)$$

$$S = \sum_i \sum_j y_{ij}^2 + \sum_i \sum_j (\hat{\alpha}_i^2 + \hat{\alpha}_j^2 + \hat{\beta}_i^2 + 2\hat{\alpha}_i \hat{\alpha}_j + 2\hat{\alpha}_i \hat{\beta}_i + 2\hat{\alpha}_j \hat{\beta}_i) - 2 \sum_i \sum_j y_{ij} (\hat{\alpha}_i + \hat{\alpha}_j + \hat{\beta}_i)$$

$$S = \sum_i \sum_j y_{ij}^2 + r k \hat{\alpha}^2 + r \sum_j \hat{\alpha}_j^2 + k \sum_i \hat{\beta}_i^2 + 2 \hat{\alpha} r \sum_j \hat{\alpha}_j + 2 \hat{\alpha} k \sum_i \hat{\beta}_i + 2 \sum_j \hat{\alpha}_j \sum_i \hat{\beta}_i - 2 \hat{\alpha} \sum_i \sum_j y_{ij} - 2 \sum_j \hat{\alpha}_j \sum_i y_{ij} - 2 \sum_i \sum_j \hat{\beta}_i y_{ij}$$

$$S = \sum_i \sum_j y_{ij}^2 + \hat{\alpha} (r k \hat{\alpha} + r \sum_j \hat{\alpha}_j + k \sum_i \hat{\beta}_i) + \sum_j \hat{\alpha}_j (r \hat{\alpha} + r \hat{\alpha}_j + \sum_i \hat{\beta}_i) + \sum_i \hat{\beta}_i (k \hat{\alpha} + k \hat{\beta}_i + \sum_j \hat{\alpha}_j) - 2 \hat{\alpha} \sum_i \sum_j y_{ij} - 2 \sum_j \hat{\alpha}_j \sum_i y_{ij} - 2 \sum_i \sum_j \hat{\beta}_i y_{ij}$$

By using the results of eq (i), (ii), (iii)

$$S = \sum_i \sum_j y_{ij}^2 + \hat{\alpha} Y_{..} + \sum_j \hat{\alpha}_j Y_{.j} + \sum_i \hat{\beta}_i Y_{i.} - 2 \hat{\alpha} Y_{..} - 2 \sum_j \hat{\alpha}_j Y_{.j} - 2 \sum_i \hat{\beta}_i Y_{i.}$$

$$S = \sum_i \sum_j y_{ij}^2 - \hat{\alpha} Y_{..} - \sum_j \hat{\alpha}_j Y_{.j} - \sum_i \hat{\beta}_i Y_{i.}$$

Now the regression SS is:

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Now the variation due to fitting the reduced model is

$$\text{reg}(u, B) = u'Y_0 + \sum_i B_i Y_i$$

$$= (\bar{Y}_0)Y_0 + \sum_i [(Y_i - \bar{Y}_0)]Y_i$$

$$SS_{\text{treatment}} = \text{reg}(u, \alpha, B) - \text{reg}(u, B)$$

$$= \cancel{Y_0'} + \sum_i \alpha_i Y_i + \sum_i B_i Y_i - \cancel{Y_0'} - \sum_i B_i Y_i$$

$$= \sum_i \alpha_i Y_i$$

$$= \sum_i (Y_i - \bar{Y}_0) Y_i$$

$$= \sum_i Y_i (Y_i - \bar{Y}_0) = \sum_i Y_i^2 - \bar{Y}_0 \sum_i Y_i$$

$$SS_{\text{treatment}} = \sum_i Y_i^2 - (Y_0)^2$$

ii) Assume that block effects are the insignificant model reduces to

$$Y_i = u + \alpha_i + \epsilon_i$$

The N. eqs are:

$$Y_0 = r k \alpha_0 + \sum_i \alpha_i \quad (6)$$

$$y_i = \mu_i + \alpha_i \quad (7)$$

if we add (k) eqs in (7) we get the same eq as (6) for unique selection put $\sum \alpha_i = 0$

From eq (6)

$$y_i = \mu_i + \alpha_i$$

From eq (7)

$$y_i = \mu_i + \alpha_i$$

$$\alpha_i = y_i - \mu_i$$

Now the variation due to giving the above reduced model is

$$reg(u, \alpha) = \mu_i y_i + \sum \alpha_i y_i$$

$$SS_{blocks} = reg(u, \alpha, \beta) = reg(u, \alpha)$$

$$= \mu_i y_i + \sum \alpha_i y_i + \sum \beta_i y_i - \mu_i y_i - \sum \alpha_i y_i$$

$$= \sum \beta_i y_i$$

$$= \sum (y_i - \mu_i) y_i$$

$$= \sum y_i \left(\frac{y_i}{K} - \mu_i \right) = \frac{1}{K} \sum y_i^2 - \mu_i \sum y_i$$

$$SS_{blocks} = \sum y_i^2 - \frac{1}{K} \left(\sum y_i \right)^2$$

